

2025 Washington Metropolitan Area Water Supply Study

Demand and Resource Availability Forecast for the Year 2050



DECEMBER 2025

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PREPARED BY S.N. AHMED, K. BENCALA, S. NUMMER, C.L. SCHULTZ, AND A. SECK
SECTION FOR COOPERATIVE WATER SUPPLY OPERATIONS ON THE POTOMAC

INTERSTATE COMMISSION ON THE POTOMAC RIVER BASIN
401 N. WASHINGTON ST. STE 300 • ROCKVILLE, MD 20850

**2025 WASHINGTON METROPOLITAN AREA WATER SUPPLY STUDY:
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Preface

This report was prepared by the Interstate Commission on the Potomac River Basin (ICPRB), Section for Cooperative Water Supply Operations on the Potomac (CO-OP) with funding provided by the three major Washington, DC, metropolitan area (WMA) water suppliers: Fairfax Water; the Washington Aqueduct Division of the U.S. Army Corps of Engineers; and WSSC Water.

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List of Abbreviations/Glossary

°C	Degrees Celsius
°F	Degrees Fahrenheit
ac-ft	Acre-feet
ACS	American Community Survey
AI	Artificial Intelligence
ANN	Artificial Neural Network
AQU	Aquaculture
APS	Alleghany Power Service
ARIMA	Autoregressive Integrated Moving Average
BG	Billion Gallons
BGD	Billion Gallons per Day
cfs	Cubic Feet per Second
CMIP5	Coupled Model Intercomparison Project Phase 5
CMIP6	Coupled Model Intercomparison Project Phase 6
CO-OP	Section for Cooperative Water Supply Operations on the Potomac
CO-OP Suppliers	Fairfax Water, WSSC Water, and Washington Aqueduct
COM	Commercial
COMAR	Code of Maryland Regulations
CU	Consumptive Use
DC	District of Columbia
DC Water	District of Columbia Water and Sewer Authority
DES	Department of Environmental Services
DOE	U.S. Department of Energy
DPW	Department of Public Works
DUR	Dwelling Unit Ratio
ECDF	Empirical distribution function
EMP	Employee(s)
ENERGY STAR	(Program name)
EPACT 1992	Energy Policy Act of 1992
EPRI	Electric Power Research Institute
EQAC	Environmental Quality Advisory Council
ExtendSim	ExtendSim simulation software (ANDRITZ)
FNN	Feedforward Neural Network
Fairfax Water	Fairfax County Water Authority
FY	Fiscal Year

GIS	Geographic Information System
gpd	Gallons per Day
GRU	Gated Recurrent Unit
HH	Occupied Households
HU	Housing Units
HYE	Hydroelectric
ICPRB	Interstate Commission on the Potomac River Basin
IND	Industrial
IRR	Irrigation
JLARC	Virginia Joint Legislative Audit and Review Commission
Km	Kilometer
LFAA	Low Flow Allocation Agreement
LIV	Livestock
LSTM	Long Short-Term Memory
LW	Loudoun Water
MAE	Mean Absolute Error
MDE	Maryland Department of the Environment
ME	Mean Error
MFH	Multi-Family Household
MG	Million Gallons
MGD	Million Gallons per Day
MG/yr	Million Gallons per Year
MGS	Maryland Geological Survey
MIN	Mining
mm/yr	Millimeters per Year
MSL	Mean Sea Level
MW	Megawatt
MWCOG	Metropolitan Washington Council of Governments
NASA	National Aeronautics and Space Administration
NDAA	National Defense Authorization Act
NEX-DCP30	NASA Earth Exchange Downscaled Climate Projections
NPS	National Park Service
NWIS	National Water Information System
PJM	PJM Interconnection
PP	Thermoelectric Power Generation
PRISM	Parameter-Elevation Relationships on Independent Slopes Model
PRRISM	Potomac Reservoir and River Simulation Model

PWD	Public Works Department
PWS	Public Water Supply
R^2	Coefficient of Determination
RMSE	Root Mean Square Error
SA	Service Area
SD	Standard Deviation
SE	Standard Error
SFH	Single-Family Household
SP1	Savings Projection 1
SP2	Savings Projection 2
SP3	Savings Projection 3
SSP	Shared Socioeconomic Pathway
SSD	Self-Supplied Domestic
TAZ	Traffic Analysis Zone
UOSA	Upper Occoquan Service Authority
USACE	U.S. Army Corps of Engineers
USDA	U.S. Department of Agriculture
USEPA	U.S. Environmental Protection Agency
USGS	U.S. Geological Survey
9VAC25-780	Virginia Administrative Code Title 9, Chapter 25-780
VADEQ	Virginia Department of Environmental Quality
VAWC	Virginia-American Water Company
WCU	Withdrawal and Consumptive Use
WCRP	World Climate Research Program
WFP	Water Filtration Plant
WMA	Washington Metropolitan Area
WMA Suppliers	CO-OP Suppliers, Loudoun Water, and the City of Rockville
WRF	Water Reclamation Facility
WRRF	Water Resource Recovery Facility
WSCA	Water Supply Coordination Agreement
WSSC Water	Washington Suburban Sanitary Commission
WTP	Water Treatment Plant
WTF	Water Treatment Facility
WWTP	Wastewater Treatment Plant
XGBoost	Extreme Gradient Boosting

Executive Summary

This is the eighth in a series of studies by the Interstate Commission on the Potomac River Basin (ICPRB) Section for Cooperative Water Supply Operations on the Potomac (CO-OP) to evaluate the adequacy of the Washington, DC, metropolitan area (WMA) water supplies to meet expected water demands over the coming decades. The WMA has a unique, cooperative water supply system that was established more than 40 years ago by a set of regional agreements signed by the Fairfax County Water Authority (Fairfax Water), the Washington Suburban Sanitary Commission (WSSC Water), the Washington Aqueduct Division of the U.S. Army Corps of Engineers (Washington Aqueduct), the District of Columbia, and CO-OP. One of the requirements of the regional agreements is that every five years, such an evaluation be conducted. The current study's planning horizon is 2050. It provides decision-makers with the following:

- Forecasts of water demands for the WMA, taking into account projected demographic and societal changes that may affect future water use
- Forecasts of water availability, considering the potential impact of changes in meteorological conditions and upstream consumptive use of water on system resources
- An evaluation of the ability of existing and planned system resources to meet the forecasted demands

In addition to these core analyses, the study addresses new drivers and tools shaping the region's water future. It considers the role of emerging technologies through two lenses: the impact of new infrastructure, such as data centers supporting Artificial Intelligence (AI) and cloud services, on regional water use; and the use of AI to enhance water demand forecasting.

The current study extrapolates its forecasts through 2060, consistent with requirements of Virginia's Local and Regional Water Supply Planning regulation (9VAC25-780), and through 2085, to support an ongoing feasibility study on a "secondary source backup water supply" for the region initiated by the U.S. Army Corps of Engineers (USACE) in September 2024.

Forecasts of conditions decades in the future are beset with tremendous uncertainty. In this study, a scenario planning approach is used to investigate the future performance of the WMA water supply system in the face of such uncertainty. Three scenarios are considered for future WMA water demands and three scenarios for the impact of future meteorological change, that is, the impacts of changing temperature and precipitation on regional stream flows and demands, resulting in a combined total of nine scenarios. The evaluation of system performance is conducted with CO-OP's planning model, the Potomac Reservoir and River Simulation Model (PRRISM), for each of the future condition scenarios for two future years, 2045 and 2050, as well as for a near term year, 2030.

A map showing the existing and planned resources of the WMA system is given in Figure ES-1. The region's primary source of water is the Potomac River. The Potomac River watershed upstream of [Little Falls Dam](#) provides approximately 75 percent of the WMA's water, with the remainder supplied by off-Potomac reservoirs in the Occoquan and Patuxent watersheds. Two planned resources are also shown in the figure: Vulcan Quarry, assumed to be operational in 2040 (Fairfax Water's future Edgemon

Reservoir), and Luck Stone Quarry A, assumed to be operational in 2028 (Loudoun Water's future Milestone Reservoir). They are both existing hard rock quarries for which agreements and/or contracts are in place which pave the way for future conversion to raw (untreated) water storage reservoirs. Also shown is Beaverdam Reservoir, which is currently being used as an emergency supply by Loudoun Water.

Figure ES-1 also shows the WMA water service areas, that is, the District of Columbia and the portions of the Maryland and Virginia suburbs that are supplied water, either directly or indirectly, by Fairfax Water, Washington Aqueduct, and WSSC Water (known collectively as the CO-OP suppliers) and by Loudoun Water and the City of Rockville Department of Public Works (DPW).

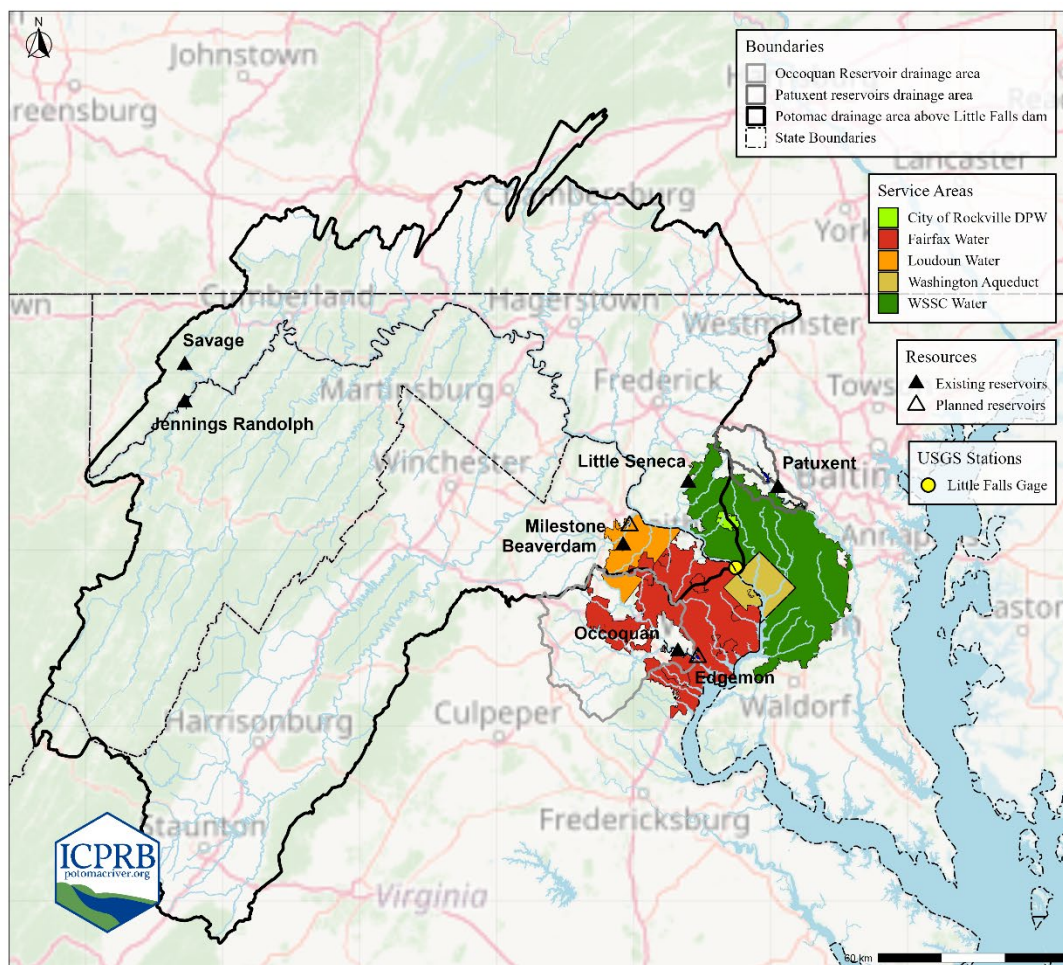


Figure ES-1: WMA supplier service areas and existing and planned water supply resources.

FORECASTS OF WATER DEMAND

Water production by the CO-OP suppliers has remained remarkably stable over the past several decades despite significant population growth in the WMA. Figure ES-2 shows annual, winter, summer, and peak day of the year water production by the CO-OP suppliers from 1990 to 2023 in units of million gallons per day (MGD). The WMA's population grew from 3.4 million to 5.0 million over this period, a 47 percent increase. While changes in individual supplier production are evident in the figure, total WMA

production, equivalent to total demand for the region, exhibits no clear trend in annual, summer, or peak-day demands. This lack of growth in demand, consistent with patterns observed in many other regions of the United States, is due to continued adoption of water saving plumbing fixtures and household appliances, among other factors. This stability in WMA water demand has benefited the region by reducing the need for costly new water supply infrastructure.

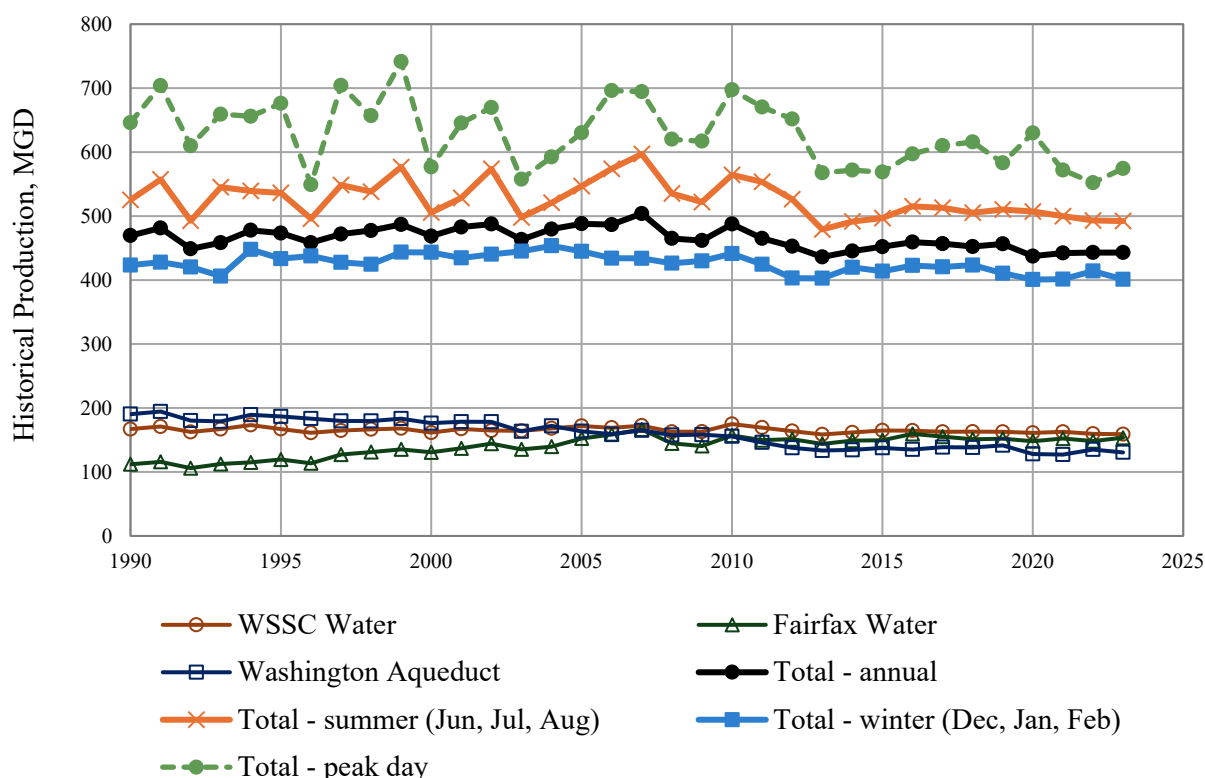


Figure ES-2: CO-OP supplier historical annual production, and combined total annual, summer, winter (by water year), and annual peak-day production.

Forecasts of average annual WMA water demand are developed with the same methodology used in past CO-OP water supply studies: by combining recent water use information derived from either billing records or withdrawal reports provided by the WMA suppliers (CO-OP suppliers, Loudoun Water, and City of Rockville) and their wholesale customers with the most recent demographic forecasts. The demand forecasts include estimates of water losses derived from recent data on unmetered and unaccounted-for water use, and adjustments are made to the forecasts to represent the impact of continued adoption of water-saving fixtures and appliances. Water use data are disaggregated into three categories for forecasting purposes: single family households, multi-family households (apartments and condominiums), and employees (including commercial, industrial, and institutional use). The Metropolitan Washington Council of Governments (MWCOC) forecasts that WMA population will grow from 5.1 million in 2023 to 6.1 million in 2050, a 20 percent increase. Average annual demand in the WMA, including Rockville, was 459 MGD in 2023 and is forecasted to increase to 538 MGD by 2050, a 17 percent increase. The estimated uncertainty in the demand forecast for 2050 (one standard error) is ± 10.4 percent.

Figure ES-3 shows forecasted WMA demands compared with results from past studies by ICPRB (Kame'enui *et al.*, 2005; Hagen and Steiner, 2000; Mullusky *et al.*, 1996; Holmes and Steiner, 1990; Ahmed *et al.*, 2010; Ahmed *et al.*, 2015; Ahmed *et al.*, 2020), by the USACE (1963; 1983), and by MWWCOG (as reported in USACE, 1975). WMA demand forecasts have tended to fall over time due to reductions in per household and per employee use. Past studies by ICPRB have consistently over-predicted overall water demand. The current study's forecasts are slightly higher than those of CO-OP's 2020 study, but the difference is small compared to the range of uncertainty.

In addition to long-term annual forecasts, this study explores methods for more accurately predicting daily demands. Several machine learning models, including neural networks and tree-based ensembles, were evaluated for forecasting daily water use. While the current linear regression approach used in PRRISM remains effective for capturing broad patterns, the machine learning models demonstrated improved accuracy in capturing nonlinear behavior, peak demands, and weather sensitivity. These models may be considered for future planning and operational applications, particularly for enhancing accuracy during high-demand periods and extreme weather events.

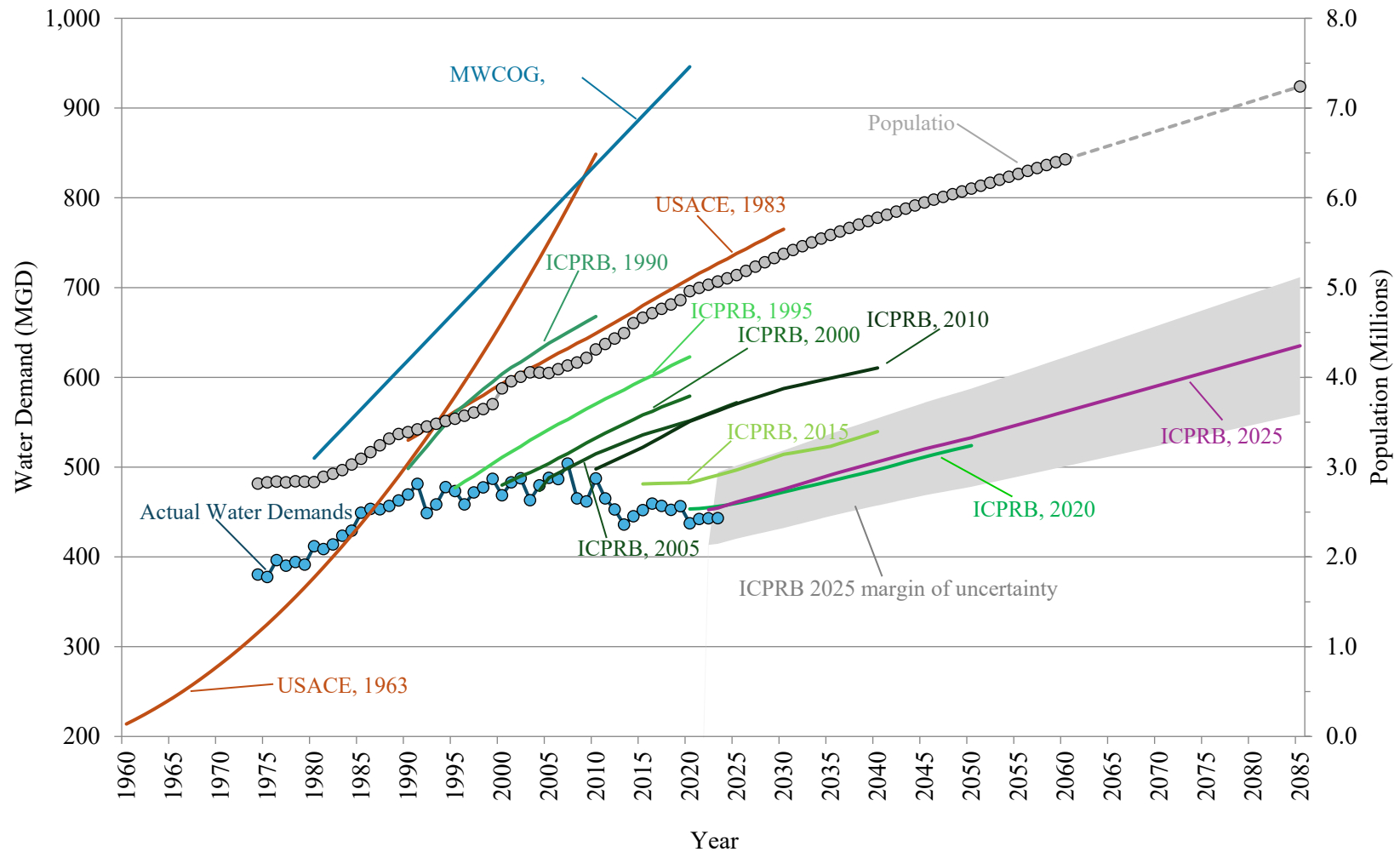


Figure ES-3: Comparison of this study's forecast of WMA total demand and range of uncertainty (grey shaded area) with forecasts from past studies (Rockville not included).

FORECASTS OF WATER AVAILABILITY

This study's forecasts of water availability take into account two factors that will have impacts on future system resources: changes in meteorological conditions, that is, precipitation and temperature, and increases in upstream consumptive use of water. Average precipitation and average temperature are both projected to rise in the Potomac basin, and significant changes are now evident in the observational record. These trends will have competing effects on stream flow, with increasing precipitation tending to increase stream flow, but higher temperatures tending to increase evaporation rates and thus reduce stream flow. Upstream consumptive use of water, that is, the portion of water that is withdrawn from the watershed's streams and groundwater aquifers and not returned, is also projected to rise, thus reducing future water availability at WMA water supply intakes.

Projected changes in meteorological conditions used in this study, The National Aeronautics and Space Administration's (NASA's) NEX-DCP30-CMIP6 ensemble, are based on the most recent compilation of global model results from the Coupled Model Intercomparison Project Phase 6 (CMIP6). Projections for the Potomac watershed call for long-term mean precipitation to rise by 10.2 percent in 2040-2069 compared to means in the historic baseline period (1950-1979) and by 13.1 percent in 2070-2099. Projections for long-term mean temperature are 2.9 °C (5.2 °F) higher for 2040-2069 than historic means and 4.0 °C (7.2 °F) higher for 2070-2099.

A major uncertainty in many regions, including the Potomac, is how stream flow will respond to increases in temperature. To represent this uncertainty, future trends in annual flows are investigated for three scenarios that reflect a range of sensitivities of flow to temperature. Results are shown in the box to the right. In the Lower Flows Scenario, the response of flows to rising temperatures is significant, resulting in a drying effect that outweighs the projected future increases in precipitation. In the Higher Flows Scenario, the response to rising temperatures is minimal, with rising precipitation dominating changes in flow. Each flow scenario is constructed from changes in long-term flow statistics computed over 30-year periods, as compared to values in the historic baseline period, 1950-1979. The use of 30-year periods is necessitated by the need for samples of ample size to compute statistics characterizing extreme years. In each scenario, changes in annual flows for extreme wet years are projected to be very different than those for extreme dry years, indicating that Potomac River flows in future extreme drought years will likely be lower than those experienced in the past even in a future where average flows rise. In this study, extreme dry years refer to years in which annual stream flow has a 99 percent or greater probability of being exceeded (percentile value ≤ 1). It should be noted that the Potomac River's drought of record, which occurred in

HOW WILL POTOMAC BASIN STREAM FLOWS RESPOND TO RISING TEMPERATURES IN 2050?

Lower Flows Scenario – significant response:

- 32% or greater decrease in extreme dry years
- 3% decrease in average years
- 3% or greater increase in extreme wet years

Medium Flows Scenario – moderate response:

- 13% or greater decrease in extreme dry years
- 4% increase in average years
- 7% or greater increase in extreme wet years

Higher Flows Scenario – minimal response:

- 7-9% increase in extreme dry years
- 12% increase in average years
- 11% or greater increase in extreme wet years

1930, has a percentile value of approximately 0.8, based on the 123-year historical record used for analyses in the current study.

New estimates of current and future consumptive water use upstream of WMA water supply intakes were developed for this study. These forecasts quantify the watershed-scale cumulative impact of upstream consumptive use on river flows and provide essential information on future water availability for the WMA water supply system. Analyses are based on data from Potomac basin state water supply agencies, the U.S. Geological Survey (USGS), and the U.S. Department of Agriculture (USDA). Water use categories considered were aquaculture, commercial, hydroelectric, industrial, irrigation, livestock, thermoelectric power generation, public water supply, and self-supplied domestic. A special separate analysis is conducted to provide estimates for a relatively new category: water use by data centers. The estimate for current annual upstream consumptive use is approximately 100 MGD, with a minimal contribution from data centers (less than 0.1 percent). Forecasted total annual average upstream consumptive use in 2050 is 117 MGD, including 4.7 MGD from data centers (based on this study's Medium Water Use scenario for data centers), which is about four percent of the total. While modest in absolute terms, projected data center consumptive use is comparable to the current upstream consumptive use shares of several established sectors, including commercial, thermoelectric, and public water supply. Data center water use estimates are based on grid-connected energy forecasts and reflect only data centers located upstream of the WMA. As the sector continues to expand, ongoing tracking of data center development, technology trends, and regulatory changes will be essential for future planning.

EVALUATIONS OF SYSTEM PERFORMANCE

CO-OP's daily system simulation model, PRRISM, is used to evaluate the performance of the WMA cooperative water supply system in two future years, 2045 and 2050, and a near-term year, 2030. The evaluations are conducted under nine scenarios that represent ranges of uncertainties in future WMA water demand and of the impact of meteorological change on water availability in the Potomac basin. The nine scenarios are formed by combining three demand scenarios, based on this study's forecasts for total WMA annual demand plus or minus the estimated standard errors, and three stream flow scenarios, described above. For comparison purposes, evaluations are also conducted using historical hydrometeorological conditions with no alterations. Qualitative results of the evaluations are given below in Table ES-1, where the colors indicate system performance during an extreme drought, comparable to the 1930 drought of record as altered by projected impacts of future meteorological conditions. GREEN denotes reliable performance; YELLOW denotes marginal performance, that is, with some emergency water use restrictions, a small chance of a shortage of water averaging one MGD or less, or both; and RED denotes system failure, that is, the inability of the system to meet combined WMA water supply needs and the 100 MGD flow-by requirement at Little Falls Dam. Quantitative results are given in Chapter 8 and Appendix A.4.

For a given scenario year, results in Table ES-1 range from complete system reliability with ample water to spare in Little Seneca and Jennings Randolph reservoirs, to system failure with many days or months when flow in the Potomac River falls far short of meeting the region's water needs, that is, combined WMA demands and the 100 MGD environmental flow-by at Little Falls Dam. For all scenario years in the Higher Flows and Lower Demands scenario, there is ample combined water supply storage in Little Seneca and Jennings Randolph even on the worst day of an extreme drought. But these reservoirs are

predicted to be empty on at least one day of an extreme drought for all scenario years in all the Lower Flows scenarios, and for some of the Medium Flows scenarios as well.

An unexpected outcome of this study is the indication of vulnerability of the WMA water supply system in the near term, as indicated by results in Table ES-1 for the 2030 scenario year. **Deficits are predicted in the event of extreme drought in 2030 for the Medium Flows and High Demands scenario and for all the Lower Flows scenarios. In four out of nine of the modeled 2030 scenarios, combined water supply storage in Little Seneca and Jennings Randolph reservoirs is predicted to fall to zero in the event of extreme drought, which would likely trigger emergency water allocations under the Low Flow Allocation Agreement.** The CO-OP suppliers entered into the Water Supply Coordination Agreement with the expressed intent to "reduce or eliminate the possibility that the Emergency Stage of the Low Flow Allocation Agreement will ever be reached," indicating that measures should be investigated in the short term to enhance the current system. A set of non-structural measures that increase system safe yield was evaluated in the 2017 water supply alternatives study (Schultz *et al.*, 2017). These may be further evaluated and refined in the ongoing USACE feasibility study or in a future CO-OP study to assess what set of alternatives can be implemented to address near-term vulnerabilities.

Table ES-1: PRRISM results reflecting WMA cooperative water supply system performance during extreme drought comparable to conditions during the drought of record (GREEN - reliable, YELLOW - marginal, RED - system failure).

	Based on historic record as altered by projected changes in hydrometeorological conditions									Based on unaltered historic hydrometeorological record		
	Higher Flows (Low sensitivity of flows to temperature)			Medium Flows (Medium sensitivity of flows to temperature)			Lower Flows (High sensitivity of flows to temperature)					
	Low Demands	Medium Demands	High Demands	Low Demands	Medium Demands	High Demands	Low Demands	Medium Demands	High Demands	Low Demands	Medium Demands	High Demands
2030												
2045												
2050												

1 Introduction

This report is the eighth in a series of regional studies led by the Interstate Commission on the Potomac River Basin (ICPRB) to help ensure a reliable water supply for the Washington, DC, metropolitan area (WMA). These studies are conducted by ICPRB's Section for Cooperative Water Supply Operations on the Potomac (CO-OP) on behalf of the area's major water suppliers.

The WMA, as defined for this study, includes the District of Columbia and parts of Maryland and Virginia that receive drinking water, either directly or indirectly, from Fairfax Water, WSSC Water, the Washington Aqueduct, and Loudoun Water. The WMA also includes the City of Rockville. The Potomac River is the primary source of the WMA's water supply, with several local reservoirs serving as supplemental secondary sources.

This study explores how the region's water needs are expected to change in the coming decades and evaluates whether the current cooperative water supply system can meet those needs under a range of future conditions. It includes forecasts of future water demand, projections of water availability from the Potomac River and local reservoirs taking into account the growth in upstream water use and the effects of rising temperatures, and an assessment of the system's ability to maintain supply and an environmental flow-by during future droughts (MDNR, 1981; Cummins *et al.*, 2010).

The study fulfills requirements of two regional agreements: the Low Flow Allocation Agreement (LFAA) and the Water Supply Coordination Agreement (WSCA), both of which require that 20-year forecasts of water demands and availability be conducted every five years. The current study provides forecasts through 2050 and extrapolations of results through 2085 to support long-term planning and regulatory needs.

As the eighth regional planning study conducted since 1990, this update builds on a long-standing collaborative process. It provides the region's major water suppliers with updated tools and insights to guide coordinated decision-making. The results will help utilities, planners, and emergency managers work together to ensure a safe and reliable water supply even during droughts and other extreme events.

The Interstate Commission on the Potomac River Basin (ICPRB) conducts studies to help ensure a reliable water supply for the Washington, DC, metropolitan area (WMA).

The study helps answer three key questions:



How much water will the region need in the future?

We project future demand through 2050 and beyond, based on expected population growth, changes in water use habits, and expected conservation trends.



How much water will be available in the future?

We assess how much water will likely be available from the Potomac River and current and planned reservoirs under a range of possible future conditions.



Will the water supply system be able to meet future demand?

We use a computer model to test whether the WMA cooperative water supply system will be able to meet water needs during future droughts.

The results will help the WMA continue to have a safe and reliable water supply.

1.1 Study Overview

The CO-OP Section of ICPRB conducts regular studies to ensure that the Washington, DC, metropolitan area will have enough water to meet future needs. These studies, which began in 1990 and are updated every five years, help water suppliers prepare for population growth, changing water use patterns, and potential changes in future stream flow. They also provide an opportunity to improve forecasts and tools as new data becomes available.

This 2025 study includes several important updates:

- Updated annual demand forecasts, incorporating a range of uncertainty based on previous study's statistical analyses (Chapter 3)
- Updated daily demand variation models, and an evaluation of the use of machine learning models in capturing daily use patterns (Chapter 4)
- Updated estimates of upstream consumptive use, including the projected impacts of data centers, (Chapter 6)
- Updated scenarios for future changes in Potomac River flows and reservoir inflows due to shifting rainfall and temperature patterns (Chapter 7)
- Use of a newly constructed version of CO-OP's planning model, the Potomac Reservoir and River Simulation Model (PRRISM) to conduct the resource analysis (Chapters 5 and 8)

Forecasts of future annual water demands are developed using recent customer billing records or withdrawal data, delineations of current and projected service areas, and the latest household and employment forecasts from the Metropolitan Washington Council of Governments (MWCOC). These forecasts extend to 2050 and are extrapolated to 2085, as described in Chapter 3.

Daily and seasonal water use patterns, such as changes in summertime demands and differences in weekday versus weekend use, are analyzed using historical daily production records and weather data. Chapter 4 explains how this variation is modeled in PRRISM to simulate the timing and magnitude of daily withdrawals from the Potomac River and other sources.

The study provides new estimates of upstream consumptive use of water, which reduces water availability at downstream WMA intakes. The upstream consumptive use estimates include the impacts of data centers (Chapter 6).

Future water availability is estimated using an approach which combines projections of regional precipitation and temperature from the Coupled Model Intercomparison Project Phase 6 (CMIP6) and statistical analyses to compute flow-dependent changes in future annual river flow (Chapter 7). Projections of meteorological conditions are also used to adjust PRRISM temperature and precipitation inputs that influence daily water use patterns in simulations of future water demands.

Results on how well the regional system, including both current resources and planned projects, can meet future water demands under a wide range of possible scenarios are presented in Chapter 8. This assessment is conducted using PRRISM, which simulates daily river flow, water demands, and reservoir

operations and incorporates simulations of real-world operating rules and of the river flow forecasts used by system managers during droughts to guide releases of water from system reservoirs.

By updating this assessment every five years, regional water managers can incorporate the latest data and evolving conditions into long-term planning. These regular updates support informed decisions and timely investments that help ensure the region's water supply system remains reliable, now and in the future (Hagen *et al.*, 2005). The current study builds on a foundation of past studies by ICPRB (Ahmed *et al.*, 2020; Ahmed *et al.*, 2015; Ahmed *et al.*, 2010; Kame'enuei *et al.*, 2005; Hagen & Steiner, 2000; Mullusky *et al.*, 1996; Holmes & Steiner, 1990), as well as earlier efforts by the U.S. Army Corps of Engineers (USACE) (1963, 1983) and MWCOG (as reported in USACE, 1975).

1.2 Water Suppliers

The Potomac River is the main source of drinking water for the Washington, DC, metropolitan area. Five public water utilities withdraw and treat water from the river to serve residents and businesses across the region:

- **Washington Aqueduct** supplies water to the District of Columbia (through DC Water), Arlington County, and to parts of Fairfax Water's service area.
- **WSSC Water** serves Montgomery and Prince George's counties in Maryland. It also provides limited or emergency water supply to Howard and Charles counties, the City of Rockville, and DC Water.
- **Fairfax Water** provides water to the City of Fairfax, the City of Falls Church, and Fairfax County, Virginia, and to many surrounding areas in northern Virginia through wholesale contracts.
- **City of Rockville**, located in Montgomery County, Maryland, operates its own water treatment plant and distribution system.
- **Loudoun Water** supplies customers in Loudoun County, Virginia, using a combination of Potomac River water treated at its Trap Rock Water Treatment Facility and treated water purchased from Fairfax Water.

Together, these five utilities rely on the Potomac River for about 75 percent of the region's water supply. The remaining 25 percent comes from reservoirs outside of the Potomac basin: the Occoquan Reservoir (used by Fairfax Water) and the Patuxent reservoirs (used by WSSC Water).

Three of the utilities, Washington Aqueduct, WSSC Water, and Fairfax Water ("CO-OP suppliers"), participate in a cooperative water supply system established by a set of agreements signed in 1982, which include the Water Supply Coordination Agreement and cost-share agreements for three upstream reservoirs: Jennings Randolph, Savage, and Little Seneca. These reservoirs can be used to augment Potomac River flow during droughts.

1.3 History of Cooperation

The Washington, DC, area began seriously planning for its long-term water supply in the 1960s, when rapid population growth and recurring droughts raised concerns about having enough water to meet future

needs. At the time, forecasts projected that the region's population could reach five million by 1985, which is a major jump from two million in 1960. Meanwhile, drought conditions were already straining the region's ability to provide a reliable supply of drinking water.

To prepare for the future, the USACE evaluated options for increasing water supply. In a 1963 study (USACE, 1963), the USACE proposed building 16 major new dams in the Potomac River watershed upstream of Washington, DC, to store water for use during dry periods. However, many of these proposals faced strong public opposition. Of all the 1963 study's proposed sites, only Jennings Randolph Reservoir, located in the northwestern portion of the watershed, was eventually built. Other ideas, like inter-basin water transfers and estuary treatment plants, were considered but not pursued.

In the 1970s and early 1980s, the region's actual population growth was slower than expected. However, water demands still regularly exceeded what the Potomac River had provided during the record Potomac River low flow year of 1966. Even though extreme shortages were avoided, it became clear that a new approach was needed that didn't rely on building a large number of new dams.

By the late 1970s, researchers at Johns Hopkins University had proposed a new strategy: instead of expanding infrastructure, water suppliers could coordinate operations to make better use of existing and planned resources. The key was to rely more on the Potomac River during wetter months and conserve reservoir storage for the dry summer season. This approach was both more cost-effective and less environmentally disruptive than constructing new reservoirs.

With this strategy in mind, the region's water suppliers and governments, with assistance from the ICPRB CO-OP Section, developed a cooperative approach through the Washington Metropolitan Area Water Supply Task Force (Sheer, 1983). The addition to regional resources of a small local reservoir, Little Seneca, was shown to provide needed operational flexibility. A set of landmark agreements were developed and signed to establish a regional cooperative water supply system:

- In 1978, the LFAA was signed by the United States, Maryland, Virginia, the District of Columbia, Fairfax Water, and WSSC Water. It established how water from the Potomac would be allocated when river flow is low, taking into account an environmental flow-by at [Little Falls Dam](#).
- In 1982, the WSCA was signed by the District of Columbia, Fairfax Water, WSSC Water, Washington Aqueduct, and CO-OP. It requires the water suppliers to coordinate their operations during droughts, with managerial and technical support provided by CO-OP.
- Also in 1982, a set of cost-share agreements were signed to fund the construction and/or operations and maintenance of three shared upstream reservoirs: Jennings Randolph, Savage, and Little Seneca.

ICPRB and its CO-OP Section were instrumental in developing these cooperative agreements. Under the WSCA, CO-OP coordinates regional water withdrawals and directs the amount and duration of water supply releases from shared upstream reservoirs (Jennings Randolph, Little Seneca, and Savage) when needed. These three reservoirs together contain about 19.6 billion gallons of storage that can be released to augment Potomac River flow during times of drought to benefit WMA water supply. Since the cooperative system was established, water supply releases from the upstream reservoirs have been needed

three times: in 1999, 2002, and 2010. In each instance, the coordinated drought response worked as planned, ensuring that water continued to flow to the millions of people in the region. The Washington area's cooperative approach to water supply is recognized as an example of how planning, coordination, and shared investment can help areas manage regional water resources efficiently, even during times of stress.

1.4 Planned and Potential New Resources

This study's water availability forecasts take into account two new reservoirs that WMA suppliers will have in place before the start of this study's planning horizon, 2050. Both planned reservoirs will make use of retired quarries. Fairfax Water has an agreement in place to convert a quarry currently being mined by the Vulcan Materials Company into a raw water storage facility which will be named the Steven T. Edgemon Reservoir. Edgemon Reservoir, Phase 1, is scheduled to be operational by 2040. Loudoun Water is in the process of converting a quarry that was acquired from the Luck Stone Company into a raw water storage facility named Milestone Reservoir, scheduled to be completed by 2028.

The WMA is highly reliant on the Potomac River as its major source of water supply. An effort is underway by regional stakeholders, including Washington Aqueduct, WSSC Water, Fairfax Water, DC Water, Arlington Department of Environmental Services (DES), ICPRB, and MWCOG, to put into place additional resources to enhance the resiliency of the WMA water supply system in the face of dual risks: spills of hazardous materials into the region's waterways and extreme drought. A spill of a toxic contaminant or an unprecedented drought due to rising temperatures could leave the region with a shortage of water. A feasibility study by the USACE was initiated in September 2024 to identify "a secondary water source and additional water storage capability" for the WMA. This study was authorized by Congress in the Water Resources Development Act of 2022 (NDAA, 2023). The planning horizon of the USACE feasibility study is approximately 2085; therefore, this study's annual demand, upstream consumptive use, and stream flow forecasts have been extrapolated to 2085 to support this effort.

Because the USACE is conducting an independent analysis of future supplemental sources, CO-OP's current study is limited to an evaluation of the adequacy of the region's existing resources and those whose development is already underway. Its findings will help define the timing, scale, and reliability needs that any future resource, such as those being considered under the USACE study, will need to meet.

2 System Overview

2.1 Introduction

Chapter 2 provides an overview of the WMA water supply system, highlighting its critical role in supporting residents, businesses, and institutions. The system relies on key water resources, including the Potomac River, and as well as on the coordinated efforts of multiple entities that withdraw, treat, and distribute water to meet regional demands. Figure 2-1 maps the service areas of WMA suppliers and details existing and planned system resources. The map also shows the U.S. Geological Survey (USGS) [stream gage at Little Falls Dam near Washington, DC](#), which monitors Potomac River flow just downstream of the WMA water supply intakes. During droughts, ICPRB staff members work to optimize system resources, meet customer water demands, and maintain Potomac River flow at Little Falls Dam above the environmental flow-by requirement of 100 million gallons per day (MGD), or 155 cubic feet per second (cfs) as a daily average. (MDNR, 1981; Cummins *et al.*, 2010).

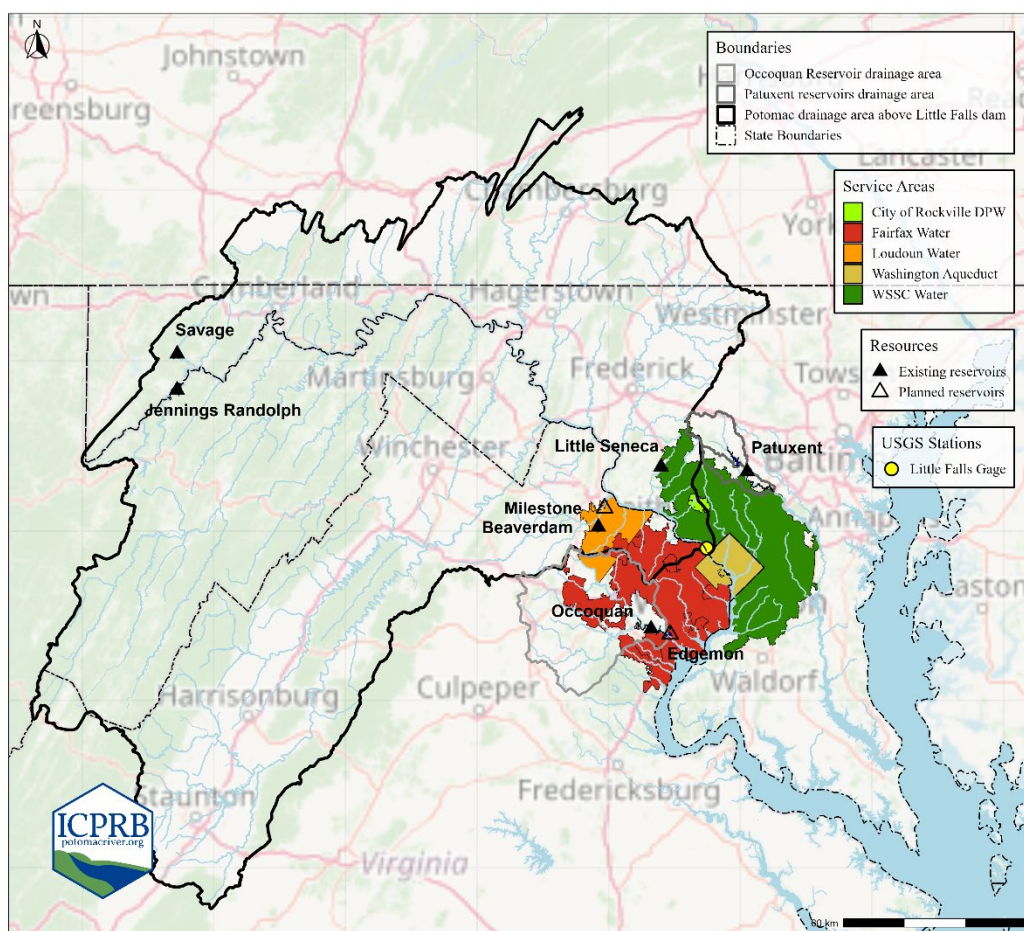


Figure 2-1: WMA supplier service areas and existing and planned water supply resources.

2.2 Customers and Demand

2.2.1 Service Areas

The WMA water supply system serves over five million residents, with suppliers withdrawing and treating water from the Potomac River and off-Potomac reservoirs before distributing it to retail and wholesale customers. Figure 2-2 shows the supplier service areas. Each supplier provided existing and anticipated (2050) service area boundaries for this study. Most reported no expected expansion by 2050, though new developments may occur within existing boundaries. This study therefore assumes that service area boundaries remain unchanged, which is important for estimating households, employees, and population within each service area, as described in Chapter 3.

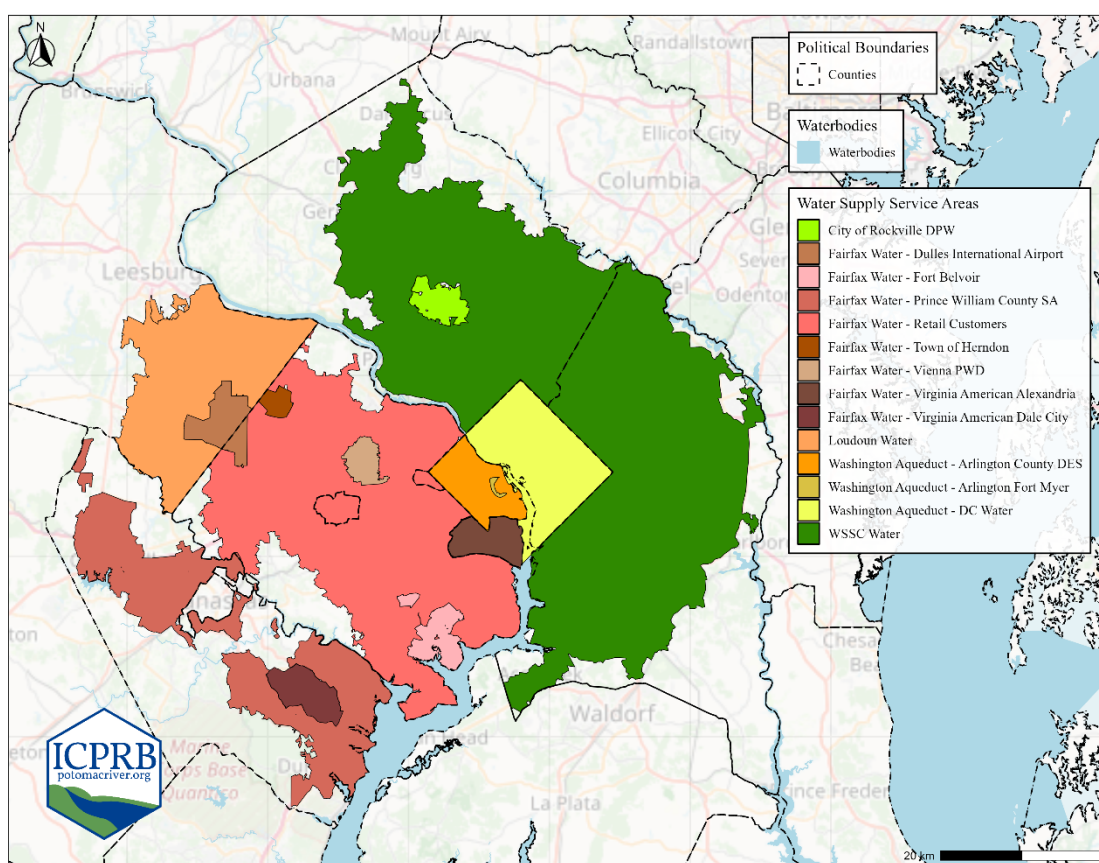


Figure 2-2: Water supplier service areas in the WMA, existing and anticipated as of 2025.

Fairfax Water, Virginia's largest water supplier, serves nearly two million residents in Fairfax County, the City of Fairfax, and the City of Falls Church, with wholesale service to Loudoun Water (Loudoun County), Prince William Water (formerly Prince William County Service Authority), Virginia American Water Company (Alexandria and Dale City), Vienna Department of Public Works (Vienna PWD), the Town of Herndon, Fort Belvoir, and the Washington Metropolitan Airports Authority at Dulles Airport.

The Washington Aqueduct, the District of Columbia's first public water supply system and the only water supply system in the nation currently operated by the U.S. Army Corps of Engineers, supplies treated water to DC Water, Arlington County, and Fairfax Water. Together, these wholesale customers provide water to nearly one million residents in the District of Columbia, Arlington County (including Fort Myer), and parts of Fairfax Water's service area.

On January 3, 2014, Fairfax Water acquired the City of Falls Church and City of Fairfax water systems. This transfer incorporated the former Falls Church system into Fairfax Water's retail service area, making Fairfax Water a wholesale customer of the Washington Aqueduct.

WSSC Water, the eighth largest combined water and wastewater utility in the United States, serves nearly two million residents in Montgomery and Prince George's counties, with additional limited or emergency service to Rockville, Charles County, Howard County, and the City of Bowie.

The City of Rockville owns and operates its water supply system, withdrawing water from the Potomac River at an intake located just downstream of WSSC Water's intake.

Loudoun Water, the newest water supplier in the WMA, has long operated its own water system but historically relied heavily on Fairfax Water. With the completion of its Potomac River intake and treatment plant, it now produces a portion of its own supply to meet the county's rapidly growing demand.

The population served by WMA water suppliers is projected to grow from nearly five million in 2020 to over six million by 2050, approximately a 20 percent increase. Fairfax Water, WSSC Water, and Washington Aqueduct collectively serve most of the region. Loudoun Water is expected to experience significant growth, with its population rising from approximately 302,624 in 2020 to approximately 399,338 by 2050, reflecting Loudoun County's rapid development. These trends highlight the need for strategic resource planning and collaboration among WMA suppliers to ensure reliable service for the growing population. These numbers are from the MWCOC Round 10.0 demographic projections (DesJardin and Goodwin, 2023), and their implications for water demand are discussed in Chapter 3.

2.2.2 Historical Water Production Trends

Trends by Water Supplier and Region: Water production by CO-OP suppliers has remained stable over the past several decades despite significant population growth in the WMA. Figure 2-3 presents annual average production for Washington Aqueduct, Fairfax Water, and WSSC Water, along with seasonal and peak-day trends from 1990-2023. Between 1990 and 2023, WMA's population grew from 3.4 million to 5.0 million, a 47 percent increase. Although production levels for individual suppliers vary from year to year, total WMA water production, which represents overall regional demand, has remained relatively stable with no clear long-term trend in annual, summer, or peak-day values.

Over the last five years of available data (2019-2023), CO-OP suppliers collectively produced an average of 445 MGD, distributed as follows:

- Washington Aqueduct: 133 MGD (30 percent of total)
- Fairfax Water: 151 MGD (34 percent of total)
- WSSC Water: 161 MGD (36 percent of total)

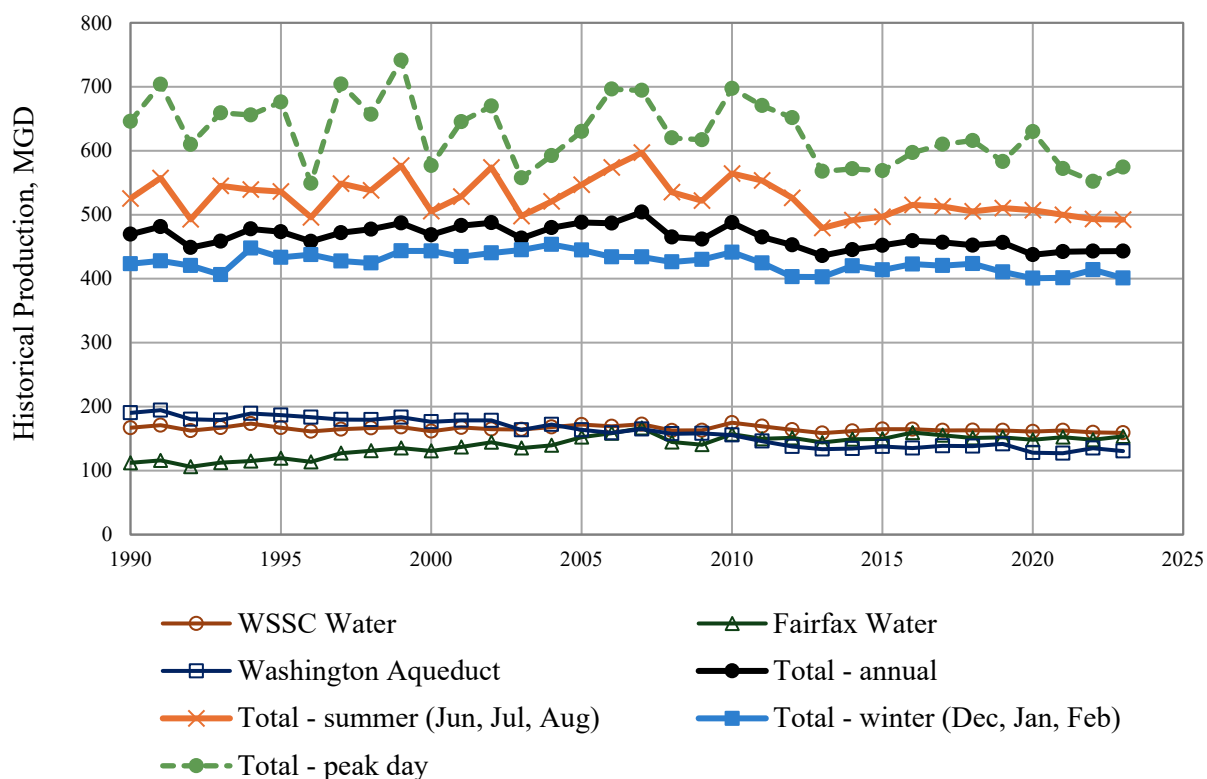


Figure 2-3: CO-OP supplier historical annual production and combined total annual, summer, winter (by water year), and annual peak-day production.

A substantial portion of water treated by WSSC Water (32 percent) and Fairfax Water (41 percent) originates from the Patuxent and Occoquan reservoirs, respectively. Over these same past five years, CO-OP suppliers' Potomac River production has averaged 331 MGD, with the following approximate distribution by supplier:

- WSSC Water 33 percent (109 MGD)
- Fairfax Water 27 percent (89 MGD)
- Washington Aqueduct 40 percent (133 MGD)

During this time, Fairfax Water has supplied 61 percent of Loudoun Water's demand, with the remainder produced directly by Loudoun Water at its Trap Rock Water Treatment Facility, which began operations in 2018.

Seasonal and Annual Variability: Total water production varies seasonally, with summer and peak-day demands exceeding annual averages due to increased outdoor use, primarily for lawn and landscape irrigation. From 2019 through 2023, summer production averaged 501 MGD, about 13 percent higher than the annual average of 445 MGD. During the same period, the peak-day production reached 630 MGD in 2020, approximately 42 percent higher than the annual average.

Peak-day production factors represent the ratio of peak-day production to annual average production. No new peak day production records were set in the years 2019-2023. Between 1990 and 2023, Fairfax Water's peak-day record was 259 MGD (July 7, 2010); WSSC Water's was 263 MGD (June 8, 1999), compared to a previous high of 267 MGD (July 8, 1988); and Washington Aqueduct's was 281 MGD (July 7, 1999). The combined historical peak-day production of the three CO-OP suppliers was 741 MGD (June 8, 1999). These peak-day productions correspond to factors that are 1.6 for WSSC Water, 1.7 for Washington Aqueduct, and 1.9 for Fairfax Water, with a combined CO-OP system factor of 1.6 and a Potomac factor of 1.8.

Although Loudoun Water is a relatively new supplier, it has become an important component of the regional water supply system. Its peak consumption, which includes both purchased and produced water, reached 47 MGD on September 6, 2023, while peak production at the Trap Rock Water Treatment Facility was 15 MGD on July 7, 2021. Over the past five years, Loudoun Water's peak-day production factor, evaluated separately from CO-OP suppliers, has averaged 1.5, with a corresponding peak-day consumption factor of approximately 1.8.

2.3 Intakes and Facilities

The WMA water supply system includes ten raw water intakes, seven major storage reservoirs, and seven water treatment plants (WTPs), also referred to as water filtration plants (WFPs) or water treatment facilities (WTFs). Thousands of miles of pipelines transport treated water from these facilities to retail and wholesale customers. Figure 2-1 shows the locations of existing and planned system resources. Figure 2-4 provides a schematic illustrating existing and planned water resources as well as inter-supplier transfers.

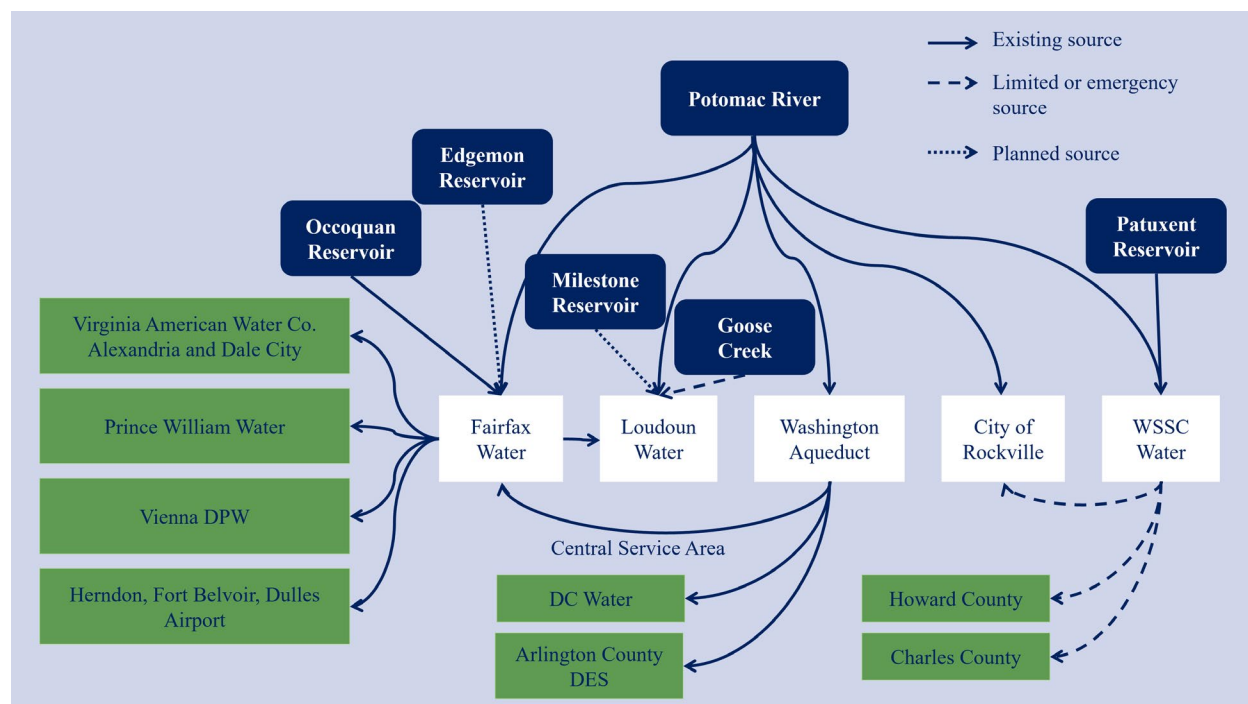


Figure 2-4: Schematic of the WMA's existing and planned water sources, five major suppliers, and their wholesale customers.

2.3.1 Fairfax Water

Fairfax Water owns and operates the first and third largest water treatment facilities in Virginia: the James J. Corbalis Jr. Water Treatment Plant (Corbalis WTP), located at the northern tip of Fairfax County, and the Frederick P. Griffith Jr. Water Treatment Plant (Griffith WTP), situated near the county's southern border. The Corbalis WTP, completed in 1982, has undergone several phased expansions and now has a treatment capacity of 225 MGD. Phase II, completed in 1996, added 75 MGD of capacity, while Phase III in 2008 added another 75 MGD along with major technological upgrades. The plant treats water from two Potomac River intakes—an onshore intake and a mid-river offshore intake. This plant serves primarily the northwestern portion of the service area (Western SA). The Griffith WTP has a current capacity of 120 MGD. It treats water from the Occoquan Reservoir and serves much of southeastern Fairfax County (Eastern SA), as well as portions of Prince William County and other nearby jurisdictions. Fairfax Water also purchases water from the Washington Aqueduct to primarily supply customers in the central parts of Fairfax Water's service area (Central SA).

Fairfax Water plans to convert the Vulcan Quarry near the Griffith WTP into a reservoir, Edgemon Reservoir, to augment its raw water storage. This new reservoir, planned to be operational by 2040, will provide additional storage capacity to support the existing water supply from the Occoquan Reservoir, as discussed in Section 2.4.4.

2.3.2 WSSC Water

The Washington Suburban Sanitary Commission (WSSC Water) is a water and wastewater utility that draws raw water from the Patuxent and Potomac rivers. It owns two water filtration plants: the Patuxent WFP near Laurel, Maryland, and the Potomac WFP adjacent to the Potomac River in Potomac, Maryland.

The Patuxent WFP began operation in 1944 and treats water stored in Triadelphia and T. Howard Duckett (also known as Rocky Gorge) reservoirs on the Patuxent River. The plant has a current treatment capacity of 72 MGD, with an emergency capacity of 110 MGD.

The Potomac WFP has been in service since 1961 and draws water directly from the Potomac River. It is WSSC Water's largest treatment facility, with a capacity of 280 MGD, and provides the majority of the utility's drinking water supply.

WSSC Water supplies treated water throughout its service area, with some portions receiving a blend of water from the Patuxent and Potomac WFPs. The distribution system is designed to allow treated water to be delivered from either treatment plant to most service points, providing a degree of redundancy during an emergency shutdown. However, the Patuxent plant alone cannot meet WSSC Water's total demand during the summer months, and if the Potomac plant is offline, large portions of the service area would be without water within several days.

2.3.3 Washington Aqueduct

Washington Aqueduct is a federally owned and operated public water supply agency managed by the Washington Aqueduct Division of the U.S. Army Corps of Engineers. Authorized by Congress in 1852, the original aqueduct conduit was constructed between 1853 and 1864 to convey water from an intake on the Potomac River at Great Falls to Dalecarlia Reservoir, located on the border between the District of Columbia and Montgomery County. Water then flowed by gravity to Georgetown Reservoir, located two miles downstream, for settling and distribution.

Today, "Washington Aqueduct" refers to the entire system, which now includes a second aqueduct conduit from Great Falls, two water treatment plants, Dalecarlia WTP and McMillan WTP, the McMillan Reservoir, and an additional intake and pumping station at Little Falls. All three of Washington Aqueduct's reservoirs are used as integral components of the treatment process (the Dalecarlia Reservoir for pre-sedimentation ahead of the Dalecarlia WTP; the Georgetown and McMillan Reservoirs for sedimentation and oxidation processes ahead of the McMillan filtration plant.) While Washington Aqueduct relies on the three reservoirs for operations, their combined storage capacity is insufficient for them to be considered storage reservoirs in this study.

The Washington Aqueduct system has a design capacity of 300 million gallons per day of treated water. Washington Aqueduct does not manage a distribution system but rather sells finished drinking water to its wholesale customers: DC Water, Arlington County, and Fairfax Water.

2.3.4 Loudoun Water

Loudoun Water serves customers in Loudoun County, Virginia, providing water and wastewater services. Until recently, most of the water it distributed came from Fairfax Water. In 2014, Loudoun Water acquired an intake and treatment plant on Goose Creek from the City of Fairfax. In 2018, Loudoun Water

began meeting a portion of its demand with water withdrawn from its new Potomac River intake and treated at its Trap Rock WTF while continuing to purchase some treated water from Fairfax Water. The Goose Creek plant was taken out of service when the Trap Rock facility became operational. In 2020, Loudoun Water received a modification of its Virginia Water Protection permit, allowing it to use the Goose Creek intake to supply its new raw water system during emergency events (Loudoun Water Correspondence, June 18, 2020).

Loudoun Water purchased a retired quarry from the Luck Stone Company. They are currently converting the quarry, formerly known as Quarry A, into the Milestone Reservoir by 2028. The reservoir, with an initial planned storage capacity of 1.25 billion gallons (BG), will be filled with raw water pumped from Loudoun Water's Potomac River intake and used to augment the supply during drought periods. Virginia Water Appropriations permit No. 10-2020 regulates Loudoun Water's Potomac River withdrawals and future use of Milestone Reservoir. The permit requires that Loudoun Water coordinate its Potomac River withdrawals with CO-OP operations during droughts, ensuring that such withdrawals do not conflict with water supply releases from the Jennings Randolph Reservoir.

As part of its 2014 purchase from the City of Fairfax, Loudoun Water also acquired Beaverdam Reservoir in the Goose Creek watershed upstream of the Goose Creek intake. Beaverdam Reservoir currently serves as a backup for Loudoun Water in emergencies and will serve as a water supply resource in the future. Loudoun Water also manages the reservoir's public access and recreational use in partnership with the Northern Virginia Regional Park Authority.

2.3.5 City of Rockville

The City of Rockville's Water Treatment Plant (Rockville WTP) began service in 1958 with a four MGD production capacity. In 1965, the plant underwent an upgrade that increased its capacity to eight MGD. Further upgrades in the mid-1990s ensured compliance with regulations set by the U.S. Environmental Protection Agency and the Maryland Department of the Environment. Since those updates, the WTP has averaged a withdrawal of five MGD of raw water from the Potomac River, which it then treats and distributes to the City's water customers.

Currently, the City supplies water to 70 percent of its population, while WSSC Water serves the remainder. When the Rockville WTP is offline, either due to necessary improvements, maintenance activities, or regional drought conditions, the City purchases water from WSSC Water. In 2018, Rockville purchased about 188 million gallons of water, which accounted for approximately 11 percent of its annual production, from WSSC Water. Since then, the metered use of water from WSSC Water has ranged between zero and two percent of Rockville's production.

2.4 Water Sources

The primary raw water source for the WMA is the Potomac River, supplying about three-quarters of the region's water needs. Suppliers withdraw water from an 18-mile stretch of the river upstream of the Little Falls Dam near Washington, DC. The area also relies on off-Potomac storage from Fairfax Water's Occoquan Reservoir, WSSC Water's two Patuxent reservoirs, and three shared reservoirs in the Potomac watershed: Jennings Randolph, Savage, and Little Seneca. In addition, agreements are in place for two

future raw water storage facilities: Loudoun Water's Milestone Reservoir, anticipated to be operational by 2028, and Fairfax Water's Edgemon Reservoir, expected by 2040.

The CO-OP suppliers rely on ICPRB's Section for Cooperative Water Supply Operations on the Potomac (CO-OP) to coordinate drought operations and determine when water supply releases are needed to supplement Potomac River flows. During drought conditions, CO-OP monitors river flow, forecasts demand, and requests releases from shared storage in the Jennings Randolph and Little Seneca reservoirs to maintain adequate flow for water supply and environmental needs. The USACE owns and operates the Jennings Randolph and Savage reservoirs, releasing water from a designated water supply account upon CO-OP's request to augment downstream flows. These coordinated operations ensure efficient use of regional storage and equitable water availability for all metropolitan suppliers.

2.4.1 Potomac River

The freshwater portion of the Potomac River extends to the head of tide, located between Little Falls Dam and Chain Bridge near Washington, DC. The watershed area draining to Little Falls Dam spans 11,560 square miles and is monitored by the [USGS gage at Station No. 01646500](#). Additionally, the USGS reports daily mean "adjusted flow" (where estimates of adjusted flow exclude major WMA withdrawals but include reservoir releases) as [Station No. 01646502](#). From 1930 to 2023, the mean adjusted flow was 7.7 billion gallons per day (BGD), with a standard deviation of 4.0 BGD. River flows exhibit a seasonal pattern with higher flows in winter and lower flows during late summer and early fall.

WMA Potomac River demands are generally below the lowest recorded minimum flows, but flow levels can drop below unrestricted demands, particularly in August and September. This seasonal variability is evident in the historical adjusted flow statistics at Little Falls Dam: minimum flow, 10th percentile flow, median flow, and 90th percentile flow calculated over 94 years (1930-2023). Figure 2-5 illustrates this point by showing 2002 conditions: for example, on September 11, the historical minimum adjusted flow was 411 MGD, while the median flow for that date is 1,897 MGD. By contrast, in 2002, demand plus the 100 MGD environmental flow-by at Little Falls reached 544 MGD. The 2002 adjusted flow data also highlights variability, with many new records for daily minimum flow set in February, March, and April of that year.

A comparison of key drought years reveals differences in minimum recorded daily adjusted flow: 505 MGD in 1930, 388 MGD in 1966 (the record low), 704 MGD in 1999, and 1,222 MGD in 2002. These values partly reflect differing drought intensities and supplemental releases from the Jennings Randolph and Little Seneca reservoirs in later years. The 1930 drought remains the most severe on record, with flow levels staying exceptionally low from mid-July to mid-December, while the 1966 drought ranks second, with adjusted flow dropping to 388 MGD on September 10. The number of days with record-low flows also illustrates each drought's severity: 41 days in 1930, 110 days in 1966, 17 days in 1999, and 42 days in 2002.

From 2019 to 2023, average summer water demands (June-August) on the Potomac River were approximately 379 MGD. However, during dry years such as 1999, 2002, 2007, and 2010, demands rose to an average of 459 MGD, peaking at 480 MGD in 2002. While withdrawals typically account for about five percent of the adjusted flow, they can surpass 50 percent during dry periods. Figure 2-5 shows that 2002-level withdrawals would have created flow deficits had they occurred during earlier droughts (e.g.,

1930 or 1960s), when reservoir releases were unavailable to offset low flows. The 1999 and 2002 droughts were especially notable as the first low-flow events supplemented substantially by upstream reservoir releases, which prevented even lower flow conditions in the Potomac River.

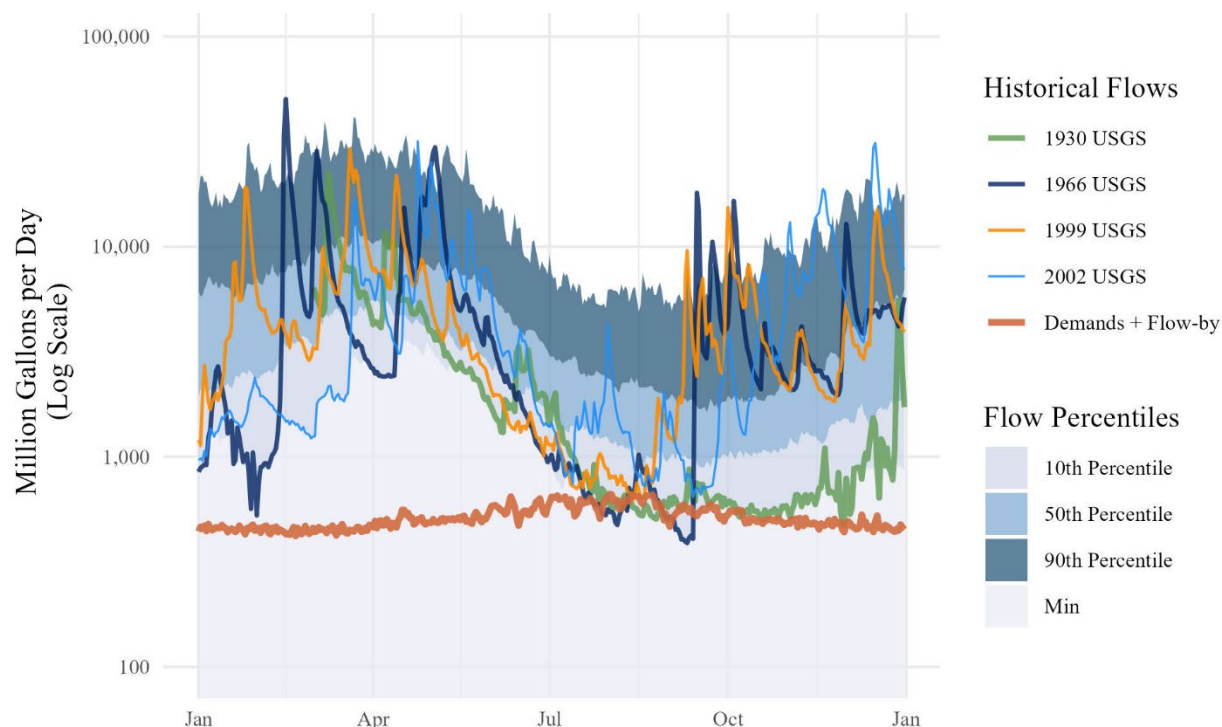


Figure 2-5: Adjusted daily flow at Little Falls for 1930, 1966, 1999, and 2002 compared to daily adjusted flow percentiles for 1930-2023 and demands for 2002 plus 100 MGD flow-by.

2.4.2 Shared Reservoirs

The three CO-OP suppliers entered into cost-sharing agreements in 1982 to jointly fund storage in two reservoirs, Jennings Randolph and Little Seneca, located upstream of their Potomac River intakes, to augment Potomac River flows during droughts. They also fund part of the operations and maintenance costs of a third upstream reservoir, Savage, which operates in conjunction with Jennings Randolph. Each reservoir is described below, along with its usable capacity, which is the volume of water that can be withdrawn or removed for supply purposes.

Jennings Randolph Reservoir: Located in the far northwest corner of the Potomac River basin, bordering Garrett County, Maryland, and Mineral County, West Virginia, this reservoir is operated by the USACE's Baltimore District Office. The reservoir has a usable storage capacity of 29.4 BG, with 13.1 BG dedicated to meeting the water supply needs of the CO-OP suppliers. The remaining 16.3 BG supports the USACE's daily operations, helping maintain downstream water quality and supporting recreational activities in the reservoir and on the North Branch of the Potomac River. ICPRB staff members direct releases from Jennings Randolph's water supply storage based on current and projected water demand, flow forecasts, and the status of other system reservoirs. Occurring about 200 miles upstream of the WMA, releases take over a week to reach the WMA intakes during low-flow conditions. The drainage area for Jennings Randolph Reservoir spans about 263 square miles.

Little Seneca Reservoir: Situated in Black Hill Regional Park in Montgomery County, Maryland, WSSC Water operates this reservoir's dam. During droughts, CO-OP staff members can request releases from Little Seneca Reservoir to augment Potomac River flows to meet water demands, including the environmental flow-by at Little Falls Dam. With a usable storage capacity of 3.9 BG, Little Seneca Reservoir is much smaller than Jennings Randolph Reservoir. However, Little Seneca releases make more efficient use of system storage since the travel time to Little Falls Dam is just about one day. The drainage area for Little Seneca Reservoir covers approximately 21 square miles.

Savage Reservoir: Located on the Savage River in the headwaters of the Potomac River basin near Jennings Randolph Reservoir, Savage Reservoir is owned by the Upper Potomac River Commission and operated under the guidance of the USACE's Baltimore District Office. The USACE coordinates release rates from Savage and Jennings Randolph reservoirs. During CO-OP drought operations, the combined releases from these two reservoirs help meet a flow target set by CO-OP at the [USGS stream flow gage \(Station ID 01598500\) in Luke, Maryland](#). Savage Reservoir has a usable storage capacity of approximately 6.3 BG and a drainage area of about 105 square miles. Additionally, it serves as the water supply source for the Town of Westernport, Maryland.

2.4.3 Other Existing Reservoirs

Three off-Potomac River reservoirs support the region's water supply, with individual suppliers managing them: Fairfax Water oversees the Occoquan Reservoir, while WSSC Water operates two Patuxent reservoirs, Triadelphia and Rocky Gorge.

Occoquan Reservoir: Fairfax Water manages this reservoir on the Occoquan River, situated within the Potomac basin but outside the freshwater drainage area serving Potomac mainstem intakes. According to a 2020 bathymetric survey (Charles P. Johnson & Associates, Inc., 2020), the reservoir has a usable storage capacity of 8.2 BG. Fairfax Water treats the reservoir's water at the Griffith WTP and distributes it to customers in the Eastern service area and Prince William County, Virginia. Fairfax Water can transfer up to 35 MGD from the Griffith WTP to the Western service area, with plans to expand this capacity to 50 MGD by 2040. The reservoir's drainage area is 592 square miles.

Patuxent Reservoirs: WSSC Water manages the Triadelphia and T. Howard Duckett (Rocky Gorge) reservoirs in the neighboring Patuxent River watershed. These reservoirs are operated in series and are treated in this study as a single source. Combined, they provide 10.4 BG of usable storage and drain a total area of 132 square miles. The Patuxent WTP treats water from these reservoirs. WSSC Water uses the treated water daily to supplement production at its Potomac plant.

2.4.4 Planned Reservoirs

As water demand continues to grow in Northern Virginia, regional water utilities are developing additional storage solutions to enhance supply reliability. Two key projects, Edgemon Reservoir (Fairfax Water) and Milestone Reservoir (Loudoun Water), aim to repurpose retired quarries for raw water storage. These reservoirs will serve as critical components of future drought management strategies, ensuring a stable water supply for their respective service areas.

Loudoun Water has plans to make two quarries owned by the Luck Stone Company available in the future as regional water supply resources. More details on these quarries are available in CO-OP's 2017 and

2020 studies (Schultz *et al.*, 2017; Ahmed *et al.*, 2020). Use of the Luck Stone quarries as part of the WMA system is not considered in the current study because regional agreements are not in place. Use by the region of the Luck Stone quarries is an alternative under consideration by the ongoing USACE feasibility study, and if this is one of the selected alternatives, its benefits will be evaluated by CO-OP in a separate study.

Edgemon Reservoir: Fairfax Water's Edgemon Reservoir is scheduled to become operational by 2040. Vulcan Materials will stop quarry activities in the northern portion (the Northern Reservoir) by 2035 and provide 1.7 BG of volume for Fairfax Water's use. Use of the Northern Reservoir, formerly referred to as Vulcan Quarry Phase 1, is planned for 2040. By 2085, it is planned that the combined storage of the Northern Reservoir and an additional volume referred to as the Main Reservoir will become available, increasing Edgemon Reservoir's total storage capacity to 15 BG (Vulcan Phase 2). Fairfax Water will utilize Edgemon Reservoir as a dual-purpose facility for raw water storage and solids disposal from the Griffith WTP. The solids waste stream, which consists of water and residuals formed when suspended solids react with coagulants and other treatment chemicals, will discharge into the reservoir for solids settling. Fairfax Water will reclaim water from the reservoir and return it to the Griffith WTP for treatment.

The 2017 alternatives study (Schultz *et al.*, 2017) included Edgemon Reservoir, which was then referred to as Vulcan Quarry, as one of ten potential storage options. The study assumed that Fairfax Water would implement system upgrades to maximize the regional benefits of the reservoir's storage. These upgrades included increasing the Griffith WTP's production capacity from 120 MGD to 160 MGD and enhancing the system to transfer treated water more easily and at higher rates to the Western service area. With these upgrades, the study estimated Vulcan Phase 1 could provide a regional benefit of 25 MGD in terms of safe summertime yield. However, with slower-than-anticipated water demand growth, the current study assumes no upgrades to the Griffith WTP, keeping its capacity at 120 MGD through 2050. Under these conditions, sensitivity tests from the 2017 study estimate Vulcan Phase 1 would provide a regional benefit of 5 MGD.

Milestone Reservoir: Loudoun Water plans to convert a retired quarry, acquired from the Luck Stone Company and formerly referred to as Quarry A, into Milestone Reservoir by 2028. This raw water storage facility will have an initial storage capacity of 1.25 BG. Loudoun Water will fill the Reservoir with raw water pumped from its Potomac River intake and use it to supplement the supply during drought conditions.

3 Annual Demand Forecast

3.1 Introduction

Chapter 3 presents the forecasts of annual water demand for the WMA, covering the period from 2020 to 2050. Forecasts are extrapolated to 2060 to support Virginia Department of Environmental Quality (VADEQ) Northern Virginia Regional Planning requirements and further extrapolated to 2085 to support the USACE feasibility study. The chapter outlines the process used to estimate future annual water demand, incorporating historical data, demographic projections, and anticipated changes in usage patterns. The forecast integrates unit use rates. Unit use rates refer to the average daily water use per household or employee, and combines them with demographic data to project total demand for each supplier and their wholesale customers. In addition, the estimation of unmetered and unaccounted-for water use is included to ensure total system demand is accurately represented.

Key factors affecting demand, such as the impacts of the Energy Policy Act of 1992 (U.S. Congress, 1992), water conservation education, and water use restrictions are also discussed. The resulting forecasts, including ranges of uncertainty, are used in CO-OP's water supply planning model, PRRISM, to help simulate daily water demand and evaluate the adequacy of available water to meet future needs.

3.2 Method

Determining current and future water demand requires detailed water use and demographic data. This process, illustrated in Figure 3-1, involves estimating demand across three use categories: single-family households (SFH), multi-family (apartment) households (MFH), and employees (EMP). The employee category represents water use in commercial, governmental, and industrial facilities. Projections for these categories rely on the MWCOG Round 10.0 Cooperative Forecast and Geographic Information System (GIS) delineations of utility service areas. Utilities either provided updated GIS data files (e.g., Loudoun Water) or confirmed no changes since the 2020 water supply study (Ahmed *et al.*, 2020). Local planning agencies supplied data on the number of single-family and multi-family homes.

Unit use rates, representing average daily water use per household (HH) or employee, are developed through utility water use surveys. Historical demands are converted into unit use rates using supplier billing data and MWCOG demographic data, modified by service area boundaries and jurisdictional dwelling unit ratios (DURs). DURs represent the number of SFH divided by the number of MFH within each jurisdiction in the WMA and are collected from local planning agencies and the U.S. Census Bureau's American Community Survey (ACS). These unit use rates are adjusted for future conditions, accounting for expected reductions in water use due to the Energy Policy Act of 1992 and the WaterSense Program.

The annual demand forecast process combines future unit use rates with MWCOG projections for number of households and employees to estimate total future demand. This approach, which employs a uniform methodology, ensures a consistent forecast for 11 wholesale and retail suppliers in the WMA.

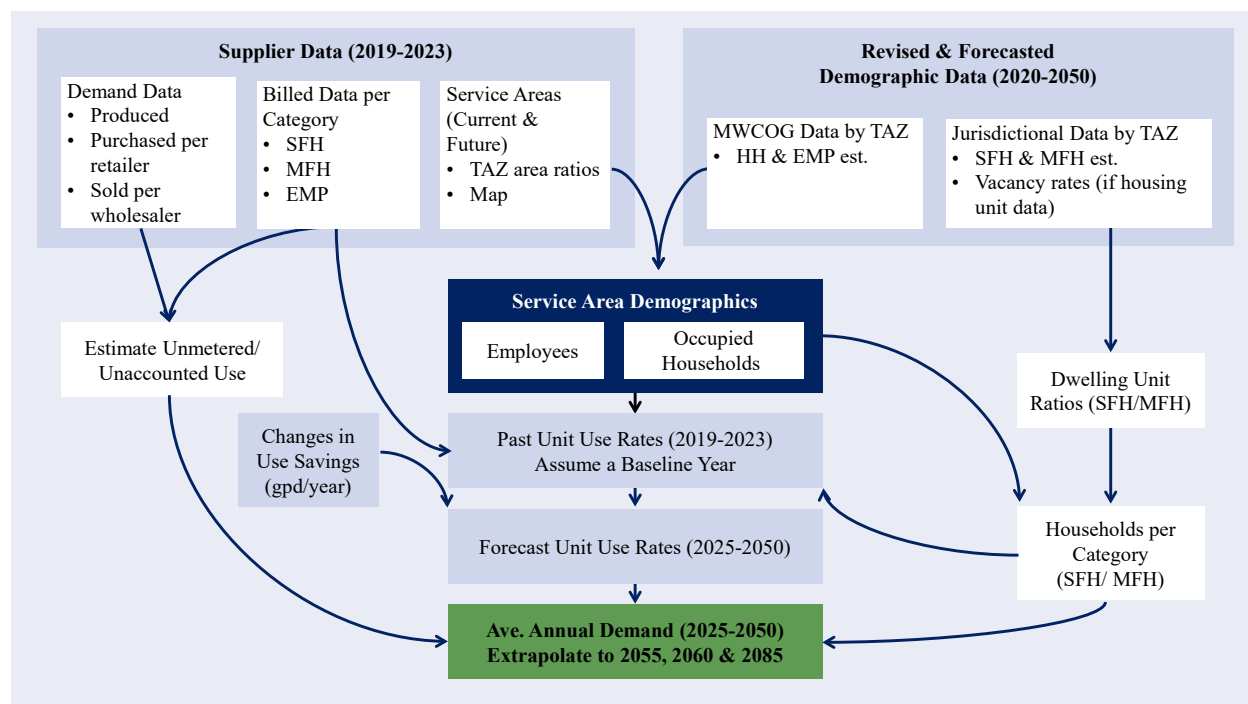


Figure 3-1: Data requirements for the annual demand forecast.

3.2.1 MWCOC Cooperative Forecast

MWCOG Round 10.0 Cooperative Forecast (Desjardin and Goodwin, 2023) provides population, household, and employee estimates for 2020-2050 in five-year increments. These forecasts, developed through collaboration with the Council of Governments, member jurisdictions, the Baltimore region, and other planning agencies, use common assumptions about future growth. Established in 1975, the Cooperative Forecasting Program enables coordinated local and regional planning.

The MWCOG forecasts are developed by combining regional and local data. Regional demographics and economic trends create a statistical benchmark, while local jurisdictions provide forecasts based on building permits, comprehensive plans, and local policies. These projections are reconciled to produce accurate, politically acceptable local forecasts distributed by traffic analysis zones (TAZ).

GIS software and python scripts extracted MWCOG Round 10.0 demographic data specific to each utility service area from the TAZ 3722 Zone System for 2020-2050. Table 3-1 through Table 3-3 summarize the populations, households, and employees by service area. Table 3-4 linearly extrapolates these values for the years 2055, 2060, and 2085 from 2040, 2045, and 2050 forecast values.

Table 3-1: MWCOG Round 10.0 population figures by supplier.

Supplier – customer (retail or wholesale)	2020	2023*	2025	2030	2035	2040	2045	2050
Fairfax Water – Retail customers	1,118,657	1,138,392	1,151,549	1,200,824	1,239,187	1,276,437	1,312,991	1,345,780
Fairfax Water – Dulles International Airport	475	598	680	920	1,117	1,254	1,346	1,438
Fairfax Water – Fort Belvoir	10,295	10,174	10,094	10,141	10,224	10,325	10,427	10,510
Fairfax Water – Town of Herndon	21,803	22,633	23,187	25,759	26,473	26,835	27,246	27,680
Fairfax Water – Prince William Water	374,500	389,980	400,300	419,700	432,900	442,300	449,000	453,500
Fairfax Water – Vienna PWD	28,121	28,351	28,504	28,674	28,934	29,213	29,470	29,693
Fairfax Water – Virginia American Alexandria	159,434	172,071	180,495	201,960	222,170	239,798	252,885	261,818
Fairfax Water – Virginia American Dale City	75,433	77,371	78,663	79,551	80,739	81,620	82,256	82,592
Fairfax Water Subtotal (no Loudoun Water)	1,788,718	1,839,570	1,873,472	1,967,529	2,041,744	2,107,782	2,165,621	2,213,011
Washington Aqueduct – Arlington County DES	236,794	241,374	244,428	258,862	271,680	283,994	296,442	309,592
Washington Aqueduct – DC Water	688,846	693,716	696,962	727,923	756,494	786,440	815,718	843,684
Washington Aqueduct – Arlington Fort Myer	1,486	1,318	1,207	1,217	1,199	1,212	1,564	1,633
Washington Aqueduct Subtotal	927,126	936,408	942,597	988,002	1,029,373	1,071,646	1,113,724	1,154,909
WSSC Water – Prince George’s County	921,989	939,657	951,435	984,797	1,030,761	1,068,865	1,103,161	1,134,914
WSSC Water – Montgomery County	967,823	979,261	986,886	1,017,346	1,048,353	1,078,750	1,106,551	1,130,298
WSSC Water Subtotal	1,889,812	1,918,917	1,938,321	2,002,143	2,079,114	2,147,615	2,209,712	2,265,212
CO-OP Supplier Subtotal	4,605,656	4,694,895	4,754,390	4,957,674	5,150,231	5,327,043	5,489,057	5,633,132
City of Rockville DPW	51,547	52,763	53,574	55,927	58,819	62,450	65,593	69,130
Loudoun Water	302,624	319,740	331,150	361,907	378,098	387,639	393,596	399,338
WMA Subtotal	4,959,827	5,067,398	5,139,114	5,375,508	5,587,148	5,777,132	5,948,246	6,101,600

* 2023 is interpolated.

Table 3-2: MWCOG Round 10.0 occupied household figures by supplier.

Supplier – customer (retail or wholesale)	2020	2023*	2025	2030	2035	2040	2045	2050
Fairfax Water – Retail customers	402,374	411,903	418,256	440,340	457,351	473,918	490,222	504,984
Fairfax Water – Dulles International Airport	185	229	259	356	440	500	541	580
Fairfax Water – Fort Belvoir	2,995	2,927	2,881	2,901	2,939	2,984	3,028	3,064
Fairfax Water – Town of Herndon	7,686	8,096	8,370	9,636	9,985	10,158	10,349	10,550
Fairfax Water – Prince William Water	121,600	127,060	130,700	138,100	143,600	147,600	150,700	152,900
Fairfax Water – Vienna PWD	9,327	9,407	9,461	9,517	9,603	9,695	9,782	9,854
Fairfax Water – Virginia American Alexandria	75,534	81,630	85,694	96,375	106,687	115,398	122,014	126,005
Fairfax Water – Virginia American Dale City	22,109	22,791	23,246	23,749	24,239	24,616	24,905	25,077
Fairfax Water Subtotal (no Loudoun Water)	641,810	664,044	678,867	720,974	754,844	784,869	811,541	833,014
Washington Aqueduct – Arlington County DES	109,372	114,432	117,806	125,854	132,979	139,754	146,402	153,101
Washington Aqueduct – DC Water	312,187	331,229	343,923	366,499	386,298	407,306	425,722	441,081
Washington Aqueduct – Arlington Fort Myer	410	321	262	268	258	265	464	502
Washington Aqueduct Subtotal	421,969	445,982	461,991	492,621	519,535	547,325	572,588	594,684
WSSC Water – Prince George’s County	326,201	358,047	337,316	350,343	367,831	381,383	393,603	404,934
WSSC Water – Montgomery County	351,766	332,870	362,237	378,119	393,515	407,094	418,245	427,319
WSSC Water Subtotal	677,967	690,917	699,553	728,462	761,346	788,477	811,848	832,253
CO-OP Supplier Subtotal	1,741,746	1,800,945	1,840,411	1,942,057	2,035,725	2,120,671	2,195,977	2,259,951
City of Rockville DPW	20,855	21,478	21,893	23,146	24,535	26,339	27,919	29,686
Loudoun Water	98,401	104,096	107,894	118,455	124,663	128,508	130,995	133,431
WMA Subtotal	1,861,002	1,926,519	1,970,198	2,083,658	2,184,923	2,275,518	2,354,891	2,423,068

* 2023 is interpolated.

Table 3-3: MWCOG Round 10.0 employee figures by supplier.

Supplier – customer (retail or wholesale)	2020	2023*	2025	2030	2035	2040	2045	2050
Fairfax Water – Retail customers	608,124	625,381	636,885	692,344	730,648	754,427	773,568	784,194
Fairfax Water – Dulles International Airport	13,237	14,286	14,986	17,171	18,765	19,531	20,245	20,909
Fairfax Water – Fort Belvoir	40,676	43,246	44,960	50,754	50,983	51,035	51,065	51,076
Fairfax Water – Town of Herndon	18,663	18,961	19,159	20,041	20,729	21,682	22,865	23,688
Fairfax Water – Prince William Water	129,200	138,860	145,300	158,600	171,700	183,300	193,700	202,900
Fairfax Water – Vienna PWD	13,885	13,905	13,919	13,945	13,977	14,016	14,055	14,057
Fairfax Water – Virginia American Alexandria	101,806	101,305	100,971	99,255	106,641	112,785	116,817	123,245
Fairfax Water – Virginia American Dale City	13,932	14,098	14,209	14,724	15,199	15,587	15,888	16,131
Fairfax Water Subtotal (no Loudoun Water)	939,523	970,042	990,389	1,066,834	1,128,642	1,172,363	1,208,203	1,236,200
Washington Aqueduct – Arlington County DES	215,918	216,882	217,524	231,084	253,601	260,752	269,624	277,975
Washington Aqueduct – DC Water	786,959	823,074	847,151	887,316	924,553	955,375	989,969	1,022,463
Washington Aqueduct – Arlington Fort Myer	4,390	4,393	4,394	4,391	4,397	4,424	4,523	4,549
Washington Aqueduct Subtotal	1,007,267	1,044,349	1,069,069	1,122,791	1,182,551	1,220,551	1,264,116	1,304,987
WSSC Water – Prince George’s County	330,924	338,810	344,067	354,136	368,202	381,942	399,340	415,375
WSSC Water – Montgomery County	416,370	432,649	443,502	463,731	483,472	501,958	520,757	540,354
WSSC Water Subtotal	747,294	771,459	787,569	817,867	851,674	883,900	920,097	955,729
CO-OP Supplier Subtotal	2,694,084	2,785,850	2,847,027	3,007,492	3,162,867	3,276,814	3,392,416	3,496,916
City of Rockville DPW	63,357	64,291	64,913	67,062	69,677	73,629	76,983	79,730
Loudoun Water	141,769	153,031	160,540	175,153	186,904	196,625	202,788	208,940
WMA Subtotal	2,899,210	3,003,172	3,072,480	3,249,707	3,419,448	3,547,068	3,672,187	3,785,586

* 2023 is interpolated.

Table 3-4: MWCOG Round 10.0 demographics extrapolated for 2055, 2060, and 2085 using official 2040, 2045, and 2050 values.

	Population			Occupied Households			Employees		
Supplier – customer (retail or wholesale)	2055	2060	2085	2055	2060	2085	2055	2060	2085
Fairfax Water – Retail customers	1,381,079	1,415,751	1,589,108	520,774	536,307	613,972	800,497	815,380	889,798
Fairfax Water – Dulles International Airport	1,530	1,622	2,082	620	660	860	21,606	22,295	25,740
Fairfax Water – Fort Belvoir	10,606	10,698	11,161	3,105	3,145	3,345	51,100	51,120	51,223
Fairfax Water – Town of Herndon	28,099	28,521	30,634	10,744	10,940	11,920	24,751	25,754	30,769
Fairfax Water – Prince William Water	459,467	465,067	493,067	155,700	158,350	171,600	212,900	222,700	271,700
Fairfax Water – Vienna PWD	29,939	30,179	31,379	9,936	10,016	10,413	14,084	14,104	14,207
Fairfax Water – Virginia American Alexandria	273,520	284,530	339,580	131,746	137,050	163,567	128,076	133,306	159,456
Fairfax Water – Virginia American Dale City	83,128	83,614	86,044	25,327	25,558	26,710	16,413	16,685	18,045
Fairfax Water Subtotal (no Loudoun Water)	2,267,367	2,319,982	2,583,054	857,953	882,026	1,002,388	1,269,426	1,301,344	1,460,937
Washington Aqueduct – Arlington County DES	322,274	335,073	399,068	159,766	166,440	199,807	286,673	295,285	338,342
Washington Aqueduct – DC Water	872,525	901,147	1,044,257	458,478	475,366	559,803	1,056,357	1,089,901	1,257,621
Washington Aqueduct – Arlington Fort Myer	1,891	2,101	3,154	647	766	1,358	4,624	4,686	4,999
Washington Aqueduct Subtotal	1,196,689	1,238,321	1,446,478	618,891	642,571	760,968	1,347,654	1,389,872	1,600,962
WSSC Water – Prince George’s County	1,168,362	1,201,387	1,366,509	416,858	428,633	487,511	432,319	449,035	532,618
WSSC Water – Montgomery County	1,156,748	1,182,522	1,311,392	437,778	447,890	498,453	559,419	578,617	674,607
WSSC Water Subtotal	2,325,110	2,383,909	2,677,901	854,635	876,523	985,963	991,738	1,027,652	1,207,225
CO-OP Supplier Subtotal	5,789,166	5,942,211	6,707,433	2,331,480	2,401,120	2,749,320	3,608,817	3,718,868	4,269,123
City of Rockville DPW	72,404	75,744	92,444	31,328	33,002	41,369	82,882	85,932	101,185
Loudoun Water	405,223	411,073	440,320	135,901	138,362	150,670	215,099	221,257	252,044
WMA Subtotal	6,266,794	6,429,028	7,240,198	2,498,709	2,572,484	2,941,359	3,906,798	4,026,057	4,622,352

The percent change between MWCOG Round 9.1 (used in the 2020 study) and Round 10.0 forecasts for 2020 and 2050 shows differences across data categories. For the year 2020, the Fairfax Water subtotal (excluding Loudoun Water) experienced a two percent increase in population and one percent in households but a five percent decrease in employees from Round 9.1 to Round 10.0. The Washington Aqueduct subtotal saw decreases in all metrics, with the population down four percent, households down two percent, and employees down five percent. WSSC Water had a three percent rise in population and one percent in households but a seven percent decline in employees. The CO-OP supplier subtotal showed a slight increase in population (one percent) and no change in households, with a six percent decrease in employees. The City of Rockville DPW and Loudoun Water decreased in all metrics, with population drops of three percent and one percent, respectively. Overall, the WMA subtotal saw a one percent increase in population, no change in households, and a five percent drop in employees.

For the year 2050 forecasts, the difference between Round 9.1 and Round 10.0 Fairfax Water subtotals show a one percent decrease in population and a three percent decrease in households, with a ten percent reduction in employees. The Washington Aqueduct subtotal saw a dramatic 14 percent population drop but a three percent increase in households and a four percent decrease in employees. Forecasts for WSSC Water show an eight percent rise in population and a five percent increase in households, with a four percent drop in employees. The CO-OP supplier subtotal has a minor one percent decrease in population, a one percent increase in households, and a six percent drop in employees. The City of Rockville DPW and Loudoun Water service areas continue to show employee declines, but Loudoun Water anticipates a 13 percent rise in population and 12 percent rise in households. The Loudoun Water values account for an expanded service area as part of the Round 10.0 projections for the year 2050 and use service area percentages provided by Loudoun Water for future growth. The WMA subtotal reflects no change in population, a two percent increase in households, and a six percent drop in employees.

These changes reflect employee forecast declines, which is a common trend across supplier service areas. Figure 3-2 and Figure 3-3 summarize differences between Round 9.1 and 10.0. In both of these figures, Loudoun Water's values reflect a service area change. It should be noted that Round 9.1 2050 values were linearly extrapolated from 2035, 2040, and 2045, while MWCOG officially published Round 10.0 estimates for that year. These differences are significant as this study extrapolates forecasts for 2055, 2060, and 2085 using similar methods (Table 3-4).

These forecasts should be interpreted in the context of broader regional shifts that are not directly captured in the Round 10.0 projections. Reductions in the Federal workforce (MWCOG, 2025; Pessin, 2025), ongoing efforts to relocate agencies outside the WMA (DC Policy Center, 2025), national immigration policy changes (*H.R. 1, 119th Cong.*, 2025) that could substantially affect population growth, and evolving employment trends across the region all represent important uncertainties. For the first time in decades, the Washington Business Journal (2025) has reported that no commercial office buildings are under construction anywhere in the WMA, underscoring structural changes in the regional economy that may further influence future demographics and employment. Recent discussions with MWCOG staff indicate that Round 10.1 revisions reflect only minor adjustments to Round 10.0 estimates, despite evidence of slower population growth and continued uncertainty in domestic and international migration (G. Goodwin, personal communication, 2025).

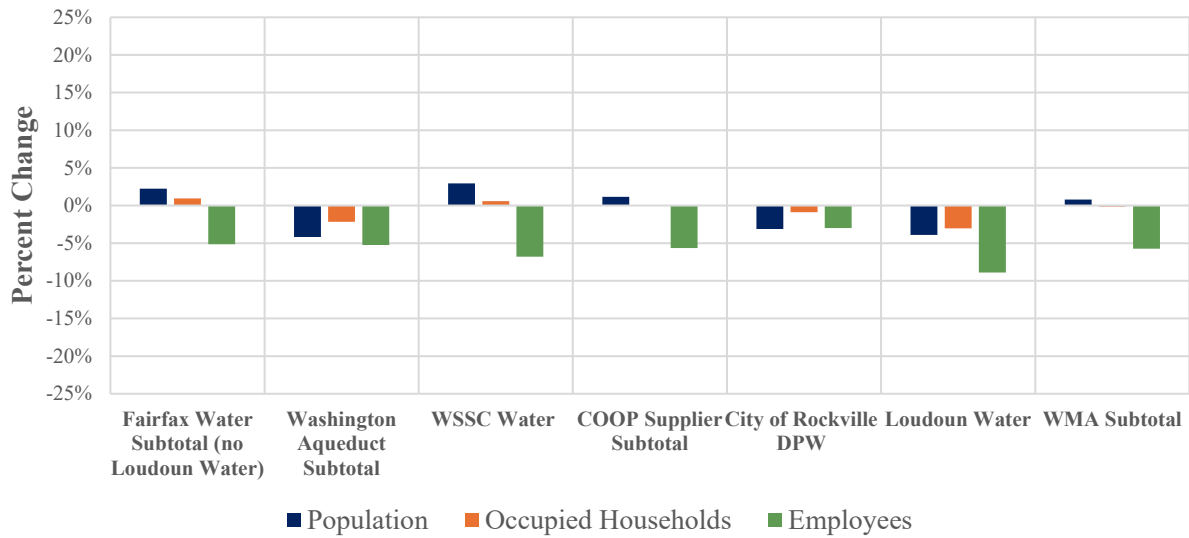


Figure 3-2: Percent change from Round 9.1 to Round 10.0 for the year 2020.

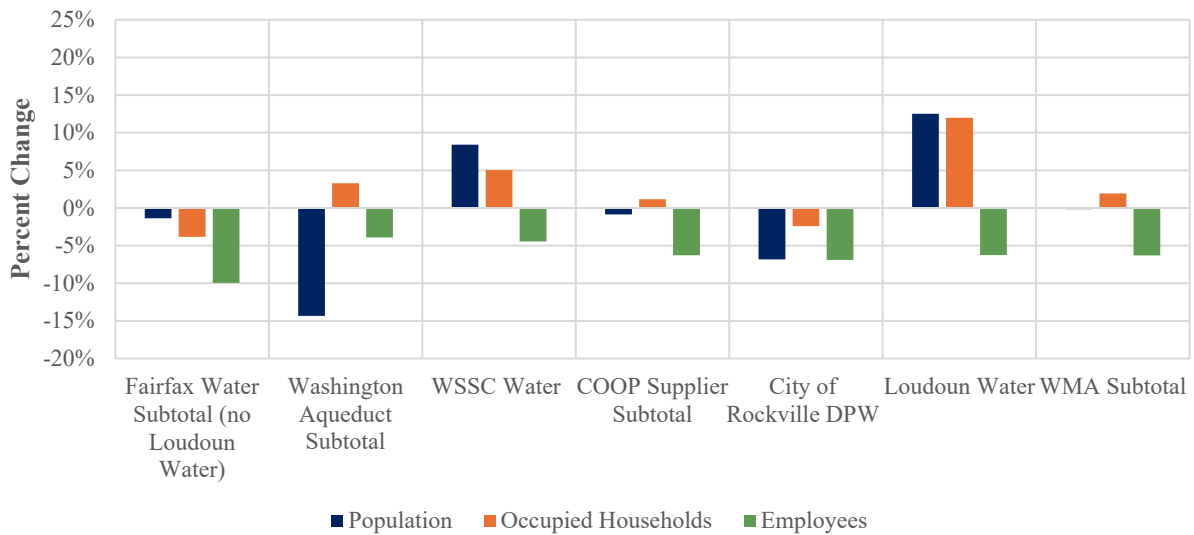


Figure 3-3: Percent change from Round 9.1 to Round 10.0 for the year 2050. Round 9.1 2050 values are linearly extrapolated from 2035, 2040, and 2045 values.

3.2.2 Dwelling Unit Ratios

DURs represent the number of SFH divided by the number of MFH. This information, necessary for calculating unit use rates, is collected from local planning agencies and demographic sources such as the U.S. Census Bureau's ACS. Appendix A.2 provides a detailed summary of the DUR ratios and the corresponding separation of household forecasts into single-family and multi-family categories.

The DURs break down MWCOG's occupied household data into SFH and MFH estimates (Figure 3-1). MWCOG supplies occupied household, employee, and population data for WMA jurisdictions to support forecasting efforts. For years not included or outside Round 10.0's 2020 to 2050 range, values are derived through interpolation or extrapolation.

Estimating DURs involves determining the number of SFH and MFH units in each jurisdiction within the suppliers' service areas. Various sources provide data, including the District of Columbia Office of Planning, Fairfax County Department of Planning and Zoning, City of Falls Church Department of Community Planning and Economic Development Services, Town of Herndon Planning Office, Prince William County Planning Office, City of Alexandria Department of Planning and Zoning, Arlington County Department of Community Planning, Housing and Development, Montgomery County Planning Department, Prince George's County Planning Department, City of Rockville Community Planning and Development Services, and Loudoun County Department of Planning and Zoning.

Since some supplier service area boundaries do not align with political jurisdiction boundaries, DURs are calculated for the service areas within each jurisdiction whenever possible, as detailed in Appendix A.2. Figure 3-4 highlights key trends in the DURs, and Table 3-5 presents a set of jurisdictional DURs in the WMA. Overall, there is an observable trend of increasing multi-family households, leading to decreased DURs across most jurisdictions. This shift towards multi-family housing is particularly pronounced in areas with initially high DUR values, such as Prince William and Loudoun counties, indicating significant changes in housing patterns towards denser, multi-family solutions. Societal changes have also influenced water use, as newer single-family housing tends to be more compact, with smaller lots and fewer occupants per household, resulting in lower average water use per household (Loudoun Water Correspondence, September 2025). Despite this trend, jurisdictions like Prince William County continue to have a substantial share of single-family homes, although their ratios are decreasing. Many areas begin to stabilize their DURs after an initial decline, likely due to either development saturation or the application of constant assumptions after a certain period.

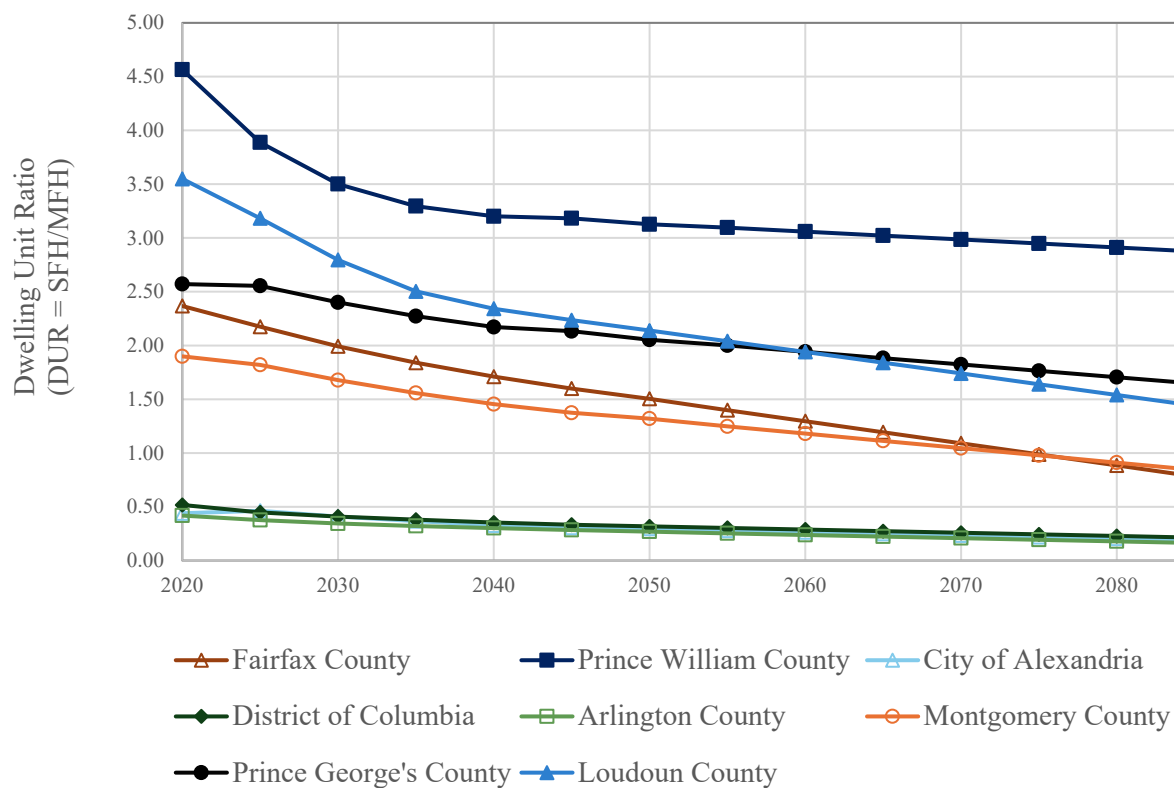


Figure 3-4: Dwelling unit ratios for select jurisdictions. Values for 2020-2050 are based on jurisdictional data collection, while projections for 2055-2085 are estimated using linear extrapolation of household data points from 2040, 2045, and 2050.

Table 3-5: Dwelling Unit Ratios by Jurisdiction and Service Area.

Jurisdiction	Service Area	MWCOG Round 10.0 Forecast Period							Extrapolation		
		2020	2025	2030	2035	2040	2045	2050	2055	2060	2085
Fairfax County	Fairfax Water Retail, Dulles International Airport, and Fort Belvoir	2.37	2.17	1.99	1.84	1.71	1.60	1.50	1.40	1.30	0.78
Town of Herndon	Town of Herndon	2.13	1.47	0.81	0.70	0.59	0.59	0.59	0.59	0.59	0.59
Prince William County	Prince William Water	4.56	3.89	3.50	3.30	3.20	3.18	3.13	3.10	3.06	2.87
Town of Vienna	Vienna Public Works Department	9.67	8.51	8.55	8.55	8.55	8.55	8.55	8.55	8.55	8.55
City of Alexandria	Virginia American Water, Alexandria	0.44	0.46	0.41	0.37	0.33	0.30	0.30	0.28	0.26	0.19
Dale City	Virginia American Water, Dale City	9.14	5.62	5.17	4.90	4.78	4.76	4.67	4.63	4.57	4.30
District of Columbia	DC Water	0.52	0.45	0.41	0.38	0.35	0.33	0.32	0.30	0.29	0.21
Arlington County	Arlington County Department of Environmental Services and Fort Myer	0.42	0.38	0.34	0.32	0.30	0.28	0.27	0.25	0.24	0.16
Montgomery County	WSSC Water	1.90	1.82	1.68	1.56	1.45	1.37	1.32	1.25	1.18	0.84
Prince George's County	WSSC Water	2.57	2.55	2.40	2.27	2.17	2.13	2.05	2.00	1.94	1.65
City of Rockville	City of Rockville Department of Public Works	1.22	1.15	1.05	0.96	0.86	0.77	0.70	0.62	0.54	0.14
Loudoun County	Loudoun Water	3.55	3.18	2.80	2.50	2.34	2.24	2.14	2.04	1.94	1.44

3.2.3 Unit Use Rates

Utilities report average daily water consumption for single-family, multi-family, and employee users in gallons per household or employee per day. Dwelling unit ratios, MWCOG household and employee data, and utility billing data determine these values (Figure 3-1). Unit use data is a primary input for the long-term water demand model, which uses current unit use values, estimates of unit use savings due to the Energy Policy Act of 1992, and forecasts of number of household and employees.

Unit use values for 2023 are summarized in Table 3-6 and further discussed in Appendix A.2. Utilities provided billing data for single-family households, multi-family households, and employee categories. Each utility's billing system determined the availability of disaggregated data, with any gaps or assumptions documented in Appendix A.2.

Regional supplier retail and wholesale customer data provided the basis for calculating total water consumption for each category by the number of households or employees. Unmetered water, defined as the difference between water produced (or purchased at wholesale) and water billed to customers, was also estimated and is presented in Appendix A.2. These estimates account for distribution system losses and other non-revenue water, consistent with supplier-reported audits. The assumed percentages vary by supplier and are documented in the appendices.

The updated record of unit use values spans the years from 2019 to 2023 and includes unit use values for 2014-2018 from the previous study. This record of unit use values provides the basis for this study's base year 2023 unit use, estimated from past data using least squares regression. Figure 3-5 illustrates unit use values from 1990 to 2023, providing a comprehensive view of the available data for the Fairfax Water retail, WSSC Water, and DC Water portions of the CO-OP supplier service areas. For each service area listed in Table 3-6, the analysis calculates a least squares regression line for unit use values from 2014 to 2023. Table 3-6 presents unit use values derived from 2023 billing data (2023 value), regression line estimates for 2023 (2023 estimate), and the standard error of the estimate (SE). The regression line estimates are employed in annual demand forecasts as they capture recent trends and are less sensitive to yearly fluctuations caused by weather or other transient conditions. The uncertainty analysis for annual demand forecasts relies on the SE estimates.

Table 3-6: Year 2023 unit use rates by service area in gallons per day per unit (gpd/unit) calculated from billing data and estimated from regression analysis.

Service Area	Single Family Unit Use (SFH UU)			Multi-family Unit Use (MFH UU)			Employee Unit Use (EMP UU)		
	2023 value	2023 estimate	SE	2023 value	2023 estimate	SE	2023 value	2023 estimate	SE
Fairfax Water – retail customers	156	156	3.2	150	150	2.1	29	28	1.7
Fairfax Water – Dulles International Airport	(156)	(156)	(3.2)	(150)	(150)	(2.1)	40	37	4.9
Fairfax Water – Fort Belvoir	(156)	(156)	(3.2)	(150)	(150)	(2.1)	24	22	3.8
Fairfax Water – Town of Herndon	105	112	10.4	105	112	10.4	62	59.8	3.9
Fairfax Water – Prince William Water	168	165	437	184	193	14.4	37	34	3.0
Fairfax Water – Vienna PWD	162	159	5.5	120	145	15.4	23	22	1.4
Fairfax Water – Virginia American Alexandria	(156)	(156)	(3.2)	(150)	(150)	(2.1)	19	19	3.7
Fairfax Water – Virginia American Dale City	(156)	(156)	(3.2)	(150)	(150)	(2.1)	65	62	8.7
Washington Aqueduct – Arlington Co. DES	149	154	8.0	66	65	1.5	38	35	3.1
Washington Aqueduct – DC Water	117	120	4.8	93	96	4.8	39	38	2.4
Washington Aqueduct – Arlington – Fort Myer	149	154	8.0	66	65	1.5	54	56	23.9
WSSC Water	132	126	8.0	152	154	4.4	41	40	3.0
City of Rockville	138	128	5.6	121	114	4.7	15	13	0.7
Loudoun Water	190	187	6.7	130	129	3.9	49	46	4.0

Note: Values in parenthesis are assumed, based on values from another supplier.

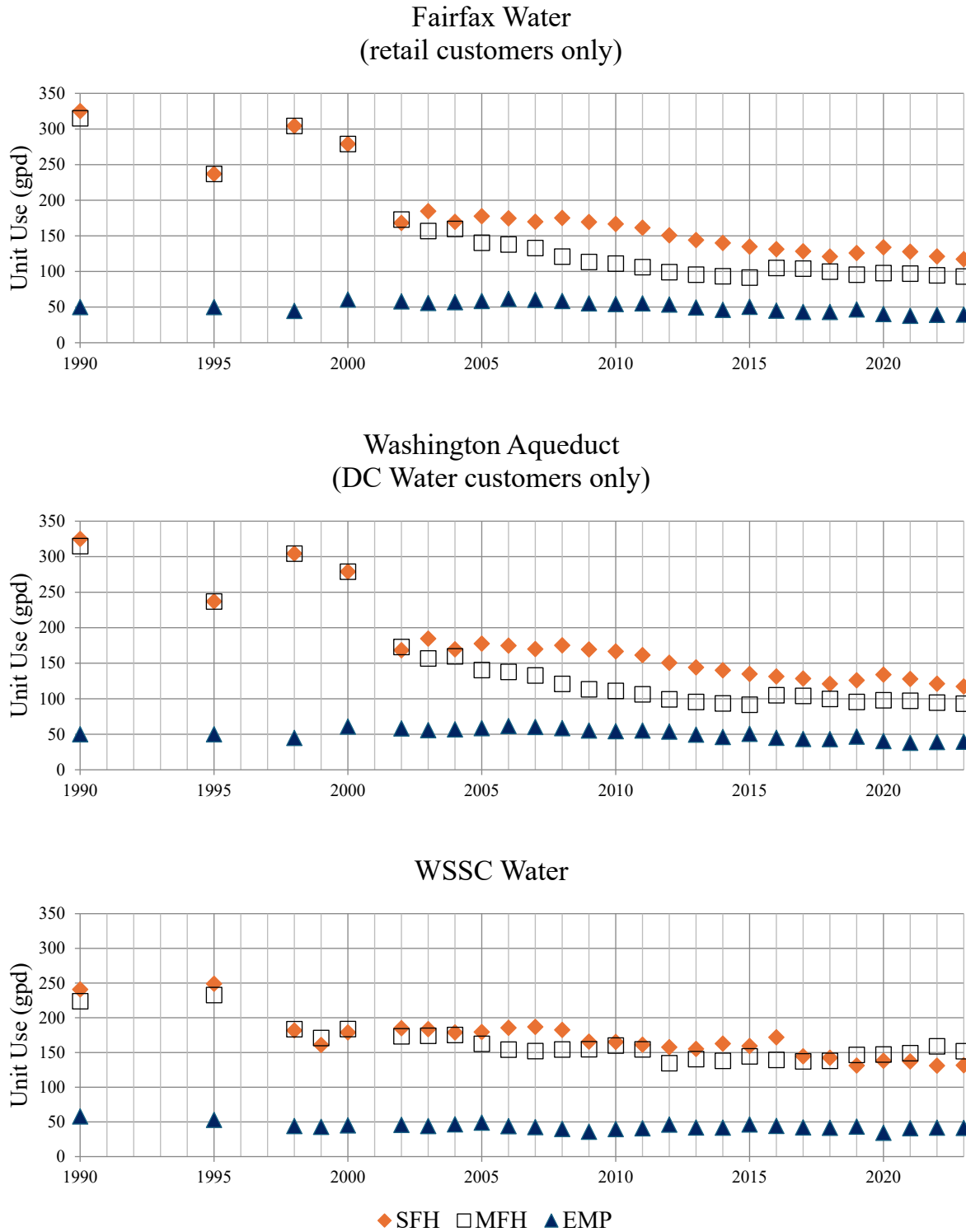


Figure 3-5: Unit use rates for the three CO-OP suppliers for 1990-2023.

3.2.4 Effects of Energy Policy Act 1992 (EPACT 1992)

Since the first WMA water supply study in 1990, per household water use has steadily declined, as shown in Figure 3-5. Predicting future demands involves estimating potential changes in household water use, influenced by weather, pricing structures, and policies like the *Energy Policy Act of 1992* (U.S. Congress, 1992) and the U.S. Environmental Protection Agency (USEPA) voluntary WaterSense program, which promotes efficient fixtures and appliances. The Energy Policy Act of 1992 established the first federal water efficiency standards for household and commercial appliances, mandating reduced water use for toilets, faucets, and showerheads (U.S. Congress, 1992). This legislation laid the foundation for long-term water savings and spurred the development of voluntary federal programs, most notably the USEPA WaterSense Program and the Department of Energy (DOE) ENERGY STAR initiatives. These programs certify products that meet rigorous water and energy efficiency criteria, verified through standardized testing and ongoing market surveillance (USEPA, 2025a; USEPA, 2025b; DOE, 2023). ENERGY STAR has helped consumers save over \$500 billion in energy costs and identify energy-efficient products for more than 30 years (Walton, 2025). Both programs played a central role in driving market adoption of efficient technologies through consumer labeling and industry participation. Many local governments and utilities have further supported adoption through rebates and other incentives (Vickers, 2014), contributing to widespread market uptake.

This section estimates future household water savings in the WMA, based on the continued implementation of current policies and programs, as well as insights from consumer behavior literature. Projected water savings are applied to reduce the baseline unit use of 2023 as shown in Section 3.3. The adjusted unit use is then multiplied by household and employee growth figures from the MWCOG Round 10.0 forecast (Desjardin & Goodwin, 2023) to project water use through 2050, with linear extrapolations extended to 2085.

The methodology used to estimate reductions follows the approach used in the 2020 water supply study (Ahmed *et al.*, 2020). Specifically, it relies on assumptions about residential water use rates; the extent of household remodeling; rates of bathroom, fixture, and appliance replacement; the construction of new houses with low-flow appliances and fixtures; and market share estimates for these products. This approach draws from two key studies: *WaterSense Program: Methodology for National Water Savings Analysis Model Indoor Residential Water Use* (Whitehead *et al.*, 2008) and *Tampa Bay Water: Water Demand Management Plan Final Report* (Hazen & Sawyer, 2013). The savings estimates represent average projections for the entire WMA system rather than individual suppliers because the analysis did not use supplier-specific data.

The methodology incorporates findings from available literature, including revised fixture replacement rates (Schein *et al.*, 2017) and use event frequencies (DeOreo *et al.*, 2016). Additionally, ranges in literature values for market shares (Whitehead *et al.*, 2008) and fixture replacement rates define three different water savings projections. These projections, denoted as SP1, SP2, and SP3, reflect varying assumptions about market shares and replacement rates, allowing for various possible outcomes and sensitivities. SP1 assumes the highest values for replacement rates and market shares of high-efficiency fixtures and appliances, SP3 assumes the lowest values, and SP2 represents a middle ground. Detailed descriptions of the assumptions are provided in Appendix A.2. Savings projections SP1 and SP3 are

computed for comparison purposes, but projection SP2 is selected for use in this study's forecasts of WMA supplier annual demands.

Table 3-7 presents household water use by appliance type for 2023 and 2050. Based on this and similar data for intermediate years, expected household savings are calculated for the period from 2023 to 2050. These savings are applied to the 2023 baseline single-family and multi-family household unit use rates for each supplier, starting in 2025. Appendix A.2 details the estimated savings for intermediate forecast years. Additionally, household water use savings are projected for 2055, 2065, and 2085 through linear extrapolation based on MWCOG Round 10.0 household forecasts for 2040, 2045, and 2050. These projections are summarized in Table 3-8.

The savings projections (SP1, SP2, and SP3) for clothes washers and dishwashers are identical because the literature does not provide market share and lifetime ranges for these appliances. Differences in the projections exist only for showerheads due to variations in available minimum, average, and maximum lifetime values. All showerhead scenarios assume low faucet market shares. Market share and lifetime data for toilets and faucets are available for all projections, contributing to the differences across SP1, SP2, and SP3.

Table 3-7: Estimated WMA household water use in 2023 & 2050 by savings category (gpd/unit).

Water Use	2023			2050		
	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)
Toilets	18.1	23.0	26.6	16.7	17.5	19.6
Clothes Washers	10.0	10.0	10.0	9.9	9.9	9.9
Dishwashers	1.2	1.2	1.2	1.2	1.2	1.2
Faucets	30.4	32.9	35.7	30.0	30.1	30.4
Showerheads	16.5	18.2	20.0	16.2	16.3	16.8
Total	76.2	85.4	93.5	74.0	74.9	78.0

Table 3-8: Extrapolated WMA household water use in 2055 through 2085 by savings category (gpd/unit).

Water Use	2055			2060			2085		
	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)
Toilets	16.7	17.2	19.1	16.7	17.1	18.7	16.7	16.8	17.5
Clothes Washers	9.9	9.9	9.9	9.9	9.9	9.9	9.9	9.9	9.9
Dishwashers	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Faucets	30.0	30.0	30.2	30.0	30.0	30.2	30.0	30.0	30.0
Showerheads	16.2	16.2	16.6	16.2	16.2	16.4	16.2	16.2	16.2
Total	74.0	74.6	77.0	74.0	74.4	76.3	74.0	74.0	74.8

The calculation for employee water savings from low-flow toilets used the same approach, but it assumes medium toilet market shares for all commercial toilet scenarios. Table 3-9 shows employee water use for commercial toilets in 2023 and 2050. The study did not perform a more detailed analysis due to the

extensive data requirements, including the number of male and female employees and the number of toilets and urinals per building. Additionally, employee water use savings are projected for 2055, 2065, and 2085 through extrapolation interpolation based on MWCOG Round 10.0 employee forecasts for 2040, 2045, and 2050 (Table 3-10).

Appendix A.2 presents the tabular results. Table 3-11 summarizes the incremental change in water use savings to forecast unit use from 2025 through 2085. These numbers are used to adjust the 2023 baseline unit use estimates shown in Table 3-6.

Figure 3-6 compares the estimated cumulative water savings from 1994 to 2085 under the three different efficiency scenarios: SP1 (High), SP2 (Middle), and SP3 (Low), as well as projections from the 2020 and 2025 demand studies. The High Scenario (SP1) achieves the most significant savings early, peaking around 2020, with diminishing returns afterward. In contrast, the Low Scenario (SP3) shows a steady increase in savings over time, with significant gains continuing through 2050. The Middle Scenario (SP2) indicates moderate, consistent savings, with a plateau effect beginning around 2025. The SP2 projection is relatively insensitive to changes to the ICPRB efficiency model since the 2015 water supply study (Ahmed *et al.*, 2015). Notably, the projections show the cumulative impact of policy interventions and market trends, with the most substantial savings realized under aggressive early adoption of high-efficiency fixtures. The missing 2020 water supply study data post-2045 values reflect differences in the two study forecast horizons. The current study uses only the Middle Scenario (SP2) for the water demand forecast, although previous studies have utilized SP1 and SP3 to inform the High and Low demand scenarios, respectively.

Table 3-9: Estimated WMA employee water use in 2023 and 2050 by commercial toilet use (gpd/unit).

Water Use	2023			2050		
	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)
Commercial Toilets	5.7	5.7	5.7	4.3	4.3	4.3

Table 3-10: Extrapolated WMA employee water use in 2055 through 2085 by commercial toilet use (gpd/unit).

Water Use	2055			2060			2085		
	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)
Commercial Toilets	4.2	4.2	4.2	4.1	4.1	4.1	3.9	3.9	3.9

Table 3-11: Water use savings by forecast increment (gpd/unit).

Year	SFH and MFH (Total Water Use)			EMP (Toilets)		
	High (SP1)	Middle (SP2)	Low (SP3)	High (SP1)	Middle (SP2)	Low (SP3)
2023-2025	0.7	2.2	2.7	0.2	0.2	0.2
2026-2030	0.9	3.8	5.1	0.4	0.4	0.4
2031-2035	0.4	2.1	3.4	0.3	0.3	0.3
2036-2040	0.2	1.2	2.1	0.2	0.2	0.2
2041-2045	0.1	0.6	1.0	0.2	0.2	0.2
2046-2050	0.0	0.5	1.3	0.1	0.1	0.1
2051-2055 estimate ¹	0.0	0.3	1.0	0.1	0.1	0.1
2056-2060 estimate ¹	0.0	0.2	0.7	0.1	0.1	0.1
2061-2065 estimate ¹	0.0	0.1	0.5	0.1	0.1	0.1
2066-2070 estimate ¹	0.0	0.1	0.4	0.0	0.0	0.0
2071-2075 estimate ¹	0.0	0.1	0.3	0.0	0.0	0.0
2076-2080 estimate ¹	0.0	0.0	0.2	0.0	0.0	0.0
2081-2085 estimate ¹	0.0	0.0	0.2	0.0	0.0	0.0

¹2055 through 2085 values are linearly extrapolated from 2040, 2045, and 2050 values.

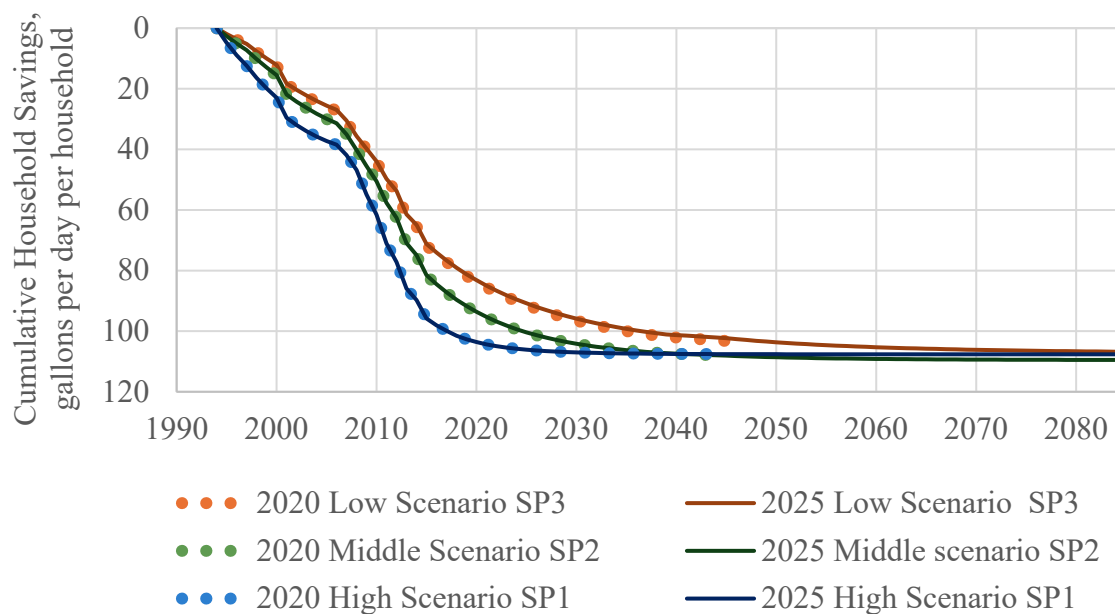


Figure 3-6: Estimated per household savings since 1994 with extrapolation from 2050 through 2080.

While this study focuses on household and appliance-based efficiency measures, future analyses could also consider the influence of pricing structures, particularly inclining block rate tiers, on household water demand. Such rate structures can provide a financial incentive for conservation by increasing the marginal cost of higher water use. A survey of local utilities would be needed to determine which jurisdictions within the WMA employ this type of pricing structure and to quantify its potential impact on demand patterns.

3.2.5 Prospects for Continuation of Water Conservation Policies

When modeling future water demand, the assumption of continued appliance efficiency improvements is a critical factor. For example, the Middle Scenario (SP2) assumes that moderate efficiency savings will begin leveling off by 2025, consistent with historical conservation trends. However, if progress stalls, particularly due to policy rollbacks, weakened standards, or shifts in federal priorities, projections change significantly. Removing the forecasted per-household unit use reduction of 10.5 million gallons per day (MGD) and per-employee reduction of 1.4 MGD by 2050 results in notably higher water demand forecasts (Figure 3-7, see the solid “ICPRB, 2025” forecast compared to the dotted “ICPRB, 2025 no savings” forecast). This scenario could strain long-term resource planning and undermine sustainability efforts within the WMA.

Recent administrative actions to relax, delay, or dismantle federal efficiency programs, most notably the proposed termination of the ENERGY STAR program, underscore growing policy uncertainty surrounding future water and energy savings (The White House, 2025; DOE, 2025; USEPA, 2025c). ENERGY STAR currently supports over 16,000 participating organizations, including more than 1,800 manufacturers, and its reach extends across both residential and commercial markets. Its certified homes accounted for approximately 25 percent of the new U.S. housing starts last year. The program has also helped utilities manage energy demand and reduce system costs. In response to the proposed rollbacks, over 1,000 stakeholders including utilities, manufacturers, and environmental groups have urged the USEPA to preserve the program, citing consumer cost savings and grid reliability benefits. Critics warn that eliminating the label could lead to market confusion, higher energy use, and long-term cost increases (Walton, 2025).

While the program’s future remains uncertain, its longstanding bipartisan support and integration into many state-level policies suggests its elimination may face political and legal challenges. In particular, any roll-back or significant alteration would likely require congressional approval to amend mandates established by the Energy Policy Act of 1992. This is because of the Act’s anti-backsliding provision, which prohibits federal agencies from weakening or rescinding existing efficiency standards once in place (Keller & Heckman LLP, 2025). As a result, proposed changes to reduce or eliminate the program would almost certainly face legal challenges without new legislation. These developments reinforce the importance of incorporating policy uncertainties into long-term demand models. Regularly reassessing

assumptions about future water and energy savings is essential to robust planning as federal and state-level regulations continue to evolve.

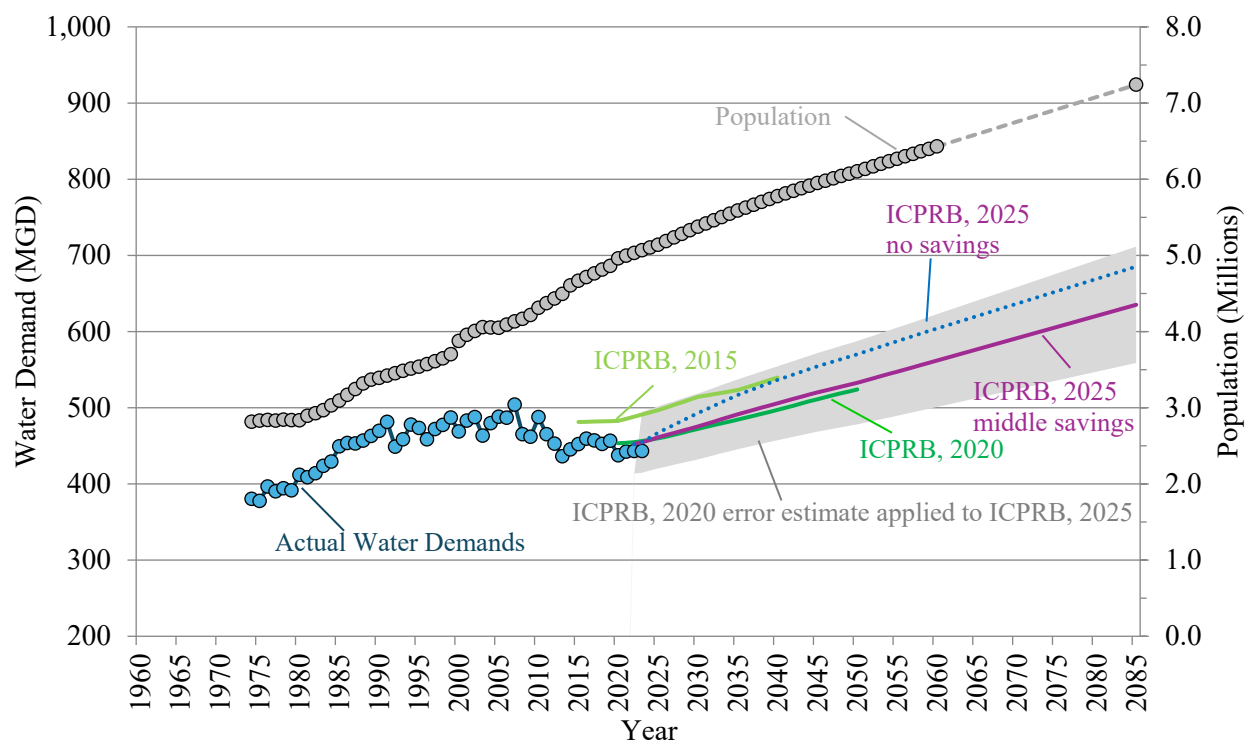


Figure 3-7: Demonstration of 2025 WMA demand forecast with no water efficiency savings applied.

3.3 Unit Use Forecast

Table 3-12 and Table 3-13 give the estimated unit use rates for each supplier and wholesale customer for the 2020 through 2050 forecast years. Table 3-14 provides an extrapolation of these results through 2085. As described in the previous sections, the forecast relies on historical information supplied by the suppliers, past demographic information, and assumptions about changing customer behavior. The baseline 2023 estimates, shaded in grey in Table 3-12, are modified using the Middle Scenario (SP2) incremental water use savings reported in Table 3-11. The unit use rates, at five-year increments for 2020 through 2050, rely on the best available data described in previous sections of this report. The 2055, 2060, and 2085 unit use rates rely on similar calculations and use demographic data linearly extrapolated from 2040, 2045, and 2050 values. This approach provides WMA suppliers with a longer-term planning horizon and supports the VADEQ water supply planning process.

Note that the forecasted unit use rates rely on a system-wide analysis of water use savings, assuming all WMA suppliers will achieve the same savings. However, assumptions used in this analysis, such as the age of the housing stock and fixture and appliance replacement rates, will differ by supplier service area. Future ICPRB water supply studies may warrant a more individualized analysis of water use savings.

Table 3-12: Unit use forecast by supplier (gpd/unit) using SP2 middle scenario water use savings assumption.

Service Area	2020			2023 Baseline			2025			2030		
	SFH	MFH	EMP	SFH	MFH	EMP	SFH	MFH	EMP	SFH	MFH	EMP
Fairfax Water - Retail customers	165	156	27	156	150	28	154	148	28	150	144	27
Fairfax Water - Dulles International Airport	(165)	(156)	29	(156)	(150)	37	(154)	(148)	37	(150)	(144)	36
Fairfax Water - Town of Herndon	120		52	111		60	109		60	105		59
Fairfax Water - Fort Belvoir	(165)	(156)	26	(156)	(150)	37	(154)	(148)	37	(150)	(144)	36
Fairfax Water - Loudoun Water	193	131	39	187	129	46	185	127	46	181	123	45
Fairfax Water – Prince William Water	171	196	32	165	193	34	163	191	34	159	187	33
Fairfax Water - Vienna PWD	168	149	22	160	145	22	158	143	22	154	139	21
Fairfax Water - Virginia American Alexandria	(165)	(156)	15	(156)	(150)	19	(154)	(148)	19	(150)	(144)	18
Fairfax Water - Virginia American Dale City	(165)	(156)	61	(156)	(150)	62	(154)	(148)	62	(150)	(144)	61
Washington Aqueduct - Arlington County DES	161	71	31	154	65	35	152	63	35	148	59	34
Washington Aqueduct - Arlington Fort Myer	(161)	(71)	61	(154)	(65)	56	(152)	(63)	56	(148)	(59)	55
Washington Aqueduct - DC Water	134	98	40	120	96	38	118	94	38	114	90	37
WSSC Water	138	147	34	126	154	40	124	152	40	120	148	39
City of Rockville DPW	132	120	14	128	114	13	126	112	13	122	108	12

Note: Values in parenthesis are assumed, based on values from another supplier.

Table 3-13: Unit use forecast by supplier (gpd/unit) using SP2 middle scenario water use savings assumption.

Service Area	2035			2040			2045			2050		
	SFH	MFH	EMP	SFH	MFH	EMP	SFH	MFH	EMP	SFH	MFH	EMP
Fairfax Water - Retail customers	148	142	27	147	141	27	146	140	27	146	140	27
Fairfax Water - Dulles International Airport	(148)	(142)	36	(147)	(141)	35	(146)	(140)	36	(146)	(140)	36
Fairfax Water - Town of Herndon	103		59	102		59	101		59	101		59
Fairfax Water - Fort Belvoir	(148)	(142)	36	(147)	(141)	36	(146)	(140)	36	(146)	(140)	36
Fairfax Water - Loudoun Water	179	121	45	178	120	45	177	119	45	177	119	45
Fairfax Water – Prince William Water	157	185	33	156	184	33	155	183	33	155	183	33
Fairfax Water - Vienna PWD	152	137	21	151	136	21	150	135	21	150	134	21
Fairfax Water - Virginia American Alexandria	(148)	(142)	18	(147)	(141)	18	(146)	(140)	18	(146)	(140)	18
Fairfax Water - Virginia American Dale City	(148)	(142)	61	(147)	(141)	61	(146)	(140)	61	(146)	(140)	61
Washington Aqueduct - Arlington County DES	146	57	34	145	56	34	144	55	34	144	55	34
Washington Aqueduct - Arlington Fort Myer	(146)	(57)	55	(145)	(56)	55	(144)	(55)	55	(144)	(55)	55
Washington Aqueduct - DC Water	112	88	37	111	87	37	110	86	37	110	86	37
WSSC Water	118	146	39	117	145	39	116	144	39	116	144	39
City of Rockville DPW	120	106	12	119	105	12	118	104	12	118	104	12

Note: Values in parenthesis are assumed, based on values from another supplier.

Table 3-14: Unit use extrapolation by supplier (gpd/unit) using SP2 middle scenario water use savings assumption.

Service Area	2055			2060			2085		
	SFH	MFH	EMP	SFH	MFH	EMP	SFH	MFH	EMP
Fairfax Water - Retail customers	145	139	26	145	139	26	145	139	26
Fairfax Water - Dulles International Airport	(145)	(139)	35	(145)	(139)	35	(145)	(139)	35
Fairfax Water - Town of Herndon	100		58	100		58	100		58
Fairfax Water - Fort Belvoir	(145)	(139)	35	(145)	(139)	35	(145)	(139)	35
Fairfax Water - Loudoun Water	176	118	44	176	118	44	176	118	44
Fairfax Water – Prince William Water	154	182	32	154	182	32	154	182	32
Fairfax Water - Vienna PWD	149	134	20	149	134	20	149	134	20
Fairfax Water - Virginia American Alexandria	(145)	(139)	17	(145)	(139)	17	(145)	(139)	17
Fairfax Water - Virginia American Dale City	(145)	(139)	60	(145)	(139)	60	(145)	(139)	60
Washington Aqueduct - Arlington County DES	143	54	33	143	54	33	143	54	33
Washington Aqueduct - Arlington Fort Myer	(143)	(54)	54	(143)	(54)	54	(143)	(54)	54
Washington Aqueduct - DC Water	109	85	36	109	85	36	109	85	36
WSSC Water	115	143	38	115	143	38	115	143	38
City of Rockville DPW	117	103	11	117	103	11	117	103	11

Note: Values in parenthesis are assumed, based on values from another supplier.

3.4 Annual Demand and Production Forecasts

The forecasts for the WMA's average annual water demand appear in Table 3-15. The estimates derive from current and forecasted numbers of single-family households, multi-family households, and employees; historic produced and billed water use; current and forecasted service areas; possible changes in water use behavior; and estimates of unmetered water use. The forecasts in Table 3-15 are presented by retail and wholesale class, grouped by each WMA supplier. Subtotals are shown for each CO-OP supplier's retail service area, as well as for Loudoun Water and Rockville. A detailed breakdown of each forecast is available in Appendix A.2.

Table 3-16 provides forecasts for WMA water production. This study treats water 'demand' and 'production' as equivalent when a single supplier provides water to a given service area. However, demand and production are distinct in cases where two or more suppliers provide water to a service area, as is evident from a comparison of Table 3-15 and Table 3-16. For example, demand and production differ for Fairfax Water's retail service area because they purchase water from Washington Aqueduct to serve a portion of demand for the Central Service Area. Similarly, Loudoun Water's demand differs from its production because it purchases a portion of the water it distributes to its customers from Fairfax Water.

The last rows of Table 3-15 provide an estimate of the uncertainty in the forecast of total WMA demand, expressed as both a percentage and an absolute value. The uncertainty is 8.5 percent in 2025 and rises to 12.8 percent in 2085. The range of uncertainty is displayed in Figure 3-8 as a grey shaded area centered around the WMA demand forecast, that is, the “medium demand” forecast in Table 3-15. In this table, the “low demand” forecast is the difference between the medium demand forecast and the uncertainty, and the “high demand” forecast is the sum of the medium demand forecast and the uncertainty. The low, medium, and high demand forecasts are used later in this study as three planning scenarios that represent a plausible range of future water demands.

Virginia water suppliers are required by regulation to ensure adequate available capacity in their systems, and water demand forecasts are developed to avoid underestimating future demands, which could result in inadequate system capacity.

Table 3-15 indicates that WMA suppliers are forecast to use an average of approximately 471 MGD annually in 2025, rising to 543 MGD by 2050 and to 647 MGD by 2085. Between 2025 and 2050, Fairfax Water's demand will increase by 27 MGD, Washington Aqueduct's by 18 MGD, and WSSC Water's by 22 MGD.

This study's forecast for total WMA demand, along with its range of uncertainty, is compared with estimates from past studies in Figure 3-8. Also included in this graph is the change over time of actual WMA demand and actual and forecasted population. The past overprediction of water demand is illustrated by this figure, indicating that systematic and statistical errors influence the results. The systematic errors not addressed in this study are clearly, at least in part, associated with the inability of water planners to predict the technological and policy changes that have reduced per-household and per-employee water use.

Table 3-15: Forecast of average annual water demand by supplier service area, 2023-2085 (MGD).

Supplier/Wholesale Customer	2023 ²	2025	2030	2035	2040	2045	2050	2055	2060	2085
Fairfax Water										
Retail customers	98.0	98.5	102.2	105.2	108.0	110.9	113.3	116.2	119.2	133.9
Dulles International Airport	0.6	0.7	0.8	0.8	0.9	0.9	0.9	1.0	1.0	1.1
Town of Herndon	2.2	2.2	2.4	2.4	2.5	2.6	2.6	2.7	2.8	3.2
Fort Belvoir	2.3	2.3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Prince William Water	26.8	27.4	28.8	29.9	30.8	31.6	32.3	33.1	33.8	37.9
Vienna PWD	2.0	2.0	2.0	1.9	1.9	1.9	2.0	2.0	2.0	2.0
Virginia American Alexandria	15.9	16.4	17.6	19.2	20.5	21.5	22.1	23.1	24.0	28.5
Virginia American Dale City	4.9	4.9	4.9	5.0	5.0	5.1	5.1	5.2	5.2	5.5
Subtotal (excluding Loudoun Water)	152.8	154.4	161.1	166.9	172.2	177.1	180.9	185.7	190.4	214.7
Washington Aqueduct										
Arlington County DES	20.0	19.7	19.7	20.8	21.3	21.8	22.5	23.1	23.7	25.0
DC Water	92.6	94.3	97.1	99.9	103.0	106.4	109.5	112.9	116.3	133.9
Arlington Fort Myer	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4
Subtotal (excluding sales to Fairfax Water)	112.8	114.4	117.1	121.0	124.6	128.6	132.3	136.3	140.4	159.3
WSSC Water										
Retail customers	156.3	156.9	159.8	164.8	169.7	174.5	179.1	184.1	189.3	216.6
Charles County ⁴	4.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Howard County ⁴		5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Subtotal	161.0	163.7	166.6	171.6	176.5	181.3	185.9	190.9	196.1	223.4
CO-OP suppliers total (excluding Loudoun Water)	426.7	432.5	444.8	459.6	473.2	487.0	499.1	513.0	526.9	597.4
Other WMA suppliers										
City of Rockville DPW	4.2	4.2	4.2	4.4	4.6	4.8	5.0	5.3	5.5	6.6
Loudoun Water	27.9	28.7	30.6	31.8	32.7	33.2	33.8	34.4	34.9	37.8
Subtotal	32.1	32.8	34.9	36.2	37.3	38.0	38.8	39.6	40.4	44.4
WMA suppliers total										
Medium demand (sum of subtotals)	458.7	465.3	479.6	495.8	510.5	525.0	537.9	552.6	567.4	641.8
Uncertainty, MGD		±39.6	±42.2	±45.1	±48.0	±50.9	±53.8	±57.5	±60.7	±82.2
Uncertainty, percent		±8.5	±8.8	±9.1	±9.4	±9.7	±10.0	±10.4	±10.7	±12.8
Low demand (Med. – uncertainty)		425.7	437.4	450.7	462.5	474.1	484.1	495.1	506.7	559.6
High demand (Med. + uncertainty)		504.9	521.8	540.9	558.5	575.9	591.7	610.1	628.1	724.0

¹ Demand is water distributed, including unmetered water.² 2023 value estimated by the same methods used in the forecasts.⁴ Charles and Howard County values taken from 2020 study assumptions.

Table 3-16: Forecast of average annual water production by supplier customers, 2020-2085 (MGD).

Supplier/wholesale customer	2023 ²	2025	2030	2035	2040	2045	2050	2055	2060	2085
Fairfax Water										
Retail customers	85.3	85.6	88.6	91.0	93.3	95.6	97.5	99.8	102.1	113.6
Dulles International Airport	0.6	0.7	0.8	0.8	0.9	0.9	0.9	1.0	1.0	1.1
Town of Herndon	2.2	2.2	2.4	2.4	2.5	2.6	2.6	2.7	2.8	3.2
Fort Belvoir	2.3	2.3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Prince William Water	26.8	27.4	28.8	29.9	30.8	31.6	32.3	33.1	33.8	37.9
Vienna PWD	2.0	2.0	2.0	1.9	1.9	1.9	2.0	2.0	2.0	2.0
Virginia American Alexandria	15.9	16.4	17.6	19.2	20.5	21.5	22.1	23.1	24.0	28.5
Virginia American Dale City	4.9	4.9	4.9	5.0	5.0	5.1	5.1	5.2	5.2	5.5
Loudoun Water Purchased	16.7	17.2	18.4	19.1	19.6	19.9	20.3	20.6	21.0	22.7
Subtotal	156.7	158.7	166.0	171.8	177.0	181.6	185.3	190.0	194.4	217.0
Washington Aqueduct										
Arlington County DES	20.0	19.7	19.7	20.8	21.3	21.8	22.5	23.1	23.7	25.0
DC Water	92.6	94.3	97.1	99.9	103.0	106.4	109.5	112.9	116.3	133.9
Arlington Fort Myer	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4
Fairfax Water (Falls Church retail sale)	12.7	12.8	13.5	14.2	14.7	15.3	15.9	16.5	17.1	20.3
Subtotal	125.5	127.2	130.6	135.2	139.3	143.9	148.2	152.8	157.5	179.6
WSSC Water										
Retail customers	156.3	156.9	159.8	164.8	169.7	174.5	179.1	184.1	189.3	216.6
Charles County ³	4.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Howard County ⁴		5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Subtotal	161.0	163.7	166.6	171.6	176.5	181.3	185.9	190.9	196.1	223.4
CO-OP suppliers total	443.4	449.5	463.2	478.6	492.8	506.7	519.4	533.7	547.9	620
Other WMA suppliers										
City of Rockville DPW	4.2	4.2	4.2	4.4	4.6	4.8	5.0	5.3	5.5	6.6
Loudoun Water ⁵	11.2	11.5	12.3	12.7	13.1	13.3	13.5	13.7	14.0	15.1
Subtotal	15.3	15.6	16.5	17.1	17.7	18.1	18.6	19.0	19.5	21.7
WMA suppliers total	458.7	465.1	479.7	495.7	510.5	524.8	538	552.7	567.4	641.7

¹ Actual water produced at the supplier water treatment plants.² 2023 value estimated by the same methods used in the forecasts.³ Charles and Howard County values taken from 2020 study assumptions.⁵ Assume 60 percent Fairfax Water purchase and 40 percent Loudoun Water production with a cap at 20 MGD (D. Geldert, private communication, August 16, 2024).

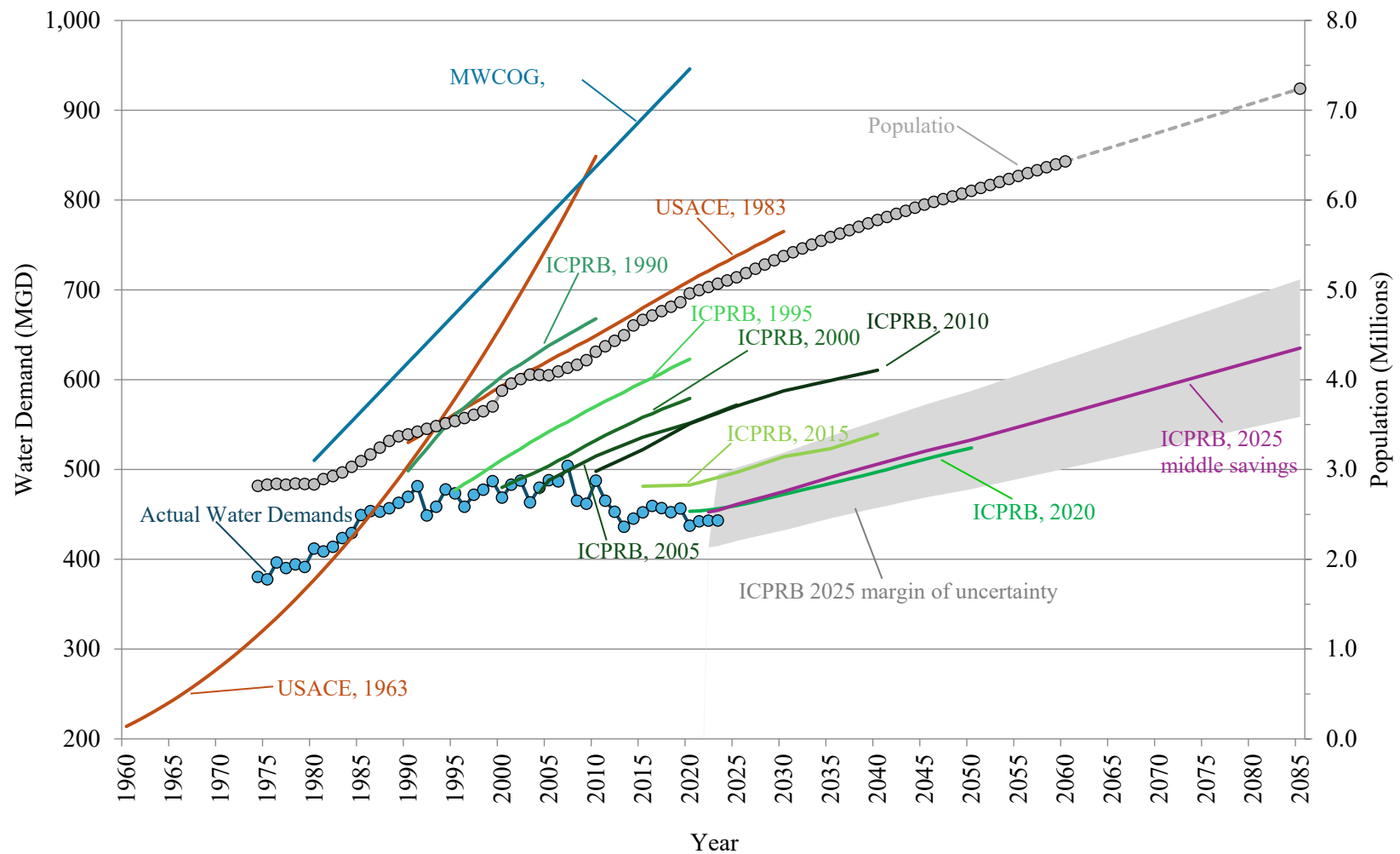


Figure 3-8: Comparison of this study's forecast of WMA total demand and range of uncertainty (grey shaded area) with forecasts from past studies (Rockville not included).

4 Modeling Daily Variations in Water Demands

4.1 Introduction

Water use in the WMA varies with season, day of the week, and hour of the day. In the summer and early fall, demand on any given day also depends on weather conditions, since people are more likely to water lawns and landscapes on hot, dry days. High summer demands often coincide with periods of low flow in the Potomac River. To assess whether the WMA water supply system will be able to meet future demands, seasonal and daily variations in demands are considered and combined with average annual demand forecasts presented in Chapter 3.

CO-OP's planning model, PRRISM, incorporates water demand models which are used to simulate the daily withdrawals of each of the three CO-OP suppliers and of Loudoun Water. The models add daily variations to the annual demand forecasts described in Chapter 3. These daily demand variations are obtained using a model developed from daily water production data for the years 2013-2023, provided by the four suppliers.

In PRRISM, daily water demands are a function of:

- simulation year
- month
- season
- weather conditions
- day of the week
- a daily error term based on an autoregressive integrated moving average (ARIMA) model

PRRISM simulates monthly variations by applying empirically derived monthly “production factors” to the annual demand forecasts to obtain monthly demand forecasts. Multiple linear regression models are used to add daily variations due to weather conditions and day of the week, with weather-related inputs derived from historical records and, when appropriate for scenario analyses, adjusted to reflect expected future meteorological changes. The sum of demand forecasts for the four suppliers is then further enhanced using an ARIMA model, which adds information to the forecast not captured by the regression model.

This chapter describes the structure of the daily demand model and the derivation from historical data of the model parameters used in PRRISM. The preparation of the raw data for the multiple regression and the ARIMA models is also discussed. Important features of the data preparation include the choice of predictor variables (independent variables), collation of data and pre-processing, detrending, and finally fitting the regression models for each supplier and season. Additionally, a range of alternative daily demand models is evaluated using machine learning methods. These include feedforward neural networks (FNN), long short-term memory networks (LSTM), gated recurrent units (GRU), a convolutional neural network–LSTM hybrid (CNN–LSTM), Random Forest, and Extreme Gradient Boosting (XGBoost). Each

model was trained using the same input data and evaluated with standard forecasting metrics to assess its potential for improving daily demand prediction in both planning and operational contexts.

4.2 Data Sources and Preprocessing

The daily demand variation models are based on daily production data (in MGD) representing water pumped from the treatment facilities of the three CO-OP suppliers, along with daily records of Loudoun Water's total water use. These data are used in place of billing records, which form the basis of the average annual demand forecasts discussed in Chapter 3, because billing data are typically only available on a monthly, quarterly, or, in some cases, annual basis. Appendix A.1 summarizes the production data as provided by each supplier. The regression models were developed using data from January 1, 2013, through December 31, 2023. This dataset reflects the most recent and complete period of record available at the time of the analysis.

National Weather Service data were used to represent weather conditions affecting water demand. These data include daily maximum temperature (°F), daily precipitation (inches), and the number of consecutive days with less than 0.15 inches of precipitation. Records were obtained from the [National Climatic Data Center](#) for the following stations: Washington Reagan Airport (USW00013743), College Park (USC00181995), Frederick Police Barracks (USC00183348), Laurel 3 W (USC00185111), and Vienna (USC00448737).

Precipitation data for each supplier were constructed as composite time series to fill gaps and ensure continuity:

- Fairfax Water and Loudoun Water: Vienna, Reagan, College Park, Laurel, Frederick Police Barracks
- WSSC Water: College Park, Laurel, Frederick Police Barracks, Reagan
- Washington Aqueduct: Reagan, College Park, Laurel, Frederick Police Barracks

Temperature data were composited using records from Reagan, College Park, and Frederick Police Barracks. The same composite temperature series was applied to all suppliers, with the order of stations listed above reflecting the priority used in assembling each series. To prepare inputs for the regression models, several feature engineering steps were applied to better represent how weather influences demand. Temperature was partitioned to capture nonlinear demand responses, with separate regression coefficients applied to temperatures above and below 85°F. Precipitation values were also transformed to limit the influence of large rainfall amounts, with values capped at 0.2 inches for WSSC Water and 0.3 inches for Washington Aqueduct, Fairfax Water, and Loudoun Water, based on empirical relationships established in previous iterations and validated through sensitivity checks in this study. Rainfall caps reflect the asymptotic relationship between rainfall and outdoor water use, recognizing that beyond a certain threshold, additional precipitation does not further reduce demand. While heavy rainfall events, including those from tropical systems, can play a significant role in replenishing soil moisture and ending extended dry periods, their immediate effect on daily water use is not directly proportional to the rainfall volume. The applied cap ensures that such events do not exert undue influence on short-term demand variability, while their broader hydrologic effects are captured elsewhere in the planning model (such as

through increased river flow and reservoir storage). These transformations help the models reflect more accurately behavioral and operational responses to varying weather conditions.

In addition to weather-related factors, seasonal and weekly variations in user behavior are incorporated into the modeling framework. To represent seasonal variation, separate regression models are developed for winter, spring, summer, and fall, rather than including season as a variable within a single model. Day-of-week effects are captured using a categorical variable in the multiple regression models. Broader societal influences, such as long-term trends in water use, are removed by detrending the raw data, as described in Section 4.3. These long-term trends are already accounted for in the annual demand forecast presented in Chapter 3.

4.3 Removing the Long-Term Time Trends

Long-term trends are evident in the historical production data of all four suppliers. These trends are due to a combination of factors, including population growth, decreases in flow rates of plumbing fixtures and household appliances, and changes in economic conditions. These factors (except for economic conditions) are included in the annual demand forecast, as described in Chapter 3. Therefore, these time trends are removed from the production data prior to the development of the daily variation regression models. This detrending procedure aims to ensure stationarity in the mean of the data. Detrending is done by subtracting a time trend from the production data to obtain a set of residuals that are then added back to the long-term stationary mean (explained later in this section). In this way, the time-dependent influences on demand that are not explicitly accounted for in the regression models (see Section 4.5) are removed.

The detrending procedure follows the method used by Steiner (1984), Kame'enui *et al.* (2005), Ahmed *et al.* (2010, 2015, 2020). A linear model is used for the time trends as represented by Equation 4-1,

$$Y(x) = mx + b \quad \text{Equation 4-1}$$

where Y is production in MGD, x is the time index of the corresponding production in days, m is the slope coefficient, and b is the intercept.

Statistics for the linear models for the long-term time trends for each utility, are reported on Table 4-1. The near zero coefficients of determination (R^2) and similarly valued standard error of estimate (SE) and standard deviation (SD) statistics show that the linear models are not significantly different from their respective mean production values ($\bar{Y}$) for the years 2013 through 2023. However, all slope coefficients, m, and intercepts, b, were significant (P-values ≤ 0.01). Based on a positive slope coefficient, m, Loudoun Water and Fairfax Water show some increase in total consumption over the past 11 years. Based on negative slope coefficients, m, WSSC Water and Washington Aqueduct show slight decreases in production although WSSC Water production remains relatively constant in comparison to Washington Aqueduct.

Table 4-1: Coefficients and goodness-of-fit statistics for the linear model for long-term trend in production for January 2013 through December 2023.

Statistic	Fairfax Water (w/o Loudoun Water)	WSSC Water	Washington Aqueduct	Loudoun Water
Intercept, b	130	162.6	137.1	21.4
Slope, m	0.0019	-0.0004	-0.0013	0.0015
Yearly Growth Rate (%)	0.51	-0.08	-0.34	2.27
Coefficient of Determination, R ²	0.0136	0.001	0.0099	0.0981
Standard Error, SE	18.4	13.2	14.7	5.4
Standard Deviation, SD	18.5	13.2	14.8	5.7
Mean, $\bar{Y}$ (2013-2023)	133.8	161.9	134.6	24.5

The last point on the long-term trend line, x' , is the long-term stationary mean, $\hat{Y}(x')$, to which all the residuals are added to form the trended (detrended) time series (see Figure 4-1). The result is a demand series that represents current conditions from which forecasts can be made and from which model parameters can be estimated. The point on the regression line corresponding to the most recent observation can be represented by Equation 4-2,

$$\hat{Y}(x') = mx' + b \quad \text{Equation 4-2}$$

where x' is the time index of the most recent observation, which is 4016 and corresponds to December 31, 2023. Table 4-2 shows that the linear formula in Equation 4-2 produces a long-term stationary mean that is close to the 2023 mean annual production from the data.

Table 4-2: Long-term Stationary Means (MGD) for 2025.

Supplier	2025 Linear $\hat{Y}(x')$ using 2013-2023	2023 Mean $\bar{Y}$ Production from data
Fairfax Water (w/o Loudoun Water)	137.5	136
WSSC Water	161.2	159
Washington Aqueduct	132	130.6
Loudoun Water	27.6	28

4.4 Deriving Daily Demand Variations

Daily variation in demand was chosen as the criterion variable (or dependent/response variable) for the multiple regression models. These variations represent short-term departures from expected monthly production levels and are used to capture the influence of temperature, precipitation, and weekly patterns on water use. After removing long-term trends from the production data (Section 4.3), the detrended daily values are compared to expected monthly means to isolate these short-term behavioral and weather-related fluctuations.

The derivation of daily demand variations involves two key intermediate quantities: monthly production factors and long-term monthly means. These are defined as follows:

Monthly production factors are calculated as the ratio of average monthly production to average annual production, using the period 2013-2023. These factors describe the typical seasonal pattern in water use and are used to disaggregate annual forecasts into monthly estimates. Values for each utility are listed in Table 4-3.

Long-term monthly means are obtained by multiplying each monthly production factor by the utility's long-term stationary mean, derived from the detrending analysis (Figure 4-1). These monthly means serve as the expected production baselines for each month under current demand conditions. Corresponding values are also shown in Table 4-3.

Daily variation in demand is calculated as the residual difference between the detrended daily production data and the corresponding long-term monthly mean for each day. This represents short-term deviations from expected production levels based on typical seasonal patterns. The detrended and monthly baseline time series are compared in Figure 4-2. Using daily variation as the dependent variable allows the PRRISM model to simulate potential daily deviations from future monthly production estimates. When these predicted daily variations are added back to the monthly disaggregation of the annual demand forecast, the result is a full-resolution daily production forecast that reflects both long-term planning and short-term dynamics. This formulation enables the generation of daily production forecasts that reflect both long-term planning assumptions and short-term variability driven by weather and behavior.

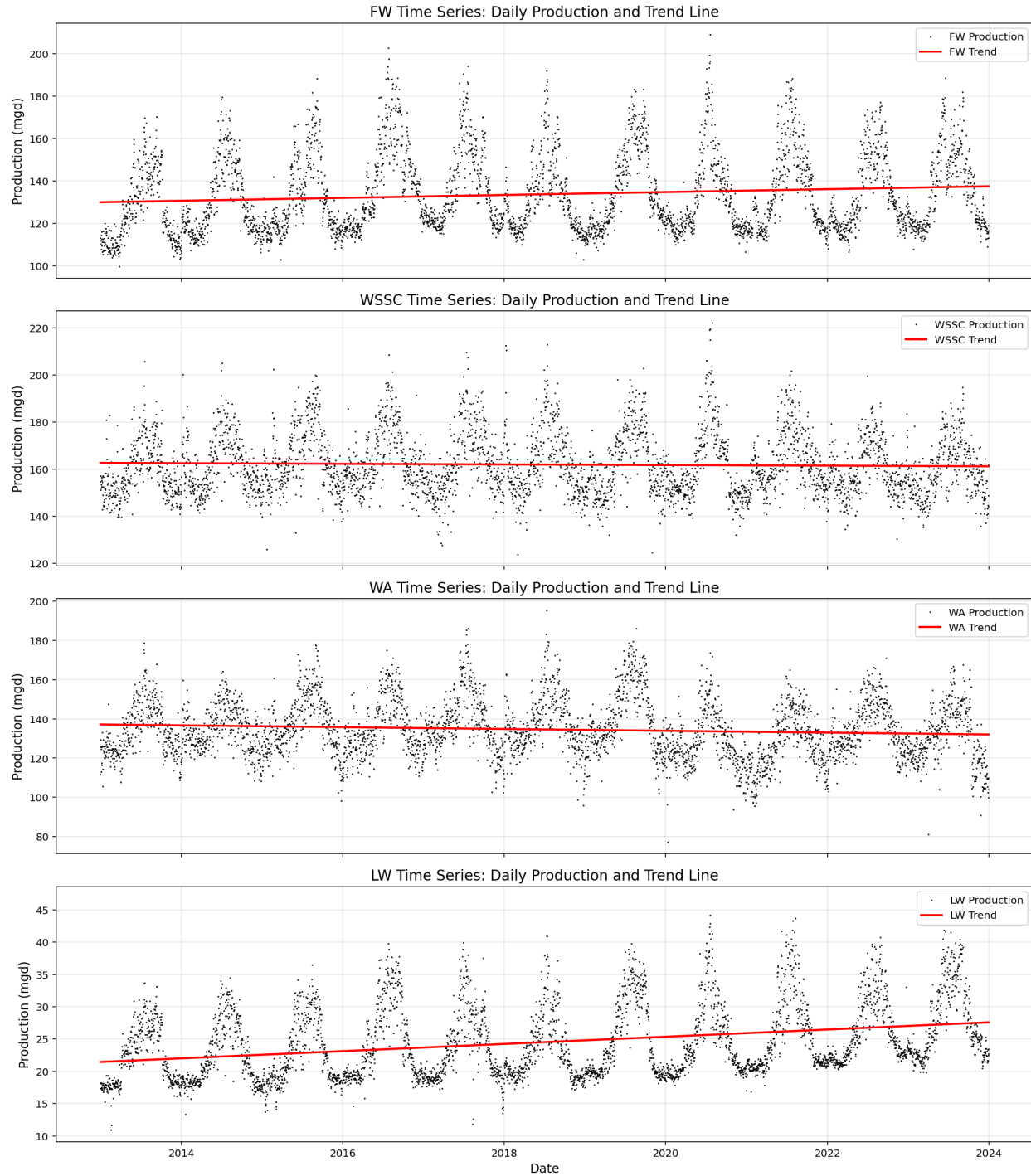


Figure 4-1: CO-OP supplier – Fairfax Water (FW), WSSC, Washington Aqueduct (WA) – and Loudoun Water (LW) production data fitted to a linear model (red line).

Table 4-3: Production factors and monthly means by supplier for years 2013-2023.

	Average Monthly Production Factors				Long-term Monthly Means, MGD			
Month	Fairfax Water	WSSC Water	Washington Aqueduct	Loudoun Water	Fairfax Water	WSSC Water	Washington Aqueduct	Loudoun Water
January	0.89	0.97	0.94	0.80	122.8	156.3	123.5	22.2
February	0.88	0.96	0.94	0.80	120.9	154.1	124.4	22.0
March	0.87	0.93	0.94	0.81	119.7	150.4	123.6	22.5
April	0.93	0.96	0.96	0.92	128.4	154.0	127.4	25.4
May	1.01	1.00	1.00	1.03	139.5	161.3	131.9	28.5
June	1.12	1.06	1.08	1.22	153.9	170.5	143.1	33.7
July	1.18	1.10	1.14	1.31	163.0	178.0	150.1	36.0
August	1.16	1.08	1.10	1.26	159.2	174.2	145.0	34.8
September	1.12	1.05	1.09	1.18	154.1	170.0	143.3	32.6
October	1.01	0.99	1.00	0.99	139.4	159.3	131.7	27.3
November	0.92	0.96	0.92	0.85	126.2	154.1	121.8	23.3
December	0.89	0.94	0.90	0.81	122.1	152.0	118.2	22.3
Annual					137.4	161.2	132.0	27.6

4.5 Regression Models

Seasonal multiple least squares regression analyses for daily variation explain the differences seen between daily demand and monthly average conditions. A generic form of a regression equation is as follows:

$$Y_t = b_0 + b_1 x_{1,t} + \dots + b_k x_{k,t} + N_t \quad \text{Equation 4-3}$$

where t is a time index representing the day of the time series, the criterion variable Y_t is the predicted daily variation in demand on day t , and the k predictor variables are $x_{1,t}$, ..., $x_{k,t}$. The residual (error) term in this equation is N_t , and the coefficients b_0 , ..., b_k , describe the fixed coefficients that modify the predictor variables.

The dependent variable Y is taken as the departure of detrended daily demand from detrended monthly average conditions. Variables examined as explanatory variables in the regression for the WMA water suppliers include temperature, both forecast and lagged by one day, precipitation, both forecast and lagged by one day, day of week (Sunday, Monday, Tuesday...etc.), and the number of days in a row without significant rainfall (defined as less than 0.15 inches).

The regression model specifications used in this analysis closely follow the approach established in previous studies for the WMA suppliers. Established non-linear relationships between water demand and key explanatory variables (temperature, precipitation, and consecutive dry days) were preserved and applied in this update.

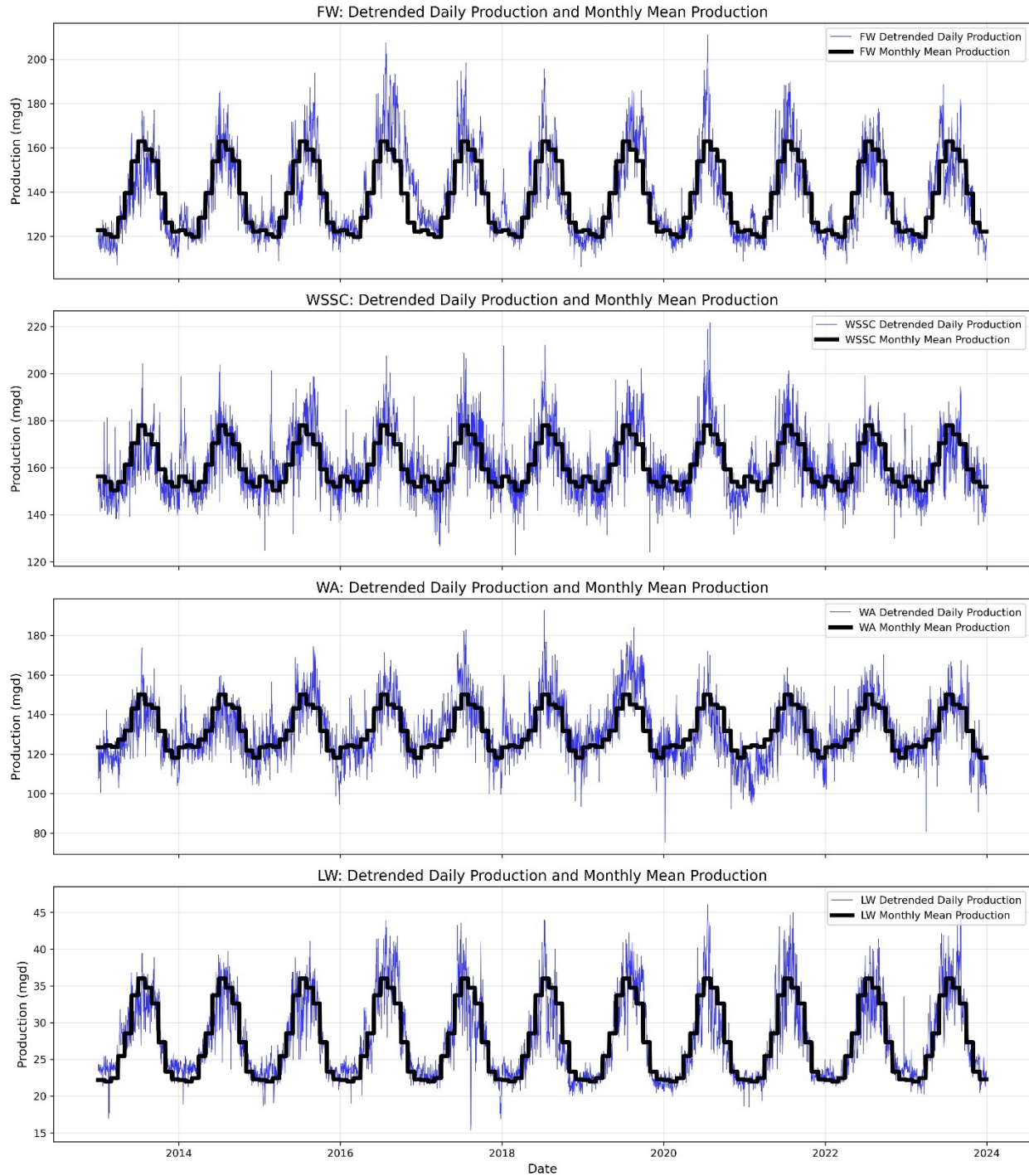


Figure 4-2: CO-OP supplier – Fairfax Water (FW), WSSC Water (WSSC), Washington Aqueduct (WA) – and Loudoun Water (LW) daily production data, detrended and compared to the monthly disaggregation of the long-term stationary mean.

Specifically, the response of demand to temperature is modeled as piecewise-linear, with a breakpoint at 85°F. This reflects the observed pattern that demand increases at a modest rate below 85°F but rises more steeply above that threshold. For this reason, forecast temperature, same-day temperature, and lagged temperature are each split at 85°F, with separate coefficients estimated for values above and below the threshold.

For precipitation, demand exhibits a saturation effect beyond certain thresholds. Following previous modeling efforts, precipitation values above 0.2 inches are capped at 0.2 inches for WSSC Water. For Fairfax Water, Washington Aqueduct, and Loudoun Water, the cap is 0.3 inches. This transformation ensures the regression model reflects the diminishing marginal impact of heavier rainfall on demand.

Backward stepwise regression methods are used to calibrate Equation 4-3. Predictor variables include forecasted and lagged temperature and precipitation, day of the week (Sunday, Monday, Tuesday, etc.), and the number of consecutive days without significant rainfall.

The set of seasonal regression models include a version of Equation 4-3 that simulates spring (March, April, and May), summer (June, July, and August), fall (September, October, November) and winter (December, January, February) daily water demand. Tables 4-4, 4-5, 4-6 and 4-7 summarize the different coefficients used in Equation 4-3 for the respective fall, summer, spring and winter seasons for this study. Figure 4-3 shows scatterplots of detrended production of the four suppliers versus predicted demand from the sets of regression models. These four graphs indicate that the regression models do a reasonable job predicting intermediate demands but tend to under-predict the highest demands and over-predict the lowest demands.

Table 4-4: Fall Regression Coefficients. Blank cells indicate variables that were not statistically significant and not included in the final model for that supplier after stepwise selection.

	Fairfax Water (w/o Loudoun Water)	WSSC Water	Washington Aqueduct	Loudoun Water
Intercept, b0	-8.90	-1.02	-1.82	-0.68
Maximum daily temperature >85 °F, one-day forecast	0.20	-0.10	-0.07	0.02
Maximum daily temperature <85 °F, one-day forecast	0.12	-0.13	-0.08	
Maximum daily temperature >85 °F		0.15	0.10	0.02
Maximum daily temperature <85 °F		0.10	0.08	0.01
Maximum daily temperature >85 °F, one day prior		0.03	0.05	
Daily precipitation, one-day forecast	-12.3	-2.92	2.13	-4.1
Daily precipitation, actual	-19.3	-2.15	4.28	-6.86
Daily precipitation, one day prior	-11.8	-17.2	-5.63	-3.61
No. of days in a row without significant precipitation	0.42	0.31	0.35	0.07
Day of week - Monday	2.28	3.00	-0.89	0.31
Day of week - Tuesday	-2.53	-0.08	0.78	-0.74
Day of week - Thursday	-2.52	-2.39	0.39	-0.80
Day of week - Saturday	-3.08	-2.31	-5.79	-1.28
Day of week - Sunday	-0.81	2.86	-8.43	-1.06
Standard Error of Estimate	9.33	7.78	8.99	2.52
Standard Deviation of Criterion Series	11.5	9.28	10.6	2.98
Coefficient of Determination (R²)	0.34	0.30	0.28	0.29

Table 4-5: Summer Regression Coefficients. Blank cells indicate variables that were not statistically significant and not included in the final model for that supplier after stepwise selection.

	Fairfax Water (w/o Loudoun Water)	WSSC Water	Washington Aqueduct	Loudoun Water
Intercept, b0	-54.4	-105	-82.7	-12.9
Maximum daily temperature >85 °F, one-day forecast	0.71	0.27		0.17
Maximum daily temperature <85 °F, one-day forecast	0.70	0.30		0.17
Maximum daily temperature >85 °F		0.60	0.45	
Maximum daily temperature <85 °F		0.61	0.45	
Maximum daily temperature >85 °F, one day prior		0.33	0.53	
Maximum daily temperature <85 °F, one day prior		0.34	0.54	
Daily precipitation, one-day forecast	-16.7	3.66	-3.41	-4.27
Daily precipitation, actual	-37.4	3.55	-0.80	-12.6
Daily precipitation, one day prior	-20.8	-41.4	-13.8	-5.44
No. of days in a row without significant precipitation	0.74	0.58	0.46	0.11
Day of week - Monday	1.16	0.98	-2.02	
Day of week - Tuesday	-6.55	-1.05		-1.82
Day of week - Thursday	-5.68	-2.95	-0.70	-1.80
Day of week - Saturday	-7.63	-2.80	-8.93	-2.79
Day of week - Sunday	-7.04	-0.80	-13.05	-2.69
Standard Error of Estimate	9.46	8.65	8.66	2.87
Standard Deviation of Criterion Series	14.27	11.44	11.52	4.02
Coefficient of Determination (R²)	0.56	0.43	0.43	0.49

Table 4-6: Spring Regression Coefficients. Blank cells indicate variables that were not statistically significant and not included in the final model for that supplier after stepwise selection.

	Fairfax Water (w/o Loudoun Water)	WSSC Water	Washington Aqueduct	Loudoun Water
Intercept, b0	-5.69	-3.15	-2.48	0.20
Maximum daily temperature >85 °F, one-day forecast	0.17		-0.05	0.04
Maximum daily temperature <85 °F, one-day forecast	0.09	-0.04	-0.07	0.01
Maximum daily temperature >85 °F	0.05	0.130	0.130	
Maximum daily temperature <85 °F		0.07	0.11	
Maximum daily temperature >85 °F, one day prior		0.07	0.08	
Daily precipitation, one-day forecast	-10.0	2.92	-2.11	-2.56
Daily precipitation, actual	-14.6	-8.61	-0.51	-3.72
Daily precipitation, one day prior	-7.54	-18.8	-7.45	-2.21
No. of days in a row without significant precipitation	0.34	0.05	0.09	0.06
Day of week - Monday	1.82	1.12	-1.79	0.06
Day of week - Tuesday	-1.66	1.62		-0.36
Day of week - Thursday	-1.63	-1.02	0.59	-0.36
Day of week - Saturday	-0.64	0.54	-3.9	-1.34
Day of week - Sunday	0.56	4.45	-6.88	-1.43
Standard Error of Estimate	6.76	7.08	7.77	1.83
Standard Deviation of Criterion Series	8.3	8.28	8.84	2.19
Coefficient of Determination (R²)	0.34	0.27	0.23	0.30

Table 4-7: Winter Regression Coefficients. Blank cells indicate variables that were not statistically significant and not included in the final model for that supplier after stepwise selection.

	Fairfax Water (w/o Loudoun Water)	WSSC Water	Washington Aqueduct	Loudoun Water
Intercept, b0	5.38	7.30	-2.26	1.50
Maximum daily temperature <85 °F, one-day forecast			0.05	
Maximum daily temperature <85 °F	-0.04			
Maximum daily temperature <85 °F, one day prior	-0.10	-0.17		-0.02
Daily precipitation, one-day forecast	-4.71		-0.86	-0.93
Daily precipitation, actual	-1.01	-5.92	-3.02	-0.21
Daily precipitation, one day prior	-0.32	-12.74	-6.54	-0.55
No. of days in a row without significant precipitation	0.09		0.15	
Day of week - Monday	2.15	2.74	0.23	0.45
Day of week - Tuesday	1.03	2.23	1.71	0.15
Day of week - Thursday	-0.23	-0.27	1.49	
Day of week - Saturday	1.77	-0.24	-3.31	0.17
Day of week - Sunday	4.34	6.04	-4.00	0.33
Standard Error of Estimate	5.39	8.09	9.46	1.22
Standard Deviation of Criterion Series	5.85	8.68	9.82	1.26
Coefficient of Determination (R²)	0.15	0.13	0.07	0.06

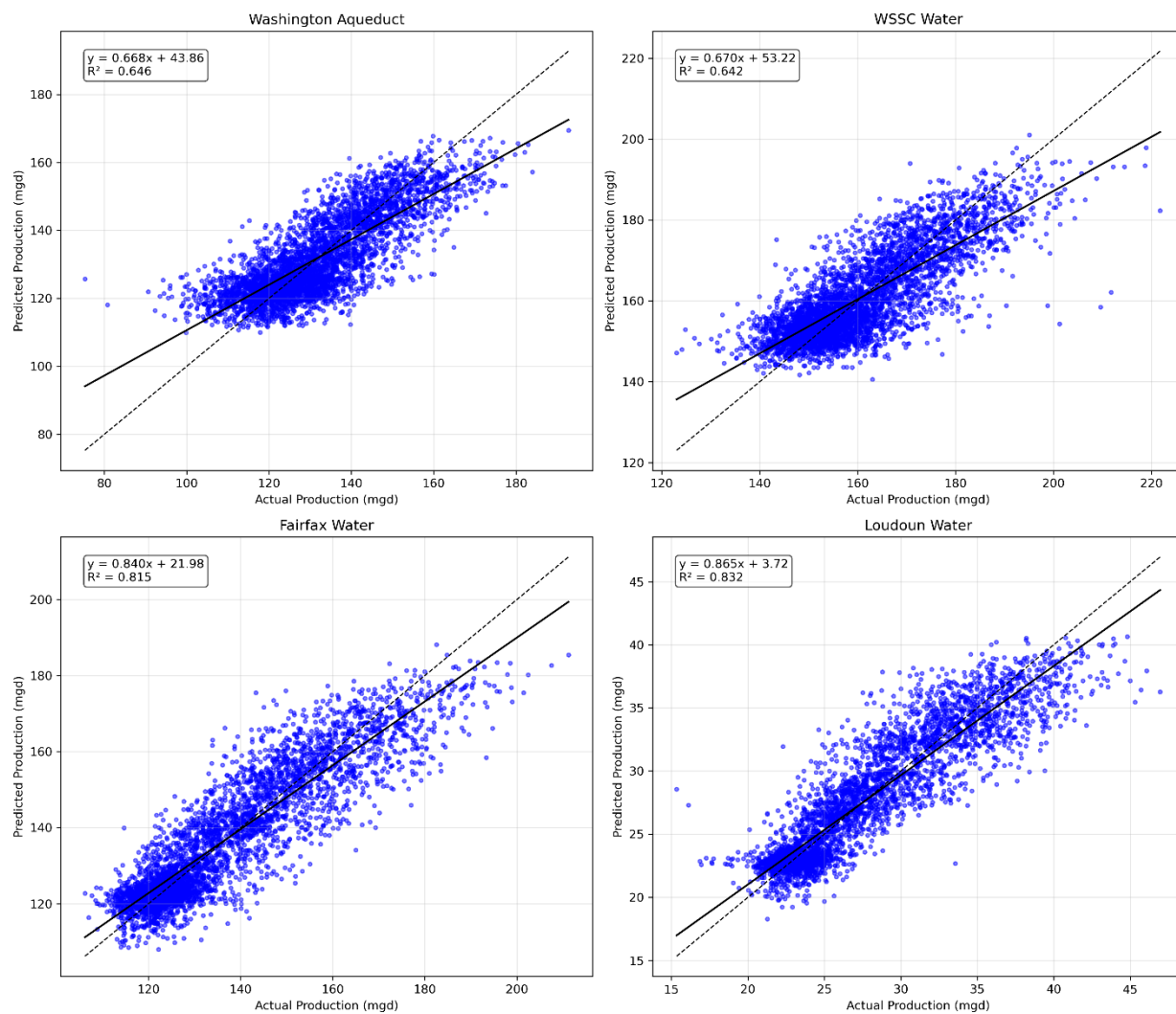


Figure 4-3: Regression model results compared with detrended actual data for the CO-OP suppliers and Loudoun Water.

4.6 ARIMA Model

The ARIMA model is used to help simulate the autocorrelations observed in demand time series because demand today is a good predictor of demand tomorrow. In Equation 4-4, the error term, N_t , from Equation 4-3 is separated into a random and a non-random component:

$$N_t = \text{ARIMA}_t + e_t \quad \text{Equation 4-4}$$

where ARIMA_t is the non-random portion of N_t calculated by the ARIMA process at time t , and e_t is the random component of N_t at time t . The non-random portion of the error term, N_t , is based on the ARIMA model for all four suppliers combined.

The data used for the ARIMA analysis covers 4016 days for the period January 1, 2013, to December 31, 2023, and is the prediction error calculated from the daily sum of Fairfax Water (without Loudoun Water), Washington Aqueduct, WSSC Water, and Loudoun Water. An ARIMA model trained on the summer days of the same period was also developed.

An optimization routine was developed to select the ARIMA model. The selected model is classified as an $\text{ARIMA}(p,d,q)$ model, where: p is the number of autoregressive terms, d is the number of non-seasonal differences needed for stationarity, and q is the number of lagged forecast errors in the prediction equation. Coefficients are fitted to the selected model such that:

$$N_t = \text{AR}_1 N_{t-1} + \text{AR}_2 N_{t-2} + \text{MA}_1 e_{t-1} + e_t \quad \text{Equation 4-5}$$

where AR_1 and AR_2 correspond to a p of two, and MA_1 corresponds to a q of one. Table 4-8 and Table 4-9 summarize the statistical significance of the terms of the chosen forecasting model for the full year trained model and the summer trained model. The p -values for the AR_1 , AR_2 , and MA_1 terms are generally less than 0.05 and are, therefore, significantly different from zero at the 95 percent confidence level. The estimated standard deviation of the input random variable, e_t , equals 13.8 for the full year model and 15.7 for the summer model.

Tables 4-8 and 4-9 also summarize the performance of the selected model in fitting the historical data. They display: (1) the root mean square error (RMSE), (2) the mean absolute error (MAE), (3) the mean error (ME). The first two statistics measure the magnitude of the errors. A better model will give a smaller value for these metrics. The last statistic measures bias. A better model will give a value closer to zero.

The ARIMA term is divided among the four suppliers based on the fraction of total detrended demand that each one has provided for the period 2013-2023. The four fractions are 0.35, 0.29, 0.30, and 0.06 for WSSC Water, Washington Aqueduct, Fairfax Water (without Loudoun Water), and Loudoun Water, respectively.

Table 4-8: Full year ARIMA(2,0,1) model for the four suppliers.

	AR1	AR2	MA1
Coefficient	1.02	-0.1	-0.58
Standard Error	0.04	0.03	0.04
T-statistic	25.94	-3.39	-16.53
p-value	<1.00E-06	6.98E-04	<1.00E-06
Root Mean Square Error (RMSE)	13.84		
Mean Absolute Error (MAE)	10.63		
Mean Error (ME)	-0.01		

Table 4-9: Summer ARIMA(2,0,1) model for the four suppliers.

	AR1	AR2	MA1
Coefficient	1.01	-0.1	-0.61
Standard Error	0.08	0.06	0.07
T-statistic	13.01	-1.67	-8.76
p-value	<1.00E-06	0.095	<1.00E-06
Root Mean Square Error (RMSE)	15.69		
Mean Absolute Error (MAE)	12.41		
Mean Error (ME)	-0.06		

4.7 Daily Demand Model Evaluation

Daily demands in PRRISM are simulated by summing the daily variation predicted by the seasonal regression models; Equation 4-3 with coefficients from Tables 4-4 through 4-7; the residual error term from the ARIMA model; Equation 4-5 with coefficients from Table 4-8 and 4-9; and the predicted monthly demands for the forecast year. The monthly demands for the forecast year are given by the product of the monthly production factors from Table 4-3 and the forecast of average annual demands from Chapter 3. The model can also be used to predict demands for historical years.

Table 4-10 summarizes some of the statistical characteristics of the production data, regression, and ARIMA model simulated time series for total system demand, including Loudoun Water, over the 2013-2023 time period, excluding the year 2020 due to pandemic-related anomalies in demand patterns. Results given for the ARIMA models (summer and full year trained) are averages of statistics from twelve different simulated time series, each of which was computed with a different random error time series, e_t .

Table 4-10: Comparison of statistics for total system demand time series for the period, 2013-2023.

Statistic	Actual	Regression	Full-Year ARIMA (12 runs)	Summer-Only ARIMA (12 runs)
Peak day factor	1.36	1.30	1.35	1.36
Maximum (MGD)	625	596	620	622
Mean (MGD)	459	459	458	458
90th Percentile (MGD)	529	526	529	529
Median (MGD)	446	442	448	448
10th Percentile (MGD)	410	413	404	402

The year 2020 was excluded from model evaluation (but retained for training) due to unique water demand patterns caused by COVID-19 pandemic behavioral changes that are not representative of normal operational conditions. While 2020 experienced extreme heat events that legitimately stress the water system, the demand response significantly exceeded what would be predicted based on meteorological conditions alone. Analysis of the top 10 peak demand days in the 2013-2023 period reveals that 2020 contributed three of the top six highest demand days, including the all-time system peak of 642.6 MGD on July 20, 2020. This peak occurred at 99°F average daily temperature, while a comparable day in 2016 (July 25) reached 100°F but lead to a demand of 604.7 MGD, a difference of 38 MGD that cannot be explained by meteorological variables alone. This excess demand likely reflects pandemic-related behavioral changes including increased residential water use from work-from-home policies, heightened lawn care and pool maintenance activities, and altered consumption patterns that are not predictable through normal demand forecasting models. Retaining 2020 in the training dataset ensures the model includes responses to extreme meteorological conditions, while excluding it from evaluation provides a more representative assessment of forecast accuracy under typical operational scenarios.

The ARIMA models, on average, do a good job in reproducing the system's peak-day factor as well as other time series statistics, including the 90th percentile demand and the 10th percentile demand. While previous demand studies have typically trained the ARIMA model on summer-only data to better capture peak demand behavior, in this study, the model is trained on full-year data. The summer-trained model more closely reproduces the observed system peak (622 MGD vs. 625 MGD actual), and both models nearly match the observed peak-day factor of 1.36. However, the full-year model demonstrates stronger overall performance, with lower error metrics (RMSE of 13.84 vs. 15.69; MAE of 10.63 vs. 12.41), more statistically significant coefficient estimates (Table 4-8), and better reproduction of the lower end of the demand distribution (e.g., 10th percentile).

Figure 4-4 shows total system demand (including Loudoun Water) in dry year 2023. In this graph, actual demands are compared to total demands predicted by the regression models with no random error term and four total demand time series predicted by the full-year ARIMA model, designated on the graph as A1 through A4. These four demand time series were produced with four different time series of random errors, e_t . In comparison, the ARIMA model designated on the graph as A0 uses an error term e_t calculated from the data and predictions. The graphs show how the ARIMA term maintains some of the qualitative characteristics of the time series, especially in the summer months.

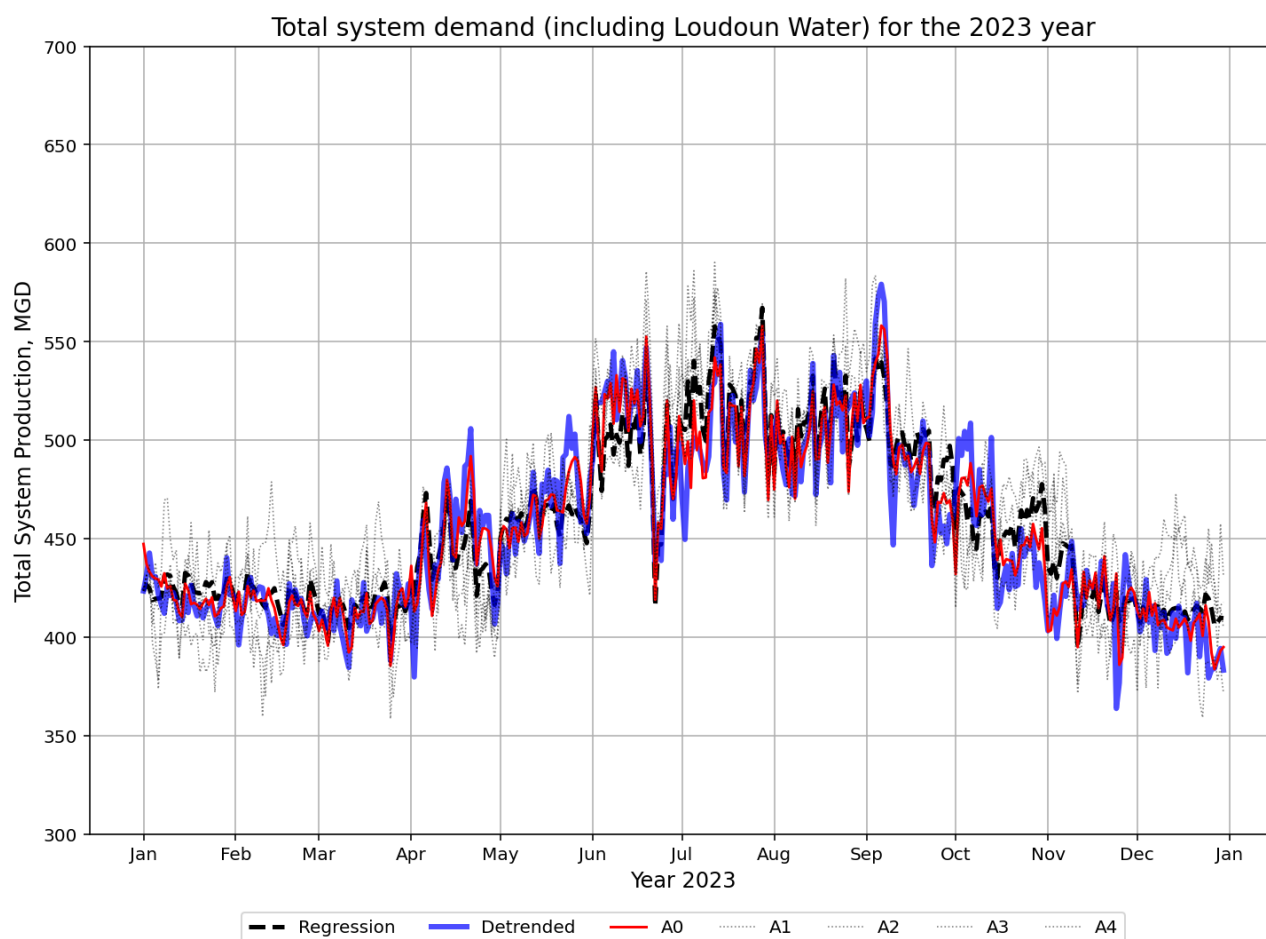


Figure 4-4: Total system demand (including Loudoun Water) for the 2023 dry year.

4.8 Evaluating the Use of Machine Learning Models

This section explores the use of alternative modeling approaches for simulating daily variations in water demand. While the PRRISM system continues to rely on the established linear regression and ARIMA method described in earlier sections, advances in machine learning have created opportunities to test newer methods that may better capture short-term variability in water demand and nonlinear relationships. Prior studies have shown that machine learning models such as artificial neural networks (ANNs), LSTM networks, and tree-based ensembles (e.g., Random Forest, XGBoost) can significantly improve forecasting accuracy compared to traditional statistical approaches, particularly for short-term demand prediction (Adamowski & Karapataki, 2010; Wang *et al.*, 2021; He *et al.*, 2023).

In this analysis, we evaluate whether tree-based ensemble models (random forests and gradient boosting), as well as neural network-based models, including feedforward networks, recurrent models (long short-term memory and gated recurrent units), and hybrid architectures, offer measurable improvements over the traditional linear regression approach. All models evaluated are trained and validated at the individual utility level to allow direct comparison with the current linear regression method. The purpose

is to benchmark emerging tools against a well-established framework and assess their potential for future modeling efforts.

As discussed in previous sections, water demand modeling in PRRISM has historically relied on a hybrid approach combining multiple linear regression and an autoregressive integrated moving average (ARIMA) model. In the current PRRISM configuration, the linear regression model captures the effects of temperature, precipitation, day-of-week, and seasonal indicators on daily variation in demand following the approach developed by Hagen and Steiner (2000) and subsequently refined by Ahmed *et al.* (2010, 2015, 2020). We applied this multiple linear regression approach to utility daily production data (response variable) and in addition, evaluated the following alternative approaches:

- **Feed-forward Neural Network (Multilayer Perceptron):** This model uses a sequence of fully connected dense layers to capture complex nonlinear interactions among predictor variables. Layers are individual processing steps that transform the input data and progressively extract useful patterns. The architecture implemented in this study includes four dense layers (128, 64, 32, and 16 units), with batch normalization and dropout regularization applied to reduce overfitting. In a dense layer, every input is connected to every unit, and the number of units (e.g., 128, 64) controls how many patterns the model can learn at that stage (larger numbers allow for more complexity, while smaller layers help refine and simplify the information). Batch normalization (a technique that standardizes the output of each layer to stabilize and speed up training) and dropout (a method that randomly disables some units during training to reduce overfitting) are included to improve training performance and generalization. The model processes one day of input at a time and does not explicitly account for time sequences. It aims to model nonlinearity and interactions not easily captured by traditional regression.
- **CNN–LSTM Hybrid Model:** This hybrid architecture combines a one-dimensional convolutional layer (Conv1D) with LSTM layers. The convolutional layer identifies short-term patterns in recent sequences (for example, abrupt weather changes), while the LSTM layers capture longer-term trends using a rolling 21-day input window. To enhance performance, we applied target standardization (rescaling the output variable), Huber loss (to reduce the influence of outliers), and increased weighting for high-demand days (to improve accuracy under peak conditions). The combined structure is designed to integrate both short- and long-range temporal signals and better reflect fluctuations in daily demand.
- **Bidirectional GRU Model:** This model is based on GRUs, a type of recurrent neural network similar to LSTMs but with fewer parameters and faster training. The GRU architecture used here includes two stacked bidirectional GRU layers with 64 and 32 units. Bidirectional layers allow the model to read the input sequence both forward and backward, enhancing its ability to capture asymmetries in the rise and fall of demand over time. Like the LSTM model, the GRU model uses a 21-day sequence window for prediction and includes preprocessing and weighting for high demands.
- **Random Forest:** This is a tree-based ensemble model that constructs multiple decision trees on bootstrapped samples of the data and aggregates their predictions. While Random Forests do not model time sequences directly, they can still be effective for short-term prediction when lagged variables and other engineered predictors are included. In this study, the Random Forest model

was trained using the same feature set as the linear regression model for comparability, including lagged precipitation and temperature features.

- **XGBoost:** This is a high-performance implementation of gradient boosted decision trees. It builds trees sequentially, where each new tree corrects errors made by the previous ensemble. XGBoost was applied using the same engineered feature set as the Random Forest and linear regression models, including lagged meteorological and demand variables. Regularization was also used here to control model complexity and reduce overfitting. XGBoost is often used for capturing complex nonlinear interactions and has proven particularly effective in structured tabular data scenarios, including short-term forecasting tasks.

All models were trained independently for each utility using the same set of initial predictor variables and consistent preprocessing steps, with refinements applied as needed to accommodate model structure.

Model performance was assessed using several standard time series evaluation metrics, including RMSE, which measures the average magnitude of prediction errors; MAE, which captures the average absolute deviation between predicted and actual values; ME, which assesses directional bias in predictions, and the Coefficient of Determination (R^2), which represents the proportion of variance explained by the model.

Metrics were calculated over the full time series from 2013-2021 for training and 2022-2023 for testing. This training/testing procedure was applied to all models, including the multiple linear regression model. In addition to the quantitative metrics, visual inspection of residual plots and time series overlays was used to assess each model's qualitative behavior and sensitivity to seasonal demand patterns and extremes.

4.8.1 Overall Model Performance

Figure 4-5 shows the performance of the different models for the overall system (summed daily totals across all utilities). The Feed-Forward Neural Network approach (referred to as Neural Network in the graph) emerged as the top performer with an R^2 of 0.894, followed closely by the Bidirectional GRU Model (referred to as GRU) ($R^2 = 0.889$) and XGBoost ($R^2 = 0.827$). The traditional Linear Regression achieved an R^2 of 0.821, while Random Forest performed similarly at 0.827. The CNN-LSTM Hybrid Model (referred to as LSTM) showed the poorest performance with an R^2 of 0.760.

RMSE values show similar rankings, with the Neural Network and GRU achieving the lowest RMSE values (14.06 and 14.20 MGD, respectively), substantially outperforming the traditional Linear Regression (18.23 MGD). Notably, LSTM exhibited the highest RMSE (20.85 MGD), indicating significant challenges both in training and generalization.

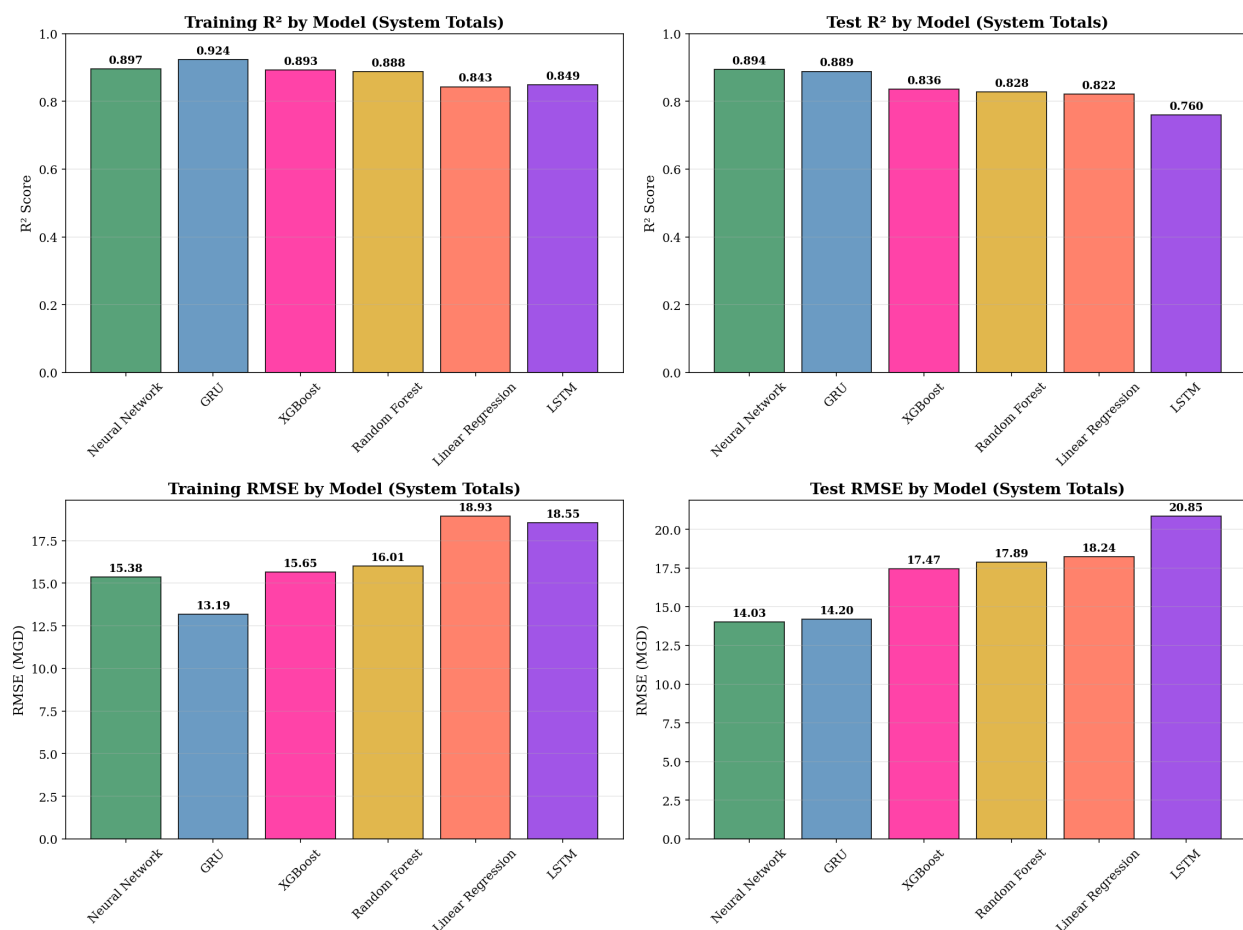


Figure 4-5: Model Performance for the overall system.

These results align with expectations based on model characteristics. The strong performance of the GRU and Feed-Forward Neural Network is consistent with their ability to capture nonlinear patterns and, in the case of GRU, sequential, time-dependencies. Tree-based models like XGBoost and Random Forest perform moderately well, as they are well-suited for structured tabular data with engineered lagged features but lack time-sequence awareness. The relatively weak performance of LSTM is not unexpected, as these models tend to require more tuning and data to generalize well, and are more prone to overfitting when data is limited or noisy.

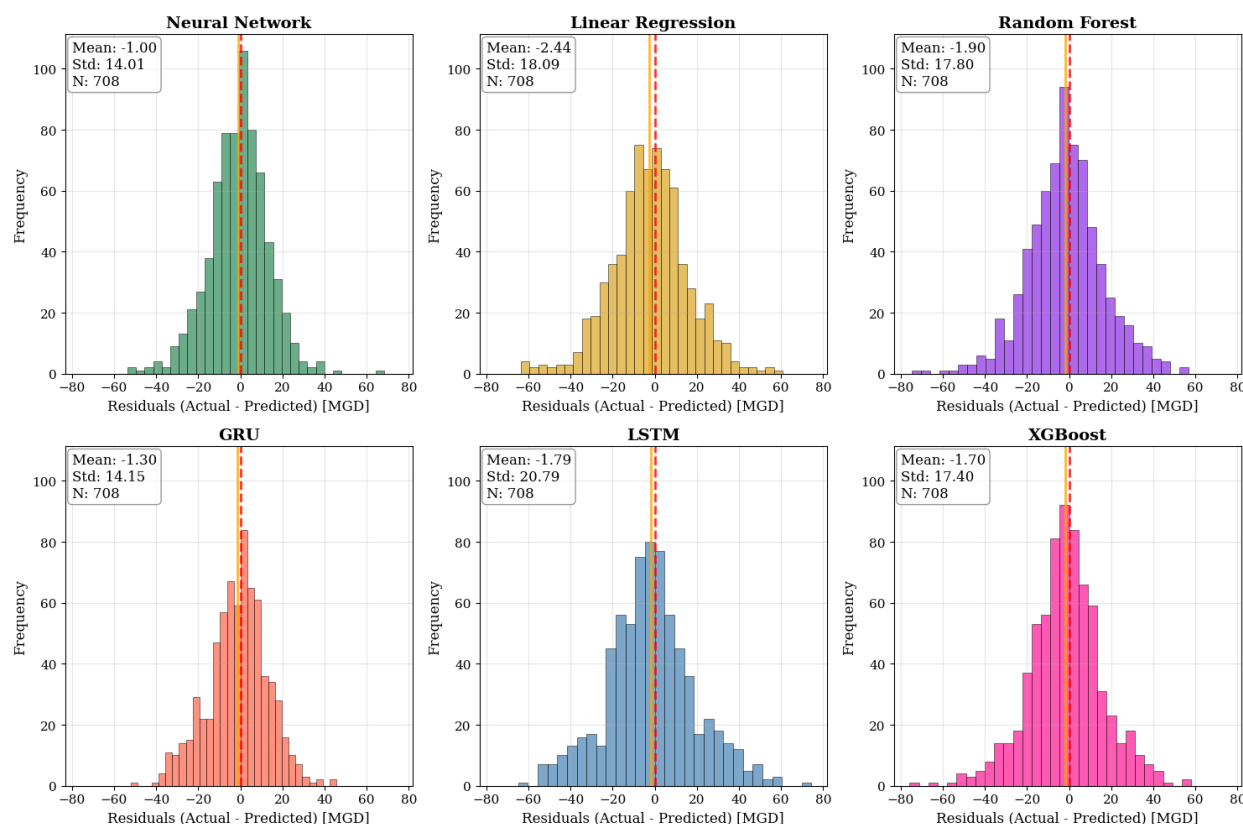


Figure 4-6: Residual distribution plots for each model, showing prediction error (actual minus predicted) across all test days.

4.8.2 Residual Analysis

Figure 4-6 presents the distribution of residuals for each model. The Feed-Forward Neural Network showed the most desirable pattern, with a near-zero mean residual (-1MGD) and the lowest standard deviation (std, 14 MGD), indicating minimal bias and tight error dispersion. The GRU model performed similarly well (mean: -1.3 MGD; std: 14.2 MGD), with both models producing near-symmetric, bell-shaped distributions centered close to zero.

In contrast, the Linear Regression model displayed a more noticeable bias (mean: -2.4 MGD), pointing to consistent under-prediction of demand. LSTM showed the widest spread (std: 20.8 MGD), suggesting difficulty generalizing to unseen data and overfitting. XGBoost had a reasonable residual profile (mean: -1.7 MGD, std: 17.4 MGD), with a well-formed normal shape, while Random Forest showed slightly higher dispersion (std: 17.8 MGD).

All models showed a tendency toward under-prediction, as reflected in negative mean residuals. This consistent bias may reflect difficulty in fully capturing high-demand days and indicate the need for additional or refined predictor variables.

4.8.3 Seasonal Performance Patterns

Figure 4-7 shows model performance by season. Winter was the most challenging period for all models, with notably lower R^2 values and higher RMSE. This may be due to several factors. In our region, winter water main breaks can lead to unaccounted-for water losses that distort observed demand. Additionally, key predictors such as temperature primarily influence outdoor water use, which is minimal in winter, reducing the strength of their relationship with demand. The Neural Network performed consistently well, maintaining an R^2 of 0.850 in fall and 0.410 in winter. GRU followed a similar pattern, with slightly better winter performance ($R^2 = 0.276$) compared to the remaining models. Most models performed best during summer, particularly GRU ($R^2 = 0.745$), neural network ($R^2 = 0.653$), and XGBoost ($R^2 = 0.641$). Spring results were moderate across all models. LSTM showed the steepest seasonal decline, including a negative R^2 in winter, indicating worse-than-mean predictions. XGBoost demonstrated the most stable seasonal performance, while neural network models were more sensitive to seasonal variation.

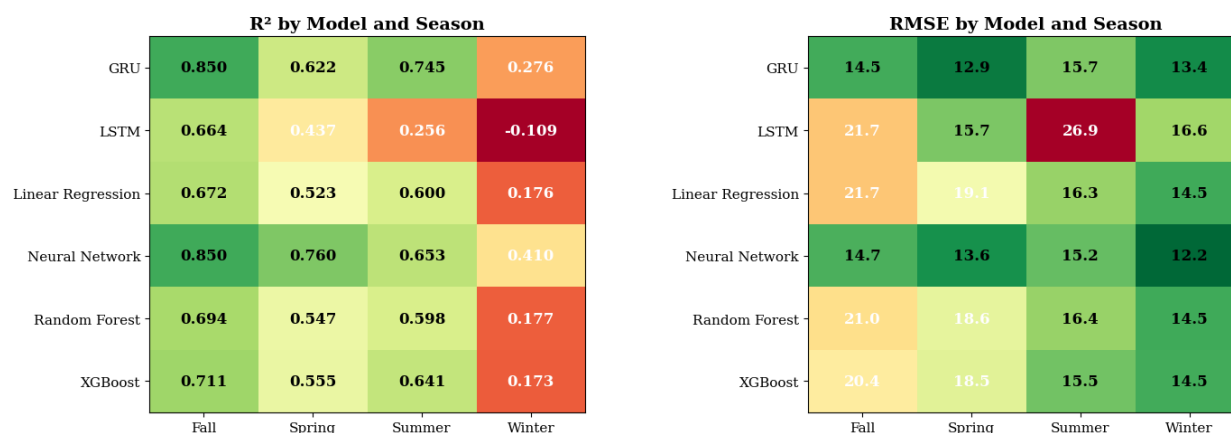


Figure 4-7: Seasonal performance.

4.8.4 Time Series Behavior and Temporal Patterns

Figure 4-8 illustrates model performance across a critical peak period (summer 2023), short-term variability (7-day rolling error), longer-term trends (30-day averages), and overall error distributions. The summer period time series (Figure 4.8a) shows that the Neural Network ($R^2 = 0.602$) tracked demand patterns more closely during peak periods, while Linear Regression ($R^2 = 0.515$) deviated more during high-demand events.

The 7-day rolling average of absolute error (Figure 4.8b) highlights the Neural Network's consistent advantage, with values generally staying below 15 MGD throughout the year. Linear Regression, in contrast, showed frequent spikes above 20 MGD. The 30-day rolling error (Figure 4.8c) confirms these differences, with the Neural Network maintaining a tighter error band (8-16 MGD) compared to Linear Regression (12-28 MGD). GRU showed intermediate performance, while LSTM had the highest variability and weakest results overall.

The error distribution comparison (Figure 4.8d) further supports these findings. The Neural Network showed the narrowest spread, centered near zero, indicating better accuracy and low bias. Linear Regression exhibited a wider distribution, and LSTM had the most erratic error pattern. These results are consistent with the residual patterns shown in Figure 4-6 and help explain why the Neural Network's R^2 advantage (0.883 vs. 0.790 full-year) translates into meaningful operational gains, typically 20 to 40 percent better daily forecasting accuracy than the traditional Linear Regression model.

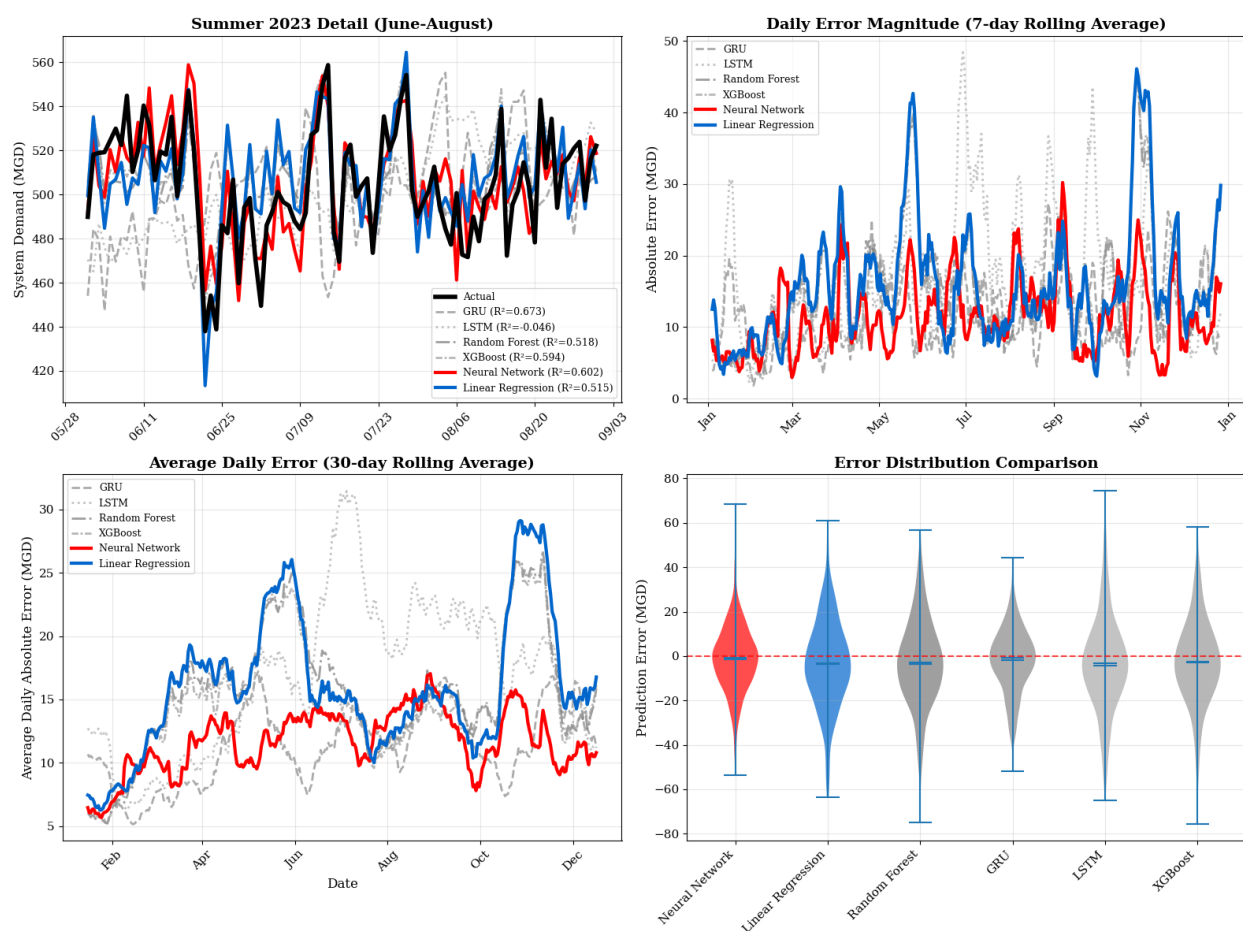


Figure 4-8: Model Performance Comparison - 2023 System Demand. Comparison showing (a) Actual vs modeled system demands during summer period (b) daily error trends, (c) average daily forecasting accuracy, and (d) error distribution patterns.

4.8.5 Summary and Recommendations

This analysis demonstrates that modern machine learning models, particularly feedforward neural networks and GRUs, offer clear improvements over the traditional linear regression approach currently used in PRRISM. The Neural Network achieved an R^2 of 0.894 and the GRU model 0.889, both notably higher than Linear Regression (0.821), along with 20 to 25 percent reductions in RMSE.

While Linear Regression remains valuable for its simplicity, transparency, and speed, deep learning models outperformed it in capturing nonlinear relationships, improving peak-period accuracy, and reducing error variability. Linear regression is currently used in both PRRISM and our real-time demand forecasting tools. Given the improved performance of the machine learning models, they are strong candidates for integration into these operational tools in the near future.

Neural networks were tested over a relatively short historical period, yet already show promise for improving daily demand forecasts, especially for operational use. Recommended next steps include expanding the training period to evaluate long-term performance, testing under severe drought, and exploring hybrid configurations with ARIMA to better capture both nonlinear patterns and autocorrelations.

5 Modeling System Resources and Operations

The primary purpose of this study is to evaluate whether the WMA water supply system will be able to meet the region's future water needs in the face of rising WMA demands, rising upstream consumptive use of water, and potential alterations to stream flows and to demands due to changing meteorological conditions. The evaluation is conducted using CO-OP planning model, PRRISM. This chapter provides an overview of the PRRISM model, which has recently been rewritten in an updated software platform. It also describes and updates information on key system components simulated by PRRISM: reservoirs and return flows from wastewater treatment plants. Also discussed are water treatment plant production capacities, production losses, and other system constraints.

5.1 PRRISM

CO-OP's planning model, PRRISM, simulates WMA system operations and key processes governing water supply and demand at a daily time step. PRRISM simulations include representation of the following:

- Potomac River and tributary flows
- Cumulative upstream consumptive demands, including estimates for data centers
- Discharges from WMA supplier WWTPs into the freshwater Potomac River and its tributaries and the Occoquan Reservoir
- Inflows, storage, and releases of each system reservoir
- Water demands, production, and withdrawals of each WMA supplier
- Transfers of treated water between WMA suppliers and within Fairfax Water's three service areas
- Forecasts of Potomac River flow at Little Falls today, tomorrow, and nine days in the future (to support decisions on a Patuxent load shift, a Little Seneca Reservoir release, and a Jennings Randolph Reservoir water supply release, respectively)
- Potential stream flow and withdrawal changes due to projected meteorological change
- Impact of state drought management decisions on upstream consumptive demand (currently Maryland and Virginia drought declarations only)

PRRISM has undergone a significant rewrite with the release of PRRISM version 4.1 (v4.1), which transitioned model implementation to the latest version of the ExtendSim simulation software package, ExtendSim 10. PRRISM v4.1 represents operational decision rules and a system structure that are almost identical to those of the version used in the 2020 water supply study, PRRISM v3.11. One minor difference is that Lake Manassas is not included in v4.1, at the request of Fairfax Water. Also, the only new resources represented in PRRISM v4.1 are Fairfax Water's Edgemon Reservoir, planned to be operational in 2040, and Loudoun Water's Milestone Reservoir, planned to be operational in 2028. When resource recommendations of the USACE feasibility study are made available, they will be added to the next version of PRRISM and likely evaluated in a special CO-OP study.

PRRISM v4.1 has several new features that enhance usability and model accuracy:

- Database-driven parameter and time series inputs, allowing for efficient selection and tracking of assumptions used in model runs
- Simplified menu-type selection of demand and meteorological scenarios, reducing the potential for user error in the setup of model runs
- Reconfiguration as a dual-purpose model, providing support for both planning studies and real-time operations
- Option for use of flow-dependent lag times for forecasts of today and tomorrow's Potomac River flows at Little Falls, providing a more realistic simulation of hydrograph travel times

In addition, PRRISM v4.1 incorporates the usual updates of input parameters and time series based on new data compilations and analyses conducted for the current study, including inputs determining annual demands (Chapter 3), daily demands (Chapter 4), upstream consumptive use (Chapter 6), the impacts of meteorological change on stream flows and demands (Chapter 7), as well as updated system constraints, WWTP return flows, and reservoir sedimentation rates (Chapter 5). It should be noted that the flow-dependent lag time option requires additional calibrations and verification and was not used in the current study. Also, PRRISM v4.1 also has a slightly revised calculation of the amount added to the Jennings Randolph Reservoir release for the purpose of keeping storage balanced in Jennings Randolph and Little Seneca reservoirs (see Schultz *et al.*, 2017). The balancing coefficient used in the current study is 500 when storage in Little Seneca is greater than or equal to 25 percent of capacity and 1000 otherwise, a change that led to a slight improvement in system performance. CO-OP's 2020 water supply study used a balancing coefficient of 1000 for all conditions. Additional optimization of the balancing algorithm should be considered for future studies.

5.1.1 Potomac River Flow

PRRISM simulates daily flows in future years based on daily conditions in historical years, with annual flow change factors applied to represent the future impacts of meteorological change (see Section 7.2). The starting point is construction of time series of Potomac River “natural” flows, that is, estimates of flows that would have occurred in the past without withdrawals, diversions, or reservoir regulations. To develop natural, historic daily Potomac River flows and reservoir inflows for PRRISM input, ICPRB staff members processed historic stream flow records (Hagen and Steiner, 1998a; Hagen *et al.*, 1998b; 1998c) from October 1, 1929, through December 22, 2009. This time period includes the extreme drought years of 1930 and 1966 and the moderate drought years of 1999 and 2002. As discussed in Section 7.1, the period, 1929 through 1979, is assumed to represent conditions before the onset of meteorological change. The impacts of meteorological change are “removed” from the daily flow time series for 1980 through 2009 using the methodology described in Section 7.2. After this step, the natural flow time series from 1929 through 2009 provide a representation of potential flows without the impacts of meteorological change for approximately 80 years.

To obtain an 80-year set of daily time series representing future Potomac River flows and reservoir inflows that reflect the potential impacts of meteorological change, annual flow-dependent scaling factors

are applied to each calendar year of the time series, as described in Section 7.2. This study uses potential flow scenarios for three future years: 2030, 2045, and 2050.

5.2 Environmental Flow-by

CO-OP's operational target during drought operations is the Potomac River environmental flow-by at Little Falls Dam. River flow at this location is measured at the [USGS stream gage Station No. 01646500](#). The environmental flow-by is based on a 1981 Maryland Department of Natural Resources study (MDNR, 1981). The USACE requested that the 1981 study provide a quantitative recommendation for a clause in the LFAA regarding the minimum flow needed downstream from Little Falls Dam to maintain environmental conditions. The 1981 study recommended a minimum daily average environmental flow-by of 100 MGD at Little Falls Dam. The study also addressed flow conditions below Great Falls, located about nine miles upstream of Little Falls Dam. It recommended that when flow below Great Falls drops below 500 MGD, Washington Aqueduct shift its withdrawal from the Great Falls intake to the Little Falls intake. This load shift would increase the flow between Great Falls and Little Falls Dam by the amount of the shift. At the time, Washington Aqueduct's summer withdrawals often approached 200 MGD, meaning the recommendation would have resulted in a minimum daily average flow of approximately 300 MGD between Great Falls and Little Falls Dam during low-flow periods.

In June of 2025, ICPRB convened a Potomac River Environmental Flow-By Task Force in response to an ICPRB Resolution of August 2023. The Resolution called for establishment of a collaborative process "to discuss and develop recommended processes, criteria and methodologies for assessing and determining the environmental and human needs to be considered in water supply operational and planning decisions for the Washington Metropolitan Area, and how to balance those needs in making operational and long-term planning decisions." Under direction of the Task Force, a preliminary assessment will be conducted to determine whether a new ecological flows study is needed, and if the answer is yes, a multi-step process will be conducted to develop a new recommendation for the environmental flow-by that balances ecological and human needs.

5.3 Reservoirs

5.3.1 Overview of Operations

This section provides an overview of operational strategies for the WMA water supply system, which relies on the Potomac River as its primary source and supplements Potomac withdrawals with daily withdrawals from the system's off-Potomac reservoirs in the Patuxent and Occoquan watersheds. When forecasts indicate that Potomac River flow will fall below combined water supply needs and the 100 MGD environmental flow-by at Little Falls Dam without operational changes, CO-OP staff evaluate the need for releases from upstream reservoirs and shifts of withdrawals to off-Potomac reservoirs to help maintain adequate flow, as described below. This section also discusses use of planned resources: the Milestone and Edgemon reservoirs.

Jennings Randolph and Little Seneca reservoirs, located upstream of the WMA water supply intakes, augment Potomac River flow when forecasted flow cannot meet combined withdrawal demands and environmental flow-by requirements. Since the CO-OP system began operating in the early 1980s, water

supply releases from these reservoirs have occurred three times: summer of 1999, summer of 2002, and fall of 2010.

WSSC Water and Fairfax Water use the Patuxent and Occoquan reservoirs daily in conjunction with the Potomac River to meet their demands. During droughts, the Patuxent and Occoquan reservoirs are operated based on CO-OP flow forecasts in coordination with Little Seneca Reservoir to maximize system-wide water supply reliability.

In normal years, system operators manage reservoirs to maintain a 95 percent probability that storage is at least 90 percent full on June 1. This strategy ensures that reservoirs provide maximum benefits during drought conditions. The following sections detail specific reservoir operations.

5.3.2 North Branch Reservoirs

Jennings Randolph and Savage reservoirs (referred to in this report as the North Branch reservoirs) are situated in the northwestern corner of the Potomac River basin on the North Branch (see Figure 2-1). These reservoirs serve four primary purposes: flood control, water quality enhancement, recreation, and water supply. The USACE's Baltimore District Office manages their operations in accordance with master operation manuals and incorporates input from the North Branch Potomac River Advisory Committee, which was established in 2005 to provide stakeholder feedback on reservoir management, surrounding public lands, and downstream flow levels (NPS, 2008).

Jennings Randolph Reservoir, the largest storage resource in the WMA system, holds 13.1 BG for CO-OP water supply storage and 16.2 BG for water quality storage. These storage segments function separately. The USACE's Baltimore District Office manages water quality releases on an ongoing basis to meet primary objectives, while also supporting whitewater boating, fishing opportunities, and recreational access at Jennings Randolph Reservoir. CO-OP staff members may initiate water supply releases on behalf of CO-OP suppliers in accordance with the WSCA and the operating rules in the WSCA's Operations Manual.

Savage Reservoir operates in coordination with Jennings Randolph Reservoir, typically releasing water at a one-to-five ratio for both water quality and water supply. Unlike Jennings Randolph, Savage Reservoir does not have designated water supply storage allocation for the WMA, though it supplies water to the Town of Westernport, Maryland, and the three major WMA water suppliers fund 80% of the Savage Reservoir operation and maintenance costs. The combined release from Jennings Randolph and Savage reservoirs is measured at the USGS stream gage ([Station ID 01598500](#)) in Luke, Maryland.

PRRISM simulation of Jennings Randolph Reservoir's water quality releases was developed by CO-OP staff in coordination with the USACE's Baltimore District Office (Ahmed *et al.*, 2010). The model incorporates all mandated storage elevation targets using a stepped rule table that adjusts releases when reservoir elevations drop below target levels. Earlier versions of PRRISM also included summer recreational elevation and environmental flow targets recommended by the North Branch Potomac River Advisory Committee. However, the USACE Baltimore District Office has recently prioritized maintaining downstream cold-water temperatures to support local fishery habitats, resulting in the removal of some previously used targets for this version of PRRISM.

During droughts, CO-OP evaluates flow forecasts and demand projections to determine when North Branch reservoir releases from water supply storage are needed to meet WMA demands and the Little Falls environmental flow-by. Because these reservoirs sit approximately 200 miles upstream of the WMA intakes, CO-OP staff members must initiate North Branch reservoir water supply releases approximately nine days in advance to account for travel time downstream.

5.3.3 Use of River Flow Forecasts

Upstream reservoir releases require significant travel time to reach the WMA, and certain operational changes involve lag times before taking effect. To optimize system management, CO-OP relies on flow and demand forecasts to determine reservoir release rates and coordinate system operations. PRRISM incorporates these forecasts to simulate operational decisions.

5.3.3.1 Nine-day forecasts of river flow

CO-OP bases water supply releases from Jennings Randolph and Savage reservoirs on nine-day forecasts of Potomac River flow at Little Falls, using an empirical equation derived from historical drought flows (Ahmed *et al.*, 2015). The PRRISM recession equation estimates natural flow at Little Falls in 9 days, excluding Jennings Randolph and Little Seneca contributions, and accounts for seasonal adjustments. The final forecast sums the estimated natural flow, water quality releases, and wastewater treatment plant discharges while subtracting managed upstream consumptive use. This forecast helps determine potential flow deficits at Little Falls in 9 days.

5.3.3.2 One-day forecasts of river flow

CO-OP uses one-day river flow forecasts to determine Little Seneca Reservoir release rates and Fairfax Water load-shifts between Potomac River and Occoquan Reservoir intakes that will meet water needs tomorrow. CO-OP estimates that it takes approximately one day for a release from Little Seneca and a Fairfax Water Potomac intake withdrawal change to reach Little Falls. These short-term adjustments help maintain system reliability by correcting errors in the nine-day forecasted releases and ensuring adequate flow at Little Falls.

- If too much water was released nine days earlier, a Fairfax Water load-shift to the Potomac can recover some excess by conserving Occoquan storage.
- If too little water was released due to an overestimated nine-day forecast, a Little Seneca release and Fairfax Water load-shift to Occoquan can compensate by augmenting Potomac River flow.

Fairfax Water Load-Shift

- A change in Potomac (Corbalis WTP) withdrawals takes approximately 15 hours to be reflected at Little Falls Dam
- A transfer from Occoquan (Griffith WTP) to Potomac (Corbalis WTP) takes 4-25 hours to implement
- Griffith WTP production is constrained to 45-120 MGD (max daily increase of 15 MGD)
- Corbalis WTP production is constrained to 60-225 MGD
- PRRISM operates on a daily time step, rounding impact times to the nearest day

PRRISM's one-day forecast of flow at Little Falls mirrors the approach used in CO-OP's drought operations tools, using an upstream flow accumulation model based on flows observed at the USGS's

Point of Rocks and Monocacy gages, and assuming a constant lag time of 2.3 days from these gages to Little Falls Dam (see Ahmed *et al.*, 2015). PRRISM v4.1 also has the option to implement more realistic flow-dependent lag times for the one-day and current day forecasts, but the flow-dependent lag time model is still undergoing calibration and was not implemented in the current study.

One-day forecasts are also used in PRRISM to simulate Edgemon Reservoir operations. Future studies may refine this assumption as new data becomes available for this planned resource.

5.3.4 Current Day River Flow

CO-OP uses forecasts of today's flow at Little Falls to determine whether to request WSSC Water to shift a portion of its withdrawals between its Potomac River and Patuxent intakes. If these "0-day" forecasts indicate that river flow on a given day will fall short of meeting WMA demands plus the 100 MGD flow-by at Little Falls, CO-OP staff will request WSSC Water to shift withdrawals from their Potomac to their Patuxent intake. This adjustment increases flow at Little Falls, helping prevent a Potomac River flow deficit and ensuring daily flow remains above 100 MGD. Conversely, if forecasts indicate that a surplus of flow is expected today, WSSC Water will be requested to shift more of its withdrawal to its Potomac intake to help conserve Patuxent reservoir storage.

WSSC Water Load-Shift

- If Little Falls flow nears 140 MGD (217 cfs), Patuxent withdrawals can increase to reduce reliance on the Potomac
- Under low-flow conditions, a change in Potomac withdrawals takes approximately 10 hours to affect Little Falls flow
- Potomac WFP production is constrained to 100-283 MGD
- Patuxent WFP production is constrained to 33-62 MGD

5.4 Reservoir Sedimentation and Capacities

Reservoir storage capacities decline over time due to sediment deposition, which varies with hydrologic conditions and is most pronounced during major storm events. PRRISM models this decline using estimated sedimentation rates for each reservoir. Table 5-1 presents current and projected reservoir storage along with the sedimentation rates assumed in this study. Since the 2020 study (Ahmed *et al.*, 2020), one new bathymetric survey has been conducted. An updated usable capacity and sedimentation rate are provided for Occoquan Reservoir based on a 2020 bathymetry study (Charles P. Johnson & Associates, Inc., 2020). The usable capacity and sedimentation rates for the Patuxent, Little Seneca, Jennings Randolph, and Savage reservoirs remain unchanged from the 2020 study.

5.4.1 Occoquan Reservoir

Sedimentation in the Occoquan Reservoir has been monitored through periodic bathymetric surveys, with the most recent comprehensive survey conducted in 2020. That survey measured the reservoir's normal pool volume at 8.50 BG, an increase of 0.17 BG from the previous survey. This increase suggests that sediment flushing may occur at times, helping preserve storage capacity. Of this total, 8.17 BG is considered usable, with 0.33 BG classified as dead storage, lying below the lowest intake elevation. This is consistent with the 2020 stage-storage curves provided by Fairfax Water (G. Prelewicz, personal

communication, August 29, 2023). For future projections, a sedimentation rate of zero million gallons per year (MG/yr) is assumed, reflecting the stable storage capacity observed over the past 25 years (Charles P. Johnson & Associates, Inc., 2020).

Table 5-1: Effects of sedimentation on reservoir storage capacities.

Reservoir	Baseline year	Usable capacity in baseline year (MG)	Projected usable capacity (MG)					Sedimentation rate (MG/yr)
			2025	2045	2050	2060	2085	
Occoquan¹	2020	8,170	8,170	8,170	8,170	8,170	8,170	0
Patuxent²	2015	10,530	10,284	9,792	9,669	9,423	8,808	24.6
Little Seneca³	2010	3,903	3,843	3,763	3,743	3,703	3,603	4
Jennings Randolph - water supply⁴	2013	13,098	12,857	12,456	12,356	12,156	11,654	45 (distributed between water supply and quality storages ⁶)
Jennings Randolph - water quality⁴	2013	16,295	15,996	15,497	15,371	15,122	14,499	
Savage⁵	2005	6,241	5,881	5,521	5,431	5,251	4,801	18

¹ Baseline usable capacity and sedimentation rate from Charles P. Johnson & Associates, Inc. (2020).

² Baseline usable capacity and sedimentation rate from Van Ryswick and Sylvia (2015).

³ Baseline usable capacity and sedimentation rate from Ortt, et al. (2011). Values will be revised upon completion of the Little Seneca Reservoir forebay dredging project.

⁴ Baseline usable capacity based on the 2013 revised stage-storage curve provided by USACE's Baltimore District Office (B. Haines, personal communication, January 16, 2014).

⁵ Baseline usable capacity and sedimentation rate from Kame'enui et al. (2005).

⁶ 44.56% for water supply and 55.44% for water quality.

5.4.2 Patuxent Reservoir

Bathymetric surveys of WSSC Water's Triadelphia and T. Howard Duckett (Rocky Gorge) reservoirs were conducted in April 2015 by the Maryland Geological Survey (MGS) using the same methods as the 2004/2005 survey to ensure consistency. In 2015, Triadelphia's volume was measured at 6.45 BG and Duckett's at 5.49 BG, for a total of 11.94 BG. Of this, WSSC Water designates 0.77 BG in Triadelphia and 0.60 BG in Duckett for flood storage, while 0.04 BG is considered unusable storage, which is a reduction from the previous estimate of 0.40 BG.

By comparing survey results, MGS estimated annual sedimentation rates of 19.6 MG for Triadelphia and 5.0 MG for Rocky Gorge, leading to a combined annual capacity loss of 24.6 MG. The resulting usable storage, or "normal storage" capacity, was 10.53 BG in 2015. These findings were incorporated into the current study's modeling as a conservative estimate of sedimentation impacts.

5.4.3 Little Seneca Reservoir

The most recent bathymetric survey of Little Seneca Reservoir was conducted in July and August of 2010 by the MGS for the Maryland Department of Natural Resources (Ortt et al., 2011). It found a total storage capacity of 3.922 BG at the mean pool elevation (385 ft MSL) and a usable capacity of 3.903 BG. This

exceeds the 3.86 BG measured in a 1996 survey, likely due to higher data resolution in 2010. Based on pre-construction topography and 2010 data, the sedimentation rate was estimated at 4 MG/yr.

The reservoir's three forebays are effectively trapping incoming sediment; however, by 2019 they were nearing full capacity, with remaining capacities estimated at 16 percent (Ten Mile Creek), 29 percent (Cabin Branch), and 34 percent (Little Seneca Creek) (WSSC Water, 2021). To maintain their function, WSSC Water is planning a dredging project, with design and permitting to be completed in Fiscal Year (FY) 2023-2025 and construction anticipated for FY 2026-2027. These plans are informed by 2019 field investigations. Following dredging, CO-OP may revise its estimate of the reservoir's sedimentation rate in consultation with WSSC Water.

5.4.4 North Branch Reservoirs

Sedimentation rates and storage capacities for Jennings Randolph and Savage reservoirs remain unchanged from the 2015 study (Ahmed *et al.*, 2015). That study included a detailed discussion of historical bathymetric surveys and storage allocations for Jennings Randolph Reservoir.

Originally, Jennings Randolph had a conservation pool storage volume of 94,700 acre-feet (ac-ft; 30,860 million gallons, MG) (USACE, 1997), with 2,700 ac-ft reserved for dead storage and long-term sediment accumulation over a 100-year period (USACE, 1986). This left 92,000 ac-ft (29,978 MG) of usable conservation pool storage, allocated into 40,995 ac-ft (13,358 MG) for water supply and 51,005 ac-ft (16,620 MG) for water quality (Future Storage Agreement, 1982).

A 2013 hydrographic survey led to a revised storage allocation in 2014 (USACE, 2014), reflecting updated measurements:

- Usable conservation storage: 90,203 ac-ft (29,397 MG)
- Water supply: 40,194 ac-ft (13,099 MG)
- Water quality: 50,009 ac-ft (16,298 MG)

These revised values maintain the original storage proportions: 44.56% for water supply and 55.44% for water quality. Agreements with the USACE include provisions for future reallocation if sedimentation reduces capacity.

5.5 Treated Wastewater Return Flows

Most of the area's wastewater is treated at DC Water's Blue Plains Advanced Wastewater Treatment Plant, which discharges into the Potomac estuary south of Washington, DC. However, several other facilities serving the WMA discharge upstream of the region's water supply intakes. These upstream discharges contribute treated wastewater that can be reused downstream.

PRRISM incorporates projected return flows from the following facilities: WSSC Water's Seneca and Damascus Wastewater Resource Recovery Facilities (WRRFs), Loudoun Water's Broad Run Water Reclamation Facility (WRF), and the Upper Occoquan Service Authority (UOSA) Wastewater Treatment Plant. The Seneca, Damascus, and Broad Run facilities discharge into the Potomac River or its tributaries; UOSA discharges upstream of the Occoquan Reservoir. Table 5-2 lists the projected annual return flows for these facilities.

Wastewater return flows vary seasonally, typically reaching their lowest levels during summer. Since reservoir releases from Jennings Randolph and Little Seneca are most critical during these low-flow periods, PRRISM adjusts for seasonal variation by applying monthly production factors.

These monthly factors are derived using return flow data from 2019 to 2023. For each facility, the ratio of monthly return flow to annual average flow is calculated for each year. To ensure a conservative estimate, the minimum ratio for each month across the five-year period is used. These factors convert annual return flow projections (Table 5-2) into monthly estimates and ensure that PRRISM reflects lower return flows during critical periods. Table 5-2 shows the final monthly production factors for each facility.

*Table 5-2: Past and projected treated wastewater return flows (MGD) from Seneca and Damascus WRRFs, the Broad Run WRF and the UOSA WWTP to the Potomac River and Occoquan Reservoir.**

Year	Seneca WRRF to Potomac River¹	Damascus WRRF to Potomac River¹	Broad Run WRF to Potomac River²	UOSA WWTP to Occoquan Reservoir³
2025	15.95	0.85	8	35.14
2030	17.20	0.85	10	35.70
2035	17.88	0.88	12	36.26
2040	18.30	0.89	13	36.82
2045	18.99	0.93	13	37.39
2050	(19.50)	(0.95)	14	37.95
2060	(21.44)	(1.04)	(16)	(39.92)
2085	(23.37)	(1.12)	(16)	(41.88)

¹ Data provided by WSSC Water (K. Six, personal communication, February 2024).

² Data provided by Loudoun Water (Loudoun Water Correspondence, September 2025).

³ Data obtained from UOSA by Fairfax Water. FY 2023 onward includes Micron Fab II, Phase II. Estimates may differ from Master Plan (A. Spiesman, personal communication, March 2024).

* Values in parentheses are linearly extrapolated from the three preceding forecasts. For Broad Run, future values are held constant at the 2050 level.

Table 5-3: Monthly production factors for treated wastewater return flows for Seneca and Damascus WRRFs, the Broad Run WRF, and the UOSA WWTP to the Potomac River and Occoquan Reservoir.

Month	Seneca WRRF Factors (average of 2019-2023)¹	Damascus WRRF Factors (average of 2019-2023)¹	Broad Run WRF Factors (minimum of 2019-2023)²	UOSA WWTP Factors (minimum of 2019-2023)³
January	1.05	0.86	0.85	1.02
February	1.05	0.88	0.84	1.03
March	1.04	0.88	0.91	0.90
April	1.00	0.83	0.94	0.94
May	1.02	0.85	0.86	0.96
June	0.98	0.77	0.83	0.91
July	0.97	0.74	0.75	0.86
August	0.97	0.73	0.80	0.88
September	0.97	0.76	0.81	0.83
October	0.95	0.73	0.78	0.88
November	0.98	0.75	0.86	0.91
December	1.02	0.85	0.97	0.89

¹ Data provided by WSSC Water (K. Six, personal communication, February 2024).

² Data provided by Loudoun Water (Loudoun Water Correspondence, March 2024).

³ Data obtained from UOSA by Fairfax Water (A. Spiesman, personal communication, March 2024).

Wastewater discharge from the Broad Run WRF is affected by Loudoun Water's non-potable reuse system, as reflected in the discharge estimates provided in Table 5-4 (Loudoun Water Correspondence, March 2024). Average annual reclaimed water demand increased from 1.69 million gallons per day (MGD) in 2019 to 2.23 MGD in 2023. July typically represents the month with the highest reclaimed use, with average demands ranging from 2.69 to 3.65 MGD during that period. However, the peak day of demand can occur anytime between June and September. As additional data center customers come online, reclaimed water demand is expected to continue increasing over the next five years. This growth results in higher overall water demand for both potable and non-potable uses, and it reduces discharge from the Broad Run WRF due to evaporative losses at data centers. Future studies may reevaluate how non-potable water reuse is represented in WMA system modeling in order to more accurately reflect these effects.

5.6 Water Production Losses

Water losses from the raw water treatment process are incorporated into PRRISM. The methodology used by ICPRB to represent these losses is described by Ahmed *et al.* (2015). These losses occur due to the removal and transport of residual solids generated during treatment, including sediment from the source water and solids produced by coagulants used in the treatment process.

Historically, water containing these residuals was returned to the source water body, resulting in minimal net loss. However, beginning in the 1990s, increasing concerns about water quality led to regulations that restricted these discharges. In 1996, WSSC Water entered into an agreement with the Maryland Department of the Environment to construct solids removal facilities at its Potomac Water Filtration

Plant. This agreement included exceptions for periods of high flow in the Potomac River. In 2015, a Consent Decree was issued requiring WSSC Water to implement short-term measures to significantly reduce solids discharges and to develop a long-term plan for meeting effluent limits, regulatory conditions, and waste load allocations (U.S. District Court MD, 2015).

To comply with new permit requirements issued by the U.S. Environmental Protection Agency, the Washington Aqueduct completed a residuals management facility in November 2012. Fairfax Water's Griffith Water Treatment Plant at the Occoquan Reservoir and its Corbalis Water Treatment Plant on the Potomac River are both relatively new and were built with residual solids management systems. Since 2012, when the Corbalis facility modified its solids handling process, the belt filter press filtrate and thickener supernatant are routed through an on-site stormwater pond before being discharged through a Virginia Pollutant Discharge Elimination System permitted outfall to a tributary of Sugarland Run. The discharge to Sugarland Run represents less than 1 percent of the total plant flow. The Griffith Water Treatment Plant discharges residuals to an abandoned quarry and is estimated at 10 percent of the total plant flow.

Loudoun Water's Trap Rock Water Treatment Facility minimizes production losses by recycling waste streams. These streams are routed through a separate treatment process and then returned to the head of the plant. During recycling, raw water withdrawals are adjusted to reflect the volume of recycled flow. For 2023, the reported treatment plant loss rate at Trap Rock was less than one percent, and this is expected to remain the same for 2025 (Loudoun Water Correspondence, July 2024).

In PRRISM, production loss rates are based on data analyses presented by Ahmed *et al.* (2015) and on additional input provided by the suppliers. A three percent loss rate is applied to all plants except for Fairfax Water's Griffith Water Treatment Plant, which uses a ten percent loss rate, Fairfax Water's Corbalis facility, which uses a one percent loss rate, and Loudoun Water's Trap Rock facility, which is also modeled using a one percent loss rate.

5.7 Water Use Restrictions

Water use restrictions are a management tool that may be used by states, local authorities, and water suppliers to temporarily reduce water demands during droughts or other serious situations. Restrictions can be voluntary or mandatory, depending on the severity of the drought. Such restrictions typically include the banning of lawn watering, filling of swimming pools, and operation of ornamental fountains. In 2000, the MWCOG board of directors endorsed a regionally coordinated public response plan that sets trigger levels for regional water use restrictions (MWCOG, 2001). Water use restrictions are simulated in PRRISM using the triggers that appear in Table 5-4, which provide an interpretation of the values in the MWCOG response plan. These values were also used in the 2017 alternatives study (Schultz *et al.*, 2017) and in CO-OP's 2020 water supply study (Ahmed *et al.*, 2020).

Estimates of demand reduction levels when restrictions are in place are based on past regional experience and are also provided in Table 5-4. A 5% voluntary reduction in demand is consistent with that achieved by Fairfax Water in March of 1993 during the Colonial Oil Company pipeline spill. Fairfax Water had to temporarily shut down its Potomac intake, taking all its water instead from the Occoquan Reservoir. Fairfax Water asked its customers to voluntarily reduce their water use. Average demand from February 1 through March 28 was 97.6 MGD. Demand fell to 92.6 MGD during March 29 through April 7, which is

equal to a 5% reduction. It is likely that even greater reductions in demand are possible during higher demand summer months with more discretionary outdoor water uses, but to be conservative a 5% reduction is assumed for summer months and 3% for other months.

An emergency summertime demand reduction of 15% is consistent with results in a study by Halich and Stephenson (2009) on the impact of water use restrictions in Virginia during the drought of 2002. They found that under mandatory restrictions, residential water use fell by 4.5% to 22.1%, with a reduction of 15.4% associated with a medium level of public information dissemination and a medium level of enforcement.

Table 5-4: Water use restriction triggers and assumed demand reductions.

Restriction status	Restriction trigger – level of combined storage in Little Seneca and Jennings Randolph reservoirs	Percent reduction in system demand, June through September	Percent reduction in system demand, October through May
Voluntary	< 60% of combined capacity	5	3
Emergency	< 5% of combined capacity	15	5

5.8 System Capacities and Operational Constraints

Table 5-5 summarizes current and planned capacities and operational assumptions used in PRRISM's simulation of the WMA water supply system. It includes information on the existing system components, including intake and treatment capacities, intersystem transfers capacities, and reservoir flow-bys. Table 5-6 gives assumed capacities and constraints of future resources that are expected to be implemented during the study period. No information is included in this table for the City of Rockville because the City's daily operations and daily variations in withdrawals are not simulated by PRRISM; rather, estimated monthly withdrawals are used to simulate Rockville's impact on Potomac River flows.

Another assumption used in PRRISM is the division of Fairfax Water's service area (SA) into three separate distribution areas - referred to in this study as the Western SA, Central SA, and Eastern SA - with total Fairfax Water daily demand apportioned according to Table 5-7 (A. Spiesman, personal communication, March 2024). The Western SA is primarily provided water that is withdrawn from the Potomac River and treated at the Corbalis WTP, the Eastern SA is primarily provided water withdrawn from the Occoquan Reservoir and the future Edgemon Reservoir and treated at the Griffith WTP, and the Central SA is primarily provided water purchased from Washington Aqueduct. But Fairfax Water has capability to transfer treated water to each of these three service areas from other plants. Constraints on such transfers are given in Table 5-5.

Table 5-5: Current WMA system capacities and constraints.

System resource/connections		Description	Current	2040-2050
Fairfax Water				
	Potomac River intakes/Corbalis WTP (primarily for Western SA)	Maximum production rate, MGD Minimum production rate, MGD	225 60	225 60
	Occoquan Reservoir intake/Griffith WTP (primarily for Eastern SA)	Maximum production rate, MGD Minimum production rate, MGD Maximum change in production, MGD per day	120 45 40	120 45 40
	Finished water purchased from Washington Aqueduct (for Central SA)	Maximum flow from Washington Aqueduct to Falls Church, MGD	31	31
	Transfer of finished water between Eastern and Western SA	Maximum transfer from Western to Eastern SA, MGD	65	65
		Maximum transfer from Eastern to Western SA, MGD Maximum change in Eastern to Western SA transfer, MGD per day	35 10	50 40
	Transfer of finished water from Central to Western SA	Transfer rate, MGD	+6 to -10	+6 to -10
Loudoun Water				
	Potomac River intake/Trap Rock WTF	Maximum pump rate, MGD Maximum treatment rate, MGD	40 20	40 30
WSSC Water				
	Potomac River intake/WFP	Maximum production rate, MGD Minimum production rate, MGD	288 100	288 100
	Patuxent reservoir intake/WFP	Maximum production rate (emergency purposes only), MGD Minimum production rate, MGD Minimum environmental flow-by over Duckett dam, MGD Storage trigger for discontinuation of load-shifting, BG (withdrawals reduced to 20 MGD via intermittent WFP shutdowns)	 72 33 10.3 1.0	 110 33 10.3 1.0
Washington Aqueduct				
	Dalecarlia WTP	Maximum production rate, MGD Minimum production rate, MGD	225 60	225 60
	McMillan WTP	Maximum production rate, MGD Minimum production rate, MGD	120 60	120 60
	Pipe from Dalecarlia to Fairfax Water’s Falls Church SA	Maximum transfer rate of finished water, MGD	35	35
Shared resources				
	Jennings Randolph Reservoir	Minimum downstream flow target for North Branch Potomac River at Luke, Maryland, MGD (cfs)	77.6 (120)	77.6 (120)
	Savage Reservoir	Minimum downstream flow target, MGD (cfs)	12.9 (20)	12.9 (20)
	Little Seneca Reservoir	Minimum environmental flow-by over dam, MGD (cfs)	1.12 (1.73)	1.12 (1.73)

Table 5-6: Planned resource assumed capacities and constraints.

Planned resource		Assumptions
	Vulcan Quarry (Phase 1)	Available in 2040 1.7 BG initial usable storage capacity Refill is from backwash of Fairfax Water's Griffith WTP (5% of plant production) Sedimentation rate based on Griffith production, from M. Pirnie (2012)
	Milestone Reservoir (formerly Quarry A)	Available in 2028 1.25 BG storage capacity and 1.12 BG usable capacity beginning in 2028 Refill is via the 40 MGD pipe from Loudoun Water's Potomac River intake 1.12 MG/year sedimentation rate ¹

Table 5-7: Fairfax Water Service Area - projected demand percentages.

Year	Western SA	Central SA	Eastern SA
2020	43.13%	9.77%	47.10%
2025	43.35%	9.95%	46.69%
2030	43.84%	10.04%	46.11%
2040	43.39%	10.19%	46.43%
2050	43.30%	10.34%	46.36%

6 Upstream Consumptive Use

This study includes new estimates of the cumulative impact on river flows of consumptive use of water in areas upstream of the WMA water supply intakes, including a separate estimate for current and future consumptive use by data centers. Communities, farms, and industries located upstream of the WMA withdraw water from the Potomac River, its tributaries, and groundwater aquifers. These upstream users have an impact on the amount of water available to meet demand in the WMA. Some of the water withdrawn upstream is subsequently returned to Potomac basin streams and aquifers. However, a portion is not returned due to evaporation, transpiration by trees and other vegetation, incorporation into products, consumption by humans or livestock, diversion to another basin, or other processes. The portion of water withdrawn that is removed and not returned for downstream use is termed “consumptive demand,” or equivalently in this study, “consumptive use.” Forecasts of watershed-scale cumulative consumptive use upstream of WMA intakes provide essential information on future water availability for the WMA water supply system.

Estimating future consumptive demand is challenging and forecast results will necessarily differ depending on assumptions used and approximations made to address the large number of uncertainties. These uncertainties include the rate of population growth, changing household water use rates, future changes in the agricultural sector in response to market forces, and changes in commercial and industrial use. For example, thermal electric generating facilities require water for cooling purposes and are large consumptive users, and needs for energy are projected to rise. Data centers, which often rely on water for cooling, have a large presence in Northern Virginia and their numbers are forecast to grow due to increasing reliance on artificial intelligence and cloud computing. Water reuse is being promoted throughout the country and is attractive to Potomac basin municipalities as a revenue source and as a means of meeting Chesapeake Bay nutrient discharge limits. However, most water reuse alternatives are highly consumptive, and state regulatory agencies will have to determine how to balance benefits and adverse impacts. The future national and state regulatory environment is also unknown and will influence the consumptive use of water. Adding to the challenge of forecasting consumptive use are the gaps in available water use data and the inherent difficulty in estimating consumptive use for public water supply systems.

This chapter presents ICPRB's most recent estimates for current and future consumptive use of water in portions of the Potomac River watershed upstream of the WMA intakes. Section 6.1, which excludes the impacts of data centers, provides an update of upstream consumptive use estimates made for CO-OP's 2015 water supply study, available in Ahmed *et al.* (2015), and also used in the 2020 water supply study (Ahmed *et al.*, 2020). Section 6.2 presents estimates of upstream consumptive use by data centers, based on a compilation of information from regional water suppliers and state agencies, cooling technology specifications, and regional energy forecasts. In Section 6.3, estimates of total cumulative upstream consumptive use in two key forecast years, 2050 and 2085, are obtained by combining results from Section 6.1 and Section 6.2. Comparisons are also provided for estimates for 2050 from the current study with those used in CO-OP's 2020 water supply study. Estimates presented in this chapter are used in PRRISM to simulate the reduction in flow at Little Falls Dam due to upstream consumptive use.

6.1 Consumptive Use, Excluding Data Centers

Results in this section were derived with the aid of ICPRB’s most recent version of the Potomac River Withdrawal and Consumptive Use (WCU) database, initially described in Ducnuigee *et al.* (2015). A brief discussion of the methods used to obtain future consumptive use estimates appears below, while a more detailed look at forecasted reported water withdrawals can be found in Moltz *et al.* (2024) and Nummer *et al.* (in preparation). Consumptive use by data centers is not considered in Section 6.1.

6.1.1 Current Upstream Consumptive Use by Use Type

Current upstream consumptive use values presented in this study represent total estimated consumptive use upstream of the WMA intakes above Little Falls. Equivalently, these values are the sum of consumptive demand in the upper Potomac River basin, that is, the 11,560 square mile watershed upstream of Little Falls Dam, excluding the Potomac River withdrawals by the Washington Aqueduct, Fairfax Water, Loudoun Water, WSSC Water, and the City of Rockville. Also excluded from the analysis are the past withdrawals of the City of Fairfax, which, beginning in January 2014, became part of the area served by Fairfax Water.

In this study, upstream consumptive use is computed at a monthly time step from available water withdrawal data and from monthly consumptive use coefficients, which are estimates of the percentage of a withdrawal that is consumptively lost. Estimates of current consumptive use in the basin upstream of the WMA suppliers’ Potomac River intakes rely on the following data and information sources:

- Historical withdrawal data
 - ICPRB’s updated database of historical withdrawals and estimated consumptive use for the upper Potomac River basin, which includes a compilation of mean monthly withdrawal time-series provided by the following Potomac River basin state agencies:
 - Maryland Department of the Environment, 1979-2019
 - Pennsylvania Department of Environmental Protection, 2005-2019
 - Virginia Department of Environmental Quality, 1982-2002 and 2004-2019
 - West Virginia Department of Environmental Protection, 2003-2019
 - The USGS’s data sets of annual average withdrawals, by county, 2015, available via the USGS’s website (<http://water.usgs.gov/watuse/data>)
 - The U.S. Department of Agriculture (USDA) 2017 Census of Agriculture county-level inventory of livestock and acres of irrigated land (USDA NASS, 2019)
 - The U.S. Census Bureau’s 2010 and 2020 Decennial Census total population counts (U.S. Census Bureau, 2010; U.S. Census Bureau, 2020)
- Monthly consumptive use coefficients, by water use type, from
 - A USGS study on consumptive use in Ohio, Indiana, and Wisconsin (Shaffer, 2009); the current study relies primarily on the coefficients calculated from Ohio withdrawal and return flow data
 - Coefficients for public water suppliers derived by ICPRB from Potomac River basin withdrawal data using the “winter base rate” method, discussed below

- Coefficients for hydroelectric water withdrawals from Thompson & Pindar (2021) and for self-supplied domestic use from Horn *et al.* (2008)

Ten water use categories are considered in this study: aquaculture (AQU), self-supplied commercial (COM), hydroelectric (HYE), self-supplied industry (IND), irrigation (IRR), livestock (LIV), mining (MIN), thermoelectric power generation (PP), public water supply (PWS), and self-supplied domestic use (SSD). Table 6-1 gives estimates of current upstream annual average withdrawals, annual average consumptive use, and average consumptive use during summer months (June, July, and August). These estimates, unless otherwise noted, are based on 2017-2019 state withdrawal data, since these are the last three full years in the current version of the WCU database.

Included in the updates made to consumptive use estimates are estimated unreported water withdrawals, and associated consumption, for the irrigation, livestock, and self-supplied domestic sectors. Recent research has synthesized basin-wide estimates of reported and unreported water withdrawals by water use sector for 2017 using the WCU database and appropriate unreported estimation methods (Wallace *et al.*, 2024). This evaluation of reported and unreported water withdrawals showed that the irrigation, livestock, and self-supplied domestic water use sectors were mainly unreported, rather than reported (Wallace *et al.*, 2024). Therefore, work to incorporate unreported water withdrawals and consumption focused on these three sectors.

Unreported consumptive use for irrigation and livestock was calculated using the unreported withdrawals reported in Wallace *et al.* (2024) and the consumptive use coefficients presented in Shaffer (2009). As these unreported consumptive use estimates are only for one year, an annual rate of change could not be calculated and therefore unreported consumption for irrigation and livestock was assumed to be constant over time. In the predicted use estimates for these sectors, the values for unreported irrigation and livestock consumption presented in Table 6-1 were used for all forecasted estimates as well (e.g. 2050 and 2085).

Unreported self-supplied domestic withdrawals in Wallace *et al.* (2024) were estimated using the 2010 Decennial Census total population counts (U.S. Census Bureau, 2010) and 2015 per capita water consumption estimates from Dieter *et al.* (2018). With the availability of the 2020 Decennial Census total population counts, 2020 self-supplied domestic unreported consumptive use estimates were calculated using the method outlined in Wallace *et al.* (2024) and the previously used per capita water consumption approximations (Dieter *et al.*, 2018). These results afforded this analysis two years' worth of estimated unreported self-supplied domestic water consumption (2015 and 2020) and subsequently enabled the annual change in self-supplied domestic consumptive use to be calculated. The calculated annual change in self-supplied domestic consumptive use was then extrapolated to forecast future consumption for this sector.

The results in Table 6-1 show that thermoelectric power facilities are the second largest water withdrawal use category and have combined annual withdrawals of approximately 1,137 MGD. Two facilities account for almost all of the thermoelectric power withdrawals: Dominion's Mount Storm Power Station in Grant County, West Virginia, and Dickerson Generating Station in Montgomery County, Maryland, now owned by GenOn Mid-Atlantic, LLC. Both facilities use water for once-through cooling systems.

The consumptive use of water by power plants is more modest, since consumptive use for once-through cooling systems is relatively small, in the vicinity of two percent (see Table 6-2).

In the upper Potomac basin, the annual average withdrawals total to approximately 2,722 MGD. However, over half of these withdrawals are for thermoelectric and hydroelectric power generation, both of which have minimal consumptive water use associated with them. Thermoelectric and hydroelectric power generating facilities withdraw almost exclusively from surface waters, therefore accounting for a large portion of the basin's surface water withdrawals. In the upper Potomac basin, approximately 96 percent of water withdrawals are from surface water. This estimate becomes much more balanced with groundwater withdrawals when the thermoelectric and hydroelectric sectors are excluded. Without withdrawals from the PP and HYE water use sectors, upstream withdrawals total approximately 400 MGD. Without hydro- and thermoelectric withdrawals, surface water withdrawals account for 61 percent and groundwater withdrawals account for 39 percent of the total average annual water withdrawals upstream of Little Falls.

Upstream consumptive use in the summer months has the greatest impact on the WMA water supply system since demands are at their highest in the summer and at the same time river flows tend to be falling. The last column of Table 6-1 shows that the four water use types with the highest summertime upstream consumptive use are irrigation, livestock, thermoelectric, and mining. Figure 6-1 gives a breakdown by percentage of upstream consumptive use by use type, including unreported consumption and excluding Dominion's Mount Storm Power Station. CO-OP analyses of the impacts of upstream consumptive use on WMA supplies exclude the Mount Storm Power Station withdrawals. This is because Mount Storm is located upstream of Jennings Randolph Reservoir, and its consumptive use from the North Branch of the Potomac River is mitigated by minimum water quality releases from the Jennings Randolph and Savage reservoirs.

For the most part, the consumptive use estimates in Table 6-1 were computed based on withdrawal data for individual sites and on the consumptive use coefficients appearing in Table 6-2. The coefficients for AQU, COM, IND, IRR, LIV, MIN, and PP in Table 6-2 were taken from the large-scale study on monthly variations in consumptive use conducted by the USGS using 1999-2004 data from Ohio, Indiana, and Wisconsin (Shaffer, 2009). For water use sectors judged to be sensitive to dry conditions in the Potomac River basin, Shaffer's 75th percentile values are used; for other sectors, average values are used. Hydroelectric consumptive use coefficients were taken from Thompson & Pindar (2021) while the coefficients for the self-supplied domestic sector were obtained from Horn *et al.* (2008). The public water supply coefficients in column "PWS_a" of Table 6-2 were computed from monthly public water supply withdrawal data from 2017, 2018, and 2019 for municipalities in the Potomac basin upstream of the WMA intakes using the winter base rate method, described in the following section. The public water supply coefficients in column "PWS_b" of Table 6-2 were computed from total monthly CO-OP system production data using the winter base rate method for 2017 through 2019.

Table 6-1: Current average estimated total withdrawals and consumptive use (CU) from 2017 to 2019 in the upper Potomac River basin, upstream of the WMA supplier intakes (MGD).

Abbreviation	Use Type	Annual avg. withdrawal ¹	Annual avg. CU ¹	Summer avg. CU ⁶
AQU	Aquaculture	27.89	1.36	1.77
COM	Commercial	5.70	3.14	2.40
HYE	Hydroelectric	1184.26	0.00	0
IND	Industrial	54.49	6.25	6.08
IRR³	Irrigation - unreported estimates	21.99	21.42	32.12
	Irrigation - reported values	8.47	8.11	9.60
LIV³	Livestock - unreported estimates	45.97	34.94	34.94
	Livestock - reported values	0.47	0.36	0.43
MIN	Mining	42.81	12.84	12.70
PP	Thermoelectric - Mount Storm ⁵	958.19	19.16	24.42
	Thermoelectric - Other	178.78	3.58	3.55
PWS⁴	Public Water Supply	131.44	4.82	11.74
SSD²	Self-Supplied Domestic	61.12	9.78	9.78
Total		2721.59	125.76	149.53
Total - excluding Mount Storm		1763.4	106.6	125.11

¹ Average of 2017-2019 reported values.

² Includes unreported estimates for 2017 added to reported average of 2017-2019 for SSD.

³ Includes unreported estimates for 2017 as a separate row for LIV and IRR.

⁴ PWS does not include the WMA suppliers (The City of Fairfax, Fairfax Water, Loudoun Water, the City of Rockville, Washington Aqueduct, or WSSC Water).

⁵ Mount Storm is upstream of Jennings Randolph Reservoir, and its consumptive demand is mitigated by water quality releases from both Jennings Randolph and Savage reservoirs.

⁶ Average of 2017-2019 reported values for summer season (June, July, and August).

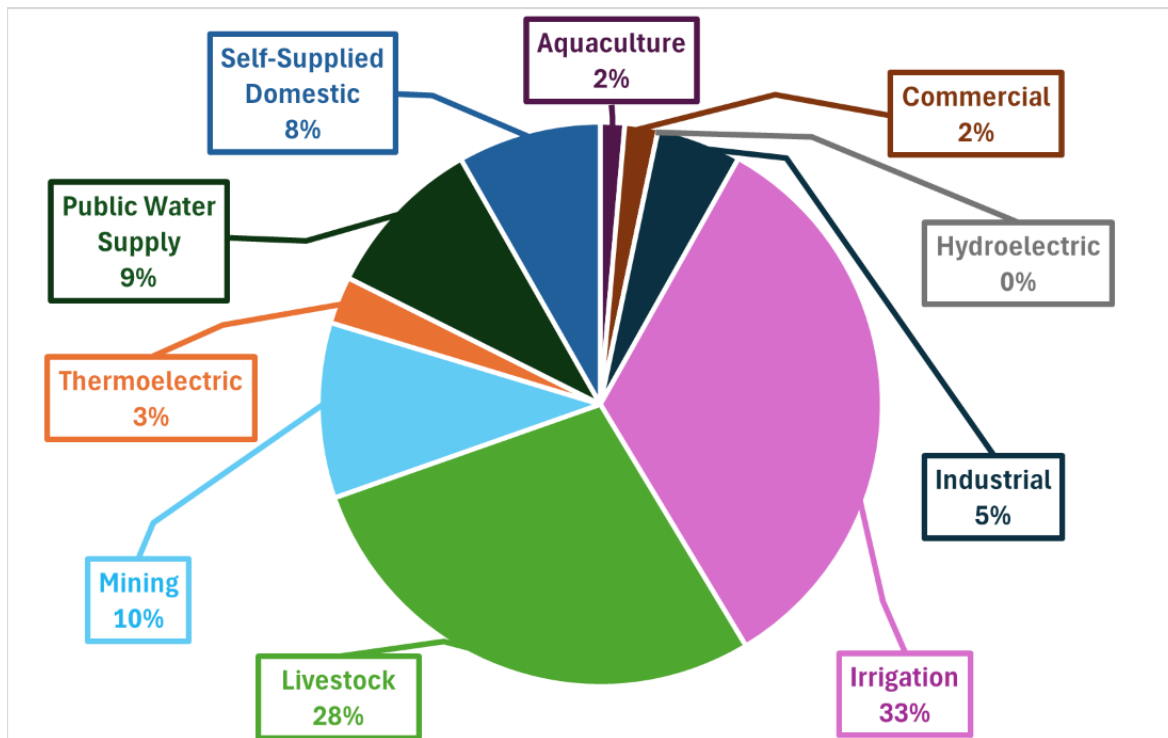


Figure 6-1: Summertime (June, July, August) upstream consumptive use by water use type, excluding WMA water suppliers and Mount Storm Power Station.

Table 6-2: Consumptive use coefficients used in the current study (percent).

Month	AQU ¹	COM ²	HYE ⁸	IND ¹	IRR ²	LIV ³	MIN ¹	PP ⁴	PWS _a ⁵	PWS _b ⁶	SSD ⁷
January	4	44	0	11	0	76	29	2	0	0	16
February	3	50	0	12	0	76	30	2	0	0	16
March	7	50	0	11	0	76	30	2	0	3	16
April	8	55	0	11	99	76	30	2	1	8	16
May	5	56	0	11	98	76	30	2	4	16	16
June	9	56	0	12	98	76	29	2	7	26	16
July	5	50	0	12	95	76	28	2	10	26	16
August	5	73	0	12	98	76	31	2	9	18	16
September	6	78	0	11	99	76	30	2	8	14	16
October	3	78	0	11	98	76	31	2	3	7	16
November	2	48	0	12	94	76	31	2	0	0	16
December	1	47	0	12	0	76	31	2	0	0	16

¹ Average monthly values for Ohio from Shaffer (2009).

² 75th percentile monthly values for Ohio from Shaffer (2009).

³ No monthly consumptive use coefficients are available; 76% is the median of average annual consumptive use coefficients for livestock based on 18 sets of withdrawal and return flow data from facilities in Ohio from Shaffer (2009).

⁴ 75th percentile of average annual coefficients computed for facilities with once-through cooling systems in Ohio (Shaffer, 2009).

⁵ Total upstream PWS consumptive use from the winter base rate method, computed by summing monthly consumptive use by individual upstream locations and then averaging – 2017, 2018, and 2019.

⁶ CO-OP system consumptive use from the winter base rate method, average for 2017 - 2019.

⁷ From Horn *et al.* (2008).

⁸ From Thompson & Pindar (2021).

6.1.2 Consumptive Demand Estimates for Public Water Suppliers

The annual average water withdrawals by public water supply systems upstream of the WMA Potomac River intakes are just under 132 MGD (Table 6-1), based on 2017-2019 data. Figure 6-2 shows the growth in total upstream withdrawals by public suppliers over the period 1990-2019. This graph is based on available data for Maryland and Virginia suppliers for the years 1990-2004 (with a data gap in 2003), and on data from all four basin states for the years 2005-2019. For years lacking data from Pennsylvania or West Virginia, it was assumed that these states together account for 31 percent of the total withdrawals. During the period 2005-2019, when complete data are available, public water supply withdrawals in Pennsylvania and West Virginia accounted for 16 and 15 percent of total public water supply withdrawals, respectively. According to these estimates, annual average upstream public water supply withdrawals grew from approximately 78 MGD in 1990 to 132 MGD in 2019, an increase of 69 percent.

After water withdrawn by a public supply system is used by the system's customers, most of it is typically treated by a wastewater treatment facility and returned to the watershed. It is difficult to quantify the portion of public water supply withdrawals that is used consumptively. The standard method to compute consumptive use, which consists of taking the difference between water withdrawn and water returned by a user, is usually not applicable because of the significant inflow and infiltration typically experienced by municipal sewer systems (LaTour, 1991; Ducnuigeen *et al.*, 2015). For this reason, an alternative method

for estimating consumptive use, the winter base rate method, is often used (LaTour, 1991; Horn *et al.*, 2008; Shaffer, 2009; Ducnuigee *et al.*, 2015). This method is applicable in regions with a temperate climate where it is reasonable to assume that consumptive use is primarily due to outdoor water use in non-winter months. The method is based on the assumptions that (1) no significant consumptive use occurs in the winter months, and (2) the observed difference between winter withdrawals and withdrawals in other months of the year is due to outdoor water use, which is completely or largely consumptive. The consumptive use (CU) coefficient from the winter base rate method is given by

$$CU \text{ coefficient (\%)} = \frac{(Monthly \text{ Withdrawal} - Winter \text{ Withdrawal})}{Monthly \text{ Withdrawal} * Outdoor \text{ Use } CU \text{ coefficient}} \quad \text{Equation 6-1}$$

This study defines the winter withdrawal rate as the average withdrawals between December and February and assumes that outdoor use is 100 percent consumptive. The winter base rate method has some obvious limitations. For example, it does not consider consumptive use that does not vary seasonally, as might occur for a public supply system that supplies a large industrial facility whose consumptive use is constant.

Monthly consumptive use coefficients for Potomac basin public water suppliers were calculated in this study using the winter base rate method, shown in Equation 1. A selection of the calculated consumptive use coefficients appears in Table 6-3, which also includes results from Shaffer's study. The coefficients for total public water supply withdrawals upstream of WMA intakes, for both an average year and a dry year, are given in columns four and five of Table 6-3.

The consumptive use coefficients for the Potomac Basin Above Little Falls – All Years (column 4 of Table 6-3) also appear in Table 6-2 as the column titled PWS_a. These coefficients were used to compute the estimates of current annual and summer upstream consumptive use for the PWS sector, given in Table 6-1, and to forecast upstream consumptive use in the public water supply sector (discussed in the next section). The consumptive use coefficients for the WMA suppliers appear in the last two columns of Table 6-3, with the average coefficients for 2017 through 2019 also appearing in Table 6-2 as the column PWS_b.

Results in Table 6-3 provide a comparison of public water supply consumptive use coefficients for various systems. Mean consumptive use coefficients for the WMA suppliers are reasonably close to 75th percentile values obtained by Shaffer (2009). Consumptive use coefficients computed for total upstream public water supply withdrawals are lower than the USGS coefficients and notably lower than coefficients computed for the WMA suppliers. These suppliers serve fairly affluent communities and have had relatively unconstrained access to water, even during low-flow periods.

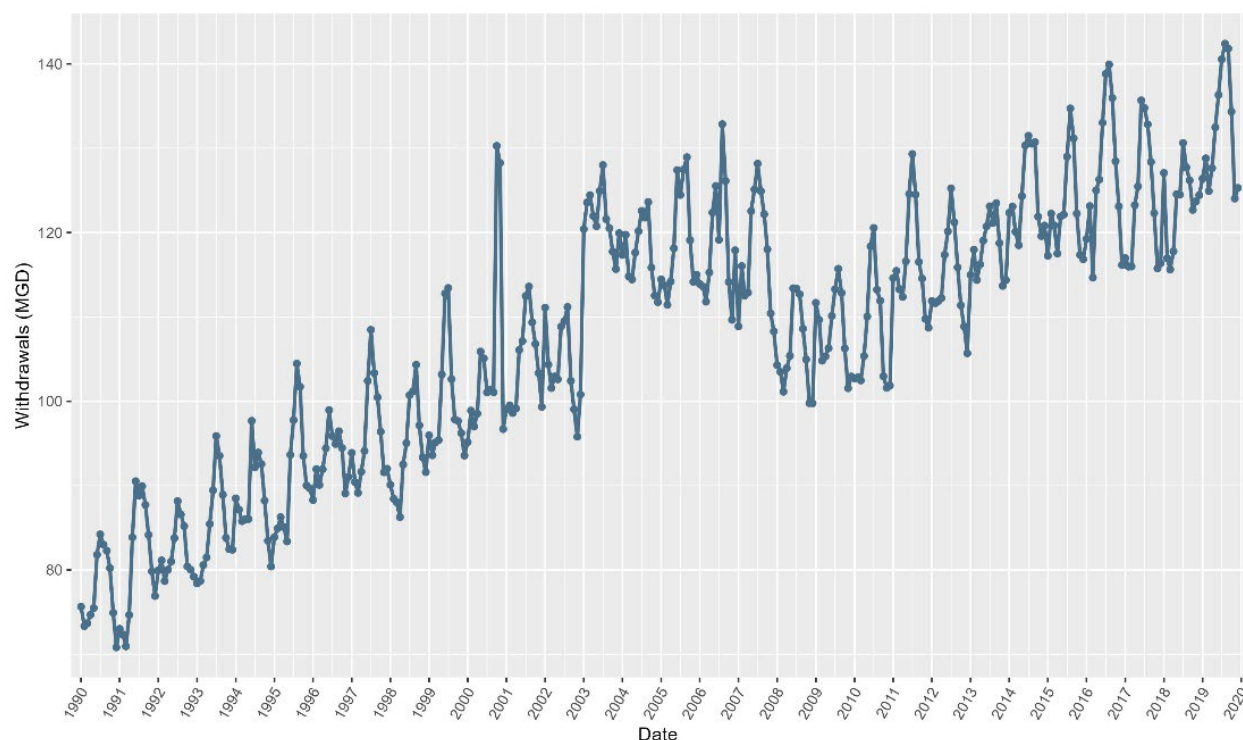


Figure 6-2: Estimated total PWS withdrawals upstream of WMA intakes.

Table 6-3: Comparison of monthly PWS consumptive use coefficients from the study by Shaffer (2009), compared with coefficients computed for Potomac basin suppliers (percent).

Month	USGS - Indiana & Wisconsin ¹		Potomac Basin Above Little Falls		WMA Suppliers	
	Median	75th percentile	Average for			
			All Years ²	Dry Years ⁴	All Years ³	Dry Years ⁴
March	--	--	0	0	3	0
April	--	--	1	3	8	7
May	6	9	4	7	16	15
June	15	17	7	16	26	32
July	20	26	10	18	26	38
August	20	22	9	11	18	30
September	15	21	8	9	14	34
October	4	6	3	0	7	5
November	--	--	0	0	0	0

¹ From Shaffer (2009).

² Total upstream PWS consumptive use from the winter base rate method, average for 2017 - 2019.

³ CO-OP system consumptive use from the winter base rate method, average for 2017 - 2019.

⁴ Dry year is calculated for 2010.

6.1.3 Forecasts of Upstream Consumptive Use

Estimates of current upstream consumptive use and annual growth rates, not including the impacts of data centers or Mount Storm Power Station, are given in Table 6-4, along with forecasts of future consumptive use for selected years. The PRRISM model uses two sets of monthly inputs to simulate the impact of upstream consumptive use on flow for a given forecast year: total upstream consumptive use in a baseline year and the annual growth rate of total upstream consumptive use. Values for these inputs were first computed by water use type and then summed to obtain totals for all users to generate estimates of upstream consumptive use for the baseline year, defined as the average of 2017-2019. The impact of dry year conditions was included in the estimates where appropriate. Total summer upstream consumptive use is estimated to be just under 123 MGD between 2017 and 2019 and approximately 140 MGD in 2050, an increase of about 14 percent. Marginal differences in the annual and summer average consumptive use estimations presented in Table 6-1 and Table 6-4 are attributed to the different order-of-operations employed in aggregating the data and calculating the averages by sector versus by month.

The forecasts of consumptive use by water use type are based on an extrapolation of historic trends in reported water withdrawals by withdrawal location, with unreported estimates incorporated for self-supplied domestic, livestock, and irrigation. Forecasted consumptive use estimates were determined by multiplying the consumptive use coefficients by predicted monthly withdrawals for each individual withdrawal location within a given sector. The forecasted reported water withdrawals for the entire Potomac basin were calculated in Wallace *et al.* (2024) and Nummer *et al.* (in preparation). The monthly forecasted withdrawals for 2025-2050 were subset to obtain the predicted withdrawals for locations upstream of Little Falls Dam, excluding the Potomac River withdrawals by Washington Aqueduct, Fairfax Water, Loudoun Water, WSSC Water, the City of Rockville, the City of Fairfax, and Mount Storm Power Station.

Chapter 3 presents a forecast of declining household indoor water use rates and discusses how this decline is incorporated into forecasts of WMA demand in future years. Because the estimates for upstream public water supply consumptive demand presented in the current chapter are based on assumed seasonal changes in outdoor use, declines in indoor use are not incorporated into the forecast for upstream consumptive use in the public water supply sector.

Table 6-4: Estimates of monthly upstream consumptive use (MGD), growth rates (MGD/yr), and forecasts for selected years (MGD) without the inclusion of water used by data centers.

Month	CU Baseline and Growth Estimates		CU Forecasts			
	2018	Annual Growth Rate	2045	2050	2060	2085
January	71.03	0.09	73	74	75	77
February	70.21	0.14	74	75	76	79
March	70.98	0.15	75	76	77	81
April	107.56	0.51	121	124	129	142
May	111.64	0.53	126	129	134	147
June	120.74	0.51	135	137	142	155
July	124.68	0.49	138	140	145	157
August	122.50	0.62	139	142	148	164
September	119.14	0.59	135	138	144	159
October	108.16	0.60	124	127	133	148
November	99.98	0.57	115	118	124	138
December	68.46	0.119	74	75	76	81
Average ¹	99.59	0.41	111	113	117	127
Summer Average ²	122.64	0.54	137	140	145	159

¹2017-2019 average.

²June, July, August average.

6.2 Consumptive Use by Data Centers

6.2.1 Introduction

The Potomac River basin is home to “Data Center Alley” in Loudoun County, Virginia, often cited as having the highest concentration of data centers in the world. Data center facilities support cloud computing, e-commerce, digital communication, and artificial intelligence. Their operation depends on reliable electricity, fast data connections, and, in some cases, large volumes of water for cooling (Siddik *et al.*, 2021; Shehabi *et al.*, 2024). As digital demand grows, so does the expansion of data centers, which raises concerns about the use of land, energy, and water resources. In addition to Loudoun County, neighboring counties like Prince William and Fairfax have seen a surge in data center development. Northern Virginia offers access to major fiber routes and power infrastructure, attracting new projects that extend data centers’ footprint in the region. Across the river, Frederick County, Maryland, is also emerging as a key siting location with large-scale developments such as the 2,100-acre Quantum

Loophole campus (Data Center Dynamics, 2024; Rowan Digital Infrastructure, 2024). These trends may create new pressures for regional infrastructure planning, including long-term water supply.

While energy consumption by data centers has been the focus of many assessments to date (e.g. JLARC, 2024), water use is increasingly recognized as a critical factor, especially for regions facing seasonal water stress or planning for long-term drought resilience (Mytton, 2021; Siddik *et al.*, 2021).

Data center water use varies significantly depending on the type of cooling technology employed. Evaporative cooling methods such as the use of cooling towers are particularly water-intensive, while air-cooled, liquid-cooled, and immersive systems may reduce water use but often have higher energy needs, infrastructure costs or requirements (Díaz *et al.*, 2020; Karimi *et al.*, 2022; Alissa *et al.*, 2025). Indirect water use through electricity generation also contributes to the overall data center water footprint (Siddik *et al.*, 2021), though the present analysis is focused on direct, on-site water use only.

In this section, we present an assessment of current and projected consumptive water use by data centers within the Potomac River basin, with a distinction between upstream consumptive use (used in PRRISM) and WMA consumptive use (which is provided for informational purposes only). The analysis is based on utility data, facility-level characteristics (such as power capacity), assumptions on cooling technologies, and regional energy forecasts. Consumptive use figures for data centers located upstream of the WMA drinking water intakes were used directly in PRRISM (in addition to consumptive use estimates presented in Chapter 6.1) to assess potential impacts on water availability. Within the WMA, data center consumptive use is assumed to be captured at least partially in existing demand forecasts and return flow estimates (Chapter 3 and Chapter 5). Adjustments to this assumption remain at the discretion of individual utilities. We summarize below the methods and assumptions used in the analysis. Additional methodological detail is available in Seck *et al.* (in preparation).

6.2.2 Approach

Consumptive water use of a data center was estimated by multiplying its Effective Power Demand by a Water Use per Unit of Power (WUP) and a consumptive use factor.

$$\text{Consumptive Use} = (\text{Effective Power Demand} \times \text{WUP}) \times \text{Consumptive Use Factor} \quad \text{Equation 6-2}$$

Where:

WUP (Water Use per Unit of Power) = Water used per unit of power demand (e.g., gallons per day per megawatt (MW))

Consumptive Use Factor = Fraction of total delivered water that is consumptive

Effective Power Demand is derived from backup generator power data often available in data center permit applications. It is expressed as follows:

$$\text{Effective (IT) Power Demand} = \frac{\text{Total Generator Power Capacity} \times \text{Redundancy} \times \text{Utilization Factor}}{\text{}} \quad \text{Equation 6-3}$$

Based on utility data, a consumptive use factor of 0.75 was applied to all estimates to reflect evaporative losses and other non-returned components. A redundancy factor of 0.5 is assumed to reflect that permitted generator capacity typically represents twice the actual IT power load (i.e., 2N backup systems), and a utilization factor of 0.8 is applied based on industry data (EPRI, 2024), indicating that data centers do not operate at full load continuously.

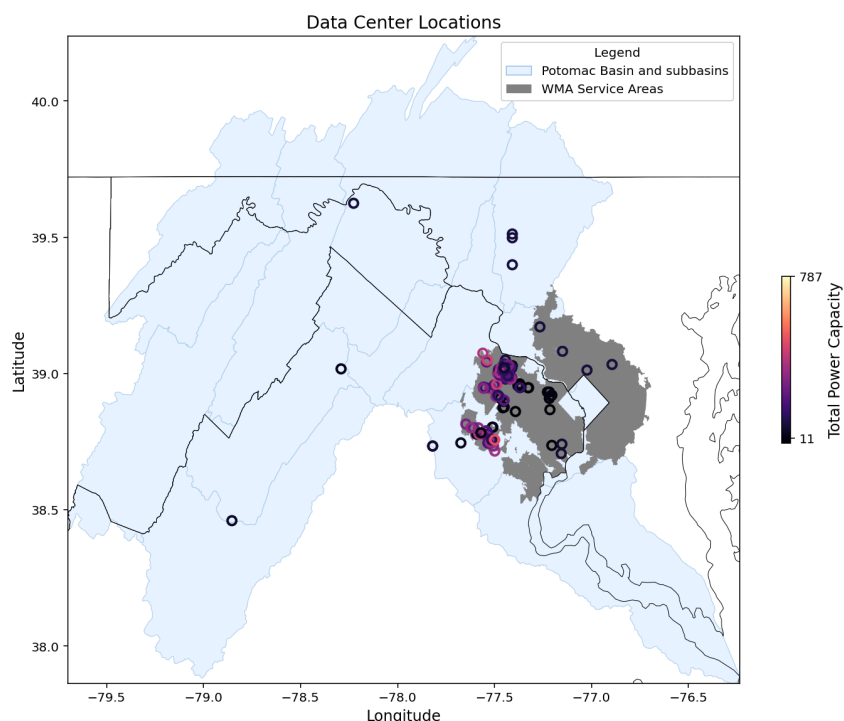


Figure 6-3: Location of data centers in the Potomac basin. Rated Power of each facility is depicted by the circle's color.

Data center locations in Virginia were obtained from a dataset developed by Virginia Joint Legislative Audit and Review Commission (JLARC) consultants using VADEQ air permits, which included facility names, operators, addresses, locality, total backup generator power capacity, building size, number of buildings, and land area. These permits are publicly available on the VADEQ Issued Air Permits for Data Centers website: <https://www.deq.virginia.gov/permits/air/issued-air-permits-for-data-centers>. Locations outside Virginia were obtained from vendor databases such as Data Center Map (<https://www.datacentermap.com/usa>). These included data centers in Frederick County that have received permit approvals and are scheduled for completion in 2025. All addresses were geocoded and overlaid on the Potomac basin and sub-basin boundaries (Figure 6-3). For centers without reported power capacity, we used a default value of 78 MW based on the average in the Virginia dataset. Data center locations and associated backup generator power are shown in Figure 6-4.

WUP values were derived using actual data center water use (based on utility reported data) and power demand (based on JLARC database) for data centers in the Loudoun Water, Fairfax Water and Prince

William Water service areas. Backup power capacities shown in Figure 6-4 were summed by water utility service area. WUP was then calculated by dividing utility reported data center water use by effective power demand. In the Loudoun Water service area, data centers are reported to have used 4.5 MGD of water annually and 10.9 MGD on the peak day in 2024. These figures include potable and reclaimed water use. In the Prince William Water service area, 0.42 MGD on average and 4.2 MGD for peak day were reported for 2023. For Fairfax County, where direct data was not available, we estimated usage at 0.27 MGD using square footage obtained from the JLARC data center database and an assumed usage rate of 90 gallons per day per 1,000 square feet (as reported by Fairfax Water, see County of Fairfax, 2024). These data yielded WUP values of 1,145 gallons/day/MW in Fairfax County, 1,006 in Loudoun for the annual average, and 2,435 for peak day, and 309 for the average and 3,060 for peak day in Prince William. Averaging the Prince William Water and Loudoun Water service areas values, we estimate a representative WUP across the basin of about 800 gallons/day/MW for annual average and about 3000 gallons/day/MW for peak day. These average WUP values were used to estimate data center consumptive use at the basin and sub-basin levels, as shown in Figure 6-4. Upstream consumptive use by data centers is currently negligible at less than 0.1 MGD (less than 0.1 percent) annually and 0.3 MGD during peak periods.

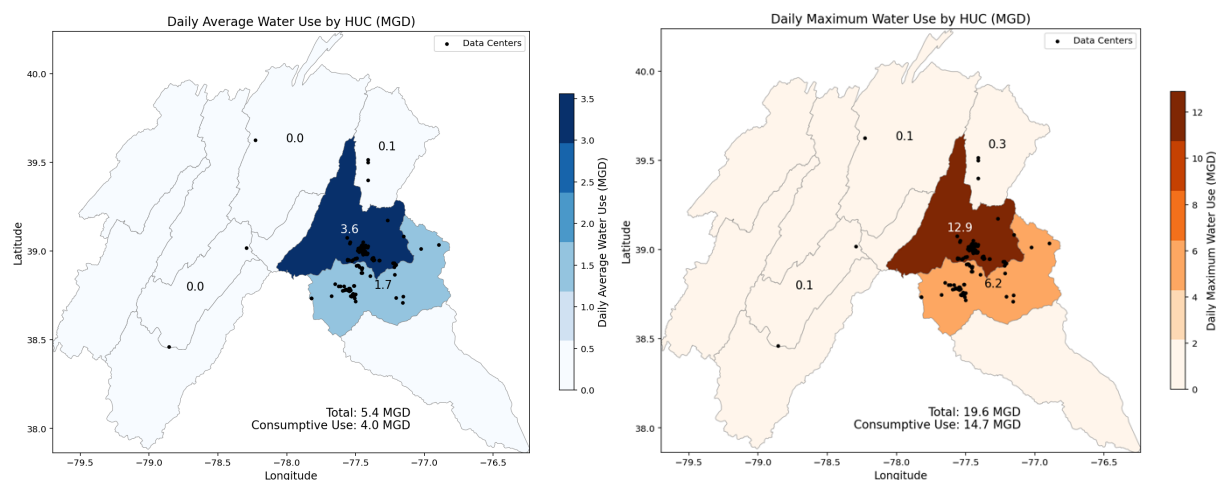


Figure 6-4: a) Average and b) Peak data center water use by sub-basin. The shading indicates variations in water use, with darker colors representing higher consumption levels. Black dots represent individual data centers, which are densely clustered in the areas of highest water use. Total use and the consumptive use portion for the basin are noted on the bottom right of each figure.

To estimate growth in data center power demand, we used PJM Interconnection forecasts for two major service territories in the basin (Allegheny Power Service and Dominion). PJM Interconnection is a Regional Transmission Organization (RTO) that manages the high-voltage electric system serving 67 million people across parts of 13 states and the District of Columbia. It is one of the largest power grids in North America and is responsible for coordinating the movement of wholesale electricity, ensuring reliability, and planning for future grid needs (see Appendix A.3). APS (Allegheny Power Service) peak load is projected to increase from 8,700 MW in 2025 to 11,700 MW in 2050, a 35% increase. Dominion's load is projected to increase from 23,000 MW to 54,000 MW over the same period, a 135% increase.

EPRI (2024) estimates that data centers currently account for 0 to 5% of APS load and 25% of Dominion's. Assuming that data centers are the primary driver of the projected growth, their share is projected to reach 27.5 percent in APS and 68 percent in Dominion by 2050. These correspond to a 14-fold increase in data center electricity use for APS and a 2.7-fold increase for Dominion. These growth rates were applied to the estimated 2025 baseline data center power demand currently operating within the basin to project future water use.

Future data center water use was estimated by applying WUP values to projected power demand under three scenarios representing different assumptions about the share of water-cooled systems. The medium scenario reflects current representative practices in the region, with an assumed WUP of 800 gallons/MW/day (average) and 2,900 gallons/MW/day (summertime peak). To derive WUPs for the low and high use scenarios, we used Loudoun- specific data, where the observed WUP was 1,006 gallons/MW/day (annual average) and 2,716 gallons/MW/day (peak) under an assumed 60 percent share of water-cooled systems. Holding the air-cooled WUP constant at 150 gallons/MW/day (value based on an estimate of 0.017 gallons/day/square foot, provided by Loudoun Water and an infrastructure density of 8,818 square feet per MW based on the JLARC database), we calculated the implied WUPs for fully water-cooled systems as 1,577 gallons/MW/day (annual average). These values were then applied at 30 percent and 90 percent water-cooled shares to construct the low and high scenarios. Table 6-5 summarizes the resulting WUP values for each scenario.

This range of WUPs was selected to reflect uncertainty in future cooling technologies and operational practices, which are evolving rapidly and often proprietary. As such, the modeled scenarios represent bounding estimates of potential future water use rather than precise forecasts.

Table 6-5: WUP values by water cooling system scenario.

Scenario	Water-Cooled Share (%)	Avg WUP (gal/MW/day)	Max WUP (gal/MW/day)
Low	30	600	2100
Medium (Observed)	~60-70 (current estimate)	800	2900
High	90	1400	5200

Annual estimates of future data center consumptive use are shown in Appendix A.3, Table A.3-1. Monthly estimates were derived by applying a set of monthly factors to the annual average water use. These factors reflect typical seasonal patterns and were developed using current data from Loudoun Water and Prince William Water. Monthly factors are also provided in Table A.3-2 of Appendix A.3.

6.2.3 Consumptive Use Forecasts

Figure 6-5 shows projected water use by data centers under three cooling technology scenarios. Each plot highlights the medium scenario (solid line) and the range of projections based on low and high water-use assumptions (shaded area).

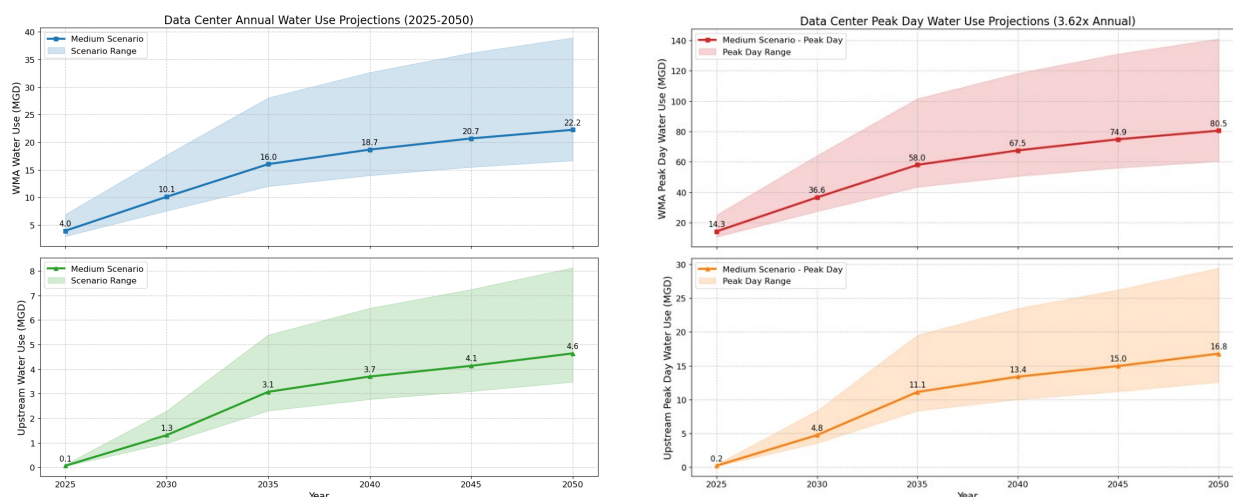


Figure 6-5: Projected annual and peak day water use by data centers in the WMA and upstream portions of the Potomac basin under low, medium, and high water-use scenarios (2025-2050). The solid line shows the medium scenario; the shaded band shows the range between low and high scenarios.

Within the WMA, data center water use is projected to increase substantially, from about 4.0 MGD in 2025 to 22.2 MGD by 2050 on average, and from 14.3 MGD to 80.5 MGD on peak days. In contrast, upstream demand remains comparatively modest, reaching 4.7 MGD annually and 16.8 MGD on peak days by 2050 under the medium scenario. This is due to the concentration of data center growth within the WMA; in the APS region upstream, installed capacity is projected to grow more gradually, from approximately 8,700 MW in 2025 to 11,700 MW in 2050. Still, this upstream growth is non-negligible and could contribute to cumulative stress on shared water resources, particularly during drought or high-demand conditions. For example, under the high-use scenario, upstream data center consumptive use reaches 8.1 MGD annually and 29 MGD during peak periods.

Table 6-6 through Table 6-8 present projected monthly upstream consumptive use in MGD from 2030 to 2050 for three scenarios: Low Water Use (30 percent water-cooled), Medium (Baseline), and High Water Use (90 percent water-cooled). Under the Medium scenario, monthly water use grows steadily over time, with the most significant increases observed during the summer months. By 2050, projected water use in August reaches 8.2 MGD, up from 2.3 MGD in 2030. Winter months, such as January and February, show lower overall use, peaking at 3.2 MGD and 2.8 MGD, respectively, in 2050. In the Low scenario, where only 30 percent of data centers use water-cooled systems, water use remains substantially lower across all months. Peak summer demand in August reaches 6.2 MGD by 2050, and winter demand in January is projected at 2.4 MGD. This scenario demonstrates the significant water-saving potential of non-water-based cooling technologies. In contrast, the High Water Use scenario assumes 90 percent adoption of water-cooled systems and yields the highest projected demands. By 2050, August usage surpasses 14.4 MGD, and several other summer months exceed 10 MGD, and 7 MGD in fall.

Table 6-6: Projected monthly consumptive use (MGD) for Low Water Use scenario - Upstream areas.

Month	2030	2035	2040	2045	2050
January	0.7	1.6	1.9	2.1	2.4
February	0.6	1.4	1.7	1.9	2.1
March	0.6	1.4	1.7	1.9	2.1
April	0.7	1.7	2	2.3	2.5
May	0.8	2	2.4	2.6	3
June	1	2.4	2.8	3.2	3.5
July	1.5	3.5	4.2	4.7	5.3
August	1.7	4.1	4.9	5.5	6.2
September	1.5	3.6	4.3	4.8	5.4
October	1	2.3	2.7	3.1	3.4
November	0.8	2	2.4	2.7	3
December	0.8	1.9	2.3	2.6	2.9

Table 6-7: Projected monthly consumptive use (MGD) for Medium Water Use scenario - Upstream areas.

Month	2030	2035	2040	2045	2050
January	0.9	2.1	2.5	2.8	3.2
February	0.8	1.8	2.2	2.5	2.8
March	0.8	1.8	2.2	2.5	2.8
April	1	2.2	2.7	3	3.4
May	1.1	2.6	3.1	3.5	3.9
June	1.3	3.1	3.8	4.2	4.7
July	2	4.7	5.7	6.3	7.1
August	2.3	5.4	6.5	7.3	8.2
September	2	4.7	5.7	6.4	7.1
October	1.3	3	3.7	4.1	4.6
November	1.1	2.6	3.2	3.6	4
December	1.1	2.6	3.1	3.5	3.9

Table 6-8: Projected monthly consumptive use (MGD) for High Water Use scenario - Upstream areas.

Month	2030	2035	2040	2045	2050
January	1.6	3.7	4.4	4.9	5.5
February	1.4	3.2	3.9	4.3	4.9
March	1.4	3.2	3.9	4.3	4.9
April	1.7	3.9	4.7	5.3	5.9
May	2	4.6	5.5	6.2	6.9
June	2.4	5.5	6.6	7.4	8.3
July	3.5	8.3	9.9	11.1	12.4
August	4.1	9.5	11.5	12.8	14.4
September	3.6	8.3	10	11.2	12.5
October	2.3	5.3	6.4	7.2	8
November	2	4.6	5.6	6.2	7
December	1.9	4.5	5.4	6.1	6.8

6.2.4 Further Considerations and Recommendations

As data center growth continues to reshape water demand patterns in the Potomac River basin, several additional considerations merit attention in future planning and policy discussions. Assessing future water use by data centers is a new and evolving area of study. CO-OP's 2030 water supply study will provide an opportunity to take advantage of new data, review assumptions, and verify or update forecasts. Key areas of further evaluation include the following:

- *Improved transparency in water and energy data is critical:*

Utilities and regulators would benefit from free access to data on data center water use, ideally reported at the daily time scale, to assess both average and peak demands. Data on peak day water use will improve understanding of potential effects on river flows and on water storage mitigation options. Energy use data with similar temporal granularity is also essential to more accurately link power demand to water use. Importantly, as some new data centers explore off-grid energy solutions (such as hydrogen-powered systems, battery-backed microgrids, or on-site natural gas generation) traditional grid-based forecasts may not fully reflect future energy use. This may create gaps in estimating associated water demands, especially where facilities operate outside of utility reporting systems (ECL, 2023; RBN Energy, 2024). Addressing this will require not only improved coordination with water and energy agencies but also greater transparency and enhanced data systems that can track evolving technologies and nontraditional infrastructure.

- *Mitigation requirements:*

As data center development continues in the Potomac River basin, particularly in Northern Virginia and in Maryland, existing state-level consumptive use regulations may become applicable. In

Virginia, the Virginia Water Protection Permit program (Va. Code § 62.1-44.15:5.02) requires that non-municipal water users withdrawing more than 500,000 gallons per day for consumptive use from the Potomac River or its tributaries (between the West Virginia border and Little Falls) implement low-flow protection measures. These may include off stream storage to augment instream flows during drought periods, purchasing storage capacity from another entity, or limiting withdrawals during designated low-flow conditions. These requirements apply to permits issued after July 1, 2007, and reflect Virginia's broader goal of safeguarding water availability during critical periods (Virginia Code Annotated, 2007).

In Maryland, most water withdrawals exceeding 10,000 gallons per day require a Water Appropriation and Use Permit issued by the Maryland Department of the Environment (MDE) (Maryland Department of the Environment, n.d.-a), with additional requirements triggered for larger or consumptive withdrawals. Under the Code of Maryland Regulations (COMAR), consumptive use is addressed through multiple provisions. Notably, COMAR 26.17.07.02 requires low-flow augmentation or usage restrictions for nonresidential surface water withdrawals in the Potomac River basin that are consumptive and exceed 1 MGD (Maryland Department of the Environment, n.d.-b). Additionally, COMAR 26.17.06.06 allows MDE to impose minimum flow requirements or curtailment conditions to protect aquatic ecosystems and downstream users during periods of low stream flow (Maryland Department of the Environment, n.d.-c).

Current state regulations often apply only to facilities withdrawing water directly from surface- or groundwater sources. In practice, however, data centers in the Potomac basin typically obtain water through municipal systems rather than direct withdrawals. As a result, data center cooling demand is generally met using existing, previously permitted municipal capacity, and the state consumptive use regulations do not apply directly to these facilities.

However, as data center development continues, the cumulative consumptive use associated with these municipally supplied facilities may have measurable effects on basin hydrology, particularly during low-flow and high-temperature periods. Assessing these potential impacts will be important to ensure that regional drought management and water allocation frameworks remain robust. Several states have begun to address this issue. For example, New York's Sustainable Data Centers Act would require operators to disclose projected water use (New York State Senate, 2025).

- *Potential need for change in demands evaluation method for WMA utilities*

This study assumes that data center water use within the WMA is implicitly captured in the current methodology for estimating demands, through employee unit use (Chapter 3) and return flow assumptions (Chapter 5). However, this assumption may warrant re-evaluation in future studies. While upstream data center consumptive use remains relatively modest in 2050, projected peak summertime use within the WMA reaches up to 140 MGD based on the PJM energy forecasts. Hence, it is important to assess whether this volume is fully reflected in current demand and return flow estimates. The current methodology (Chapter 3, and Chapter 5) may not adequately capture the full scale of data center consumptive use within the WMA, especially for future years.

- *Use of reclaimed water*

While the use of reclaimed water can reduce demand for potable supplies, it does not eliminate consumptive loss. When reclaimed water is withdrawn from a treatment plant and used for evaporative cooling, it reduces return flows and effectively removes water from the system. This can still impact water availability, especially if the reclaimed water is sourced upstream of the WMA intakes, potentially affecting both supply and environmental flow targets at Little Falls. The impacts on regional water supplies could be reduced by the siting of new data centers at locations with access to reclaimed water that would otherwise be discharged downstream of water supply intakes. A potential option is the Noman M. Cole Jr. Pollution Control Plant, which currently treats approximately 40 MGD and discharges into the Occoquan estuary (County of Fairfax, 2024; EQAC, 2024).

- *Water quality*

Water quality impacts may emerge as clusters of data centers increase pressure on existing infrastructure. Potential issues include thermal discharges and chemical contaminants from cooling systems, particularly salts, biocides, and corrosion inhibitors present in cooling tower blowdown. If not properly managed, these discharges can degrade downstream water quality and place added strain on wastewater treatment systems (Müller *et al.*, 2024). Salt drift from evaporative cooling towers has also been observed to settle on nearby soils and surface waters, potentially harming vegetation and increasing salinity in aquatic systems (Liang *et al.*, 2023). This concern is especially pressing given that salt pollution is already a growing issue in many freshwater ecosystems (Kaushal *et al.*, 2025) and in certain water supply reservoirs, including the Occoquan.

- *Water-energy nexus*

Air-cooled systems can reduce direct water use at data centers, but they often increase electricity demand. This added electricity use can raise indirect water consumption through thermoelectric generation, including nuclear. As meteorological change drives higher average temperatures and more frequent heat waves, peak electricity demand is expected to rise, placing additional strain on both energy (PJM, 2025) and water systems. To fully understand data center impacts, it is important to consider both direct and indirect water use. Policymakers should evaluate these tradeoffs and support the development of cooling technologies that minimize both water and energy use under future climate conditions.

- *Impact of technological innovations*

Emerging technologies like direct-to-chip cooling offer promise, especially for high-performance AI workloads. Amazon Web Services (2025) reports that its closed-loop liquid cooling system circulates engineered fluids directly across chip surfaces. While this approach may improve energy and water efficiency, its broader adoption warrants further evaluation. Questions remain about the long-term performance of engineered fluids, system resilience under heavy loads, waste handling, and potential health and safety concerns. These issues might limit the widespread adoption of this technology.

6.3 Total Consumptive Use

In this section, the two forecasts presented in Sections 6.1 and 6.2 are combined to provide a look at the total amount of consumptive use upstream of the WMA intakes assumed in the current study, including consumptive use by data centers. Table 6-9 shows monthly forecasts of upstream consumptive use for 2050, as well as extrapolated values for 2085. It also compares the current study's forecasts for 2050 with those developed for CO-OP's 2015 water supply study and used in CO-OP's 2020 water supply study. The monthly forecasts for data centers are from the Medium Water Use scenario, which is also assumed for the resource analysis conducted for this study (see Chapter 8). From a review of the values in Table 6-9, average annual 2050 total upstream consumptive use forecasts used in the current study are slightly higher than those used in the 2020 study, but average summer (June, July, August) consumptive use is three percent lower. Also, the current study's 2050 forecasts for October and November are 35 to 40 MGD higher than those of the 2020 study.

This is significant because low flows and reservoir releases would occur in October and November during a future drought resembling the 1930 drought of record. A graph of the 2050 monthly upstream consumptive use forecasts from Table 6-9 appears in Figure 6-6. The higher upstream consumptive use estimates in this study are driven primarily by the inclusion of estimated unreported use, representing a more complete accounting of upstream demands.

While the data center component is newly added in this study, its contribution to total upstream consumptive use is relatively modest, as is evident from Figure 6-6. This reflects lower projected data centers growth in upstream areas as compared to developments in the WMA. Table 6-9 and Figure 6-6 make use of the Medium Water Use forecasts for data centers, with a 2050 monthly maximum of 8.2 MGD in August, which is just five percent of the projected consumptive use in August. Even in the case of the High Water Use scenario, with a 2050 monthly maximum of 14.4 MGD, forecasted water use by data centers is only about nine percent of total August upstream consumptive use. The forecasted annual average upstream consumptive use by data centers in 2050 is 4.7 MGD, which is four percent of the total. While modest, these shares are comparable to those of several established sectors, such as commercial (currently two percent of upstream consumptive use), thermoelectric (currently three percent), industrial (currently five percent), and public water supply (currently nine percent), highlighting the emerging significance of data centers in the region's water demand profile.

It should be noted that these figures reflect only upstream data center use and are not intended to represent the footprint of data centers across the entire basin, which is evaluated in Seck *et al.* (in preparation). Additionally, the projections are based on grid-connected energy demand forecasts, which represent the best available basis for estimating water use. As a result, it will be important to regularly re-evaluate assumptions, improve data transparency, and continue monitoring this evolving sector as part of future planning cycles. CO-OP's 2030 water supply study will provide a timely opportunity to revisit forecasts and incorporate new insights into data center water use trends, technologies, and regulatory considerations.

To reflect the impacts of future upstream consumptive use on river flows in PRRISM simulations of system performance, PRRISM's Potomac River flow inputs, based on historical data, are first modified to eliminate the effects of historical consumptive use. This is done by adding an estimate of historical consumptive use to every daily flow record in the flow time series. The assumption is made that

consumptive use grew in a linear fashion, from a starting value of zero in the year 1929 to the 2017-2019 average baseline values appearing in Table 6-4. This linear growth model assumes no historical knowledge of past upstream water use, and the development of a more realistic time series of historical upstream consumptive use should be considered for future CO-OP water supply studies. Forecasts from Sections 6.1 and 6.2 are implemented in PRRISM to simulate, at the monthly time scale, future monthly consumptive use. The 2017-2019 average baseline values and annual growth rates from Table 6-4 are used to simulate monthly upstream consumptive use, not including the impacts of data centers. The impacts of data centers are added using the monthly forecasts in Table 6-7, which is based on the Medium Water Use scenario. PRRISM includes simulation of reductions in upstream water withdrawals due to state drought management decisions, which provides some mitigation of the impacts of upstream consumptive use, as discussed in the 2020 water supply study (Ahmed *et al.*, 2020).

Table 6-9: 2050 and 2085 results for total upstream consumptive use, including use by data centers, and a comparison of 2050 results from the current study with those from the 2020 water supply study.

	2050 forecasts - without data centers	2050 - data center forecasts Medium Water Use scenario	2050 forecasts - total	2085 extrapolations - without data centers	2085 - data center extrapolations - Medium Water Use scenario	2085 extrapolations - total	2020 study forecasts for 2050, MGD	Difference, percent
January	74	3.2	77	77	5.4	82	71	9
February	74	2.8	77	79	4.7	84	72	7
March	76	2.8	79	81	4.8	86	71	11
April	124	3.4	128	142	5.8	148	83	54
May	129	4.0	133	148	6.8	154	105	26
June	137	4.8	142	155	8.1	163	143	-1
July	141	7.1	148	158	12.1	170	155	-5
August	143	8.2	151	165	14.1	179	156	-3
September	138	7.2	145	159	12.3	171	128	13
October	127	4.6	132	148	7.9	156	97	35
November	118	4.0	122	138	6.9	145	82	49
December	72	3.9	76	76	6.7	83	74	2
Average	113	4.7	117	127	8.0	135	103	14
Summer Average	140	6.7	147	159	11.4	171	151	-3

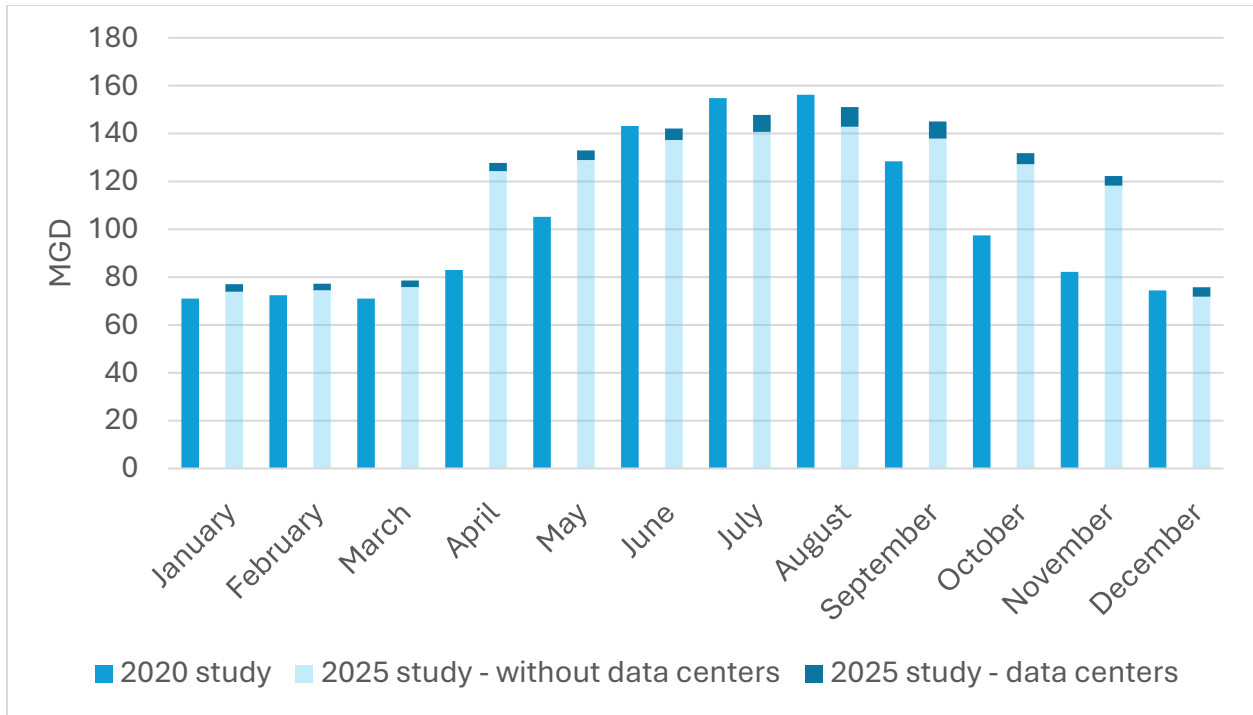


Figure 6-6: Comparison of 2050 upstream CU estimates for 2020 study and current study.

7 Impacts of Meteorological Change

In this chapter, the approach used to estimate the impacts of meteorological change on regional stream flow and on water demands is described. Stream flow is largely determined by precipitation and to a lesser degree by temperature. In the Potomac, approximately 35% of the precipitation that falls on the watershed becomes stream flow, with most of the remainder lost to evapotranspiration.

Evapotranspiration is the return of precipitation to the atmosphere by evaporation from land and water surfaces and by transpiration by trees and other vegetation. Higher temperatures increase evaporation rates and thus tend to reduce stream flow.

Water supply planning decisions for the WMA are primarily driven by the risk of future extreme drought, since reservoir storage is sufficient to augment river flow to meet forecasted supply needs and environmental requirements during moderate or short-term drought. In this study, future trends in mean annual stream flow, including flows in extreme drought years comparable to the drought of record, are investigated by estimating changes in the probability distribution function of annual stream flow under future meteorological conditions. This allows estimation of the disparate effects of meteorological change on high flow versus low flow years. The approach is similar to that used in the 2020 WMA water supply study (Ahmed *et al.*, 2020), and it is described in detail by Schultz *et al.* (2025). Results from the analyses described below are used to construct Potomac River flow and reservoir inflow time series inputs for the PRRISM planning model. Projections from this analysis are also used to adjust temperature and precipitation time series used to simulate daily water demands for future years.

There are limitations to the approach described below to investigate trends in hydrometeorological conditions. One limitation is the coarseness of the time scale used in the analysis. Hydrometeorological conditions in the Potomac are characterized below by six successive 30-year time periods: 1950-1979, 1980-2009, 2010-2039, 2040-2069, and 2070-2099. Values of statistics of interest for one of these time periods are assumed to be uniform throughout the 30 years, even though this is clearly not expected to be the case - significant meteorological change is projected to occur not only between 30-year periods but also within each 30-year period. The coarseness of the time scale is necessitated by the need for samples of ample size to compute statistics characterizing extreme years. A second limitation is the weak match between annual temperature and precipitation in the Potomac watershed computed from observed data and computed from the downscaled projections for meteorologic change that are used in the analysis. Results of any forecast should be verified and/or updated as new data become available. But in the case of the current study's hydrometeorological projections for the "near-term" period, that is, 2010-2039, results should be viewed as preliminary and subject to further validation.

7.1 Precipitation and Temperature Trends

Trends in precipitation and temperature in the Potomac River watershed above Little Falls Dam are primarily characterized in this study by looking at changes in long-term statistics calculated for successive 30-year time intervals (or, in a couple of cases, somewhat shorter periods due to data availability).

7.1.1 Historical Trends

Significant changes in historical long-term means of meteorological variables have not been evident in trend analyses done for past CO-OP water supply studies. However, with the addition of 2018-2023 data for the current study, statistically significant increases in both temperature and precipitation are now detected, as described below.

7.1.1.1 Data Sources

Historical meteorological data used in this study were from the Oregon State University's Parameter-elevation Relationships on Independent Slopes Model (PRISM), available at <http://www.prism.oregonstate.edu/>. PRISM uses weather observations from a wide range of monitoring networks and a series of regression models (Daly *et al.*, 2008) to develop maps over the conterminous U.S. for grid cells at 800-meter (m) to 4-kilometer (km) spatial resolution. For the current study, PRISM 4-km by 4-km gridded data for monthly precipitation and temperature for 1896-2023 were downloaded, spatially averaged over the Potomac River drainage area upstream of the [USGS stream gage at Little Falls Pump station near Washington, DC - Station No. 01646500](#) (38.9375 degrees north and - 77.1875 degrees west), and temporally averaged by calendar year.

Potomac River flows used in this study are “natural” annual flows at Little Falls based on a monthly reconstruction which makes use of stream flow data available from the USGS's National Water Information System website.¹ The natural flow time series are estimates of what flow at Little Falls would have been in the absence of WMA withdrawals or diversions to the C&O Canal, without the effects of the North Branch reservoirs, Jennings Randolph and Savage, and without upstream consumptive use. Details on the methods used to estimate Potomac River natural flows are given in Ahmed *et al.* (2020).

7.1.1.2 Results

Historical long-term mean precipitation, temperature, and river flow in the watershed are given in Table 7-1. The period, 1896-1979, is defined as the study's baseline, and no trends in long-term mean values are evident over this period. Subsequent 30-year means do not differ significantly from baseline values except in the case of the most recent period, 2010-2023. For 2010-2023, the mean temperature of 11.72 °C is significantly higher than the baseline mean of 11.19 °C, based on Welch's two-sample t-test, at the 1 percent significance level (p-value = 0.01). This test does not indicate an increase in precipitation during this period.

Trends in historic data were also investigated using a non-parametric method: the Mann-Kendall trend test. This statistical test does not rely on assumptions about the underlying probability distribution of the variable of interest. The Mann-Kendall trend test does not indicate a trend in annual mean precipitation, temperature, or flow over the historical baseline period, 1896-1979. However, over the period extending from 1950-2023, the Mann-Kendall test indicates a rising trend in annual precipitation at the 5 percent significance level (p-value = 0.044) and a rising trend in temperature at the 1 percent significance level (p-value = 0.005). No trend is detected in annual mean flow values.

¹ USGS National Water Information System (NWIS): <https://waterdata.usgs.gov/nwis/>

Table 7-1: Past long-term means of precipitation, temperature, and river flow in the Potomac River watershed above Little Falls Dam.

Time interval	Precipitation		Temperature		Natural river flow	
	Mean, mm/yr	Change from study base period	Mean, °C	Change from study base period, °C (°F)	Mean, cfs	Change from study base period
Historical baseline period: 1896-1979	990	0%	11.19	0	11488	0%
1896-1919	998	+1%	11.04	-0.15 (-0.27)	11759	+2%
1920-1949	983	-1%	11.46	+0.27 (+0.49)	10945	-5%
1950-1979	992	0%	11.04	-0.15 (-0.27)	11815	+3%
1980-2009	1014	+2%	11.18	-0.01 (-0.02)	12758	+11%
2010-2023	1085	+10%	11.72	+0.53 (0.95)	12460	+8%

7.1.2 Projected Trends

Though forecasts are beset by uncertainties, the best information on future precipitation and temperature in the Potomac basin is based on results from computer models that predict future global meteorological conditions. These global models have been constructed at scientific centers around the world to simulate the physical, chemical, and other processes that govern flows and energy exchanges in the Earth's atmosphere, oceans, and land surfaces. The current study is based on the latest set of projections from these models, available in a compilation called the Coupled Model Intercomparison Project Phase 6 (CMIP6). CO-OP's 2020 water supply study (Ahmed *et al.*, 2020) made use of results derived from the previous compilation of global model projections, CMIP5.

Temperature and precipitation projections from global models have a spatial resolution that is fairly coarse for use in regional studies, so they are commonly refined by techniques referred to as "downscaling." For the current study, an ensemble of 54 downscaled CMIP6 projections made available by the National Aeronautics and Space Administration (NASA) was accessed and processed, as described below.

7.1.2.1 Data Sources

Projections for future precipitation and temperature in the Potomac basin were derived from CMIP6 results. These projections have been compiled, statistically downscaled, and made available by NASA via its Data Portal at <https://data.nas.nasa.gov/nex-dcp30-cmip6/data.php>. The downscaling methodology is described by Thrasher *et al.* (2024). It includes a bias-correction step that uses a quantile mapping approach to adjust projected meteorological data based on observed data from PRISM for the historical period, 1981-2014, and also a spatial disaggregation step. The NEX-DCP30-CMIP6 data extend from 1950 through 2100 on a grid of 30 arc seconds by 30 arc seconds, providing a spatial resolution of approximately 1 km by 1 km. For the current study, NEX-DCP30-CMIP6 gridded annual data were downloaded and then clipped and spatially averaged over the drainage area of the Potomac basin upstream of the USGS stream gage at Little Falls. The following parameters were downloaded:

- Total annual precipitation, millimeters per year (mm/yr)
- Annual average of minimum daily near surface air temperature, °C
- Annual average of maximum daily near surface air temperature, °C

For each year, average daily temperature was estimated as the mean of average daily minimum and maximum temperatures.

Available NEX-DCP30 CMIP6 projections represent four distinct Shared Socioeconomic Pathways (SSPs), that is, socioeconomic narratives of potential future global conditions and greenhouse gas emissions. The current study uses two of these: SSP2-4.5, which is described as a "middle of the road" scenario representing a continuation of historical patterns and trends, and SSP3-7.0, which is described as the "regional rivalry" scenario, which represents a future with resurgent nationalism, declining investments in education and technology, and significant emissions. These two scenarios provide an ensemble of 54 time series simulations of future annual precipitation and temperature, based on model estimates from groups of scientists around the world. In this study, each of the 54 ensemble members is treated as a potential realization of future meteorological conditions in the Potomac basin.

7.1.2.2 Verification/filtering step

A filtering process was applied to verify that the NEX-DCP30-CMIP6 ensemble member simulations provide a reasonably good match to observed data over the period, 1950-2009. The filtering process used the nonparametric Kolmogorov-Smirnov test to compare the empirical cumulative distribution functions of simulated versus observed annual precipitation and annual temperature in the Potomac basin, where the observed distributions were calculated from PRISM data. All members passed the filtering test over the period 1980-2009, which is consistent with the fact that the bias correction process used to construct the downscaled projections was based on 1981-2014 PRISM data. However, for the period 1950-1979 the Kolmogorov-Smirnov test indicated significant differences in observed versus simulated probability distributions of annual precipitation and/or annual temperature for 23 of the 54 ensemble members at the $p = 0.01$ significance level. After discarding these, 31 members remained in the filtered NEX-DCP30-CMIP6 ensemble: 17 SSP2-4.5 members and 14 SSP3-7.0 members. The results presented in this study are computed using only the 31-member filtered subsets of ensemble members, which are listed in Table 7-2 by the World Climate Research Program (WCRP) source id.²

It should be noted that the downscaled CMIP6 projections used in this study did not match historical annual temperature and precipitation values as well as the previously used downscaled CMIP5 projections. When the Kolmogorov-Smirnov test was applied to 231 CMIP5 ensemble members using a more stringent criterion, the $p = 0.05$ significance level, 145 members remained after this filtering step (Schultz *et al.*, 2025). At the $p = 0.05$ significance level, none of the 54 CMIP6 ensemble members remain. This appears to be at least partly because the rise in Potomac CMIP6 temperatures between the two periods, 1950-1979 and 1980-2009, is higher than observed and higher than CMIP5 values. The CMIP6 models, however, include new and improved representations of physical, chemical, and biological

² See https://wcrp-cmip.github.io/CMIP6_CVs/docs/CMIP6_source_id.html.

processes as well as higher spatial resolution, compared to CMIP5, and are thus believed to provide us with the most useful projections of future conditions.

7.1.2.3 Results

Plots of projected precipitation and temperature in the Potomac watershed are given in Figure 7-1 and Figure 7-2. These graphs show five selected statistics computed by year from the 31-member ensemble of filtered projections and include graphs of observed values. Most of the projected values fall within the dark gray bands, which are bounded by the 10th to 90th percentile values for each given year. The light gray bands indicate the full range of values, from minimum to maximum, for each year from this set of 31 projections. The graphs indicate that future precipitation and future temperature are both projected to rise over the course of the 21st century. In both graphs, the width of the bands tends to rise with time, indicating increasing variance in global model results.

Trends in long-term means of selected climate statistics for the Potomac River drainage area above Little Falls are given in Table 7-3. Means and standard deviations of precipitation and temperature are computed for each of the five successive 30-year intervals of the simulation period for each of the 31 ensemble members and then results are averaged over the entire ensemble. Thus, the results in Table 7-3 are averages over the two SSPs considered in this study.

Table 7-2: Filtered ensemble members used in this study, by WCRP source id. The availability of the SSPs is given in the last two columns, with x indicating available, and blank indicating unavailable.

Model	Variant	SSP245	SSP370
ACCESS-CM2	rlilplfl	x	x
ACCESS-ESM1-5	rlilplfl	x	x
BCC-CSM2-MR	rlilplfl	x	x
CNRM-CM6-1	rlilplf2	x	x
EC-Earth3	rlilplfl	x	x
EC-Earth3-Veg-LR	rlilplfl	x	x
FGOALS-g3	r3ilplfl	x	x
GFDL-ESM4	rlilplfl	x	x
HadGEM3-GC31-LL	rlilplf3	x	
INM-CM4-8	rlilplfl	x	x
INM-CM5-0	rlilplfl	x	x
KACE-1-0-G	rlilplfl	x	x
KIOST-ESM	rlilplfl	x	
MPI-ESM1-2-HR	rlilplfl	x	x
MRI-ESM2-0	rlilplfl	x	x
NESM3	rlilplfl	x	
TaiESM1	rlilplfl	x	x

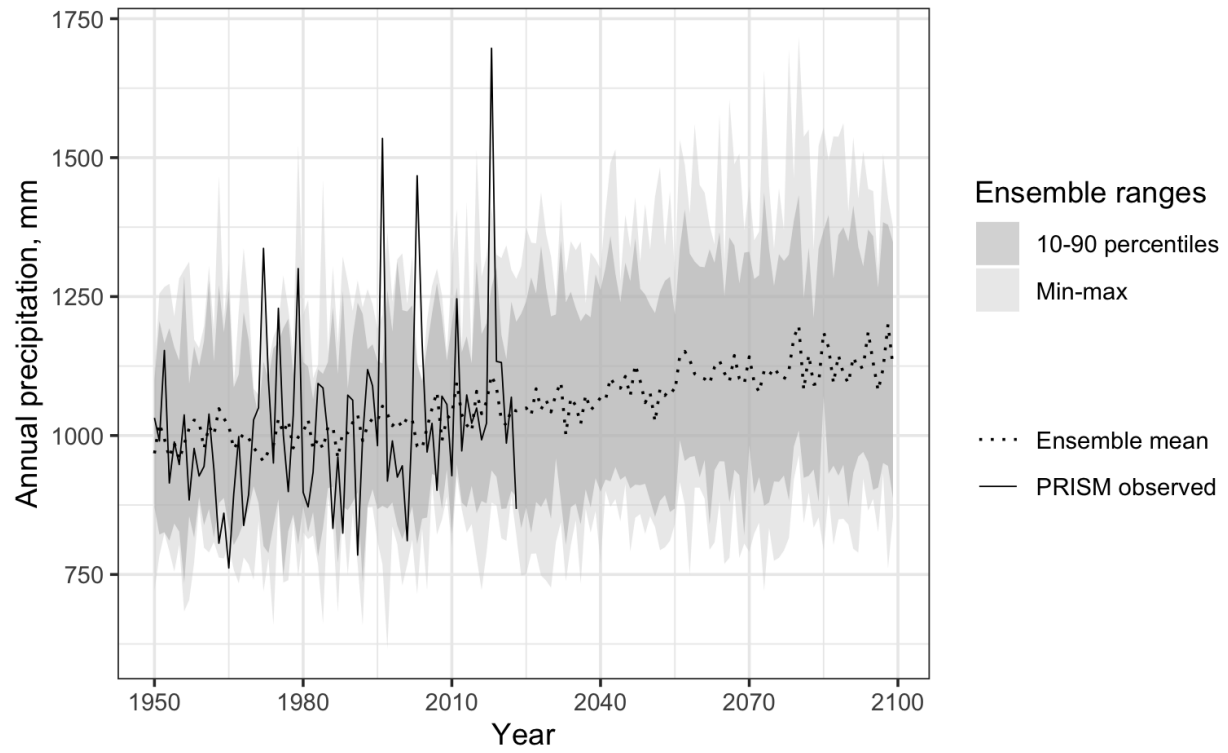


Figure 7-1: Statistics, by year, for ensemble of projected annual precipitation in the Potomac watershed.

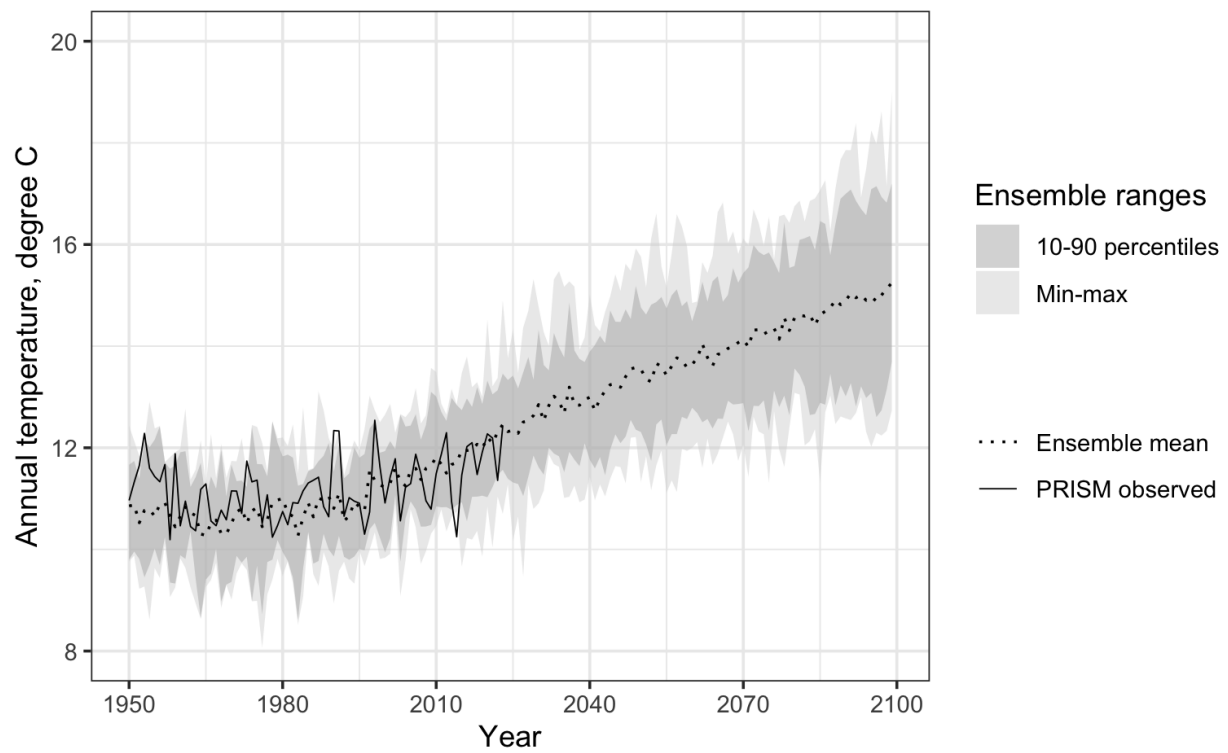


Figure 7-2: Statistics, by year, for ensemble of projected annual temperature in the Potomac watershed.

Table 7-3: Projected precipitation and temperature for the Potomac basin – long-term means and standard deviations from the 31-member combined SSP245 and SSP370 CMIP6 ensemble.

		Precipitation			Temperature		
	Time interval	Mean, mm/year (inches/year)	Change in mean from study base period	Standard deviation, mm/year (inches/year)	Mean, °C (°F)	Change in mean from study base period, °C (°F)	Standard deviation, °C (°F)
From historical data (PRISM)							
	1896-1979	990		129	11.19		0.60
	1950-1979	992		132	11.04		0.54
From ensemble of 54 CMIP6 projections							
	1950-1979 (study base period)	996 (39.2)	0.0%	120 (4.7)	10.65 (51.2)	0.0 (0.0)	0.73 (1.3)
	1980-2009	1015 (40.0)	1.9%	120 (4.7)	11.07 (51.9)	0.4 (0.7)	0.71 (1.3)
	2010-2039	1052 (41.4)	5.7%	125 (4.9)	12.35 (54.2)	1.7 (3.1)	0.67 (1.2)
	2040-2069	1098 (43.2)	10.2%	134 (5.3)	13.54 (56.4)	2.9 (5.2)	0.66 (1.2)
	2070-2099	1126 (44.3)	13.1%	143 (5.6)	14.64 (58.4)	4.0 (7.2)	0.65 (1.2)

The CMIP6 projections of both precipitation and temperature in the Potomac basin steadily rise over the 150-year simulation period, 1950-2099. Long-term precipitation means in Table 7-3 change from 996 millimeters per year (mm/year) (39.2 inches/year) for the base period, 1950-1979, to 1098 mm/year (43.2 inches/year) for 2040-2069 and 1126 mm/year (44.3 inches/year) for 2070-2099, increases of 10.2% and 13.1% respectively. Temperatures increase from 10.65 °C (51.2 °F) in the base period to 13.54 °C (56.4 °F), and 14.64 °C (58.4 °F) for the future periods, 2040-2069 and 2070-2099, increases of 2.9 °C (5.2 °F), and 4.0 °C (7.2 °F), respectively.

It should be noted that the projected mean temperature rise in Table 7-3 of 1.7 °C (3.1 °F) for the near-term period, 2010-2039, is considerably higher than the mean rise that has been observed for the period, 2010-2023: 0.53 °C (0.95 °F). This may be a reflection of the weak match between the CMIP6 projections and observed data, discussed in Section 7.1.2.2, and also of the coarseness of the time scale used for this study's analyses. Comparing Potomac basin trends from CMIP6 with those from CMIP5, reported in Ahmed *et al.*, 2020, CMIP6 predicts mixed results for changes in precipitation but higher increases in temperature, on average, with CMIP5 precipitation increasing 10.6% and 12.7% for the periods, 2040-2069 and 2070-2099, respectively, and CMIP5 temperature rising by 2.6 °C (4.7 °F) and 3.5 °C (6.2 °F), respectively, for the same two periods, in comparison with 1950-1979.

7.2 Potomac Basin Stream Flow

7.2.1 Stream Flow Response Function

The response of annual stream flow to changes in precipitation and temperature was determined using natural Potomac River flow at Little Falls, derived from historical observations, and the PRISM dataset of observed climate, described above. Multiple least-squares regression analysis was applied to develop an equation to predict mean annual flow, Q_i , from mean annual temperature, T_i , and mean annual precipitation, P_i , in a given year, i , for the 123-year historical time period, 1896 - 2018, as described by Schultz *et al.* (2025). A nonlinear precipitation term is included in the regression equation because it is found to improve predictions in very high-flow and low-flow years. The flow and precipitation variables are written as fractional changes from historical means in the historical base period, 1896-1979, and the temperature term is written as a difference of annual temperature from average temperature over the historical base period. To allow easy comparison between annual flow and annual precipitation, uniform units of mm/year are used for both variables in Equation 7-1. Annual flow for the 11560 square mile Potomac drainage area above Little Falls is converted from cfs to mm/year by using the multiplicative factor, 0.02983 mm/(year*cfs).

The regression equation is

$$\frac{(Q_i - \bar{Q})}{\bar{Q}} = \beta_1 (T_i - \bar{T}) + \beta_2 \frac{(P_i - \bar{P})}{\bar{P}} + \beta_3 \left(\frac{P_i - \bar{P}}{\bar{P}} \right)^2 + \varepsilon_i \quad (\text{Equation 7-1})$$

where

- i = calendar year, from 1896 through 2018
- Q_i = mean annual natural Potomac River flow at Little Falls in year, i (mm/year)
- $\bar{Q}$ = mean annual natural flow in the historical base period, 1896-1979
= 342.7 (mm/year)
- T_i = mean annual temperature in the Potomac watershed in year, i (°C)
- $\bar{T}$ = mean annual temperature in the historical base period, 1896-1979
= 11.19 (°C)
- P_i = mean annual precipitation in the Potomac watershed in year, i (mm/year)
- $\bar{P}$ = mean annual precipitation in the historical base period, 1896-1979
= 990 (mm/year)

Regression coefficients, β_1 through β_3 , along with statistics from the regression analysis, are given in Table 7-4. The coefficient of determination for the regression model is 0.75, and the standard error of the regression is 0.1757. The p-values indicate that the regression coefficients are all significant at the 95% level.

Table 7-4: Regression coefficients for stream flow response function based on historical data.

	β_1	β_2	β_3
Coefficient	-0.059	1.793	0.778
Standard Error	0.027	0.123	0.286
P-value	0.031	2 e-16	0.007

The simple stream flow response function, Equation 7-1, serves as this study's hydrologic model. Its predictions of annual flow are reasonably good, as indicated by the regression statistics above and by a graphical comparison of observed and predicted annual flows over the historical record, shown in Figure 7-3. This figure has time series plots of historical and predicted annual Potomac River flow along with annual precipitation and temperature used in Equation 7-1 to calculate the predicted flow. The time series extends from 1896 through 2023. The sensitivity of annual flow to change in annual temperature is represented by the value of the temperature coefficient in Table 7-4, $\beta_1 = -0.059$. Thus, Equation 7-1 predicts that if annual temperature changes by 1°C, annual flow will decrease by 5.9 percent.

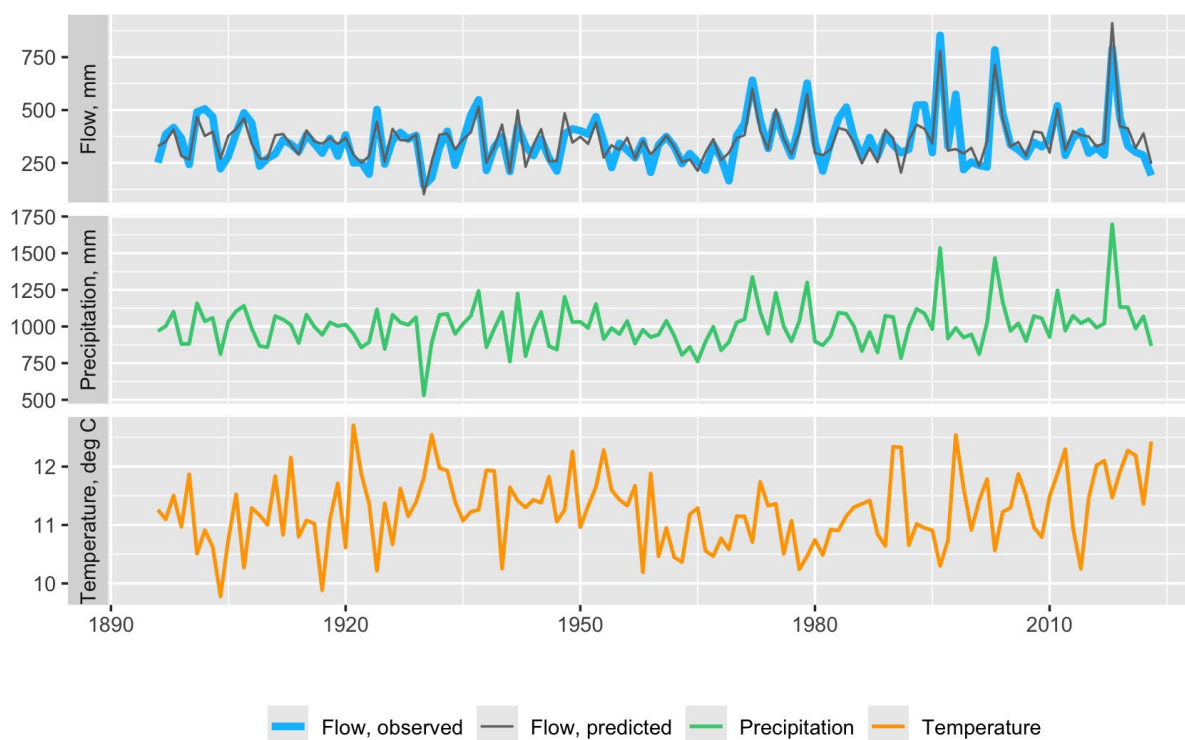


Figure 7-3: Annual values of observed and predicted natural Potomac River flow and observed precipitation and temperature for 1896 - 2023.

7.2.2 Projected Flow

Projected natural flow in the Potomac River above Little Falls is computed using the stream flow response function, Equation 7-1. This provides a time series of annual flows for each of the 31 members of the filtered ensemble. A graph of flow ensemble statistics, computed by year, is given in Figure 7-4.

This graph shows that annual flows in very low flow years, indicated by the bottom edges of the 10th to 90th percentile band and the minimum to maximum value band, are projected to be noticeably lower in the second half of the 21st century according to some projections. Long-term mean flows are also projected to decrease, as can be seen from Table 7-5, which gives long-term mean flows computed for each of the 31 ensemble members, then averaged over the ensemble.

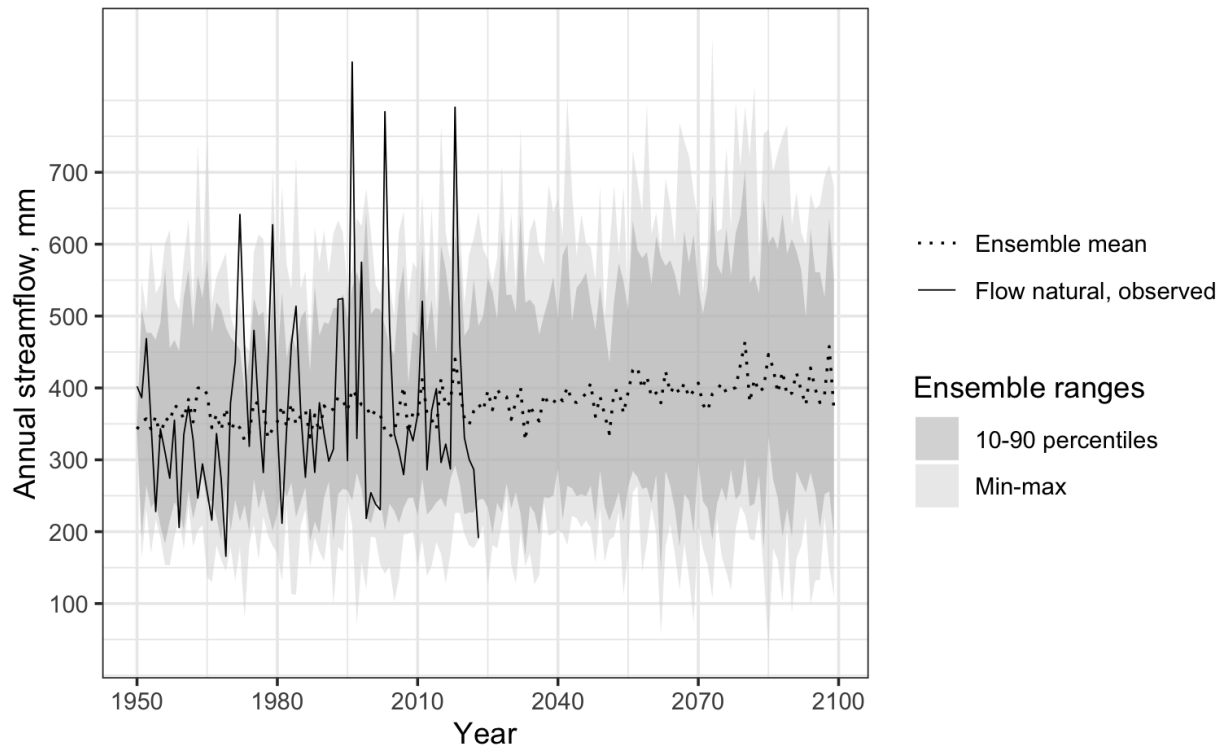


Figure 7-4: Statistics, by year, for ensemble of projected annual stream flow in the Potomac watershed.

Table 7-5: Ensemble means of time series long-term means and standard deviations of projected flows in the Potomac watershed.

		Flow		
	Time interval	Mean, cfs (mm/year)	Change in mean from study base period	Standard deviation, cfs (mm/year)
From historical data				
	1896-1979	11,488 (343)		3,336 (99)
	1950-1979	11,815 (352)		3,702 (110)
From Eq. 6-1, with ensemble of precipitation and temperature projections				
	1950-1979 (study base period)	12,348 (368)	0.0%	3,315 (99)
	1980-2009	12,289 (367)	-0.5%	3,468 (103)
	2010-2039	11,843 (353)	-4.1%	3,711 (111)
	2040-2069	11,746 (350)	-4.9%	3,763 (112)
	2070-2099	11,386 (340)	-7.8%	4,119 (123)

7.2.3 Developing Flow Inputs for Water Supply Planning Models

A nonparametric approach is used to investigate future trends in extreme values of annual climatic and hydrologic variables. Long synthetic "pooled" time series of annual climate representative of shorter 30-year time windows of interest are formed by selecting and combining segments of climate projections from different global models using the K-nearest neighbor method, where a projected climate time series from a single global model is treated as just one possible realization of the future (see Schultz *et al.*, 2025). The long time series are used to compute the empirical cumulative distribution function (ecdf) of annual flow for time periods of interest, providing a more complete picture of future changes of interannual variations in stream flow and their impact on water resources than projections of future means and variability. The resulting ecdfs are spatially and temporally disaggregated to obtain daily flow inputs needed for CO-OP's PRRISM planning model using a quantile scaling methodology.

A major uncertainty in many regions is the response of stream flow to changes in temperature. Rising temperatures will affect stream flow by increasing evaporative losses from soils and water surfaces and by changing the timing and magnitude of snowmelt. Hydrological droughts may become more severe due to increasing temperatures coupled with natural variability in precipitation. To reflect this uncertainty, future trends in annual flows are investigated for three scenarios that reflect a range of sensitivities of flow to temperature, defined by plus or minus one standard error of the stream flow response function temperature coefficient, β_1 , that is, ± 0.027 , from Table 7-4. Table 7-6 provides annual flow scaling factors for these three scenarios, labeled Low, Medium, and High. Scaling factors are given for three 30-year time windows, 2010-2039, 2040-2069, and 2070-2099. The baseline period used to compute the scaling factors is 1950-1979. The sets of scaling factors in Table 7-6 are used in the current study to construct

daily times series of Potomac River flows and reservoir inflows representative of potential current and future hydrologic conditions following the approach of Schultz *et al.* (2025).

The values of the annual flow scaling factors in Table 7-6 demonstrate the disparate impacts of changing meteorological conditions on stream flows in low flow versus high flow years. For high flow years, represented by flow quantiles ≥ 0.90 (equivalently, annual flow percentiles ≥ 90), most flow scaling factors are greater than one, indicating that flows in future high flow years will tend to increase. Conversely, for low flow years, represented by flow quantiles ≤ 0.10 (equivalently, annual flow percentiles ≤ 10), most scaling factors are less than one, indicating that flows in low flow years will tend to decrease. Results in Table 7-6 depend significantly on the temperature sensitivity scenario, demonstrating the competing effects of rising temperature and rising precipitation on future stream flows. Annual flows are predicted to rise in almost all cases for the low temperature sensitivity scenario, indicating that rising precipitation will be the dominant factor determining future stream flow. Conversely, annual flows are predicted to fall in almost all cases for the high temperature sensitivity scenario, indicating that rising precipitation is not enough to stem the decrease in flows due to rising temperatures.

Noteworthy in Table 7-6 are the flow scaling factors for the 0.01 quantile value, equivalent to the 1st percentile value of annual flow, that is, annual flow that has a 99 percent probability of being exceeded. The 1st percentile value of annual stream flow is indicative of a hydrological drought which in the past, based on the assumption of stationarity, would have been associated with a 100-year return period and referred to as a "100-year drought". Assuming medium sensitivity of flow to temperature change, the 0.01 quantile value flow scaling factors are 0.89, 0.87, and 0.71 for the time periods, 2010-2039, 2040-2069, and 2070-2099, respectively. These imply that in such a drought year, Potomac basin flows are projected to decrease by 11 percent, 13 percent, and 29 percent, respectively. These results can be compared with those from CO-OP's 2020 water supply study (based on CMIP5 projections), where projected decreases in 1st percentile flows were 2 percent, 16 percent, and 33 percent for the same periods, 2010-2039, 2040-2069, and 2070-2099. As discussed above, results for the near-term period, 2010-2039, are based on projections for temperature rise that are considerably higher than observed for 2010-2023. Therefore, the near-term annual flow scaling factors used in the current study should be viewed as preliminary and subject to further validation as more data become available.

In the current study, years with annual flow quantiles of 0.01 or less (equivalently, with annual flow percentiles ≤ 1) are referred to as extreme drought years. They include the hydrologic drought of record for the Potomac basin, which occurred in 1930 and has an annual flow quantile value of approximately 0.008 based on the 123-year period of record to develop the response function, Equation 7-1.

Key inputs of the PRRISM model are historical daily time series of natural Potomac River flows and reservoir inflows which extend from October 1, 1929, through December 22, 2009. For each 30-year time period and scenario of interest, temporal and spatial disaggregation of the projected impacts of meteorological change to historical daily flows time series is achieved by multiplying a given historical year's set of daily flows by the annual flow scaling associated with the historical year's annual flow quantile value (see Schultz *et al.*, 2025). For example, the historical year 1941 has an annual flow quantile value of approximately 0.05, so from Table 7-6 for the medium temperature sensitivity scenario, daily flows in 1941 are multiplied by the annual flow scaling factor of 0.93 to obtain the projected impact of

meteorological change for both of the periods, 2010-2039 and 2040-2069, and are multiplied by 0.85 to obtain the projected impact of meteorological change for the period, 2070-2099. The historical year 1930 is the drought year of record and has an annual flow quantile value of approximately 0.008. For the medium temperature sensitivity scenario, daily flows in 1930 are multiplied by 0.87, 0.86, and 0.69 to obtain the projected impact of meteorological change for the periods 2010-2039, 2040-2069, and 2070-2099, respectively, where interpolation was required to compute annual flow scaling factors for a flow quantile between 0.005 and 0.01. The years, 1929 through 1979, are assumed in this study to be a period prior to the onset of hydrometeorological change, as discussed above in Section 7.1.1. Before applying the scaling factors to flows in the years, 1980-2009, the projected impacts of meteorological change present in the historical time series are removed from flows in this time period using an inverse process: that is, by dividing flows by the scaling factors for the period, 1980-2009.

Table 7-6: Annual flow scaling factors, by temperature sensitivity and by 30-year time windows - 31 NEX-DCP30-CMIP6 filtered ensemble member means, baseline period of 1950-1979.

Flow quantile	Low temperature sensitivity (Higher Flows) $\beta_1 = -0.032$			Medium temperature sensitivity (Medium Flows) $\beta_1 = -0.059$			High temperature sensitivity (Lower Flows) $\beta_1 = -0.086$		
	2010-2039	2040-2069	2070-2099	2010-2039	2040-2069	2070-2099	2010-2039	2040-2069	2070-2099
0.005	0.99	1.07	0.97	0.83	0.83	0.63	0.66	0.59	0.27
0.01	1.03	1.09	1	0.89	0.87	0.71	0.75	0.68	0.44
0.02	1.03	1.08	1.02	0.93	0.92	0.78	0.83	0.76	0.55
0.05	1.01	1.07	1.03	0.93	0.93	0.85	0.86	0.8	0.67
0.10	1.03	1.09	1.05	0.96	0.97	0.89	0.9	0.86	0.73
0.25	1.03	1.09	1.08	0.99	1	0.95	0.94	0.92	0.82
0.35	1.04	1.1	1.09	0.99	1.02	0.97	0.95	0.94	0.86
0.50	1.04	1.12	1.11	1	1.04	1	0.96	0.97	0.89
0.65	1.04	1.12	1.12	1	1.05	1.02	0.97	0.99	0.93
0.75	1.05	1.13	1.13	1.01	1.07	1.04	0.98	1.01	0.95
0.90	1.05	1.15	1.15	1.02	1.09	1.07	0.99	1.04	0.99
0.95	1.05	1.15	1.15	1.02	1.09	1.07	0.99	1.04	1
0.98	1.04	1.13	1.14	1.02	1.08	1.07	0.99	1.03	1
0.99	1.02	1.11	1.12	1	1.07	1.06	0.97	1.03	1
0.995	1	1.09	1.12	0.98	1.05	1.07	0.96	1.02	1.01

7.3 Impact of Meteorological Change on WMA Demands

PRRISM's representation of daily demands includes a component based on the seasonal regression models described in Chapter 4. Because daily demand in these models depends on precipitation and temperature, simulated demand responds to changes in meteorological conditions. From Table 7-3, mean precipitation for the periods, 2010-2039, 2040-2069, and 2070-2099, is projected to increase by 5.7, 10.2, and 13.1 percent, respectively. Mean temperature for these three periods is projected to increase by 3.1, 5.2, and 7.2 °F. These changes are applied to the historical daily precipitation and temperature time series used in PRRISM to simulate daily WMA demands according to the models presented in Chapter 4. The increased precipitation tends to decrease demand for water, but the increase in temperature tends to increase water demand. Table 7-7 below summarizes total mean annual and total mean July WMA demand, as simulated by PRRISM, for the 30-year forecast periods, 2010-2039, 2040-2069, and 2070-2099. Results in Table 7-7 indicate that future meteorological change will have a modest effect on average annual WMA demands, with increases ranging from 5 to 7 MGD, or approximately one percent. There is a more significant impact on summertime demands, primarily due to the projected rise in temperatures, which will tend to increase outdoor water use activities such as lawn and landscape watering. In Table 7-7, the increase in mean July demand due to meteorological change is predicted to range from 21 to 33 MGD, which is approximately a four percent increase over predicted July demands without considering the impact of meteorological change.

The projected changes in Table 7-7 are based on assumptions and approximations that may have limited validity in future years. The changes in WMA demands are obtained from the daily demand regression models developed from recent observed data and do not include potential impacts of future changes in composition of water use types, including increases in water use by emerging industrial uses. Projected changes in demands for the current-day period, 2010-2039, may be high because they are based on an assumed uniform increase in annual temperature of 3.1 °F over this 30-year period, which is considerably higher than the temperature rise of 1 °F that has been observed to date. In addition, the region may develop new strategies for implementation of water use restrictions during periods of drought that can help mitigate future increases in July demands, by curtailing outdoor water use and potentially by managing daily water use fluctuations by data centers. Sensitivity of system reliability to these factors is a potential topic for investigation in an upcoming WMA system resource assessment.

Table 7-7: Simulated impact of meteorological change on WMA demands.

Forecast period	Assumed changes in Potomac basin climate		Projected changes in total WMA demand	
	Change in precipitation	Change in temperature, °F	Change in annual demand, MGD	Change in July demand, MGD
2010-2039	5.7%	3.1	+4.6 (1%)	+21 (4%)
2040-2069	10.2%	5.2	+6.5 (1%)	+28 (4%)
2070-2099	13.1%	7.2	+7.4 (1%)	+33 (4%)

8 Resource Analysis

This chapter presents the evaluation of the ability of existing and planned WMA system resources to meet future water demands and the environmental flow-by at [Little Falls Dam](#). The evaluation is conducted using PRRISM version 4.1, a new version of CO-OP's water supply planning model rewritten in the latest version of the ExtendSim software platform (see Section 5.1). It has updated values of parameters that determine demands and other changing system characteristics, but it represents system components, configuration, and operational decision rules that are almost identical to those in the previous version of PRRISM, version 3.11, which was used for ICPRB's 2020 water supply study (Ahmed *et al.*, 2020).

The two new system resources assumed for the resource analysis are listed below. They are both existing hard rock quarries for which agreements and/or contracts are in place that pave the way for future conversion to raw water storage reservoirs:

- Vulcan Quarry, assumed to be operational in 2040 (Fairfax Water's future Edgemon Reservoir)
- Luck Stone Quarry A, assumed to be operational in 2028 (Loudoun Water's future Milestone Reservoir)

8.1 Scenarios

A scenario planning approach is used to investigate system performance under a plausible range of future conditions. Scenario planning is a tool often applied to inform decisions regarding future infrastructure when significant uncertainties exist. Scenarios enable planners to test the robustness of alternatives and may assist in achieving stakeholder consensus. In the current study, system performance evaluations, which appear in Table 8-1, are provided for three scenario years under nine scenarios for water demand and availability.

The three scenario years considered are a near-term year, 2030, and two years farther in the future, 2045 and 2050. The year 2045 was selected because a 20-year horizon for the forecasts is specified by the two regional agreements, the WSCA and the LFAA. The year 2050 was selected because it is the last year for which MWCOG demographic forecasts are available. The values of WMA water withdrawals, upstream consumptive use, reservoir storage capacities, and other simulated quantities that change over time are all determined by the scenario year.

Three WMA water demand scenarios are considered: Low, Medium, and High. These are given in the last five rows of Table 3-15 by scenario year and listed below for reference. The Medium Demands scenario is based on the forecasts described in Chapter 3, and the Low and High scenarios are the Medium scenario demands plus or minus one standard error ($\pm SE$), where the percent error was derived in ICPRB's 2020 water supply study (Ahmed *et al.*, 2020) and depends on the number of years in the future of the forecast. Under the assumption that a forecast is a random variable with a normal distribution, this range represents approximately a 68 percent confidence interval for the forecast value. The range, $\pm SE$, was selected for this study rather than, for example, a 95 percent confidence interval, to avoid demand scenarios that might be viewed as unrealistic by stakeholders.

- Low Demands: total WMA demand is 437.4 MGD in 2030, 474.1 MGD in 2045, and 484.1 MGD in 2050
- Medium Demands: total WMA demand is 479.6 MGD in 2030, 525.0 MGD in 2045, and 537.9 MGD in 2050
- High Demands: total WMA demand is 521.8 MGD in 2030, 575.9 MGD in 2045, and 591.7 MGD in 2050

Daily demands are also influenced by meteorological conditions. The impacts of future changes in precipitation and temperature are captured in PRRISM simulations via the daily demand least squares regression models from Section 4.5, where assumed changes in long-term mean precipitation and temperature are taken from Table 7-7.

Uncertainties in water availability are represented by three scenarios for Potomac River flows and reservoir inflows: Lower Flows, Medium Flows, and Higher Flows. They are based, by scenario year, on the annual flow scaling factors in Table 7-6 that were developed from the 31 filtered sets of CMIP6 temperature and precipitation projections for the Potomac basin following the approach of Schultz *et al.* (2025) and similar to the approach used in the 2020 water supply study. The three flow scenarios represent a range of potential sensitivities of regional stream flow to changes in temperature. The descriptions below of the three flow scenarios are based on flow scaling factors for three annual flow quantile values from Table 7-6, where “extremely wet” represents changes in the 0.99 quantile, “typical” the 0.50 quantile, and where “extremely dry” represents changes to quantile values of annual flow that are ≤ 0.01 . The accuracy of these results is subject to the limitations of the analysis discussed in Chapter 7. In particular, results for the near-term period, 2010-2039, should be viewed as preliminary and should undergo further validation as more data becomes available.

- Lower Flows: assuming that stream flows are quite sensitive to rising temperatures ($\beta_1 = -0.086$)
 - Flows in extremely dry years in 2010-2039 and 2040-2069 are lower by 25 percent or more and 32 percent or more, respectively.
 - Flows in extremely wet years in the two periods, 2010-2039 and 2040-2069, are three percent lower and three percent higher, respectively.
 - Flows in a typical year in the periods between 2010-2039 and 2040-2069 are lower by four percent and three percent, respectively.
- Medium Flows: assuming that stream flows are moderately sensitive to rising temperatures ($\beta_1 = -0.059$)
 - Flows in extremely dry years in 2010-2039 and 2040-2069 are lower by 11 percent or more and 13 percent or more, respectively.
 - Flows in extremely wet years in 2010-2039 and 2040-2069 are unchanged and seven percent higher, respectively.
 - Flows in a typical year in 2010-2039 and in 2040-2069 are unchanged and four percent higher, respectively.
- Higher Flows: assuming that stream flows are not greatly affected by rising temperatures ($\beta_1 = -0.032$)

- Flows in extremely dry years in 2010-2039 and 2040-2069 are higher by three percent or less and nine percent or less, respectively.
- Flows in extremely wet years in 2010-2039 and 2040-2069 are higher by two percent and 11 percent, respectively.
- Flows in a typical year in 2010-2039 and 2040-2069 are higher by four percent and 12 percent, respectively.

8.2 Performance Metrics

CO-OP's PRRISM model is used to evaluate the ability of the future WMA system to meet current and future water demands plus the 100-MGD environmental flow-by at Little Falls Dam. A single PRRISM "run" for a given forecast year simulates WMA system daily operations over the 29,302 days of an 80-year simulation period, providing daily values of quantities of interest including reservoir storages, inflow, and release rates, river flow at Little Falls Dam, and WMA withdrawals (see Section 5.1). WMA withdrawals are based on simulated demands, with a weather-dependent daily variation, as discussed in Chapter 4. Daily river flows and reservoir inflows are based on values from historical data, which may be altered as described in Chapter 7 to reflect projected changes in meteorological conditions.

PRRISM's 80-year simulation period is constructed from daily hydrometeorological data from the period, October 1929 through December 2009, where conditions in each historic year, altered by projected changes in hydrometeorological conditions, represent possible conditions in the future scenario year. Therefore, a single PRRISM run evaluates the performance of the WMA system under a wide range of conditions similar to those that have occurred in the past, including conditions during the prolonged and extreme 1930 "drought of record" and the record Potomac River daily low flows experienced during the drought of 1966. Two additional years, 1999 and 2002, represent conditions during moderate drought. For each run, PRRISM provides a suite of summary statistics calculated from the run's sample of 29,302 days in the simulation period, including simulated values of quantities such as lowest daily reservoir storage and highest daily Potomac River deficit. Any two individual PRRISM simulations, each consisting of approximately 80 years, are different because of the random component of the daily demand model. Results reported in this study are calculated by averaging the summary statistics from a set of 100 PRRISM runs.

WMA system performance evaluations are conducted for each of the nine future conditions scenarios for each of the three scenario years, 2030, 2045, and 2050, resulting in a total of 27 PRRISM run sets. Additional evaluations are conducted for a future conditions scenario based on historical data assuming no future alterations due to changing hydrometeorological conditions. Five PRRISM summary statistics, listed below, were selected as performance metrics for this study. The values of these performance metrics for each of the evaluated years and conditions scenarios appear below in Table 8-1. Tables containing the full set of PRRISM summary statistics appear in Appendix A.4 along with a list and description of each of the statistics.

The five selected performance metrics appearing in Table 8-1 are:

1. Percent years with no deficits: The percentage of years over the simulation period in which flow in the Potomac River at Little Falls is above 100 MGD (the Little Falls Dam flow-by) on every day of the year, that is, years in which combined WMA Potomac water supply needs and the

environmental flow-by at Little Falls Dam are always met. This statistic provides a measure of system reliability.

2. Percent of years with emergency restrictions: The percentage of years over the simulation period in which emergency water use restrictions are implemented on one or more days of the year. In this study, emergency restrictions are assumed to be implemented when combined water supply storage in Jennings Randolph and Little Seneca reservoirs is below five percent of the combined capacity. This statistic provides a measure of system reliability.
3. Number of days with deficits: The number of days over the 29,302-day simulation period in which there is a Potomac River deficit. This statistic provides a measure of system vulnerability.
4. Maximum Potomac deficit: The maximum shortfall, in MGD, in meeting combined WMA Potomac water supply needs and the Little Falls Dam environmental flow-by on any single day of the simulation period. This statistic provides a measure of system vulnerability.
5. Minimum Combined Little Seneca and Jennings Randolph water supply storage (BG): The minimum ("worst day") combined storage experienced over the course of the simulation period, which reflects the impacts of reservoir releases.

8.3 Results

PRRISM was used to assess the performance of the WMA water supply system under all nine future conditions scenarios for three scenario years by conducting sets of 100 simulation runs for each scenario and by averaging performance metric values over each set of 100 runs. Additional run sets were conducted assuming no alterations in historical hydrometeorological conditions. Summaries of results appear below in Table 8-1. In this table, values are given for the five performance metrics described above. Tables with a complete set of PRRISM summary statistics for each run set are given in Appendix A.4. It should be noted that values of the performance metrics shown in Table 8-1 are primarily a reflection of performance during the most severe drought year of the simulation, which represents conditions similar to those that occurred during the drought of 1930, as altered by impacts of projected meteorological change.

Color coding is used in Table 8-1 to indicate WMA system performance. In the current study, the definition of "reliable" WMA system performance is based on values of the first two of the PRRISM system performance metrics above and follows the numerical criteria used in the 2017 alternatives study (Schultz *et al.*, 2017). That is, the system is considered reliable if the percent of years with no Potomac River deficits is greater than or equal to 99.88 percent and the percent of years with emergency water use restrictions is less than or equal to 0.06 percent. Using these criteria, the color-coding in Table 8-1 is as follows:

- GREEN – denotes reliable performance, with both reliability criteria met
- YELLOW – denotes marginal performance, if one or both of the reliability criteria are not met but the maximum Potomac deficit on a single day is very small (averaging 1 MGD or less)
- RED – denotes system failure, that is, an inability of the system to meet combined WMA water supply and environmental flows in the event of severe drought

For a given scenario year, results in Table 8-1 range from complete system reliability with ample water to spare in Little Seneca and Jennings Randolph reservoirs even in an extreme drought year, to system failure during an extreme drought, with many days or months when flow in the Potomac River falls far short of meeting the region's water needs, that is, combined WMA demands and the 100 MGD environmental flow-by at Little Falls Dam. For all scenario years in the Higher Flows and Lower Demands scenario, there is ample combined water supply storage in Little Seneca and Jennings Randolph even on the worst day of an extreme drought: 9.9, 8.6, and 7.9 BG for 2030, 2045, and 2050, respectively. But these reservoirs are predicted to be empty on at least one day of an extreme drought for all scenario years in all of the Lower Flows scenarios, and for some of the Medium Flows scenarios as well.

For the Medium Flows and Medium Demands scenario, the results for 2030 indicate that the system is currently reliable, but for the scenario years 2045 and 2050 there is a possibility of Potomac River deficits in the event of extreme drought. In 2045, Table 8-1 shows that there is a 98.77 percent likelihood that no deficit occurs, or conversely, a 1.23 percent likelihood that an extreme drought will occur with Potomac River deficits of up to 62 MGD during five days of the drought, and with no water supplies remaining in the two upstream reservoirs, Little Seneca and Jennings Randolph, on at least one day of the simulation. In 2050, there is also a 98.77 percent likelihood that no deficit occurs, or a 1.23 percent likelihood of deficits of up to 100 MGD during 10 days if an extreme drought occurs, with combined storage in Little Seneca and Jennings Randolph falling to zero reservoirs on at least one day of an extreme drought.

Results for scenario year 2050 are directly comparable to results from the 2020 CO-OP water supply study's "baseline" scenario for future system configuration, because both assume the addition to the current system of the Edgemon and Milestone reservoirs. The 2020 study results on system performance in 2050 are qualitatively similar to those of the current study, in that the system is predicted not to be reliable in an extreme drought year in the Medium Flows and Medium Demands scenario and marginally reliable in the Higher Flows and High Demands scenario. At the same time, performance metric values in the 2020 study indicate that the WMA system will be somewhat more reliable and less vulnerable in 2050.

According to results in the 2020 study, the 2050 Medium Flows and Medium Demands scenario has a 99.29 percent likelihood that no deficit occurs, or conversely, a 0.71 percent likelihood of deficits. Deficits of up to 21 MGD are expected on just one day during an extreme drought. Also, for the 2050 Higher Flows and Lower Demands scenario, the current study indicates that the worst day combined Little Seneca and Jennings Randolph water supply storage during an extreme drought is 7.3 BG, but according to results in the 2020 study, combined storage does not fall below 8.6 BG.

These differences in 2050 results are primarily due to the current study's higher forecasts for both WMA demands and upstream consumptive use. The 2050 annual demand forecast in the current study is 538 MGD (from Medium Demands scenario in Table 3-15), which is 10 MGD higher than that of the 2020 study. The current study's summer average upstream consumptive use forecast for 2050 is 117 MGD, which is 14 MGD higher than in the 2020 study, and October and November forecasts are 35 to 40 MGD higher than in the 2020 study. These differences in October and November consumptive use will have an impact on system performance because during the drought of record, low flow and reservoir releases persisted throughout the months of autumn and early winter. The updated forecasts for the impacts of meteorological change on stream flow in the current study are slightly less harsh for 2050. The predicted

change of annual flow in an extreme drought year is -13 percent for the medium temperature sensitivity scenario for 2040-2069 (from the flow quantile = 0.01 annual flow scaling factor of 0.87 in Table 7-6). The corresponding change in the 2020 study is -16 percent.

An unexpected outcome of this study is the predicted vulnerability of the WMA water supply system in the near-term, as indicated by results in Table 8-1 for the 2030 scenario year. Deficits are predicted in the event of extreme drought in 2030 for the Medium Flows and High Demands scenario and for all of the Lower Flows scenarios. This is primarily due to the use in the analysis of the updated projections for meteorological change. The current study's predicted change in annual flow in an extreme drought year for the period 2010-2039, based on the filtered downscaled CMIP6 projections, is -11 percent for the medium temperature sensitivity scenario (from the flow quantile = 0.01 annual flow scaling factor of 0.89 in Table 7-6). It was noted in Chapter 7 that projected changes in flows for the near-term period, 2010-2039, should be viewed as preliminary and should undergo further validation as more data becomes available. This is because the CMIP6 ensemble of meteorological projections used in the current study provides a somewhat poor match to available observed data. In addition, values of statistics of interest for study time periods, such as the annual flow scaling factors of Table 7-6, are assumed to be uniform throughout the 30 years, even though this is clearly not expected to be the case. It should be noted that the 2020 study's corresponding change in annual flow in an extreme drought year, based on the filtered downscaled CMIP5 projections, is much smaller: only -2 percent.

In four out of nine of the modeled 2030 scenarios, combined water supply storage in Little Seneca and Jennings Randolph reservoirs is predicted to fall to near zero in the event of extreme drought, which would likely trigger emergency water allocations under the Low Flow Allocation Agreement. The CO-OP suppliers entered into the Water Supply Coordination Agreement with the express intent to "reduce or eliminate the possibility that the Emergency Stage of the Low Flow Allocation Agreement will ever be reached," indicating that measures should be investigated to enhance the current system in the short term. A set of non-structural measures that increase system safe yield was explored and assessed in the water supply alternatives study (Schultz *et al.*, 2017). These may be further evaluated and refined in the ongoing USACE feasibility study or in a future CO-OP study to assess what set of alternatives can be implemented to address current day vulnerabilities.

Table 8-1: PRRISM results reflecting WMA water supply system performance during extreme drought. Values are 100-run averages of the following five performance metrics: percent years with no deficits, percent years with emergency restrictions, number of days with deficits, maximum Potomac deficit (MGD), and minimum combined Little Seneca and Jennings Randolph water supply storage (BG).*

	Based on historic record as altered by projected changes in hydrometeorological conditions									Based on unaltered historic hydrometeorological record		
Year	Higher Flows (Low sensitivity of flows to temperature)			Medium Flows (Medium sensitivity of flows to temperature)			Lower Flows (High sensitivity of flows to temperature)					
	Low Demands	Medium Demands	High Demands	Low Demands	Medium Demands	High Demands	Low Demands	Medium Demands	High Demands	Low Demands	Medium Demands	High Demands
2030	100.00% 0.00% 0 days 0 MGD 9.9 BG	100.00% 0.00% 0 days 0 MGD 7.8 BG	100.00% 0.00% 0 days 0 MGD 5.4 BG	100.00% 0.00% 0 days 0 MGD 6.3 BG	99.91% 0.00% 0 days 1 MGD 2.2 BG	98.77% 2.50% 2 days 54 MGD 0.0 BG	98.77% 2.50% 3 days 57 MGD 0.1 BG	98.77% 2.50% 24 days 102 MGD 0.0 BG	98.60% 2.50% 50 days 142 MGD 0.0 BG	100.00% 0.00% 0 days 0 MGD 9.1 BG	100.00% 0.00% 0 days 0 MGD 6.8 BG	100.00% 0.00% 0 days 0 MGD 3.5 BG
2045	100.00% 0.00% 0 days 0 MGD 8.6 BG	100.00% 0.00% 0 days 0 MGD 5.9 BG	100.00% 0.00% 0 days 0 MGD 2.7 BG	99.81% 0.29% 0 days 1 MGD 1.2 BG	98.77% 2.50% 5 days 62 MGD 0.0 BG	98.35% 3.75% 27 days 148 MGD 0.0 BG	98.75% 2.50% 61 days 146 MGD 0.0 BG	97.63% 3.75% 82 days 211 MGD 0.0 BG	95.06% 3.75% 103 days 256 MGD 0.0 BG	100.00% 0.00% 0 days 0 MGD 6.8 BG	100.00% 0.00% 0 days 0 MGD 2.9 BG	98.73% 2.76% 2 days 40 MGD 0.0 BG
2050	100.00% 0.00% 0 days 0 MGD 6.0 BG	100.00% 0.00% 0 days 0 MGD 4.8 BG	100.00% 0.16% 0 days 0 MGD 1.2 BG	99.05% 2.10% 1 days 11 MGD 0.3 BG	98.77% 2.51% 10 days 100 MGD 0.0 BG	97.63% 3.75% 38 days 159 MGD 0.0 BG	98.75% 2.51% 67 days 155 MGD 0.0 BG	97.26% 3.75% 87 days 228 MGD 0.0 BG	95.06% 4.66% 117 days 273 MGD 0.0 BG	100.00% 0.00% 0 days 0 MGD 6.2 BG	100.00% 0.36% 0 days 0 MGD 1.2 BG	98.49% 3.74 % 3 days 48 MGD 0.0 BG

*GREEN denotes reliable performance, YELLOW denotes marginal performance, and RED denotes system failure in the event of extreme drought.

9 Conclusion

This is the eighth in a series of studies that are conducted every five years by ICPRB's CO-OP Section on behalf of the WMA water suppliers to evaluate the adequacy of the region's cooperative water supply system over the coming decades. Forecasts are provided for water demands and water availability through the year 2050, with extrapolations through 2060, consistent with requirements of the VADEQ water supply planning program, and through 2085, to support an ongoing feasibility study on a secondary source backup water supply for the region by the USACE. Future system performance is evaluated for the years 2030, 2045, and 2050 using a newly upgraded version of CO-OP's planning model, PRRISM v4.1. The study aims to provide decision-makers with information to assist them in assessing future resource needs.

9.1 Summary of Water Demand Forecasts

Forecasts of average annual water demand follow the methodology used in past studies. They are developed by combining recent water use information derived from billing data provided by the suppliers and their wholesale customers with the most recent MWCOG demographic forecasts (Round 10.0). Average total annual demand of the WMA suppliers, including Rockville, was 459 MGD in 2023, and this is projected to rise to 525 MGD by 2045 (a 14% increase) and to 538 MGD by 2050 (a 17% increase). Results include an estimate of uncertainty based on the standard error of the forecasts developed in CO-OP's 2020 study. The standard errors for the 2045 and 2050 demand forecasts are 9.7% and 10.0%, respectively. The forecasts are slightly higher than those of CO-OP's previous study (Ahmed *et al.*, 2020), but the difference is small compared to the range of uncertainty.

In addition to long-term annual forecasts, this study explores methods for improving daily demand prediction. A range of machine learning models, including neural networks, recurrent models, and tree-based ensembles, were evaluated for forecasting daily water demands. While the current linear regression approach used in PRRISM remains a strong performer, particularly for capturing general patterns and ease of implementation, the machine learning models demonstrated improved accuracy in capturing nonlinear relationships, peak usage, and weather-driven variability. These models may be considered for future planning and operational applications, particularly to improve accuracy during peak demand periods and extreme weather conditions.

9.2 Summary of Water Availability Forecasts

This study's forecasts of water availability take into account two factors that will have impacts on future system resources: changes in meteorological conditions, that is, precipitation and temperature, and increases in cumulative consumptive use of water in the portion of the Potomac River watershed upstream of WMA water supply intakes. Projected changes in meteorological conditions are based on CMIP6, the most recent compilation of projections from global models. A climate response function, developed from the historical record using multiple least squares regression, is used to predict annual mean Potomac River flow from projected annual precipitation and temperature. A quantile scaling approach is applied to obtain flow-dependent scaling factors for future annual stream flow for the entire simulation period of the climate projections, 1950 through 2099.

A major uncertainty in many regions, including the Potomac, is how stream flow will respond to increases in temperature. To represent this uncertainty, three flow scenarios are constructed that reflect a range of sensitivities of flow to temperature, based on one standard error of the temperature coefficient of the response function. Annual flow scaling factors computed for each of the three scenarios demonstrate the disparate impacts of changing meteorologic conditions on stream flows in low flow versus high flow years. For low flow years, the scaling factors are less than one for both the medium and high temperature sensitivity scenarios, indicating that flows in future drought years will decrease. Conversely, for high flow years, most flow scaling factors are greater than one, indicating that flows in future high flow years will increase. Results depend significantly on the temperature sensitivity scenario, demonstrating the competing effects of rising temperature and rising precipitation on future stream flows. Annual flows are predicted to rise in almost all cases for the low temperature sensitivity scenario, indicating that rising precipitation will be the dominant factor determining future stream flow. Conversely, annual flows are predicted to fall in almost all cases for the high temperature sensitivity scenario, indicating that rising precipitation is not enough to stem the decrease in flows due to rising temperatures. Noteworthy are the scaling factors for years with annual flow percentile values that are ≤ 1 , referred to in this study as extreme drought years. Annual flows in an extreme drought year, which includes the 1930 drought of record, have a 99 percent probability or greater of being exceeded. Assuming medium sensitivity of flow to temperature change, the flow scaling factors from the current study project that in the event of an extreme drought, Potomac basin flows will be lower than historical values by 11 percent or more, 13 percent or more, and 29 percent or more for the time periods 2010-2039, 2040-2069, and 2070-2099, respectively.

New estimates have been developed for upstream consumptive use of water, which result in reductions in Potomac River flow at WMA intakes. These estimates reflect a more complete accounting of upstream demands through the inclusion of estimated unreported uses. They also include a new data center component. While the current contribution of data centers to total upstream consumptive use is negligible (less than 0.1%), it is projected to rise to an estimated 4.6 MGD by 2050, or four percent of the total annual average of 117 MGD. The highest monthly average upstream consumptive use for 2050 is 151 MGD in August, with 8.2 MGD, or five percent, attributed to data centers. While modest in absolute terms, these values are comparable to the current share of upstream consumptive use by several established sectors, including commercial, thermoelectric, and public water supply, and signal the growing significance of data centers in the region's water demand profile. As these estimates are based on grid-connected energy forecasts and reflect only upstream use, ongoing tracking of data center development, technology trends, and regulatory changes will be essential.

9.3 Resource Analysis Results

Evaluations of the ability of the WMA cooperative water supply system to meet future needs are conducted using a newly constructed version of CO-OP's planning model, PRRISM v4.1. The analyses assume that two new resources will be in place by 2045: Fairfax Water's Edgemon Reservoir, planned to be operational by 2040, and Loudoun Water's Milestone Reservoir, planned to be operational by 2028. The evaluations are conducted under nine planning scenarios that represent ranges of uncertainties in future WMA water demand and in the impact of meteorological change on water availability in the Potomac basin. The nine scenarios are formed by combining three demand scenarios and three stream

flow scenarios. Evaluations are also conducted based on historical meteorological conditions, assuming no future alterations.

Each PRRISM run simulates 29,302 days, or approximately 80 years, of the processes and operational decisions that determine daily Potomac River flow at WMA intakes and at the [USGS stream gage at Little Falls dam](#). PRRISM's 80-year simulation period is constructed from daily hydrometeorological data from the period, October 1929 through December 2009, where conditions in each historic year, altered by projected changes in hydrometeorological conditions, represent possible conditions in the future scenario year. Therefore, a single PRRISM run evaluates the performance of the WMA system under a wide range of conditions similar to those that have occurred in the past, including conditions during the prolonged and extreme 1930 drought of record.

PRRISM provides a suite of summary statistics calculated from this sample of over 29,302 days in the simulation period. Results reported in this study are calculated by averaging the summary statistics from a set of 100 PRRISM runs. Two of these summary statistics - percent of years with no Potomac River flow deficit and percent of years with emergency water use restrictions - are selected to be this study's performance metrics for system reliability.

For a given scenario year (2030, 2045, or 2050), results range from complete system reliability with ample water to spare in Little Seneca and Jennings Randolph reservoirs, to system failure with many days or months when flow in the Potomac River falls far short of meeting the region's water needs, that is, combined WMA demands and the 100 MGD environmental flow-by at Little Falls Dam. For all scenario years in the Higher Flows and Lower Demands scenario, there is ample combined water supply storage in Little Seneca and Jennings Randolph even on the worst day of an extreme drought. But these reservoirs are predicted to be empty on at least one day of an extreme drought for all scenario years in all of the Lower Flows scenarios, and for some of the Medium Flows scenarios as well. For the Medium Flows/Medium Demands scenario, the results for 2030 indicate that the system is currently reliable, but for the scenario years 2045 and 2050 there is a possibility of Potomac River deficits in the event of extreme drought. In 2045, there is a 98.77% likelihood that no deficit occurs, or conversely, a 1.23% likelihood that an extreme drought will cause Potomac River deficits of up to 62 MGD during 5 days, and with no water supplies remaining in the two upstream reservoirs, Little Seneca and Jennings Randolph, on at least one day. In 2050, there is again a 98.77% likelihood that no deficit occurs, or conversely, a 1.23% likelihood of deficits of up to 100 MGD during 10 days if an extreme drought occurs, with no water supplies remaining in Little Seneca and Jennings Randolph reservoirs on at least one day.

An unexpected outcome of this study is the indication of vulnerability of the WMA water supply system in the near term from results for the 2030 scenario year. Deficits are predicted in the event of extreme drought in 2030 for the Medium Flows and High Demands scenario and for all of the Lower Flows scenarios. This is primarily due to the use in the PRRISM model of the updated projections for meteorological change. It was noted in Chapter 7 that projected changes in flows for the near-term period, 2010-2039, should be viewed as preliminary and should undergo further validation as more data becomes available. To date, analyses of Potomac River flow data have not detected a long-term trend in average flow or in low flows, but as additional hydrometeorological data becomes available, it will be possible to refine and update the analyses. A set of non-structural measures that increase system safe yield were explored and assessed in the 2017 water supply alternatives study. These may be further evaluated and

refined in the ongoing USACE feasibility study or in a future CO-OP study to investigate what measures can be implemented to address potential near-term vulnerabilities.

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Appendices

A.1 Production Data

Table A.1-7: Fairfax (Total Production).

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Ave.
Ave. annual production, MGD	150	159	155	151	152	148	152	148	153	152
Monthly ave. production, MGD										
January	130	134	142	147	127	131	133	133	131	134
February	133	134	140	136	129	125	135	130	130	133
March	131	135	137	136	129	130	131	130	128	132
April	135	147	146	145	138	134	142	135	148	141
May	161	148	154	163	150	146	159	144	163	154
June	162	177	183	167	163	173	176	170	183	173
July	165	190	186	189	178	202	191	174	172	183
August	183	191	170	174	186	168	178	176	180	179
September	181	186	169	153	186	156	166	170	176	171
October	146	165	160	144	162	147	153	147	158	154
November	135	156	137	130	141	134	133	135	139	138
December	132	149	137	127	133	131	129	133	133	134
Peak 1-day production, MGD										
January	147	143	150	166	137	141	145	146	140	166
February	158	144	147	143	143	136	147	149	138	158
March	142	144	144	143	136	157	141	142	135	157
April	143	170	165	159	149	142	164	149	174	174
May	190	168	181	184	172	161	201	175	183	201
June	195	209	221	201	180	209	215	194	216	221
July	199	234	222	228	208	241	219	198	195	241
August	212	214	192	201	209	197	207	203	214	214
September	215	216	200	186	208	172	190	200	222	222
October	161	194	190	169	196	169	173	161	178	196
November	148	166	145	139	152	148	139	147	149	166
December	143	162	148	134	142	141	133	148	145	162
Ave. Jul.-Oct. production, MGD	169	183	171	165	178	168	172	167	172	172

Note: Includes water sold to Loudoun Water; See Table A.1-5 for Fairfax Water portion of Loudoun Water usage.

Table A.1-8: Washington Aqueduct (Total Production).

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Ave.
Ave. annual production, MGD	138	135	139	138	142	128	127	135	131	135
Monthly ave. production, MGD										
January	127	123	132	136	127	123	113	124	122	125
February	134	129	132	129	131	124	110	126	122	126
March	129	126	130	128	130	124	117	128	124	126
April	132	131	140	132	136	119	119	131	133	130
May	135	128	142	140	142	126	129	135	130	134
June	148	147	157	146	153	142	139	146	140	146
July	151	152	160	159	163	152	146	150	150	154
August	153	152	150	153	162	139	140	149	143	149
September	157	150	144	147	162	131	139	148	143	147
October	138	133	134	139	146	127	130	136	129	135
November	130	126	124	127	124	115	126	125	118	124
December	119	125	119	123	124	115	118	123	111	120
Peak 1-day production, MGD										
January	145	138	146	163	142	141	123	145	131	163
February	161	153	146	143	147	152	133	155	134	161
March	139	143	142	139	145	141	128	137	139	145
April	141	156	154	143	153	127	145	154	155	156
May	150	141	158	160	165	153	147	155	148	165
June	173	166	175	166	175	167	162	165	154	175
July	165	175	186	195	179	174	165	166	167	195
August	166	171	168	173	186	157	156	166	151	186
September	178	166	158	170	172	147	155	171	168	178
October	149	146	157	156	168	139	147	147	165	168
November	146	140	147	140	145	124	143	138	137	147
December	139	143	130	140	139	132	132	136	126	143
Ave. Jul.-Oct. production, MGD	150	147	147	149	158	137	139	146	141	146

Table A.1-9: Washington Suburban Sanitary Commission (WSSC Water, Total Production).

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Ave.
Ave. annual production, MGD	165	165	163	163	163	161	163	160	159	162
Monthly ave. production, MGD										
January	155	157	154	169	154	153	152	158	154	156
February	161	159	150	152	153	151	162	155	151	155
March	159	154	148	149	149	152	153	149	146	151
April	157	161	155	154	153	151	157	153	155	155
May	170	161	160	167	163	160	163	157	161	162
June	170	172	178	167	171	174	174	169	169	172
July	176	181	182	183	177	194	180	173	170	180
August	182	182	174	177	177	178	175	171	170	176
September	179	173	174	169	182	168	162	165	168	171
October	161	160	166	162	163	156	164	154	160	161
November	157	158	157	154	155	148	156	153	156	155
December	151	158	155	151	155	148	153	157	149	153
Peak 1-day production, MGD										
January	169	186	171	213	167	166	164	179	178	213
February	202	176	162	171	177	161	174	173	163	202
March	172	166	165	164	163	160	162	167	165	172
April	164	176	177	174	169	160	172	171	170	177
May	185	183	182	190	198	179	191	179	177	198
June	195	189	199	202	188	196	194	199	184	202
July	191	209	210	213	198	222	202	187	188	222
August	200	201	191	197	194	197	196	188	186	201
September	200	187	193	189	203	179	178	180	195	203
October	176	176	182	178	192	187	184	174	174	192
November	170	191	175	167	175	158	169	163	169	191
December	166	175	171	164	168	155	162	183	164	183
Ave. Jul.-Oct. production, MGD	174	174	174	173	175	174	171	166	167	172

Table A.1-10: Loudoun Water (Total Use).

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Ave.
Ave. annual use, MGD	23	25	24	23	25	25	26	27	28	25
Monthly ave. use, MGD										
January	17	19	19	20	20	20	21	22	23	20
February	19	19	19	19	20	19	21	21	22	20
March	18	20	19	20	20	21	21	22	22	20
April	20	23	22	22	22	23	23	24	26	23
May	25	23	24	25	26	25	28	27	29	26
June	28	30	32	28	30	30	33	31	35	31
July	27	32	31	32	33	37	35	33	35	33
August	30	33	27	29	34	29	33	34	36	32
September	28	32	27	24	33	27	28	32	33	29
October	22	25	25	22	27	24	26	26	27	25
November	19	20	20	19	20	21	22	23	25	21
December	18	19	18	19	20	20	21	23	23	20
Peak 1-day use, MGD										
January	20	21	22	22	22	22	22	24	26	26
February	22	20	21	22	22	21	22	23	23	23
March	22	22	22	24	23	25	22	24	26	26
April	23	28	26	25	25	24	29	27	33	33
May	32	28	32	31	32	31	36	33	34	36
June	34	37	40	35	36	38	42	36	42	42
July	32	40	38	41	40	44	43	40	42	44
August	36	38	34	37	38	35	44	41	43	44
September	33	38	37	34	37	31	34	40	47	47
October	24	28	32	27	35	29	30	30	33	35
November	21	23	23	21	23	23	23	25	27	27
December	20	21	21	20	21	21	23	33	25	33
Ave. Jul.-Oct. use, MGD	27	30	28	27	32	29	31	31	33	30

Note: Trap Rock Water Treatment Facility (Trap Rock WTF) production started on September 4, 2018. The average Potomac River withdrawal has been about 9 MGD, with an average finished water amount of 8 MGD. The Trap Rock WTF has not solidified its operational plan and the total Loudoun Water usage (production plus purchase amount) is a better representation of their water needs.

Table A.1-11: Loudoun Water (Fairfax Water Portion).

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Ave.
Ave. annual use, MGD	18	19	19	19	15	15	17	15	17	17
Monthly ave. use, MGD										
January	13	16	19	19	12	10	15	12	11	14
February	14	16	19	19	11	10	12	12	11	14
March	14	17	18	19	12	11	14	13	10	14
April	15	16	16	18	13	12	15	14	14	15
May	20	17	17	22	16	15	17	16	16	17
June	22	24	22	23	19	20	21	20	22	21
July	22	23	21	27	22	26	24	19	21	23
August	24	22	17	23	22	20	22	19	27	22
September	23	23	21	15	22	17	20	17	27	20
October	17	15	20	14	17	16	15	13	18	16
November	16	14	16	12	10	14	11	11	19	14
December	16	19	17	11	10	13	11	11	13	14
Peak 1-day use, MGD										
January	16	18	22	21	13	17	21	15	17	22
February	18	18	20	22	13	12	14	22	13	22
March	18	20	21	21	16	19	21	23	13	23
April	20	21	19	22	15	14	25	22	24	25
May	30	21	25	28	21	20	25	26	21	30
June	28	32	33	30	24	27	30	25	28	33
July	26	32	27	36	28	34	31	25	27	36
August	30	30	23	31	26	32	32	26	38	38
September	27	33	30	23	26	20	30	25	42	42
October	20	22	27	18	25	21	19	17	23	27
November	20	20	19	14	13	22	14	22	22	22
December	17	21	20	13	11	20	13	20	25	25
Ave. Jul.-Oct. use, MGD	22	21	20	20	20	20	20	17	23	20

Table A.1-12: CO-OP System (Total Production).

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Ave.
Ave. annual production, MGD	452	459	457	452	456	437	442	443	443	449
Monthly ave. production, MGD										
January	412	414	428	452	408	408	398	416	408	416
February	429	422	422	417	412	401	407	412	402	414
March	418	415	416	413	408	407	400	407	398	409
April	423	439	440	431	427	404	418	419	435	426
May	467	437	455	471	456	432	451	436	454	451
June	480	497	517	480	487	488	489	485	492	491
July	492	523	528	531	518	548	517	497	491	516
August	519	525	494	503	525	484	493	497	493	504
September	516	509	488	469	530	454	468	482	487	489
October	445	457	461	445	471	430	447	436	446	449
November	422	440	419	411	420	396	416	413	413	417
December	402	432	411	400	411	394	399	414	393	406
Peak 1-day production, MGD										
January	449	465	456	533	444	431	417	459	430	533
February	497	454	438	440	458	426	428	445	420	497
March	442	431	432	428	436	422	423	439	419	442
April	441	492	482	474	462	419	459	443	492	492
May	525	485	516	516	531	478	529	503	499	531
June	549	552	578	559	532	559	567	552	533	578
July	547	597	610	616	572	630	572	543	545	630
August	566	576	537	557	562	529	545	548	537	576
September	569	559	534	532	583	485	518	537	574	583
October	475	501	528	487	547	470	494	460	499	547
November	441	478	453	431	443	421	440	433	443	478
December	421	480	436	425	436	413	421	454	420	480
Ave. Jul.-Oct. production, MGD	493	504	493	487	511	479	482	478	479	489

Note: Includes water provided to Loudoun Water by Fairfax Water.

A.2 Annual Demand Forecasts

This appendix explains the calculation of historic and forecasted unit use rates and their application in generating the average annual demand forecast, as discussed in Chapter 3.

A.2.1 Historic and Forecast Periods

The 2025 ICPRB study covers the historic period from 2014 to 2023 and forecasts from 2023 to 2050. Results are extrapolated to 2060 to support VADEQ Northern Virginia Regional Planning requirements, and results are extrapolated to 2085 to support the USACE feasibility study.

A.2.2 Demographic Data

The study uses data from the Metropolitan Washington Council of Governments (MWCOG) Round 10.0 forecast, adopted on June 14, 2023, which includes households (HH) and employees (EMP). This data is adjusted for total occupied households without having to account for vacancy rates.

Household Categories: Local jurisdictions provide data on single-family households (SFH) and multi-family households (MFH). Where local data is unavailable, either the five-year American Community Survey data from the U.S. Census Bureau is used or an assumption is based on a related water supply service area.

ACS Data: This includes total housing units (HU), types of units, and vacancy rates. SFHU is assumed to be the sum of 1-unit detached and 1-unit attached units, while MFHU includes all other units. The vacancy rate does not affect the dwelling unit ratio (DUR) calculations.

A.2.3 Supplier Data

CO-OP Suppliers provide water use data, including:

- Service area GIS shape files
- Billed water use (SFH, MFH, Employee)
- Retail sales by category (SFH, MFH, Employee)
- Wholesales total and by customer
- Purchased or produced water use by source
- Unmetered use (water leaving the system without being measured)

Wholesale customer data includes:

- Service area GIS shape files
- Purchase or produced water use by supplier
- Billed water use (SFH, MFH, Employee)
- Unmetered use (water leaving the system without being measured)

A.2.4 Methods

Step-by-Step Annual Demand Forecast Methods is shown in Figure 3-1 and described in detail below:

1. Service Area Processing: Service areas are processed by clipping the 3722 TAZ Zone System GIS layer and calculating area ratios. In this study, the GIS layers were processed and calculated using a set of python scripts in QGIS that semi-automated the process.
2. Population, Employees, and Households Collection:
 - a. MWCOG Round 10.0 forecast provides 2020-2050 values at five-year intervals.
 - b. Values are interpolated between 2020-2025 for historic years 2020, 2021, 2022, and 2023. Year 2024 was left out because data from suppliers was not provided.
3. Dwelling Unit Ratios (DURs) Calculation:
 - a. Jurisdictional SFH and MFH data by TAZ are used when available, or ACS data is used.
 - b. Vacancy rates are applied to convert HUs to HHs as necessary.
 - c. DUR for each jurisdiction is assigned to a service area.
 - d. Future DURs are estimated in five-year increments through 2050.
 - e. SFH and MFH estimates are derived from the MWCOG Round 10.0 household forecast.
 - f. The last three years of household data (2040, 2045, and 2050) are used to extrapolate SFH and MFH for the years 2055, 2060, and 2085.
4. Historic Unit Use Rate Calculation:
 - a. A water use balance for 2020-2023 is calculated.
 - b. Historic unit use factors are generated by dividing water use by SFH, MFH, and EMP units.
5. Unit Use Forecasts:
 - a. Trends in historic unit use are reviewed, and unit use rates for 2023 are calculated.
 - b. Future changes in water use behavior and end-use savings are estimated.
 - c. End-use savings are extrapolated to estimate 2050 unit use rates.
 - d. This process is extended to forecast 2055, 2060, and 2085 unit use rates.
6. Annual Demand Forecast Calculation:
 - a. Unit use rate forecasts are applied to SFH, MFH, and EMP units to yield annual water use.
 - b. Unmetered water use is estimated, assuming rates based on historic data.
 - c. Total annual demand forecast is calculated by summing SFH, MFH, EMP, unmetered use, and wholesale customer sales.

Completing these steps requires data from multiple sources, including billing, produced, and purchased data from water suppliers and demographic information from MWCOG, local jurisdictions, and the U.S. Census Bureau.

A.2.5 Unit Use Forecasts - End Use Savings

The end use savings model used in this study applied the methods used in the 2015 demand study with updated literature values where available to estimate the reduction in water use from low flow appliances and fixtures. This study, as with the 2015 and 2020 demand studies, considered household toilets, showerheads, clothes washers, dishwashers, and faucets. A simple estimate is also made for savings from low-flow toilets in commercial buildings.

The methodology includes assumptions about the market share for conventional, low flow (EPA Act), and high efficiency (WaterSense or Energy Star) fixtures and appliances. The tables below summarize the assumptions used for the flow rates, natural replacement rates (non-incentivized), frequency of household use, and market share (Table A.2-1 through Table A.2-7). This method is adapted from information and the method described in a 2013 study completed for Tampa Bay Water on demand management (Hazen and Sawyer, 2013).

Table A.2-1: Mechanical efficiency ratings (flow rates) compiled from EPA Act and WaterSense information, and from Hazen and Sawyer (2013).

Fixture	Unit	Flow Rate		
		Conventional (average)	Low Flow	High Efficiency
Residential Toilets	Gallon per flush (gpf)	4.25	1.6	1.28
Commercial Toilets	Gallon per flush (gpf)	3.5	1.6	1.28
Showerheads	Gallon per minute (gpm)		2.2	1.15
Faucets	Gallon per minute (gpm)		2.2	1.15
Washing machines	Gallon/cycle/ft ³	40.0 – 15.1		29.7 – 12.7
Dishwasher	Gallon/load	8.7 – 6.0		4.5

Notes: Flow rate may change over time with evolving standards. Showerhead and faucets rates are difficult to estimate due to potential user adjustments. As of the publication, EPA's review of pre-2012 WaterSense criteria under the 2018 Water Infrastructure Act has not been released.

Table A.2-2: Natural replacement rate of fixtures. Natural replacement rates of washing machines and dishwashers from Hazen and Sawyer (2013). Values for all other fixtures are from Schein et al. (2017).

Fixture	Expected Life (Years)		
	Minimum	Average	Maximum
Residential Toilets	10	20	30
Commercial Toilets	10	20	30
Showerheads	5	10	15
Residential faucets	5	10	15
Washing machines	12	12	12
Dishwasher	8	8	8

Table A.2-3: Frequency of fixture and appliance use (DeOreo et al., 2016; Hazen and Sawyer, 2013).

Fixture	Event	Event Frequencies	Total Use
Residential Toilets	flushes/person/day	5.0	13 total flushes
Commercial Toilets	flushes/employee/day	3.0	n/a
Showerheads	minutes/person/day	5.4	14.04 total minutes
Faucets	minutes/person/day	10.0	25.5 total minutes
Washing machines	loads/person/day	0.3	0.78 total loads
Dishwasher	loads/person/day	0.1	0.26 total loads

Note: Assumes 2.26 people per household.

Table A.2-4: Percent market share of EPAct and WaterSense-rated toilets (McNeil, 2008).

Year	WaterSense			EPAct		
	Low	Medium	High	Low	Medium	High
2020	28%	75%	99%	72%	25%	1%
2023	39%	97%	99%	61%	3%	1%
2025	47%	99%	99%	53%	1%	1%
2030	66%	99%	99%	34%	1%	1%
2035	66%	99%	99%	34%	1%	1%
2040	66%	99%	99%	34%	1%	1%
2045	66%	99%	99%	34%	1%	1%
2050	66%	99%	99%	34%	1%	1%
2055	66%	99%	99%	34%	1%	1%
2060	66%	99%	99%	34%	1%	1%
2085	66%	99%	99%	34%	1%	1%

Notes: Gray values were extrapolated. Medium scenario market shares were assumed for commercial toilets.

Table A.2-5: Market share of EPAct and WaterSense-rated faucets (Michael McNeil, 2008).

Year	WaterSense			EPAct		
	Low	Medium	High	Low	Medium	High
2020	44%	55%	65%	56%	45%	35%
2023	56%	70%	95%	44%	30%	5%
2025	64%	80%	99%	36%	20%	1%
2030	84%	99%	99%	16%	1%	1%
2035	84%	99%	99%	16%	1%	1%
2040	84%	99%	99%	16%	1%	1%
2045	84%	99%	99%	16%	1%	1%
2050	84%	99%	99%	16%	1%	1%
2055	84%	99%	99%	16%	1%	1%
2060	84%	99%	99%	16%	1%	1%
2085	84%	99%	99%	16%	1%	1%

Notes: Gray values were extrapolated. Low scenario faucet market shares were assumed for showerheads.

Table A.2-6: Market share of conventional and high efficiency clothes washers (Hazen and Sawyer, 2013).

Year	Conventional	High Efficiency
2020	33%	67%
2023	30%	70%
2025	30%	70%
2030	30%	70%
2035	30%	70%
2040	30%	70%
2045	30%	70%
2050	30%	70%
2055	30%	70%
2060	30%	70%
2085	30%	70%

Notes: Gray values were extrapolated.

Table A.2-7: Market share of conventional and high efficiency dishwashers (Hazen and Sawyer, 2013).

Year	Conventional	High Efficiency
2020	76%	24%
2023	67%	33%
2025	62%	38%
2030	47%	53%
2035	33%	67%
2040	30%	70%
2045	30%	70%
2050	30%	70%
2055	30%	70%
2060	30%	70%
2085	30%	70%

Notes: Gray values were extrapolated.

The results from the model used for each fixture or appliance are in Table A.2-8. The end use savings rates applied to the 2025 unit use rates are in Table A.2- 9.

Table A.2-8: Low, middle, and high scenario estimated household fixture and appliance water use (gallons/household/day).

Year			2020	2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Water Use	Toilets	Low	28.0	26.6	25.7	23.8	22.3	21.2	20.3	19.6	19.1	18.7	17.5
		Middle	24.7	23.0	22.1	20.3	19.2	18.4	17.8	17.5	17.2	17.1	16.8
		High	18.9	18.1	17.7	17.2	16.9	16.8	16.8	16.7	16.7	16.7	16.7
	Clothes Washers	Low	10.0	10.0	10.0	10.0	10.0	9.9	9.9	9.9	9.9	9.9	9.9
		Middle	10.0	10.0	10.0	10.0	10.0	9.9	9.9	9.9	9.9	9.9	9.9
		High	10.0	10.0	10.0	10.0	10.0	9.9	9.9	9.9	9.9	9.9	9.9
	Dish-washers	Low	1.3	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
		Middle	1.3	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
		High	1.3	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
	Faucets	Low	37.8	35.7	34.7	32.8	31.7	31.0	30.6	30.4	30.2	30.2	30.0
		Middle	34.5	32.9	32.1	31.0	30.5	30.2	30.1	30.1	30.0	30.0	30.0
		High	30.9	30.4	30.2	30.1	30.0	30.0	30.0	30.0	30.0	30.0	30.0
	Shower-heads	Low	21.3	20.0	19.2	18.0	17.3	17.0	17.3	16.8	16.6	16.4	16.2
		Middle	19.4	18.2	17.7	16.9	16.5	16.4	16.4	16.3	16.2	16.2	16.2
		High	17.0	16.5	16.3	16.2	16.2	16.2	16.2	16.2	16.2	16.2	16.2
	Total	Low	98.5	93.5	90.8	85.8	82.4	80.3	79.3	78.0	77.0	76.3	74.8
		Middle	90.0	85.4	83.2	79.4	77.3	76.1	75.5	74.9	74.6	74.4	74.0
		High	78.0	76.2	75.5	74.6	74.3	74.1	74.1	74.0	74.0	74.0	74.0
	Commercial Toilets	Low	6.0	5.7	5.5	5.1	4.8	4.6	4.4	4.3	4.2	4.1	3.9
		Middle	6.0	5.7	5.5	5.1	4.8	4.6	4.4	4.3	4.2	4.1	3.9
		High	6.0	5.7	5.5	5.1	4.8	4.6	4.4	4.3	4.2	4.1	3.9

Table A.2- 9: Savings applied to 2025 unit use rates to estimate the unit use rate in the given forecast year (gallons/day).

Year	SFH and MFH Savings			EMP Savings		
	Low	Middle	High	Low	Middle	High
2023	--	--	--	--	--	--
2025	0.7	2.2	2.7	0.2	0.2	0.2
2030	1.6	6.0	7.7	0.6	0.6	0.6
2035	1.9	8.1	11.1	0.9	0.9	0.9
2040	2.1	9.3	13.2	1.1	1.1	1.1
2045	2.2	9.9	14.2	1.3	1.3	1.3
2050	2.2	10.5	15.5	1.4	1.4	1.4
2055	2.2	10.8	16.5	1.5	1.5	1.5
2060	2.2	11.0	17.2	1.6	1.6	1.6
2085	2.2	11.4	18.7	1.8	1.8	1.8

A.2.6 Fairfax Water – Retail

Fairfax Water Retail: Fairfax Water provides retail service in Fairfax County, the City of Fairfax (served by Fairfax Water since 2014), the City of Falls Church, and parts of Arlington. Falls Church receives water purchased from the Washington Aqueduct. Total demand includes Washington Aqueduct purchases for Falls Church and wholesale sales to Loudoun Water.

Service Area: No changes since the last update by Niffy Saji (Fairfax Water, Aug 13, 2019). Updated West, Central, East service fractions included (A. Spiesman, Feb 2, 2024).

Billing Data: Billed and purchased data for 2019–2023 from A. Spiesman (Feb 2, 2024), includes SAP quarterly retail data and wholesale data for Virginia American, Dulles, Fort Belvoir, Prince William, Vienna, Herndon, and Loudoun Water. Falls Church purchases from Washington Aqueduct included in retail sales. Categories: SFH (Single Family, Townhouse), MFH (Apartment), EMP (Commercial, Municipal, Hydrant).

Demographic Data: POP, HH, EMP data for 2020–2050 use MWCOC Round 10.0 forecasts. Categories: SFH (detached, townhouse, duplex, mobile), MFH (low-, mid-, high-rise).

Table A.2-10: Historic data for unit use calculations and results – Fairfax Water – Retail.

			Historic Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Produced (Corbalis + Griffith WTP, MGD)			149.02	149.59	159.88	155.30	151.10	152.03	148.13	152.23	148.26	153.45
Purchased (MGD)			14.01	14.08	11.44	14.77	15.06	15.35	11.39	12.55	13.5	12.26
Demand (MGD)	Retail Sales	Single Family Household	47.40	47.18	47.94	46.79	44.54	46.08	46.78	46.18	43.99	44.31
		Multi-Family Household	16.92	17.00	17.27	17.50	17.53	17.85	18.60	18.54	18.54	19.00
		Employee	19.22	19.56	20.52	19.84	19.22	20.62	16.33	17.03	17.81	18.09
	Wholesales		67.14	67.00	69.73	68.87	68.79	67.46	65.64	69.87	68.30	71.68
	Unmetered/Non-Revenue		12.36	12.92	15.86	17.07	16.08	15.37	11.83	12.47	12.6	12.33
	Percent Unmetered/Non-Revenue		8%	8%	9%	10%	10%	9%	7%	8%	8%	7%
	Total		163.03	163.67	171.32	170.07	166.16	167.38	159.53	164.78	161.75	165.7
	Demographics	Population (units)		1,063,611	1,071,499	1,079,233	1,086,966	1,094,700	1,106,679	1,118,657	1,125,235	1,131,814
Employees (units)		600,391	604,241	612,603	620,966	629,329	618,727	608,124	613,876	619,628	625,381	
Occupied Households (units)		385,513	388,677	390,986	393,295	395,605	398,990	402,374	405,550	408,727	411,903	
DUR (dimensionless)		2.6	2.5	2.5	2.4	2.4	2.4	2.4	2.3	2.3	2.2	
Single Family Households (units)		277,474	276,638	278,066	278,806	279,218	281,066	282,867	283,592	284,254	284,852	
Multi-Family Households (units)		108,039	112,039	112,920	114,490	116,386	117,923	119,507	121,959	124,472	127,051	
Unit Use (gpd/unit)	Single Family Household		171	171	172	168	160	164	165	163	155	156
	Multi-Family Household		157	152	153	153	151	151	156	152	149	150
	Employee		32	32	34	32	31	33	27	28	29	29

Table A.2-11: Forecast data for unit use calculations and results – Fairfax Water – Retail.

			Forecast Data									
			2023*	2025	2030	2035	2040	2045	2050	2055	2060	2085
Produced (Corbalis + Griffith WTP, MGD)			157.28	159.12	166.25	172.13	177.33	182.00	185.73	190.36	194.94	218.32
Purchased (MGD)			17.48	17.68	18.47	19.13	19.70	20.22	20.64	21.15	21.66	24.26
Demand (MGD)	Retail Sales	Single Family Household	44.50	44.11	44.04	43.88	43.94	44.12	44.21	44.44	44.69	39.02
		Multi-Family Household	19.07	19.48	21.20	22.87	24.62	26.43	28.15	29.96	31.78	47.80
		Employee	17.52	17.72	18.98	19.81	20.28	20.67	20.86	21.21	21.54	23.35
	Wholesales		76.2	77.8	82.0	85.6	88.5	90.8	92.5	94.7	96.9	108.1
	Unmetered/Non-Revenue		17.48	17.68	18.47	19.13	19.70	20.22	20.64	21.15	21.66	24.26
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		174.76	176.80	184.72	191.25	197.03	202.22	206.37	211.51	216.61	242.58
	Demographics	Population (units)		1,138,392	1,151,549	1,200,824	1,239,187	1,276,437	1,312,991	1,345,780	1,381,079	1,415,751
Employees (units)		625,381	636,885	692,344	730,648	754,427	773,568	784,194	800,497	815,380	889,798	
Occupied Households (units)		411,903	418,256	440,340	457,351	473,918	490,222	504,984	520,774	536,307	613,972	
DUR (dimensionless)		2.2	2.2	2.0	1.8	1.7	1.6	1.5	1.4	1.3	0.78	
Single Family Households (units)		284,852	286,486	293,197	296,286	299,062	301,616	303,359	305,643	307,792	269,368	
Multi-Family Households (units)		127,051	131,770	147,143	161,065	174,856	188,606	201,625	215,131	228,515	344,604	
Unit Use (gpd/unit)	Single Family Household		156	154	150	148	147	146	146	145	145	145
	Multi-Family Household		150	148	144	142	141	140	140	139	139	139
	Employee		28	28	27	27	27	27	27	26	26	26

A.2.7 Fairfax Water – Loudoun Water – Wholesaler

Fairfax Water Wholesaler: Loudoun Water’s system includes the Trap Rock Water Treatment Facility (WTF) with a Potomac River intake and an emergency connection to Goose Creek Reservoir. The Goose Creek WTF is decommissioned and excluded from modeling. Goose Creek intake can pump to Trap Rock WTF for emergency use only. Production at Trap Rock WTF began on September 4, 2018.

Service Area: The updated central service area boundary (D. Geldert, March 14, 2024) reflects 2020 adjustments and includes the Leesburg JLMA, which may be served by Leesburg or Loudoun Water. The TAZ data table provide in the data submission reflects presumed splits, using Loudoun Water’s best estimates for current and future service ratios by TAZ. Growth in the Central Service Area is assumed to be served by Loudoun Water, except for TAZ 2290, where 20 percent of growth is estimated to be served by Leesburg.

Billing Data: Data includes billed SF, MF, and EMP use types (EMP includes industrial, e.g., data centers). There are no wholesale accounts. Residential billing is quarterly, which may introduce minor annual discrepancies. Reclaimed usage is excluded; “Reclaimed Data Centers” reflect potable use. Assumed 60 percent of total consumption is purchased from Fairfax Water and 40 percent is produced by Loudoun Water (Loudoun Water Correspondence, August 16, 2024). An emergency connection to Leesburg can supply 2–3 MGD but has only been tested, not used.

Table A.2-13 shows non-residential water use that excludes future data center demands. Loudoun Water’s June 2025 projections estimate data center use at 7.8 MGD in 2030 and 11.5 MGD in 2035, based on Loudoun County’s permit and application data. These values are additional to the employee-based non-residential demands that are shown in table A.2-13. The annual demand forecast used in this study was not adjusted to explicitly include projected data center water use. While the forecast likely captures some of the associated increases through broader economic and development trends, it does not fully represent the magnitude of expected data center expansion. However, the high-demand scenario, which is 42.2 MGD above the baseline in 2030 and 82.2 MGD above the baseline in 2050, likely encompasses potential future data center growth. The estimated data center demand within the WMA, presented in Section 6.2, is approximately 30 MGD by 2050, suggesting that the high-demand scenario reasonably captures this additional use. Future studies should consider incorporating data center demands more explicitly within the WMA demand forecast framework.

Demographic Data: Household data is divided into SFH and MFH using Loudoun County dwelling unit ratios (B. Hilkemeyer, May 21, 2024). Housing categories: SFD (detached), SFA (attached), MFA (multi-family attached), MFST (stacked). Data is by planning subarea, with service assumed in Ashburn, Dulles, Potomac, and Sterling. The eastern Leesburg planning area is excluded, as it is outside the CO-OP system.

Table A.2-12: Historic data for unit use calculations and results – Fairfax Water – Loudoun Water – Wholesaler

			Historic Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Fairfax Water Purchased (MGD)			19.38	18.44	18.92	19.03	18.85	15.43	15.47	16.52	14.83	17.47
Trap Rock WTF Produced (MGD)			0	0	0	0	2.58	10.03	9.32	9.53	11.71	10.56
Goose Creek WTF Produced (MGD)			3.17	5.26	6.09	5.05	2.17	0	0	0	0	0
Produced/Purchased Total (MGD)			22.56	23.29	24.74	23.73	23.27	25.58	24.96	26.55	26.84	28.18
Demand (MGD)	Retail Sales	Single Family Household	13.16	13.82	14.74	14.17	13.64	14.41	14.85	15.3	14.44	15.25
		Multi-Family Household	2.09	2.3	2.5	2.47	2.5	2.66	2.83	2.89	3	3.13
		Employee	4.77	5.15	5.23	4.82	5.86	6.34	5.58	5.79	7.45	7.47
	Unmetered/Non-Revenue		2.54	2.02	2.27	2.27	1.27	2.17	1.7	2.57	1.95	2.33
	Percent Unmetered/Non-Revenue		11%	9%	9%	10%	5%	8%	7%	10%	7%	8%
	Total		22.56	23.29	24.74	23.73	23.27	25.58	24.96	26.55	26.84	28.18
Demographics	Population (units)		250,777	260,617	269,670	278,723	287,776	295,200	302,624	308,329	314,034	319,740
	Employees (units)		108,848	112,386	117,024	121,662	126,300	134,034	141,769	145,523	149,277	153,031
	Occupied Households (units)		82,019	84,853	87,604	90,353	93,103	95,752	98,401	100,299	102,198	104,096
	DUR (dimensionless)		4.1	4.0	4.0	3.9	3.8	3.7	3.6	3.5	3.4	3.3
	Single Family Households (units)		65,828	68,030	69,983	71,929	73,870	75,358	76,774	77,911	78,971	80,056
	Multi-Family Households (units)		16,191	16,823	17,621	18,424	19,233	20,395	21,627	22,388	23,227	24,041
Unit Use (gpd/unit)	Single Family Household		199.9	203.1	210.6	197.0	184.6	191.2	193.4	196.4	182.9	190.5
	Multi-Family Household		129.1	136.7	141.9	134.1	130.0	130.4	130.9	129.1	129.2	130.2
	Employee		43.8	45.8	44.7	39.6	46.4	47.3	39.4	39.8	49.9	48.8

Table A.2-13: Forecast data for unit use calculations and results – Fairfax Water – Water – Loudoun Water – Wholesaler

			Forecast Data									
			2023*	2025	2030	2035	2040	2045	2050	2055	2060	2085
Fairfax Water Purchased (MGD)			16.74	17.19	18.39	19.11	19.62	19.94	20.27	20.61	20.96	22.68
Trap Rock WTF Produced (MGD)			11.2	11.5	12.3	12.7	13.1	13.3	13.5	13.7	14.0	15.1
Goose Creek WTF Produced (MGD)			0	0	0	0	0	0	0	0	0	0
Produced/Purchased Total (MGD)			27.90	28.65	30.65	31.85	32.70	33.24	33.78	34.36	34.94	37.81
Demand (MGD)	Retail Sales	Single Family Household	14.97	15.16	15.80	15.93	16.00	16.04	16.05	16.07	16.07	15.62
		Multi-Family Household	3.10	3.27	3.83	4.31	4.61	4.81	5.04	5.28	5.55	7.26
		Employee	7.04	7.35	7.95	8.43	8.82	9.06	9.31	9.57	9.82	11.15
	Unmetered/Non-Revenue		2.79	2.87	3.06	3.18	3.27	3.32	3.38	3.44	3.49	3.78
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		27.9	28.7	30.6	31.8	32.7	33.2	33.8	34.4	34.9	37.8
Demographics	Population (units)		319,740	331,150	361,907	378,098	387,639	393,596	399,338	405,223	411,073	440,320
	Employees (units)		153,031	160,540	175,153	186,904	196,625	202,788	208,940	215,099	221,257	252,044
	Occupied Households (units)		104,096	107,894	118,455	124,663	128,508	130,995	133,431	135,901	138,362	150,670
	DUR (dimensionless)		3.3	3.2	2.8	2.5	2.3	2.2	2.1	2.0	1.9	1.4
	Single Family Households (units)		80,056	82,082	87,282	89,045	90,032	90,564	90,937	91,197	91,300	88,920
	Multi-Family Households (units)		24,041	25,812	31,172	35,618	38,475	40,430	42,494	44,704	47,062	61,750
Unit Use (gpd/unit)	Single Family Household		187	185	181	179	178	177	177	176	176	176
	Multi-Family Household		129	127	123	121	120	119	119	118	118	118
	Employee		46	46	45	45	45	45	45	44	44	44

Note: Percent Fairfax Water Purchased, percent Trap Rock WTP Produced, and Percent Unmetered/Non-Revenue are assumed.

A.2.8 Fairfax Water – Prince William Water – Wholesaler

Fairfax Water Wholesaler: The Prince William Water (formally known as Prince William County Service Authority) operates four systems: East, West, Hoadly Manor, and Bull Run Mountain/Evergreen. East, West, and Hoadly Manor receive water from Fairfax Water; West is also supplemented by Lake Manassas. Bull Run Mountain/Evergreen relies on six groundwater wells. Prince William Water purchase capacity agreements include 62.4 MGD from Fairfax Water and 5 MGD from the City of Manassas.

Service Area: The 2015 study map is used for 2020 and 2025 studies. A 2019 pressure zone boundary map (M. Knight, May 16; D. Guerra and N. Griffin, October 18) showed no significant changes.

Billing Data: Billed and purchased data for 2019–2023 was provided by D. Guerra (July 24, 2024). Billed categories SFH (Single-family attached and detached, mobile homes), MFH (Apartments and condominiums), EMP (Commercial, office, industrial). All meters are read and billed monthly.

Demographic Data: MWCOG Round 10.0 TAZ-level projections provided by F. Hunt (May 31, 2024) include households by type, population, and employment. Housing units are categorized as SF (single-family and mobile homes, detached), TH (townhomes, attached), and MF (multi-unit housing, apartments, condos). For DUR calculations, SF and TH are combined into an SFHU category.

Table A.2-14: Historic data for unit use calculations and results – Fairfax Water – Prince William Water – Wholesaler.

			Historic Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Fairfax Water Purchased (MGD)			21.01	21.20	23.36	22.75	23.05	24.44	23.95	26.70	26.16	26.99
City of Manassas Purchased (MGD)			4.71	4.74	3.78	5.51	4.63	4.02	4.78	4.90	4.64	5.22
Demand (MGD)	Retail Sales	Single Family Household	15.17	15.28	16.04	15.93	15.43	16.24	17.10	17.28	16.46	17.24
		Multi-Family Household	3.57	3.66	3.84	3.88	3.85	3.91	4.28	4.40	4.51	4.53
		Employee	4.83	5.09	5.26	5.13	4.89	5.12	4.09	4.74	4.84	5.13
	Manassas Park Sales		0.90	0.90	0.89	0.91	0.89	0.94	1.03	2.83	2.35	2.25
	Unmetered/Non-Revenue		2.15	1.91	2.00	3.32	3.51	2.25	2.23	2.35	2.64	3.06
	Percent Unmetered/Non-Revenue		8%	7%	7%	12%	13%	8%	8%	7%	9%	10%
	Total		25.72	25.94	27.14	28.26	27.68	28.46	28.73	31.60	30.80	32.21
Demographics	Population (units)		323,322	330,914	334,353	337,791	341,229	357,865	374,500	379,660	384,820	389,980
	Employees (units)		104,026	107,542	111,024	114,506	117,988	123,594	129,200	132,420	135,640	138,860
	Occupied Households (units)		106,295	108,584	110,228	111,872	113,515	117,558	121,600	123,420	125,240	127,060
	DUR (dimensionless)		3.5	3.3	3.2	3.2	3.1	3.8	4.6	4.4	4.3	4.2
	Single Family Households (units)		82,804	83,353	84,210	85,042	85,848	93,231	99,749	100,689	101,583	102,429
	Multi-Family Households (units)		23,491	25,231	26,017	26,829	27,668	24,326	21,851	22,731	23,657	24,631
Unit Use (gpd/unit)	Single Family Household		183.2	183.3	190.5	187.3	179.7	174.2	171.4	171.6	162.0	168.3
	Multi-Family Household		152.0	145.1	147.6	144.6	139.2	160.7	195.9	193.6	190.6	183.9
	Employee		46.4	47.3	47.4	44.8	41.4	41.4	31.7	35.8	35.7	36.9

Table A.2-15: Forecast data for unit use calculations and results – Fairfax Water – Prince William Water – Wholesaler.

			Forecast Data									
			2023*	2025	2030	2035	2040	2045	2050	2055	2060	2085
Fairfax Water Purchased (MGD)			26.8	27.1	27.4	28.8	29.9	30.8	31.6	32.3	33.1	33.8
City of Manassas Purchased (MGD)			4.70	4.70	4.70	4.70	4.70	4.70	4.70	4.70	4.70	4.70
Demand (MGD)	Retail Sales	Single Family Household	16.90	16.92	17.08	17.28	17.51	17.78	17.90	18.15	18.38	19.56
		Multi-Family Household	4.75	5.10	5.74	6.18	6.45	6.60	6.76	6.93	7.10	8.05
		Employee	4.72	4.91	5.30	5.68	6.02	6.33	6.61	6.91	7.22	8.75
	Manassas Park Sales		2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
	Unmetered/Non-Revenue		3.15	3.21	3.35	3.46	3.55	3.63	3.70	3.78	3.85	4.26
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		31.53	32.15	33.46	34.61	35.54	36.34	36.97	37.76	38.55	42.62
	Demographics	Population (units)		389,980	400,300	419,700	432,900	442,300	449,000	453,500	459,467	465,067
Employees (units)		138,860	145,300	158,600	171,700	183,300	193,700	202,900	212,900	222,700	271,700	
Occupied Households (units)		127,060	130,700	138,100	143,600	147,600	150,700	152,900	155,700	158,3500	171,600	
DUR (dimensionless)		4.2	3.9	3.5	3.3	3.2	3.2	3.1	3.1	3.1	2.9	
Single Family Households (units)		102,429	103,959	107,417	110,170	112,466	114,670	115,852	117,715	102,429	127,306	
Multi-Family Households (units)		24,631	26,741	306,83	33,430	35,134	36,030	37,048	38,011	24,631	44,294	
Unit Use (gpd/unit)	Single Family Household		165.4	162.8	159.0	156.9	155.7	155.1	154.5	154.2	154.0	153.6
	Multi-Family Household		192.9	190.8	187.0	184.9	183.7	183.1	182.5	182.2	182.0	181.6
	Employee		33.7	33.8	33.4	33.1	32.9	32.7	32.6	32.5	32.4	32.2

Note: City of Manassas Purchased, Manassas Park Sales, and Percent Unmetered/Non-Revenue are assumed.

A.2.9 Fairfax Water – Town of Herndon – Wholesaler

Fairfax Water Wholesaler: The Town of Herndon, located within Fairfax County, purchases finished water from Fairfax Water.

Service Area: The service area was last updated in the 2020 study with a shapefile reflecting minor boundary changes.

Billing Data: Billing data was provided by C. Bednarek (May 29, 2024). Data are grouped as residential (combined SFH and MFH) and EMP (commercial), with residential unit use applied instead of separate SFH and MFH categories.

Demographic Data: The demographic data contact was A. Zaki (May 14, 2024, via E. Gilleran, May 10, 2024). Past DUR values (2010–2020) were sourced from D. Heiberg memos, while 2024 DUR values are from A. Zaki’s “Snapshot of 2024 Housing and Population Estimate” (February 9, 2024), with values of 2.13 (2024), 0.81 (2030), and 0.59 (2040), interpolated between years and assumed constant thereafter. Housing data are categorized as SFH (single-family detached and townhomes) and MFH (apartments and condos), using data sourced with guidance from F. Khaja (Fairfax County). The Town of Herndon is analyzed separately from Fairfax County.

Table A.2-16: Historical data for unit use calculations and results – Fairfax Water – Town of Herndon – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			2.10	2.10	2.22	2.23	2.26	2.26	2.02	2.03	2.00	2.02
Billed (Town of Herndon Reported, MGD)			1.87	1.90	1.93	1.93	1.94	1.96	1.88	1.89	1.93	1.96
Demand (MGD)	Retail Sales	Residential	1.09	1.09	1.15	1.16	1.17	1.05	0.92	0.90	0.95	0.85
		Employee	0.82	0.82	0.87	0.87	0.88	0.95	0.97	0.99	1.13	1.18
	Unmetered/Non-Revenue		0.19	0.19	0.2	0.2	0.2	0.25	0.14	0.14	0.07	0.06
	Percent Unmetered/Non-Revenue		0.09	0.09	0.09	0.09	0.09	13%	7%	7%	4%	3%
	Total		2.10	2.10	2.22	2.23	2.26	2.25	2.03	2.03	2.15	2.09
Demographics	Population (units)		22,308	22,911	22,593	22,311	22,028	21,916	21,803	22,080	22,357	22,633
	Employees (units)		22,191	22,520	22,943	23,423	23,903	21,283	18,663	18,762	18,861	18,961
	Occupied Households (units)		7,423	7,508	7,512	7,526	7,540	7,613	7,686	7,823	7,960	8,096
	DUR (dimensionless)		2.2	2.2	2.2	2.2	2.2	2.2	2.1	2.1	2.1	2.1
	Single Family Households (units)		5,066	5,125	5,135	5,145	5,162	5,196	5,230	5,324	5,417	5,510
	Multi-Family Households (units)		2,356	2,384	2,377	2,382	2,379	2,417	2,456	2,499	2,543	2,587
Unit Use (gpd/unit)	Residential		147.00	145.30	153.60	154.00	155.80	137.84	119.87	115.03	119.40	104.98
	Employee		36.90	36.40	37.80	37.20	36.90	44.45	52.01	52.90	59.91	62.02

Table A.2-17: Forecast data for unit use calculations and results – Fairfax Water – Town of Herndon – Wholesaler.

			Forecast Data									
			2023*	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			2.2	2.2	2.4	2.4	2.5	2.6	2.6	2.7	2.8	3.2
Billed (Town of Herndon Reported, MGD)			2.0	2.1	2.2	2.3	2.3	2.4	2.4	2.5	2.6	3.0
Demand (MGD)	Retail Sales	Residential	0.90	0.91	1.01	1.03	1.03	1.05	1.06	1.08	1.09	1.19
		Employee	1.14	1.15	1.19	1.22	1.28	1.34	1.39	1.45	1.50	1.79
	Unmetered/Non-Revenue		0.15	0.15	0.17	0.17	0.17	0.18	0.18	0.19	0.20	0.22
	Percent Unmetered/Non-Revenue		7%	7%	7%	7%	7%	7%	7%	7%	7%	7%
	Total		2.19	2.21	2.37	2.42	2.48	2.57	2.63	2.71	2.79	3.20
Demographics	Population (units)		22633	23187	25759	26473	26835	27246	27680	28099	28521	30634
	Employees (units)		18961	19159	20041	20729	21682	22865	23688	24751	25754	30769
	Occupied Households (units)		8096	8370	9636	9985	10158	10349	10550	10744	10940	11920
	DUR (dimensionless)		2.13	1.47	0.81	0.70	0.59	0.59	0.59	0.59	0.59	0.59
	Single Family Households (units)		5510	4981	4312	4111	3769	3840	3915	3987	4060	4423
	Multi-Family Households (units)		2587	3389	5324	5874	6389	6509	6635	6757	6881	7,497
Unit Use (gpd/unit)	Residential		111.00	108.75	104.99	102.88	101.70	101.06	100.53	100.20	99.99	99.64
	Employee		60.00	59.80	59.39	59.09	58.87	58.70	58.57	58.48	58.40	58.22

Note: Percent Unmetered/Non-Revenue is assumed.

A.2.10 Fairfax Water – Town of Vienna – Wholesaler

Fairfax Water Wholesaler: The Town of Vienna purchases finished water from Fairfax Water, which can supply water from its own treatment facilities or the Washington Aqueduct. Since 2013, Vienna has been solely a Fairfax Water customer, simplifying analysis from 2014 onward.

Service Area: A. Spiesman and G. Prelewicz (Fairfax Water) confirmed the service area validity in 2024. The service area slightly overlaps with Fairfax Water’s service area but has minimal impact.

Billing Data: Billed and purchased data for CY 2019–2023 was collected with the help of Marvin Lawrence (Marvin.Lawrence@viennava.gov) with Emily Goodman copied. Detailed values were provided for single-family households (SFH), multi-family households (MFH), and employment (EMP) categories. For forecast years, it is assumed that the billed amount is 0.90 times the purchased amount from Fairfax Water.

Demographic Data: Demographic data collection followed guidance from F. Khaja (Fairfax County, April 30, 2019), though she was not contacted for the current study. The Town of Vienna is analyzed separately from Fairfax County. A Fairfax County shapefile from the Geospatial Data page was clipped to the Town of Vienna boundary to extract SF and MF home values. Data for 2020–2022 relied on Demographic Reports, while data for 2023–2050 was derived from the clipped shapefile.

Table A.2-18: Historical data for unit use calculations and results – Fairfax Water – Town of Vienna – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Vienna Reported, MGD)			2.17	2.12	1.93	2.10	2.09	2.13	2.02	2.21	2.05	2.10
Fairfax Water Purchased (Fairfax Water Reported, MGD)			2.17	2.08	2.03	2.34	2.07	2.11	2.01	2.20	2.01	2.02
Demand (MGD)	Retail Sales ⁴	Single Family Household	1.51	1.57	1.55	1.54	1.47	1.38	1.42	1.45	1.33	1.36
		Multi-Family Household	0.09	0.09	0.09	0.09	0.09	0.13	0.13	0.13	0.13	0.12
		Employee	0.31	0.33	0.32	0.32	0.31	0.36	0.30	0.29	0.31	0.32
	Unmetered/Non-Revenue		0.26	0.09	0.06	0.39	0.21	0.25	0.16	0.33	0.24	0.22
	Percent Unmetered/Non-Revenue		12%	4%	3%	17%	10%	12%	8%	15%	12%	11%
	Total		2.17	2.08	2.03	2.35	2.07	2.11	2.01	2.20	2.01	2.02
Demographics	Population (units)		27394	27533	27596	27659	27721	27921	28121	28198	28274	28351
	Employees (units)		13049	12976	13127	13279	13430	13658	13885	13892	13899	13905
	Occupied Households (units)		9199	9239	9238	9238	9237	9282	9327	9354	9381	9407
	DUR (dimensionless)		11.03	11.03	11.02	11.01	11.01	10.34	9.67	9.67	9.73	8.39
	Single Family Households (units)		8434	8471	8470	8469	8468	8463	8453	8477	8507	8406
	Multi-Family Households (units)		765	768	768	769	769	819	874	877	874	1002
Unit Use (gpd/unit)	Single Family Household		178.50	184.90	183.30	181.90	173.30	162.47	167.99	171.05	156.35	161.79
	Multi-Family Household		116.60	120.80	119.60	118.60	112.94	157.43	148.72	148.26	148.73	119.80
	Employee		24.10	25.20	24.70	24.20	22.80	26.23	21.61	20.88	22.30	23.01

Table A.2- 19: Forecast data for unit use calculations and results – Fairfax Water – Town of Vienna – Wholesaler.

			Forecast Data									
			2023*	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Vienna Reported, MGD)			2.01	1.99	1.95	1.94	1.94	1.95	1.95	1.96	1.97	2.03
Fairfax Water Purchased (Fairfax Water Reported, MGD)			2.01	1.99	1.95	1.94	1.94	1.95	1.95	1.96	1.97	2.03
Demand (MGD)	Retail Sales	Single Family Household	1.34	1.34	1.31	1.31	1.31	1.31	1.32	1.33	1.34	1.39
		Multi-Family Household	0.15	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.15
		Employee	0.31	0.30	0.30	0.29	0.29	0.29	0.29	0.29	0.29	0.29
	Unmetered/Non-Revenue		0.22	0.21	0.20	0.20	0.20	0.20	0.20	0.20	0.21	0.21
	Percent Unmetered/Non-Revenue		11%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		2.00	1.98	1.95	1.93	1.93	1.94	1.94	1.95	1.96	2.02
Demographics	Population (units)		28351	28504	28674	28934	29213	29470	29693	29939	30179	31379
	Employees (units)		13905	13919	13945	13977	14016	14055	14057	14084	14104	14207
	Occupied Households (units)		9407	9461	9517	9603	9695	9782	9854	9936	10016	10413
	DUR (dimensionless)		8.39	8.51	8.55	8.55	8.55	8.55	8.55	8.55	8.55	8.55
	Single Family Households (units)		8406	8466	8520	8597	8679	8757	8822	8895	8967	9322
	Multi-Family Households (units)		1002	995	997	1006	1016	1025	1032	1041	1049	1091
Unit Use (gpd/unit)	Single Family Household		160.00	157.75	153.99	151.88	150.70	150.06	149.53	149.20	148.99	148.64
	Multi-Family Household		145.00	142.75	138.99	136.88	135.70	135.06	134.53	134.20	133.99	133.64
	Employee		22.00	21.80	21.39	21.09	20.87	20.70	20.57	20.48	20.40	20.22

Note: Percent Unmetered/Non-Revenue is assumed.

A.2.11 Fairfax Water – Virginia American Water – City of Alexandria – Wholesaler

Virginia American Water Company (VAWC), a Fairfax Water wholesale customer, serves as the sole water distributor for the City of Alexandria, purchasing potable water from Fairfax Water, which sources from the Potomac River and Griffith Water Treatment Plant.

Service Area: Provided by Fairfax Water and confirmed by A. Spiesman and G. Prelewicz in 2024. A TAZ layer from C. Banacka (June, 2024) confirmed the existing shapefile, with minor boundary differences overlapping the Potomac River.

Billing Data: Billing data is not collected from VAWC. Fairfax Water provides the volume sold to the City of Alexandria.

Demographic Data: Collected from the Alexandria Department of Planning and Zoning with assistance from C. Beach (June, 2024) and provided by C. Banacka (June, 2024). The 2020 starting values were assumed in the 2020 demand study. Differences in single-family homes (SFH) and multi-family homes (MFH) were added for each forecast year to calculate DUR ratios. The forecast is not highly sensitive to the DUR values; for instance, a change from a DUR of 0.3 to 0.54 results in a + 0.10 million gallon per day difference in the annual forecast.

Table A.2- 20: Historical data for unit use calculations and results – Fairfax Water – Virginia American Water, City of Alexandria – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			15.42	15.72	15.87	15.13	15.35	15.63	14.97	15.51	16.18	15.90
Demand (MGD)	Retail Sales	Single Family Household	3.85	3.82	3.86	3.80	3.66	3.77	3.82	3.89	3.82	3.96
		Multi-Family Household	7.52	7.40	7.60	7.69	7.67	7.82	8.16	8.16	8.18	8.40
		Employee	2.51	2.93	2.82	2.13	2.48	2.47	1.49	1.91	2.56	1.95
	Unmetered/Non-Revenue		1.54	1.57	1.59	1.51	1.53	1.56	1.50	1.55	1.62	1.59
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		15.42	15.72	15.87	15.13	15.35	15.63	14.97	15.51	16.18	15.90
Demographics	Population (units)		146,096	147,622	149,924	152,226	154,528	156,981	159,434	163,646	167,858	172,071
	Employees (units)		105,557	106,225	107,002	107,778	108,554	105,180	101,806	101,639	101,472	101,305
	Occupied Households (units)		70,556	71,174	72,068	72,961	73,855	74,695	75,534	77,566	79,598	81,630
	DUR (dimensionless)		0.47	0.46	0.45	0.45	0.45	0.45	0.44	0.44	0.45	0.45
	Single Family Households (units)		22,559	22,425	22,366	22,643	22,921	23,003	23,080	23,870	24,667	25,472
	Multi-Family Households (units)		47,997	48,749	49,702	50,318	50,935	51,692	52,454	53,696	54,931	56,158
Unit Use (gpd/unit)	Single Family Household		170.80	170.60	172.40	167.80	159.50	163.95	165.38	162.84	154.76	155.55
	Multi-Family Household		156.60	151.70	152.90	152.90	150.60	151.37	155.64	152.02	148.95	149.55
	Employee		26.60	33.70	29.70	32.60	28.50	23.49	14.66	18.78	25.24	19.20

Note: Percent Unmetered/Non-Revenue is assumed.

Table A.2-21: Forecast data for unit use calculations and results – Fairfax Water – Virginia American Water, City of Alexandria – Wholesaler.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			15.92	16.37	17.65	19.17	20.48	21.49	22.15	23.07	23.95	28.45
Demand (MGD)	Retail Sales ⁴	Single Family Household	3.98	4.17	4.20	4.24	4.17	4.17	4.20	4.18	4.16	3.77
		Multi-Family Household	8.43	8.66	9.85	11.08	12.25	13.11	13.57	14.35	15.08	19.08
		Employee	1.92	1.90	1.83	1.93	2.02	2.07	2.17	2.24	2.32	2.76
	Unmetered/Non-Revenue		1.59	1.64	1.76	1.92	2.05	2.15	2.21	2.31	2.40	2.85
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		15.92	16.37	17.65	19.17	20.48	21.49	22.15	23.07	23.95	28.45
Demographics	Population (units)		172071	180495	201960	222170	239798	252885	261818	273520	284530	339580
	Employees (units)		101305	100971	99255	106641	112785	116817	123245	128076	133306	159456
	Occupied Households (units)		81630	85694	96375	106687	115398	122014	126005	131746	137050	163567
	DUR (dimensionless)		0.45	0.46	0.41	0.37	0.33	0.30	0.30	0.28	0.26	0.19
	Single Family Households (units)		25472	27105	27988	28637	28403	28500	28811	28753	28637	26000
	Multi-Family Households (units)		56158	58589	68387	78050	86995	93514	97194	102993	108413	137567
Unit Use (gpd/unit)	Single Family Household		156.21	153.96	150.20	148.10	146.92	146.27	145.74	145.41	145.20	144.85
	Multi-Family Household		150.08	147.84	144.07	141.97	140.79	140.14	139.61	139.28	139.07	138.72
	Employee		19.00	18.80	18.39	18.09	17.87	17.70	17.57	17.48	17.40	17.30

Note: Percent Unmetered/Non-Revenue is assumed.

A.2.12 Fairfax Water – Virginia American Water – Dale City – Wholesaler

Dale City, in Prince William County, gets its water from VAWC, which buys water from Fairfax Water and sells it to local residents and businesses.

Service Area: Confirmed by Fairfax Water via A. Spiesman and G. Prelewicz in 2024.

Billing Data: Not collected from Virginia American Water. Fairfax Water provided the amount of water sold to Dale City.

Demographic Data: The TAZ layer was received from Prince William County via F. Hunt (May, 2024). Household data was provided at the TAZ level for both Prince William County and Dale City. Data for Dale City TAZ was extracted to calculate DUR for the VAWC service area. Single-family households were estimated by combining the "SF HH" and "TH HH" categories, while "MF HH" was used for multi-family households.

Table A.2-22: Historic data for unit use calculations and results – Fairfax Water – Virginia American Water, Dale City – Wholesaler.

			Historic Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			4.73	4.73	4.76	4.68	4.67	4.59	4.99	4.93	4.94	4.94
Demand (MGD)	Retail Sales	Single Family Household	3.29	3.29	3.36	3.31	3.18	3.27	3.30	3.25	3.09	3.09
		Multi-Family Household	0.37	0.37	0.39	0.41	0.43	0.37	0.34	0.36	0.38	0.42
		Employee	0.60	0.60	0.53	0.49	0.60	0.48	0.86	0.83	0.97	0.93
	Unmetered/Non-Revenue		0.47	0.47	0.48	0.47	0.47	0.46	0.50	0.49	0.49	0.49
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		4.73	4.73	4.76	4.68	4.67	4.59	4.99	4.93	4.94	4.94
Demographics	Population (units)		68600	69086	69852	70617	71383	73408	75433	76079	76725	77371
	Employees (units)		11069	11312	11518	11724	11930	12931	13932	13987	14043	14098
	Occupied Households (units)		21597	21691	22044	22397	22751	22430	22109	22336	22564	22791
	DUR (dimensionless)		8.14	7.99	7.68	7.37	7.06	8.10	9.14	8.43	7.73	7.03
	Single Family Households (units)		19235	19278	19504	19720	19926	19965	19928	19968	19979	19952
	Multi-Family Households (units)		2362	2413	2540	2677	2824	2465	2181	2368	2585	2839
Unit Use (gpd/unit)	Single Family Household		170.80	170.60	172.40	167.80	159.50	163.95	165.38	162.84	154.76	154.76
	Multi-Family Household		156.60	151.70	152.90	152.90	150.60	151.37	155.64	152.02	148.95	148.95
	Employee		26.60	33.70	29.70	32.60	28.50	37.48	61.43	59.01	68.89	66.15

Note: Percent Unmetered/Non-Revenue is assumed.

Table A.2-23: Forecast data for unit use calculations and results – Fairfax Water – Virginia American Water, Dale City – Wholesaler.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			4.89	4.93	4.94	4.99	5.04	5.09	5.12	5.16	5.21	5.47
Demand (MGD)	Retail Sales	Single Family Household	3.10	3.04	2.99	2.98	2.99	3.01	3.01	3.03	3.05	3.14
		Multi-Family Household	0.42	0.52	0.55	0.58	0.60	0.61	0.62	0.63	0.64	0.70
		Employee	0.87	0.88	0.90	0.93	0.95	0.96	0.98	0.99	1.01	1.09
	Unmetered/Non-Revenue		0.49	0.49	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.55
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		4.92	4.94	4.99	5.04	5.09	5.12	5.16	5.21	5.47	4.94
Demographics	Population (units)		77371	78663	79551	80739	81620	82256	82592	83128	83614	86044
	Employees (units)		14098	14209	14724	15199	15587	15888	16131	16413	16685	18045
	Occupied Households (units)		22791	23246	23749	24239	24616	24905	25077	25327	25558	26710
	DUR (dimensionless)		7.03	5.62	5.17	4.90	4.78	4.76	4.67	4.63	4.57	4.30
	Single Family Households (units)		19952	19735	19901	20132	20357	20582	20655	20826	20972	21669
	Multi-Family Households (units)		2839	3511	3848	4107	4259	4323	4422	4501	4586	5041
Unit Use (gpd/unit)	Single Family Household		155.55	153.96	150.20	148.10	146.92	146.27	145.74	145.41	145.20	144.85
	Multi-Family Household		149.55	147.84	144.07	141.97	140.79	140.14	139.61	139.28	139.07	138.72
	Employee		62.00	61.80	61.39	61.09	60.87	60.70	60.57	60.48	60.40	60.30

Note: Percent Unmetered/Non-Revenue is assumed.

A.2.13 Fairfax Water – Fort Belvoir – Wholesaler

The U.S. Government (Fort Belvoir) is a wholesale customer of Fairfax Water.

Service Area: Confirmed by Fairfax Water (A. Spiesman and G. Prelewicz, 2024).

Billing Data: Not collected from Fort Belvoir. Fairfax Water provided the amount of water sold to Fort Belvoir.

Demographic Data: None received from the jurisdictional group. Fairfax County dwelling unit ratios are used.

Table A.2-24: Historical data for unit use calculations and results – Fairfax Water – Fort Belvoir – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			1.54	1.92	1.78	1.93	1.75	1.83	1.72	1.34	1.48	1.64
Demand (MGD)	Retail Sales	Single Family Household	0.32	0.33	0.34	0.33	0.31	0.33	0.35	0.34	0.32	0.31
		Multi-Family Household	0.11	0.12	0.12	0.12	0.12	0.13	0.14	0.14	0.13	0.13
		Employee	0.96	1.28	1.15	1.28	1.14	1.18	1.06	0.73	0.88	1.03
	Unmetered/Non-Revenue		0.15	0.19	0.18	0.19	0.18	0.18	0.17	0.13	0.15	0.16
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		1.54	1.92	1.78	1.93	1.75	1.83	1.72	1.34	1.48	1.64
Demographics	Population (units)		8900	9360	9427	9494	9561	9928	10295	10255	10215	10174
	Employees (units)		36003	37888	38629	39370	40110	40393	40676	41533	42390	43246
	Occupied Households (units)		2590	2713	2737	2761	2785	2890	2995	2972	2949	2927
	DUR (dimensionless)		2.60	2.50	2.50	2.40	2.40	2.38	2.37	2.33	2.28	2.24
	Single Family Households (units)		1864	1931	1947	1957	1965	2036	2105	2078	2051	2024
	Multi-Family Households (units)		726	782	791	804	819	854	890	894	898	903
Unit Use (gpd/unit)	Single Family Household		170.80	170.60	172.40	167.80	159.50	163.95	165.38	162.84	154.76	155.55
	Multi-Family Household		156.60	151.70	152.90	152.90	150.60	151.37	155.64	152.02	148.95	149.55
	Employee		26.60	33.70	29.70	32.60	28.50	29.31	26.09	17.62	20.86	23.73

Note: Percent Unmetered/Non-Revenue is assumed.

Table A.2-25: Forecast data for unit use calculations and results – Fairfax Water – Fort Belvoir – Wholesaler.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			2.3	2.3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Demand (MGD)	Retail Sales	Single Family Household	0.32	0.30	0.29	0.28	0.28	0.27	0.27	0.26	0.26	0.21
		Multi-Family Household	0.14	0.13	0.14	0.15	0.16	0.16	0.17	0.18	0.19	0.26
		Employee	1.60	1.65	1.85	1.84	1.83	1.82	1.82	1.81	1.81	1.81
	Unmetered/Non-Revenue		0.23	0.23	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		2.28	2.33	2.53	2.52	2.51	2.51	2.51	2.51	2.51	2.53
Demographics	Population (units)		10174	10094	10141	10224	10325	10427	10510	10606	10698	11161
	Employees (units)		43246	44960	50754	50983	51035	51065	51076	51100	51120	51223
	Occupied Households (units)		2927	2881	2901	2939	2984	3028	3064	3105	3145	3345
	DUR (dimensionless)		2.24	2.17	1.99	1.84	1.71	1.60	1.50	1.40	1.30	0.78
	Single Family Households (units)		2024	1973	1932	1904	1883	1863	1841	1811	1775	1468
	Multi-Family Households (units)		903	908	969	1035	1101	1165	1223	1294	1370	1877
Unit Use (gpd/unit)	Single Family Household		156.21	153.96	150.20	148.10	146.92	146.27	145.74	145.41	145.20	144.85
	Multi-Family Household		150.08	147.84	144.07	141.97	140.79	140.14	139.61	139.28	139.07	138.72
	Employee		37.00	36.80	36.39	36.09	35.87	35.70	35.57	35.48	35.40	35.30

Note: Percent Unmetered/Non-Revenue is assumed.

A.2.14 Fairfax Water – Dulles International Airport – Wholesaler

Metropolitan Washington Airports Authority (Dulles International Airport) is a wholesale customer of Fairfax Water.

Service Area: Confirmed by Fairfax Water (A. Spiesman and G. Prelewicz, 2024).

Billing Data: Not collected from Virginia American Water. Fairfax Water provided the amount of water sold to Dulles International Airport.

Demographic Data: None received from the jurisdictional group. Fairfax County dwelling unit ratios are used.

Table A.2-26: Historical data for unit use calculations and results – Fairfax Water – Dulles Airport – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			0.81	0.81	0.79	0.77	0.79	0.81	0.51	0.64	0.70	0.70
Demand (MGD)	Retail Sale	Single Family Household	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02
		Multi-Family Household	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01
		Employee	0.72	0.72	0.70	0.68	0.70	0.71	0.38	0.51	0.56	0.57
	Unmetered/Non-Revenue		0.08	0.08	0.08	0.08	0.08	0.08	0.05	0.06	0.07	0.07
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		0.81	0.81	0.79	0.77	0.79	0.81	0.51	0.64	0.70	0.70
Demographics	Population (units)		109	111	141	172	202	339	475	516	557	598
	Employees (units)		16211	15826	15989	16152	16314	14776	13237	13587	13937	14286
	Occupied Households (units)		45	46	56	68	78	132	185	200	215	229
	DUR (dimensionless)		2.57	2.47	2.46	2.44	2.40	2.38	2.37	2.33	2.28	2.24
	Single Family Households (units)		32	33	40	48	55	93	130	140	149	159
	Multi-Family Households (units)		13	13	16	20	23	39	55	60	65	71
Unit Use (gpd/unit)	Single Family Household		170.80	170.60	172.40	167.80	159.50	163.95	165.38	162.84	154.76	155.55
	Multi-Family Household		156.60	151.70	152.90	152.90	150.60	151.37	155.64	152.02	148.95	149.55
	Employee		44.51	45.59	43.88	42.22	42.83	47.91	28.70	37.40	40.45	39.75

Note: Percent Unmetered/Non-Revenue is assumed.

Table A.2-27: Forecast data for unit use calculations and results – Fairfax Water – Dulles Airport – Wholesaler.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			0.63	0.66	0.75	0.82	0.86	0.89	0.92	0.95	0.98	1.14
Demand (MGD)	Retail Sales	Single Family Household	0.02	0.03	0.04	0.04	0.05	0.05	0.05	0.05	0.06	0.05
		Multi-Family Household	0.01	0.01	0.02	0.02	0.03	0.03	0.03	0.04	0.04	0.07
		Employee	0.53	0.55	0.62	0.68	0.70	0.72	0.74	0.77	0.79	0.91
	Unmetered/Non-Revenue		0.06	0.07	0.08	0.08	0.09	0.09	0.09	0.10	0.10	0.11
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		0.63	0.66	0.75	0.82	0.86	0.89	0.92	0.95	0.98	1.14
Demographics	Population (units)		598	639	680	920	1117	1254	1346	1438	1530	1622
	Employees (units)		14286	14636	14986	17171	18765	19531	20245	20909	21606	22295
	Occupied Households (units)		229	244	259	356	440	500	541	580	620	660
	DUR (dimensionless)		2.24	2.21	2.17	1.99	1.84	1.71	1.60	1.50	1.40	1.30
	Single Family Households (units)		159	168	177	237	285	316	333	348	365	382
	Multi-Family Households (units)		71	76	82	119	155	184	208	232	255	279
Unit Use (gpd/unit)	Single Family Household		156.21	155.09	153.96	150.20	148.10	146.92	146.27	145.74	145.41	145.20
	Multi-Family Household		150.08	148.96	147.84	144.07	141.97	140.79	140.14	139.61	139.28	139.07
	Employee		37.00	36.90	36.80	36.39	36.09	35.87	35.70	35.57	35.48	35.40

Note: Percent Unmetered/Non-Revenue is assumed.

A.2.15 Washington Aqueduct – DC Water (District of Columbia Water and Sewer Authority) – Wholesaler

DC Water purchases treated drinking water from the Washington Aqueduct and distributes it to their customers.

Service Area: The 2015 study map is used in the 2020 and 2025 studies.

Billed data: The purchase data was received through A. Litz (March, 2024). Received the billing data from M. Goddard of DC Water (May, 2024). Fairfax Water (A. Spiesman, 2024) provided the purchased data. DC Water’s billing data are reported in the following categories:

- Residential – buildings with one to three dwelling units
- Commercial
- Multi-family – buildings with 4 or more dwelling units

District Government – combines previous categories labeled Exempt and Municipal, which includes meters exempt from billing and DC Government, excluding housing, respectively

- Federal Government
- Housing – DC Housing Authority

The DC Water categories are grouped into the categories needed for this study as follows:

- SFH: Residential
- MFH: Housing, Multi-family
- EMP: Commercial, District Government, Federal Government
- Other: DC Water and Washington Aqueduct

Demographic data: Collected from the Planning and Data Visualization Division with the help of T. Pate (June, 2024).

Table A.2-28: Historical data for unit use calculations and results – Washington Aqueduct – DC Water – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (MGD)			95.89	104.13	100.95	94.83	95.12	96.41	92.16	95.12	95.24	94.38
Billed (MGD)			69.53	73.28	72.11	70.82	69.94	68.68	66.77	65.80	66.53	67.39
Demand (MGD)	Retail Sales	Single Family Household	15.51	15.23	14.88	14.57	14.27	13.92	14.31	13.57	12.95	12.60
		Multi-Family Household	16.84	16.87	19.78	20.04	19.21	19.20	20.11	20.63	20.63	20.83
		Employee	36.67	40.34	36.42	35.4	35.7	34.90	31.74	30.31	31.74	32.48
		Other	0.51	0.84	1.03	0.81	0.76	0.65	0.61	1.29	1.20	1.48
	Unmetered/Non-Revenue		26.36	30.85	28.84	24.01	25.18	27.73	25.39	29.32	28.72	26.99
	Percent Unmetered/Non-Revenue		27%	30%	29%	25%	26%	29%	28%	31%	30%	29%
	Total		95.89	104.13	100.95	94.83	95.12	96.41	92.16	95.12	95.24	94.38
Demographics	Population (units)		658,137	672,230	683,684	695,138	706,593	697,720	688,846	690,469	692,092	693,716
	Employees (units)		658,137	672,230	683,684	695,138	706,593	746,776	786,959	798,997	811,036	823,074
	Occupied Households (units)		291,031	297,112	301,548	305,983	310,419	311,303	312,187	318,534	324,881	331,229
	DUR (dimensionless)		0.61	0.61	0.60	0.59	0.58	0.55	0.52	0.50	0.49	0.48
	Single Family Households (units)		110,729	112,903	113,245	113,529	117,959	110,462	106,801	106,178	106,840	107,425
	Multi-Family Households (units)		180,302	184,209	188,302	192,454	192,460	200,841	205,386	212,356	218,041	223,803
Unit Use (gpd/unit)	Single Family Household		140.10	134.90	131.40	128.30	121.00	126.04	133.95	127.78	121.21	117.28
	Multi-Family Household		93.40	91.60	105.00	104.10	99.80	95.59	97.90	97.17	94.63	93.07
	Employee		46.10	50.50	45.10	43.30	43.20	46.74	40.34	37.93	39.13	39.47

Table A.2-29: Forecast data for unit use calculations and results – Washington Aqueduct – DC Water – Wholesaler.

			Forecast Data									
			2023*	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (MGD)			92.57	94.35	97.08	99.94	103.02	106.42	109.48	112.91	116.34	133.92
Billed (MGD)			66.65	67.93	69.90	71.96	74.17	76.62	78.82	81.30	83.76	96.42
Demand (MGD)	Retail Sales	Single Family Household	12.89	12.57	12.15	11.90	11.69	11.63	11.71	11.65	11.60	10.69
		Multi-Family Household	21.49	22.24	23.39	24.60	26.16	27.55	28.58	29.97	31.36	39.05
		Employee	31.28	32.02	33.18	34.29	35.22	36.33	37.40	38.53	39.67	45.55
		Other	1.00	1.10	1.19	1.17	1.10	1.12	1.14	1.14	1.13	1.13
	Unmetered/Non-Revenue		25.92	26.42	27.18	27.98	28.84	29.80	30.65	31.62	32.57	37.50
	Percent Unmetered/Non-Revenue		28%	28%	28%	28%	28%	28%	28%	28%	28%	28%
	Total		92.57	94.35	97.08	99.94	103.02	106.42	109.48	112.91	116.34	133.92
	Demographics	Population (units)		693,716	696,962	727,923	756,494	786,440	815,718	843,684	872,525	901,147
Employees (units)		823,074	847,151	887,316	924,553	955,375	989,969	1,022,463	1,056,357	1,089,901	1,257,621	
Occupied Households (units)		331,229	343,923	366,499	386,298	407,306	425,722	441,081	458,478	475,366	559,803	
DUR (dimensionless)		0.48	0.45	0.41	0.38	0.35	0.33	0.32	0.30	0.29	0.21	
Single Family Households (units)		107,425	106,735	106,571	106,372	105,598	105,630	106,929	106,705	106,389	98,427	
Multi-Family Households (units)		223,803	237,188	259,928	279,926	301,708	320,092	334,152	351,773	368,977	461,376	
Unit Use (gpd/unit)	Single Family Household		120.00	117.75	113.99	111.88	110.70	110.06	109.53	109.20	108.99	108.64
	Multi-Family Household		96.00	93.75	89.99	87.88	86.70	86.06	85.53	85.20	84.99	84.64
	Employee		38.00	37.80	37.39	37.09	36.87	36.70	36.57	36.48	36.40	36.22

A.2.16 Washington Aqueduct – Arlington County DES – Wholesaler

Service Area: No change from the 2015 study service area boundary is made by Arlington County DES.

Billed data: The purchase data was received through A. Litz (March, 2024). Received Arlington County billed data via K. Abele (May, 2024), which is reported in the following categories:

- Apartments – multifamily duplexes and apartment buildings
- Commercial
- County Agencies
- Residential – single family homes

The Arlington County categories are grouped into the categories needed for this study as follows:

- SFH: Residential taken from cycles 1, 2, and 3
- MFH: Apartments taken from cycle 5 (and 6)
- EMP: Commercial (commercial and large apartments, county agencies) taken from cycles 4,7, and 8

Demographic data: Received Arlington County demographic data via E. Hardy, Principal Planner & Demographer (May, 2024).

Table A.2-30: Historical data for unit use calculations and results – Washington Aqueduct – Arlington County DES – Wholesaler.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			22.13	21.97	21.41	21.28	21.83	21.59	19.67	20.18	20.27	20.84
Billed with Fort Myer (MGD)			19.13	19.55	19.92	19.18	19.07	19.48	17.78	17.80	18.246	18.38
Billed (MGD)			18.86	19.25	19.66	18.97	18.73	18.95	17.44	17.55	18.08	18.46
Demand (MGD)	Retail Sales ⁴	Single Family Household	5.22	5.23	5.23	5.12	4.91	5.08	5.21	5.13	4.81	4.88
		Multi-Family Household	5.39	5.47	5.63	5.45	5.33	5.33	5.46	5.36	5.37	5.39
		Employee	8.25	8.55	8.8	8.4	8.49	8.54	6.77	7.06	7.90	8.19
	Wholesales (Fort Myer)		0.27	0.30	0.26	0.21	0.34	0.53	0.34	0.25	0.17	0.30
	Unmetered/Non-Revenue		3.0	2.4	1.5	2.1	2.8	2.1	1.9	2.4	2.0	2.1
	Percent Unmetered/Non-Revenue		14%	11%	7%	10%	13%	9.8%	9.6%	11.8%	10%	10%
	Total		22.13	21.97	21.41	21.28	21.83	21.59	19.62	20.18	20.27	20.84
Demographics	Population (units)		217,102	219,759	223,235	226,710	230,186	233,490	236,794	238,321	239,848	241,374
	Employees (units)		207,620	205,417	206,851	208,292	209,734	212,826	215,918	216,239	216,560	216,882
	Occupied Households (units)		102,389	103,531	105,188	106,845	108,502	108,937	109,372	111,059	112,746	114,432
	DUR (dimensionless)		0.57	0.56	0.55	0.54	0.53	0.48	0.42	0.41	0.40	0.40
	Single Family Households (units)		37,173	37,165	37,325	37,465	37,586	35,081	32,349	32,294	32,213	32,695
	Multi-Family Households (units)		65,216	66,366	67,863	69,380	70,916	73,855	77,023	78,765	80,533	81,737
Unit Use (gpd/unit)	Single Family Household		140.42	140.72	140.12	136.66	130.64	144.81	161.05	158.85	149.32	149.26
	Multi-Family Household		82.65	82.42	82.96	78.55	75.16	72.17	70.89	68.05	66.68	65.94
	Employee		39.74	41.62	42.54	40.33	40.48	40.13	31.35	32.65	36.48	37.76

Table A.2-31: Forecast data for unit use calculations and results – Washington Aqueduct – Arlington County DES – Wholesaler.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			20.27	20.03	20.00	21.09	21.55	22.14	22.84	23.43	24.07	25.39
Billed with Fort Myer (MGD)			18.24	18.03	18.00	18.98	19.40	19.93	20.56	21.08	21.67	22.85
Billed (MGD)			17.94	17.73	17.71	18.69	19.11	19.62	20.24	20.76	21.33	22.45
Demand (MGD)	Retail Sales ⁴	Single Family Household	5.04	4.92	4.73	4.70	4.67	4.61	4.67	4.62	4.58	4.00
		Multi-Family Household	5.31	5.24	5.04	5.34	5.61	5.92	6.24	6.53	6.88	7.21
		Employee	7.59	7.57	7.95	8.64	8.83	9.09	9.33	9.60	9.86	11.24
	Wholesales (Fort Myer)		0.30	0.29	0.29	0.29	0.29	0.31	0.31	0.33	0.34	0.40
	Unmetered/Non-Revenue		2.03	2.00	2.00	2.11	2.16	2.21	2.28	2.34	2.41	2.54
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		20.27	20.03	20.00	21.09	21.55	22.14	22.84	23.43	24.07	25.39
Demographics	Population (units)		241,374	244,428	258,862	271,680	283,994	296,442	309,592	322,274	335,073	399,068
	Employees (units)		216,882	217,524	231,084	253,601	260,752	269,624	277,975	286,673	295,285	338,342
	Occupied Households (units)		114,432	117,806	125,854	132,979	139,754	146,402	153,101	159,766	166,440	199,807
	DUR (dimensionless)		0.40	0.38	0.34	0.32	0.30	0.28	0.27	0.25	0.24	0.16
	Single Family Households (units)		32,695	32,439	31,933	32,237	32,251	32,025	32,549	32,293	32,034	28,053
	Multi-Family Households (units)		81,737	85,367	93,921	100,742	107,503	114,377	120,552	127,473	134,406	171,754
Unit Use (gpd/unit)	Single Family Household		154.00	151.75	147.99	145.88	144.70	144.06	143.53	143.20	142.99	142.64
	Multi-Family Household		65.00	62.75	58.99	56.88	55.70	55.06	54.53	54.20	53.99	53.64
	Employee		35.00	34.80	34.39	34.09	33.87	33.70	33.57	33.48	33.40	33.22

A.2.17 Washington Aqueduct – Arlington County DES – Fort Myer – Wholesaler

Service Area: No change from the 2015 study service area boundary is made by Arlington County DES.

Billed data: Received Fort Myer annual usage via K. Abele (May, 2024). Assumes Arlington County unit use rates for SFH and MFH categories to back calculate EMP unit use rates.

Demographic data: Assumes Arlington County DES dwelling unit ratios.

Table A.2-32: Historical data for unit use calculations and results – Washington Aqueduct – Arlington County DES, Fort Myer – Wholesale.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (MGD)			0.27	0.30	0.26	0.21	0.34	0.53	0.34	0.25	0.17	0.30
Demand (MGD)	Retail Sales	Single Family Household	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.01	0.01
		Multi-Family Household	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.02
		Employee	0.22	0.25	0.22	0.17	0.29	0.45	0.27	0.19	0.12	0.24
	Unmetered/Non-Revenue		0.03	0.03	0.03	0.02	0.03	0.05	0.03	0.03	0.02	0.03
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		0.27	0.30	0.26	0.21	0.34	0.53	0.34	0.25	0.17	0.30
Demographics	Population (units)		1,003	1,003	1,003	1,003	1,003	1,244	1,486	1,430	1,374	1,318
	Employees (units)		8,441	4,246	4,246	4,246	4,246	4,318	4,390	4,391	4,392	4,392
	Occupied Households (units)		175	175	175	175	175	292	410	380	351	321
	DUR (dimensionless)		0.57	0.56	0.55	0.54	0.53	0.48	0.42	0.41	0.40	0.40
	Single Family Households (units)		64	63	62	61	61	94	121	111	100	92
	Multi-Family Households (units)		111	112	113	114	114	198	288	270	250	229
Unit Use (gpd/unit)	Single Family Household		140.42	140.72	140.12	136.66	130.64	144.81	161.05	158.85	149.32	149.26
	Multi-Family Household		82.65	82.42	82.96	78.55	75.16	72.17	70.89	68.05	66.68	65.94
	Employee		26.64	59.33	50.85	40.43	68.18	104.21	60.81	43.87	26.84	54.27

Table A.2-33: Forecast data for unit use calculations and results – Washington Aqueduct – Arlington County DES, Fort Myer – Wholesale.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (MGD)			0.27	0.30	0.26	0.21	0.34	0.53	0.34	0.25	0.17	0.30
Demand (MGD)	Retail Sales	Single Family Household	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.01	0.01
		Multi-Family Household	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.02
		Employee	0.22	0.25	0.22	0.17	0.29	0.45	0.27	0.19	0.12	0.24
	Unmetered/Non-Revenue		0.03	0.03	0.03	0.02	0.03	0.05	0.03	0.03	0.02	0.03
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		0.27	0.30	0.26	0.21	0.34	0.53	0.34	0.25	0.17	0.30
Demographics	Population (units)		1,003	1,003	1,003	1,003	1,003	1,244	1,486	1,430	1,374	1,318
	Employees (units)		8,441	4,246	4,246	4,246	4,246	4,318	4,390	4,391	4,392	4,392
	Occupied Households (units)		175	175	175	175	175	292	410	380	351	321
	DUR (dimensionless)		0.57	0.56	0.55	0.54	0.53	0.48	0.42	0.41	0.40	0.40
	Single Family Households (units)		64	63	62	61	61	94	121	111	100	92
	Multi-Family Households (units)		111	112	113	114	114	198	288	270	250	229
Unit Use (gpd/unit)	Single Family Household		140.42	140.72	140.12	136.66	130.64	144.81	161.05	158.85	149.32	149.26
	Multi-Family Household		82.65	82.42	82.96	78.55	75.16	72.17	70.89	68.05	66.68	65.94
	Employee		26.64	59.33	50.85	40.43	68.18	104.21	60.81	43.87	26.84	54.27

A.2.18 Washington Aqueduct – Fairfax Water – Wholesaler

Falls Church has been a retail customer of Fairfax Water since January 3, 2014, following its transition from a wholesale customer of the Washington Aqueduct. Fairfax Water now purchases water from the Aqueduct on Falls Church's behalf, using the same infrastructure (A. Spiesman, personal communication, August 23, 2019). Falls Church use is included in Fairfax Water's retail demand forecasts and Washington Aqueduct's production forecasts.

Service Area: Confirmed by Fairfax Water via A. Spiesman and G. Prelewicz in 2024.

Billing Data: Washington Aqueduct provided the amount of water sold to Fairfax Water for Falls Church. Fairfax Water also provided the amount of water purchased from Washington Aqueduct for Falls Church.

Demographic Data: No updated values were received from the jurisdictional group. Dwelling unit use rates from the 2020 study were used, with values for 2020–2025 interpolated for historical years. Linear extrapolation of the 2040, 2045, and 2050 single-family and multi-family units was used to estimate values to calculate rates for 2055, 2060, and 2085.

Table A.2-34: Historical data for unit use calculations and results – Washington Aqueduct – Wholesale – Fairfax Water (Fairfax Water – Retail - Falls Church).

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Purchased (Fairfax Water Reported, MGD)			14.24	14.12	11.46	14.79	15.08	15.33	11.36	12.52	13.48	12.24
Demand (MGD)	Retail Sales	Single Family Household	5.30	5.32	5.38	5.23	4.96	5.04	5.05	4.93	4.64	4.61
		Multi-Family Household	3.52	3.48	3.64	3.77	3.85	3.98	4.24	4.29	4.36	4.54
		Employee	4.00	3.90	1.30	4.32	4.77	4.77	0.94	2.05	3.13	1.87
	Unmetered/Non-Revenue		1.42	1.41	1.15	1.48	1.51	1.53	1.14	1.25	1.35	1.22
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		14.24	14.12	11.46	14.79	15.08	15.33	11.36	12.52	13.48	12.24
Demographics	Population (units)		137,618	139,252	141,272	143,292	145,312	148,514	149,577	150,640	151,703	152,766
	Employees (units)		129,993	130,405	132,542	134,679	136,815	135,730	137,719	139,708	141,698	143,687
	Occupied Households (units)		53,467	54,158	54,978	55,799	56,619	57,031	57,762	58,493	59,224	59,955
	DUR (dimensionless)		1.38	1.36	1.31	1.26	1.22	1.17	1.12	1.07	1.02	0.98
	Single Family Households (units)		31,002	31,210	31,199	31,153	31,069	30,725	30,516	30,263	29,963	29,613
	Multi-Family Households (units)		22,465	22,948	23,780	24,646	25,550	26,306	27,246	28,230	29,261	30,342
Unit Use (gpd/unit)	Single Family Household		(170.80)	(170.60)	(172.40)	(167.80)	(159.50)	(163.95)	(165.38)	(162.84)	(154.76)	(155.55)
	Multi-Family Household		(156.60)	(151.70)	(152.90)	(152.90)	(150.60)	(151.37)	(155.64)	(152.02)	(148.95)	(149.55)
	Employee		30.78	29.91	9.77	32.06	34.88	35.18	6.79	14.68	22.11	13.02

Note: Used DURs from 2020 study, interpolated between 2020 and 2025. Used Fairfax Water SFH and MFH unit use rates and back calculated EMP.

Table A.2-35: Forecast data for unit use calculations and results – Washington Aqueduct – Wholesale – Fairfax Water (Fairfax Water – Retail - Falls Church).

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Purchased (Fairfax Water Reported, MGD)			12.70	12.82	13.53	14.17	14.72	15.31	15.87	16.48	17.10	20.28
Demand (MGD)	Retail Sales	Single Family Household	4.63	4.32	4.02	3.90	3.87	3.87	3.77	3.75	3.71	3.53
		Multi-Family Household	4.55	4.93	5.67	6.23	6.75	7.26	7.86	8.41	8.98	11.86
		Employee	2.25	2.29	2.49	2.62	2.63	2.65	2.65	2.68	2.70	2.86
	Unmetered/Non-Revenue		1.27	1.28	1.35	1.42	1.47	1.53	1.59	1.65	1.71	2.03
	Percent Unmetered/Non-Revenue		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Total		12.70	12.82	13.53	14.17	14.72	15.31	15.87	16.48	17.10	20.28
Demographics	Population (units)		152,766	154,892	163,849	172,160	180,182	188,037	195,447	203,154	210,787	248,950
	Employees (units)		143,687	147,666	165,478	177,374	181,097	184,183	186,231	188,971	191,538	204,374
	Occupied Households (units)		59,955	61,417	66,093	70,245	74,263	78,272	82,183	86,160	90,119	109,919
	DUR (dimensionless)		0.98	0.84	0.68	0.60	0.55	0.51	0.46	0.43	0.40	0.29
	Single Family Households (units)		29,613	28,038	26,752	26,342	26,351	26,436	25,893	25,769	25,540	24,394
	Multi-Family Households (units)		30,342	33,379	39,341	43,903	47,912	51,836	56,290	60,391	64,580	85,525
Unit Use (gpd/unit)	Single Family Household		(156.21)	(153.96)	(150.20)	(148.10)	(146.92)	(146.27)	(145.74)	(145.41)	(145.20)	(144.85)
	Multi-Family Household		(150.08)	(147.84)	(144.07)	(141.97)	(140.79)	(140.14)	(139.61)	(139.28)	(139.07)	(138.72)
	Employee		15.68	15.48	15.07	14.77	14.55	14.38	14.25	14.16	14.08	13.98

Note: Used Fairfax Water unit use rates for SFH and MFH and then used EMP water savings assumptions for EMP forecast years. Percent Unmetered/Non-Revenue is assumed.

A.2.19 WSSC Water – Retail

WSSC Water finished drinking water is distributed to their retail customers in Prince George's and Montgomery counties.

Service Area: No changes were made to the service area for this study. The WSSC Water service area was last updated in July 2020 with minor additions in Montgomery and Prince George's counties (P. Flores). A pipeline in southern Prince George's is excluded, as it serves only as redundancy. The service area is defined by pipelines and pressure zones, excluding Rockville and Bowie, which operate independently. Only Montgomery and Prince George's TAZs are included in the analysis; other jurisdictions make up about 3 percent and are excluded.

Billing Data: WSSC Water provided DAC data by K. Six (March, 2024). DAC is a rolling average based on metered use, and annual use is estimated from monthly DAC averages.

Wholesale DAC data was also provided for DC Water, Rockville, Bowie, Howard, and Charles counties. The City of Bowie lacks DAC data.

WSSC Water's FY2017 Water Loss Reduction Plan reports water losses ranging from 15.7 to 20.9 percent since 2010, with FY2017 losses at 15.7 percent. The average unmetered use rate from 2010–2018 is 17.8 percent.

Table A.2-36: Historical data for unit use calculations and results – WSSC Water – Retail.

			Historical Data									
			2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Produced (MGD)			161.78	164.89	164.68	162.80	163.05	162.80	161.13	162.80	159.65	159.04
Billed (MGD)			134.66	138.64	142.77	131.09	130.75	126.05	124.15	130.47	131.20	132.18
Demand (MGD)	Retail Sales	Single Family Household	71.73	70.91	76.92	64.99	64.36	60.20	64.25	64.12	61.60	62.10
		Multi-Family Household	27.7	29.17	28.58	28.60	29.20	31.07	31.33	32.05	34.49	33.30
		Employee	32.17	35.68	34.34	33.07	32.82	33.21	25.67	31.08	31.91	32.01
	Wholesales		3.06	2.88	2.93	4.43	4.37	1.57	2.90	3.22	3.20	4.77
	Unmetered/Non-Revenue		27.12	26.25	21.91	31.71	32.30	36.75	36.98	32.33	28.45	26.86
	Percent Unmetered/Non-Revenue		17%	16%	13%	19%	20%	23%	23%	20%	18%	17%
	Total		161.78	164.89	164.68	162.80	163.05	162.80	161.13	162.80	159.65	159.04
	Demographics	Population (units)		1,767,102	1,782,467	1,792,564	1,802,662	1,812,760	1,851,286	1,889,812	1,899,514	1,909,216
Employees (units)		761,732	762,240	768,647	775,053	781,459	764,377	747,294	755,349	763,404	771,459	
Occupied Households (units)		640,177	645,931	651,334	656,737	662,140	670,053	677,967	682,284	686,601	690,919	
Montgomery County Households (units)		338,461	341,072	344,002	346,932	349,862	350,814	351,766	353,860	355,954	358,049	
Prince George’s County Households (units)		301,716	304,859	307,332	309,805	312,278	319,240	326,201	328,424	330,647	332,870	
MoCo DUR (dimensionless)		1.99	2.01	1.98	1.94	1.91	1.91	1.90	1.88	1.87	1.85	
PGCo DUR (dimensionless)		2.47	2.45	2.45	2.44	2.43	2.50	2.57	2.57	2.56	2.56	
Single Family Households (units)		439,968	444,367	446,650	448,880	451,056	458,160	465,269	467,487	469,689	471,874	
Multi-Family Households (units)		200,209	201,565	204,684	207,857	211,084	211,894	212,698	214,797	216,913	219,045	
Unit Use (gpd/unit)	Single Family Household		163.03	159.58	172.22	144.78	142.69	131.40	138.09	137.16	131.15	131.60
	Multi-Family Household		138.36	144.72	139.63	137.59	138.33	146.63	147.30	149.21	159.00	152.02
	Employee		42.23	46.81	44.68	42.67	42.00	43.45	34.35	41.15	41.80	41.49

Table A.2-37: Forecast data for unit use calculations and results – WSSC Water – Retail.

			Forecast Data									
			2023	2025	2030	2035	2040	2045	2050	2055	2060	2085
Produced (MGD)			161.02	163.71	166.58	171.62	176.48	181.35	185.86	190.94	196.11	223.38
Billed (MGD)			128.82	130.97	133.26	137.29	141.18	145.08	148.69	152.76	156.89	178.70
Demand (MGD)	Retail Sales	Single Family Household	59.46	58.93	58.10	58.36	58.63	59.20	59.54	60.02	60.42	60.90
		Multi-Family Household	33.73	33.90	36.15	38.84	41.40	43.48	45.48	47.78	50.21	64.86
		Employee	30.86	31.34	32.22	33.29	34.35	35.61	36.87	38.16	39.46	46.14
	Wholesales		4.77	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80
	Unmetered/Non-Revenue		32.20	32.74	33.32	34.32	35.30	36.27	37.17	38.19	39.22	44.68
	Percent Unmetered/Non-Revenue		20%	20%	20%	20%	20%	20%	20%	20%	20%	20%
	Total		161.02	163.71	166.58	171.62	176.48	181.35	185.86	190.94	196.11	223.38
	Demographics	Population (units)		1,918,917	1,938,321	2,002,143	2,079,114	2,147,615	2,209,712	2,265,212	2,325,110	2,383,909
Employees (units)		771,459	787,569	817,867	851,674	883,900	920,097	955,729	991,738	1,027,652	1,207,225	
Occupied Households (units)		690,919	699,553	728,462	761,346	788,477	811,848	832,253	854,635	876,523	985,963	
Montgomery County Households (units)		358,049	362,237	378,119	393,515	407,094	418,245	427,319	437,778	447,890	498,453	
Prince George’s County Households (units)		332,870	337,316	350,343	367,831	381,383	393,603	404,934	416,858	428,633	487,511	
MoCo DUR (dimensionless)		1.85	1.82	1.68	1.56	1.45	1.37	1.32	1.25	1.18	0.84	
PGCo DUR (dimensionless)		2.56	2.55	2.40	2.27	2.17	2.13	2.05	2.00	1.94	1.65	
Single Family Households (units)		471,874	476,191	484,216	495,102	502,366	510,051	515,401	520,968	525,395	531,263	
Multi-Family Households (units)		219,045	223,362	244,246	266,244	286,111	301,797	316,852	333,667	351,128	454,700	
Unit Use <i>(cond./unit)</i>	Single Family Household		126.00	123.75	119.99	117.88	116.70	116.06	115.53	115.20	114.99	114.64
	Multi-Family Household		154.00	151.75	147.99	145.88	144.70	144.06	143.53	143.20	142.99	142.64
	Employee		40.00	39.80	39.39	39.09	38.87	38.70	38.57	38.48	38.40	38.22

A.3 Projected Consumptive Water Use by Data Centers in the Potomac River Basin

Figure A.3-1 and Figure A.3-2 show PJM Energy Demand Forecasts (through 2045) for Dominion and Allegheny Power Systems, which served as the basis for estimating projected consumptive water use by data centers in the Potomac River basin.

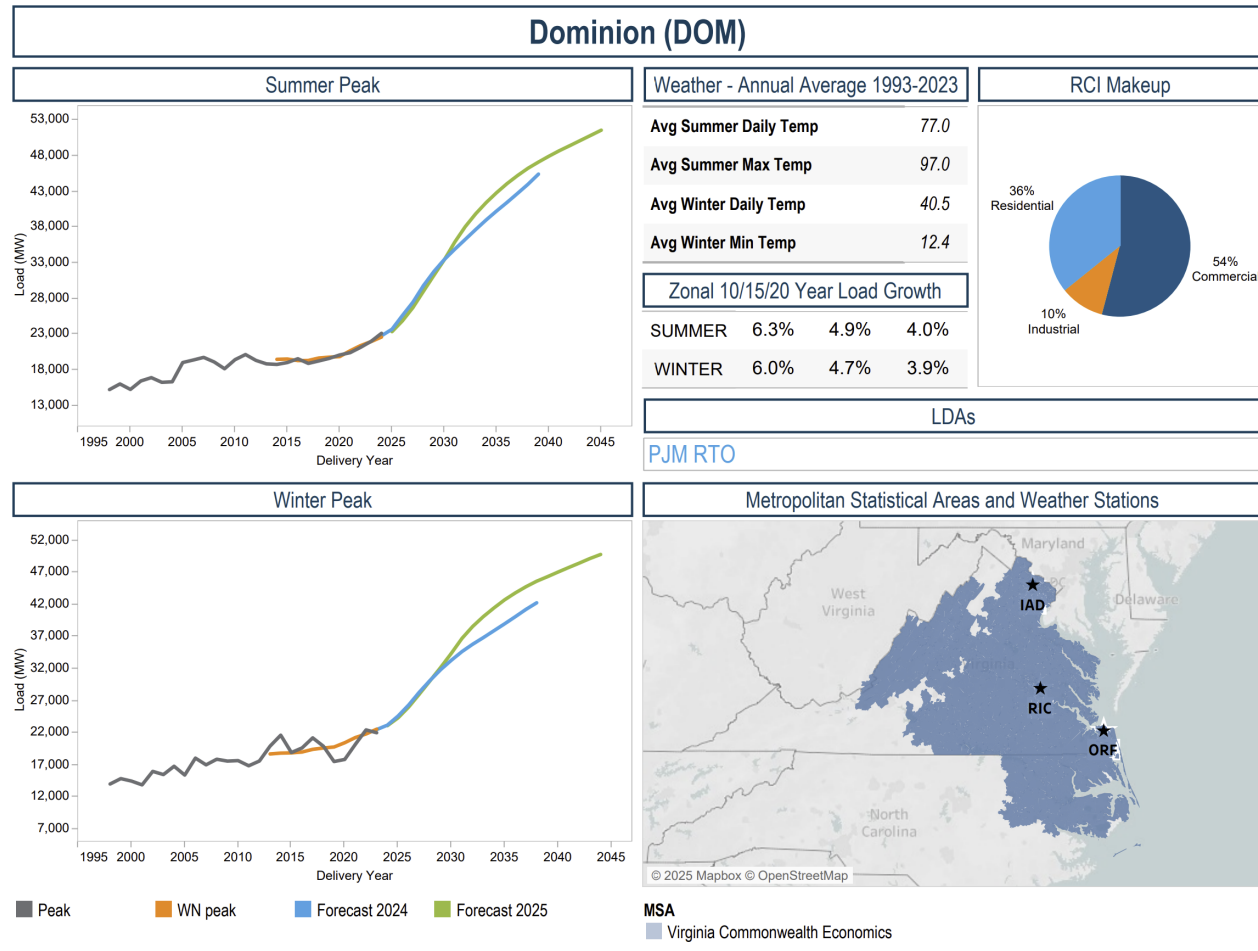


Figure A.3-1: PJM Energy Demand Forecasts (Through 2045) for Dominion Power Systems.

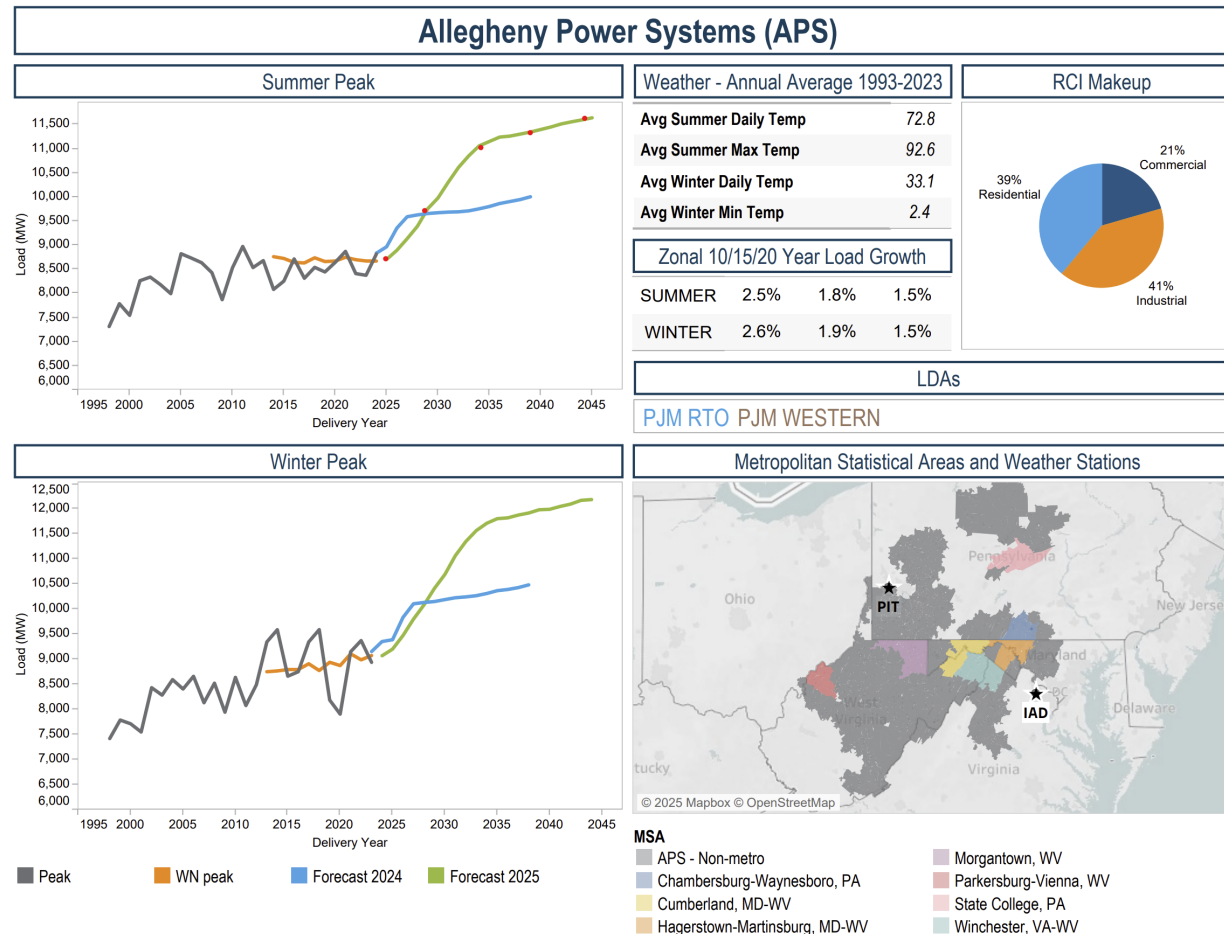


Figure A.3-2: PJM Energy Demand Forecasts (Through 2045) for Allegheny Power Systems.

Projected upstream consumptive water use by data centers in the Potomac River basin increases modestly through 2050, with values varying under different assumptions for cooling technologies (Table A.3-1).

Table A.3-1: Forecasted Upstream Consumptive Use by Data Centers (MGD), 2025–2050, Under Low, Medium, and High Scenarios

Year	Low	Medium	High
2025	0.0	0.1	0.1
2030	1.0	1.3	2.3
2035	2.3	3.1	5.4
2040	2.8	3.7	6.5
2045	3.1	4.1	7.2
2050	3.5	4.7	8.1

Seasonal variation in data center water use is determined by applying monthly factors to the annual averages, based on observed usage patterns in Loudoun and Prince William Counties (Table A.3-2).

Table A.3-2: Monthly Factors for Seasonal Variation in Data Center Water Use.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Factor	0.7	0.6	0.6	0.7	0.9	1.0	1.5	1.8	1.5	1.0	0.9	0.8

A.4 PRRISM Simulation Performance Metrics

Table A.4-1: 2030 Historic Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	22	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	2.59%	13.83%	3.27%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.56	1.92	1.56
Jennings Randolph water supply account	7.50	7.77	7.54
Jennings Randolph water quality account	3.04	4.72	3.63
Patuxent Reservoir	0.00	2.07	1.79
Occoquan Reservoir	2.58	2.58	3.04
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	0.98	0.98	1.06
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	9.09	9.69	9.10
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	14.65	14.66	15.63
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	15.77	15.78	16.69
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	439	443	442
WMA July demand, no restrictions, with climate change impact, MGD	515	539	529
WMA annual demand, no restrictions, without climate change impact, MGD	437	--	--
WMA July demand, no restrictions, without climate change impact, MGD	503	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	349	171	166
Minimum average fall flow downstream of intakes, MGD	280	158	171
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	67 (84.18%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	79 (98.75%)	1 (100%)	1 (100%)

Table A.4-2: 2030 Historic Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	31.82%	8.98%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.23	1.37	1.25
Jennings Randolph water supply account	5.57	5.62	5.77
Jennings Randolph water quality account	3.04	4.79	3.72
Patuxent Reservoir	0.00	1.93	1.67
Occoquan Reservoir	2.19	2.20	2.48
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	0.77	0.77	1.05
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	6.84	6.99	7.02
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	11.41	11.41	12.87
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	12.53	12.53	13.92
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	481	482	483
WMA July demand, no restrictions, with climate change impact, MGD	564	589	578
WMA annual demand, no restrictions, without climate change impact, MGD	480	--	--
WMA July demand, no restrictions, without climate change impact, MGD	551	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	332	154	150
Minimum average fall flow downstream of intakes, MGD	258	145	173
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	65 (81.09%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	79 (98.75%)	1 (100%)	1 (100%)

Table A.4-3: 2030 Historic Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.84%	36.39%	19.92%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.57	0.57	0.95
Jennings Randolph water supply account	2.90	2.90	3.93
Jennings Randolph water quality account	3.04	4.92	3.70
Patuxent Reservoir	0.00	1.95	1.67
Occoquan Reservoir	0.85	0.85	2.24
Savage Reservoir	0.63	0.63	0.66
Milestone Reservoir	0.88	0.88	1.02
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	3.48	3.48	4.88
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	6.56	6.56	10.50
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	7.68	7.68	11.54
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	523	521	523
WMA July demand, no restrictions, with climate change impact, MGD	612	634	627
WMA annual demand, no restrictions, without climate change impact, MGD	522	--	--
WMA July demand, no restrictions, without climate change impact, MGD	600	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	310	142	149
Minimum average fall flow downstream of intakes, MGD	247	140	174
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78.5 (98.13%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	65 (81.25%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	79 (98.75%)	1 (100%)	1 (100%)

Table A.4-4: 2045 Historic Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	31.57%	8.87%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.27	1.47	1.27
Jennings Randolph water supply account	5.46	5.54	5.61
Jennings Randolph water quality account	2.67	4.42	3.37
Patuxent Reservoir	0.00	1.70	1.61
Occoquan Reservoir	2.03	2.73	2.94
Savage Reservoir	0.62	0.65	0.65
Milestone Reservoir	0.94	0.94	1.08
Edgemon Quarry	0.16	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	6.78	7.01	6.88
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	11.83	11.84	13.12
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	13.15	13.15	15.17
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	476	476	477
WMA July demand, no restrictions, with climate change impact, MGD	557	582	571
WMA annual demand, no restrictions, without climate change impact, MGD	474	--	--
WMA July demand, no restrictions, without climate change impact, MGD	545	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	333	153	149
Minimum average fall flow downstream of intakes, MGD	257	145	174
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-5: 2045 Historic Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	4.25%	37.07%	23.68%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.54	0.55	0.82
Jennings Randolph water supply account	2.38	2.40	3.18
Jennings Randolph water quality account	2.67	4.55	3.35
Patuxent Reservoir	0.00	1.75	1.60
Occoquan Reservoir	1.55	1.62	2.51
Savage Reservoir	0.62	0.62	0.65
Milestone Reservoir	0.91	0.91	1.04
Edgemon Quarry	0.16	0.19	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	2.93	2.94	4.00
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	6.73	6.73	9.82
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	8.05	8.05	11.84
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	526	524	525
WMA July demand, no restrictions, with climate change impact, MGD	616	640	630
WMA annual demand, no restrictions, without climate change impact, MGD	525	--	--
WMA July demand, no restrictions, without climate change impact, MGD	603	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	314	142	152
Minimum average fall flow downstream of intakes, MGD	242	139	171
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78.54 (98.18%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-6: 2045 Historic Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.73%	99.49%	100.00%
Maximum number of days in a row of Potomac deficits	2	2	0
Number of days in which Potomac deficits must be allocated	2	2	0
Maximum amount of deficit allocated in a single day, MGD	-40	-40	0
Average amount of deficit allocated, MGD	-27	-27	0
Total amount of deficit allocated full simulation period, MG	-49	-49	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	1	0	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	7.50%	17.76%	36.36%
Emergency restrictions	2.76%	21.35%	0.12%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.30
Jennings Randolph water supply account	0.00	0.00	0.82
Jennings Randolph water quality account	2.60	4.54	3.26
Patuxent Reservoir	0.00	1.58	2.02
Occoquan Reservoir	0.50	0.50	2.59
Savage Reservoir	0.41	0.41	0.65
Milestone Reservoir	0.76	0.76	1.02
Edgemon Quarry	0.14	0.19	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	1.11
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	2.46	2.46	7.39
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	3.77	3.77	9.03
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	577	571	573
WMA July demand, no restrictions, with climate change impact, MGD	674	697	689
WMA annual demand, no restrictions, without climate change impact, MGD	576	--	--
WMA July demand, no restrictions, without climate change impact, MGD	662	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	290	140	147
Minimum average fall flow downstream of intakes, MGD	221	60	172
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77.96 (97.45%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-7: 2050 Historic Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	33.26%	10.14%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.17	1.34	1.18
Jennings Randolph water supply account	4.96	5.02	5.17
Jennings Randolph water quality account	2.54	4.30	3.22
Patuxent Reservoir	0.00	1.69	1.56
Occoquan Reservoir	2.08	2.68	2.89
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	0.97	0.97	1.06
Edgemon Quarry	0.15	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	6.20	6.36	6.35
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	11.22	11.22	12.50
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	12.53	12.53	14.48
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	482	482	483
WMA July demand, no restrictions, with climate change impact, MGD	564	588	580
WMA annual demand, no restrictions, without climate change impact, MGD	480	--	--
WMA July demand, no restrictions, without climate change impact, MGD	552	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	328	151	150
Minimum average fall flow downstream of intakes, MGD	254	146	172
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-8: 2050 Historic Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	6.19%	36.16%	28.92%
Emergency restrictions	0.36%	1.78%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.10	0.10	0.62
Jennings Randolph water supply account	1.09	1.09	2.33
Jennings Randolph water quality account	2.54	4.42	3.26
Patuxent Reservoir	0.00	1.72	1.55
Occoquan Reservoir	1.31	1.32	2.76
Savage Reservoir	0.55	0.55	0.65
Milestone Reservoir	0.89	0.89	1.04
Edgemon Quarry	0.16	0.17	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	1.19	1.19	2.95
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	4.60	4.60	8.98
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	5.92	5.92	10.93
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	539	537	537
WMA July demand, no restrictions, with climate change impact, MGD	631	655	646
WMA annual demand, no restrictions, without climate change impact, MGD	538	--	--
WMA July demand, no restrictions, without climate change impact, MGD	618	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	303	141	150
Minimum average fall flow downstream of intakes, MGD	240	138	170
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-9: 2050 Historic Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.49%	99.22%	100.00%
Maximum number of days in a row of Potomac deficits	2	2	0
Number of days in which Potomac deficits must be allocated	3	3	0
Maximum amount of deficit allocated in a single day, MGD	-48	-48	0
Average amount of deficit allocated, MGD	-28	-29	0
Total amount of deficit allocated full simulation period, MG	-85	-82	0
Number of Patuxent water supply shortfalls over simulation period	19	0	0
Number of Occoquan water supply shortfalls over simulation period	1	0	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	7.50%	14.92%	37.50%
Emergency restrictions	3.74%	24.86%	2.75%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.10
Jennings Randolph water supply account	0.00	0.00	0.05
Jennings Randolph water quality account	2.55	4.42	3.21
Patuxent Reservoir	0.00	0.86	2.22
Occoquan Reservoir	0.26	0.26	2.48
Savage Reservoir	0.30	0.30	0.65
Milestone Reservoir	0.78	0.78	1.02
Edgemon Quarry	0.15	0.18	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.15
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.48	1.48	6.48
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	2.80	2.80	8.08
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	593	587	587
WMA July demand, no restrictions, with climate change impact, MGD	692	717	709
WMA annual demand, no restrictions, without climate change impact, MGD	592	--	--
WMA July demand, no restrictions, without climate change impact, MGD	680	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	285	140	147
Minimum average fall flow downstream of intakes, MGD	204	52	170
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77 (96.83%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-10: 2030 High Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	9	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	0.77%	0.00%	0.43%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.60	2.20	1.60
Jennings Randolph water supply account	8.25	8.96	8.25
Jennings Randolph water quality account	3.05	4.78	3.68
Patuxent Reservoir	0.13	2.09	1.92
Occoquan Reservoir	2.65	2.65	3.15
Savage Reservoir	0.62	0.65	0.66
Milestone Reservoir	1.02	1.02	1.06
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	9.85	11.16	9.85
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	16.21	16.23	16.83
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	17.33	17.35	17.89
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	442	447	446
WMA July demand, no restrictions, with climate change impact, MGD	524	551	540
WMA annual demand, no restrictions, without climate change impact, MGD	437	--	--
WMA July demand, no restrictions, without climate change impact, MGD	503	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	707	465	475
Minimum average natural flow fall, MGD	607	466	391
Minimum average late summer flow downstream of intakes, MGD	359	173	176
Minimum average fall flow downstream of intakes, MGD	294	173	167
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	67.86 (84.83%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	79 (98.75%)	1 (100%)	1 (100%)

Table A.4-11: 2030 Higher Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	0	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	26.40%	7.14%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.26	1.74	1.26
Jennings Randolph water supply account	6.48	6.65	6.52
Jennings Randolph water quality account	3.05	4.89	3.68
Patuxent Reservoir	0.08	2.03	1.76
Occoquan Reservoir	2.57	2.68	2.57
Savage Reservoir	0.62	0.65	0.66
Milestone Reservoir	1.00	1.00	1.06
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	7.77	8.39	7.78
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	13.42	13.43	14.09
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	14.54	14.55	15.15
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	484	486	486
WMA July demand, no restrictions, with climate change impact, MGD	572	598	589
WMA annual demand, no restrictions, without climate change impact, MGD	480	--	--
WMA July demand, no restrictions, without climate change impact, MGD	551	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	707	465	475
Minimum average natural flow fall, MGD	607	466	391
Minimum average late summer flow downstream of intakes, MGD	341	164	151
Minimum average fall flow downstream of intakes, MGD	270	149	169
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	65 (81.69%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	79 (98.75%)	1 (100%)	1 (100%)

Table A.4-12: 2030 Higher Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	0	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	34.90%	13.33%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.99	1.09	1.02
Jennings Randolph water supply account	4.31	4.35	4.84
Jennings Randolph water quality account	3.05	4.97	3.75
Patuxent Reservoir	0.10	2.08	1.76
Occoquan Reservoir	1.49	1.49	2.32
Savage Reservoir	0.62	0.65	0.65
Milestone Reservoir	0.97	0.97	1.04
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	5.35	5.44	5.86
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	9.32	9.32	11.94
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	10.44	10.44	12.99
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	526	525	526
WMA July demand, no restrictions, with climate change impact, MGD	621	645	635
WMA annual demand, no restrictions, without climate change impact, MGD	522	--	--
WMA July demand, no restrictions, without climate change impact, MGD	600	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	707	465	475
Minimum average natural flow fall, MGD	607	466	391
Minimum average late summer flow downstream of intakes, MGD	319	142	148
Minimum average fall flow downstream of intakes, MGD	255	146	171
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	65 (81.25%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	79 (98.75%)	1 (100%)	1 (100%)

Table A.4-13: 2030 Medium Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	47	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	31.74%	6.26%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.99	1.00	1.29
Jennings Randolph water supply account	5.31	5.31	6.97
Jennings Randolph water quality account	3.02	4.32	3.62
Patuxent Reservoir	0.00	1.67	1.67
Occoquan Reservoir	1.52	1.52	2.77
Savage Reservoir	0.62	0.63	0.65
Milestone Reservoir	0.68	0.68	1.04
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	6.31	6.31	8.26
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	9.72	9.72	13.90
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	10.84	10.84	14.96
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	442	442	444
WMA July demand, no restrictions, with climate change impact, MGD	524	546	538
WMA annual demand, no restrictions, without climate change impact, MGD	437	--	--
WMA July demand, no restrictions, without climate change impact, MGD	503	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	652	399	442
Minimum average natural flow fall, MGD	521	400	365
Minimum average late summer flow downstream of intakes, MGD	328	149	160
Minimum average fall flow downstream of intakes, MGD	248	150	170
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.35%)	0.75 (75%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-14: 2030 Medium Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	99.91%	99.98%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	-1	-1	0
Average amount of deficit allocated, MGD	-1	-1	0
Total amount of deficit allocated full simulation period, MG	-1	-1	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	36.78%	14.46%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.03	0.03	0.99
Jennings Randolph water supply account	2.13	2.13	5.01
Jennings Randolph water quality account	3.02	4.43	3.63
Patuxent Reservoir	0.00	1.49	1.60
Occoquan Reservoir	0.35	0.35	2.28
Savage Reservoir	0.59	0.59	0.65
Milestone Reservoir	0.57	0.57	1.03
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	2.18	2.18	6.00
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	4.14	4.14	11.06
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	5.20	5.20	12.11
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	484	483	485
WMA July demand, no restrictions, with climate change impact, MGD	573	598	589
WMA annual demand, no restrictions, without climate change impact, MGD	480	--	--
WMA July demand, no restrictions, without climate change impact, MGD	551	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	652	399	442
Minimum average natural flow fall, MGD	521	400	365
Minimum average late summer flow downstream of intakes, MGD	312	141	153
Minimum average fall flow downstream of intakes, MGD	238	137	172
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	63.12 (78.9%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-15: 2030 Medium Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.77%	99.45%	100.00%
Maximum number of days in a row of Potomac deficits	2	2	0
Number of days in which Potomac deficits must be allocated	2	2	0
Maximum amount of deficit allocated in a single day, MGD	-54	-54	0
Average amount of deficit allocated, MGD	-38	-38	0
Total amount of deficit allocated full simulation period, MG	-75	-75	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	38	38	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	5.27%	19.17%	28.30%
Emergency restrictions	2.50%	19.68%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.64
Jennings Randolph water supply account	0.01	0.01	2.89
Jennings Randolph water quality account	3.02	4.43	3.65
Patuxent Reservoir	0.00	1.33	1.60
Occoquan Reservoir	0.00	0.00	2.21
Savage Reservoir	0.49	0.49	0.65
Milestone Reservoir	0.46	0.46	1.02
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.01	0.01	3.54
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.55	1.55	8.53
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	2.45	2.45	9.56
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	526	522	525
WMA July demand, no restrictions, with climate change impact, MGD	621	644	638
WMA annual demand, no restrictions, without climate change impact, MGD	522	--	--
WMA July demand, no restrictions, without climate change impact, MGD	600	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	652	399	442
Minimum average natural flow fall, MGD	521	400	365
Minimum average late summer flow downstream of intakes, MGD	292	140	153
Minimum average fall flow downstream of intakes, MGD	218	46	172
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-16: 2030 Lower Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.77%	99.19%	100.00%
Maximum number of days in a row of Potomac deficits	3	3	0
Number of days in which Potomac deficits must be allocated	3	3	0
Maximum amount of deficit allocated in a single day, MGD	-57	-57	0
Average amount of deficit allocated, MGD	-41	-41	0
Total amount of deficit allocated full simulation period, MG	-119	-119	0
Number of Patuxent water supply shortfalls over simulation period	63	0	0
Number of Occoquan water supply shortfalls over simulation period	7	7	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	5.00%	19.16%	10.30%
Emergency restrictions	2.50%	19.37%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	1.08
Jennings Randolph water supply account	0.05	0.05	5.82
Jennings Randolph water quality account	3.04	3.81	3.59
Patuxent Reservoir	0.00	1.14	1.47
Occoquan Reservoir	0.03	0.03	2.49
Savage Reservoir	0.33	0.33	0.66
Milestone Reservoir	0.55	0.55	1.04
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.05	0.05	6.90
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.36	1.39	11.36
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	2.17	2.17	12.42
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	442	440	443
WMA July demand, no restrictions, with climate change impact, MGD	524	551	539
WMA annual demand, no restrictions, without climate change impact, MGD	437	--	--
WMA July demand, no restrictions, without climate change impact, MGD	503	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	543	332	416
Minimum average natural flow fall, MGD	433	333	344
Minimum average late summer flow downstream of intakes, MGD	299	140	156
Minimum average fall flow downstream of intakes, MGD	209	43	176
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	64 (80.44%)	0.01 (1%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	75 (93.75%)	0 (0%)	1 (100%)

Table A.4-17: 2030 Lower Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.77%	93.49%	100.00%
Maximum number of days in a row of Potomac deficits	18	18	0
Number of days in which Potomac deficits must be allocated	24	24	0
Maximum amount of deficit allocated in a single day, MGD	-102	-102	0
Average amount of deficit allocated, MGD	-45	-45	0
Total amount of deficit allocated full simulation period, MG	-1076	-1076	0
Number of Patuxent water supply shortfalls over simulation period	56	0	0
Number of Occoquan water supply shortfalls over simulation period	55	55	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	5.05%	15.06%	26.96%
Emergency restrictions	2.50%	25.10%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.73
Jennings Randolph water supply account	0.00	0.00	3.47
Jennings Randolph water quality account	3.04	3.81	3.61
Patuxent Reservoir	0.00	0.43	1.39
Occoquan Reservoir	0.00	0.00	2.00
Savage Reservoir	0.17	0.17	0.65
Milestone Reservoir	0.42	0.42	1.02
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	4.20
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.56	0.56	8.07
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.16	1.16	9.12
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	484	480	483
WMA July demand, no restrictions, with climate change impact, MGD	572	596	586
WMA annual demand, no restrictions, without climate change impact, MGD	480	--	--
WMA July demand, no restrictions, without climate change impact, MGD	551	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	543	332	416
Minimum average natural flow fall, MGD	433	333	344
Minimum average late summer flow downstream of intakes, MGD	288	139	157
Minimum average fall flow downstream of intakes, MGD	164	-2	175
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	62 (77.6%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	75 (93.75%)	0 (0%)	1 (100%)

Table A.4-18: 2030 Lower Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.60%	86.22%	100.00%
Maximum number of days in a row of Potomac deficits	35	35	0
Number of days in which Potomac deficits must be allocated	50	50	0
Maximum amount of deficit allocated in a single day, MGD	-142	-142	0
Average amount of deficit allocated, MGD	-55	-55	0
Total amount of deficit allocated full simulation period, MG	-2789	-2788	0
Number of Patuxent water supply shortfalls over simulation period	95	5	0
Number of Occoquan water supply shortfalls over simulation period	69	69	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	6.96%	12.24%	37.93%
Emergency restrictions	2.50%	29.13%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.33
Jennings Randolph water supply account	0.00	0.00	1.59
Jennings Randolph water quality account	3.04	3.82	3.52
Patuxent Reservoir	0.00	0.00	1.39
Occoquan Reservoir	0.00	0.00	2.17
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.29	0.29	1.01
Edgemon Quarry	0.00	0.00	0.00
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	1.93
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.10	0.10	5.96
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.52	0.52	6.99
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	526	520	522
WMA July demand, no restrictions, with climate change impact, MGD	621	647	637
WMA annual demand, no restrictions, without climate change impact, MGD	522	--	--
WMA July demand, no restrictions, without climate change impact, MGD	600	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	543	332	416
Minimum average natural flow fall, MGD	433	333	344
Minimum average late summer flow downstream of intakes, MGD	274	139	154
Minimum average fall flow downstream of intakes, MGD	116	-42	174
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77.26 (96.58%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	60.82 (76.03%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	75 (93.75%)	0 (0%)	1 (100%)

Table A.4-19: 2045 Higher Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	0	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	1.64%	4.07%	4.12%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.46	2.05	1.46
Jennings Randolph water supply account	7.10	7.83	7.10
Jennings Randolph water quality account	2.71	4.72	3.36
Patuxent Reservoir	0.44	1.97	1.92
Occoquan Reservoir	2.17	2.86	3.10
Savage Reservoir	0.63	0.65	0.66
Milestone Reservoir	1.06	1.06	1.08
Edgemon Quarry	0.18	0.19	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	8.56	9.88	8.56
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	15.12	15.12	15.97
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	16.44	16.44	18.07
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	481	486	483
WMA July demand, no restrictions, with climate change impact, MGD	572	597	586
WMA annual demand, no restrictions, without climate change impact, MGD	474	--	--
WMA July demand, no restrictions, without climate change impact, MGD	545	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	745	495	499
Minimum average natural flow fall, MGD	646	496	411
Minimum average late summer flow downstream of intakes, MGD	365	170	173
Minimum average fall flow downstream of intakes, MGD	299	164	168
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	70 (87.1%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	80 (100%)	1 (100%)	1 (100%)

Table A.4-20: 2045 Higher Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	0	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	33.41%	10.86%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.03	1.59	1.03
Jennings Randolph water supply account	4.88	5.44	4.89
Jennings Randolph water quality account	2.71	4.78	3.37
Patuxent Reservoir	0.49	2.04	1.91
Occoquan Reservoir	1.80	2.80	2.60
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	1.04	1.04	1.06
Edgemon Quarry	0.12	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	5.92	7.03	5.92
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	12.24	12.27	12.89
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	13.59	13.59	14.97
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	532	531	533
WMA July demand, no restrictions, with climate change impact, MGD	631	655	648
WMA annual demand, no restrictions, without climate change impact, MGD	525	--	--
WMA July demand, no restrictions, without climate change impact, MGD	603	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	745	495	499
Minimum average natural flow fall, MGD	646	496	411
Minimum average late summer flow downstream of intakes, MGD	343	143	144
Minimum average fall flow downstream of intakes, MGD	270	145	168
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	69 (86.25%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	80 (100%)	1 (100%)	1 (100%)

Table A.4-21: 2045 Higher Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	0	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	4.92%	37.36%	27.53%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.50	0.72	0.51
Jennings Randolph water supply account	2.17	2.36	2.33
Jennings Randolph water quality account	2.71	4.78	3.44
Patuxent Reservoir	0.49	2.06	1.91
Occoquan Reservoir	1.39	1.84	2.78
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	1.00	1.00	1.06
Edgemon Quarry	0.10	0.19	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	2.71	3.08	2.85
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	7.49	7.49	10.09
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	8.81	8.81	12.14
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	583	581	581
WMA July demand, no restrictions, with climate change impact, MGD	690	714	705
WMA annual demand, no restrictions, without climate change impact, MGD	576	--	--
WMA July demand, no restrictions, without climate change impact, MGD	662	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	745	495	499
Minimum average natural flow fall, MGD	646	496	411
Minimum average late summer flow downstream of intakes, MGD	316	141	146
Minimum average fall flow downstream of intakes, MGD	249	141	165
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	69 (86.25%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	80 (100%)	1 (100%)	1 (100%)

Table A.4-22: 2045 Medium Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	99.81%	99.96%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	-1	-1	0
Average amount of deficit allocated, MGD	-1	-1	0
Total amount of deficit allocated full simulation period, MG	-1	-1	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	35.69%	15.84%
Emergency restrictions	0.29%	2.10%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.04	0.04	0.89
Jennings Randolph water supply account	1.19	1.19	4.60
Jennings Randolph water quality account	2.72	3.94	3.29
Patuxent Reservoir	0.00	1.38	1.57
Occoquan Reservoir	1.24	1.24	2.72
Savage Reservoir	0.43	0.43	0.65
Milestone Reservoir	0.82	0.82	1.05
Edgemon Quarry	0.16	0.19	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	1.23	1.23	5.49
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	4.09	4.09	11.05
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	5.41	5.41	13.08
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	481	480	481
WMA July demand, no restrictions, with climate change impact, MGD	572	598	587
WMA annual demand, no restrictions, without climate change impact, MGD	474	--	--
WMA July demand, no restrictions, without climate change impact, MGD	545	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	642	392	445
Minimum average natural flow fall, MGD	512	393	367
Minimum average late summer flow downstream of intakes, MGD	315	150	152
Minimum average fall flow downstream of intakes, MGD	235	128	173
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-23: 2045 Medium Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.77%	98.68%	100.00%
Maximum number of days in a row of Potomac deficits	2	2	0
Number of days in which Potomac deficits must be allocated	5	5	0
Maximum amount of deficit allocated in a single day, MGD	-62	-62	0
Average amount of deficit allocated, MGD	-31	-31	0
Total amount of deficit allocated full simulation period, MG	-137	-137	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	5.64%	15.46%	31.61%
Emergency restrictions	2.50%	24.42%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.41
Jennings Randolph water supply account	0.00	0.00	2.13
Jennings Randolph water quality account	2.72	4.07	3.27
Patuxent Reservoir	0.00	0.81	1.57
Occoquan Reservoir	0.40	0.40	2.46
Savage Reservoir	0.27	0.27	0.65
Milestone Reservoir	0.64	0.64	1.04
Edgemon Quarry	0.14	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	2.53
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.47	1.47	7.82
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	2.71	2.71	9.83
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	532	527	529
WMA July demand, no restrictions, with climate change impact, MGD	631	656	647
WMA annual demand, no restrictions, without climate change impact, MGD	525	--	--
WMA July demand, no restrictions, without climate change impact, MGD	603	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	642	392	445
Minimum average natural flow fall, MGD	512	393	367
Minimum average late summer flow downstream of intakes, MGD	299	139	151
Minimum average fall flow downstream of intakes, MGD	197	38	170
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-24: 2045 Medium Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.35%	92.60%	100.00%
Maximum number of days in a row of Potomac deficits	14	14	0
Number of days in which Potomac deficits must be allocated	27	27	0
Maximum amount of deficit allocated in a single day, MGD	-148	-148	0
Average amount of deficit allocated, MGD	-40	-40	0
Total amount of deficit allocated full simulation period, MG	-1089	-1084	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	23	22	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	8.73%	12.11%	37.93%
Emergency restrictions	3.75%	29.11%	3.07%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.02
Jennings Randolph water supply account	0.00	0.00	0.02
Jennings Randolph water quality account	2.72	4.08	3.22
Patuxent Reservoir	0.00	0.15	2.13
Occoquan Reservoir	0.00	0.00	2.44
Savage Reservoir	0.06	0.06	0.65
Milestone Reservoir	0.52	0.52	1.02
Edgemon Quarry	0.12	0.18	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.04
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.33	0.33	5.82
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.38	1.38	7.44
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	582	575	577
WMA July demand, no restrictions, with climate change impact, MGD	689	713	706
WMA annual demand, no restrictions, without climate change impact, MGD	576	--	--
WMA July demand, no restrictions, without climate change impact, MGD	662	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	642	392	445
Minimum average natural flow fall, MGD	512	393	367
Minimum average late summer flow downstream of intakes, MGD	276	140	149
Minimum average fall flow downstream of intakes, MGD	149	-48	166
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77.11 (96.39%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-25: 2045 Lower Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.75%	83.42%	100.00%
Maximum number of days in a row of Potomac deficits	37	37	0
Number of days in which Potomac deficits must be allocated	61	61	0
Maximum amount of deficit allocated in a single day, MGD	-146	-146	0
Average amount of deficit allocated, MGD	-58	-58	0
Total amount of deficit allocated full simulation period, MG	-3502	-3502	0
Number of Patuxent water supply shortfalls over simulation period	133	3	0
Number of Occoquan water supply shortfalls over simulation period	6	6	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	6.39%	11.18%	36.55%
Emergency restrictions	2.50%	30.50%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.31
Jennings Randolph water supply account	0.00	0.00	1.90
Jennings Randolph water quality account	2.53	3.23	3.10
Patuxent Reservoir	0.00	0.02	1.09
Occoquan Reservoir	0.01	0.01	2.65
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.68	0.68	1.05
Edgemon Quarry	0.15	0.17	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	2.21
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.10	0.10	5.95
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.13	1.13	7.93
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	481	474	479
WMA July demand, no restrictions, with climate change impact, MGD	572	599	587
WMA annual demand, no restrictions, without climate change impact, MGD	474	--	--
WMA July demand, no restrictions, without climate change impact, MGD	545	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	490	300	396
Minimum average natural flow fall, MGD	392	301	327
Minimum average late summer flow downstream of intakes, MGD	266	140	159
Minimum average fall flow downstream of intakes, MGD	97	-46	177
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77 (96.54%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	79 (98.75%)	0 (0%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	74 (92.5%)	0 (0%)	1 (100%)

Table A.4-26: 2045 Lower Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	97.63%	77.85%	100.00%
Maximum number of days in a row of Potomac deficits	60	60	0
Number of days in which Potomac deficits must be allocated	82	81	0
Maximum amount of deficit allocated in a single day, MGD	-211	-211	0
Average amount of deficit allocated, MGD	-94	-95	0
Total amount of deficit allocated full simulation period, MG	-7717	-7693	0
Number of Patuxent water supply shortfalls over simulation period	160	26	0
Number of Occoquan water supply shortfalls over simulation period	45	44	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	8.75%	8.39%	37.75%
Emergency restrictions	3.75%	34.51%	3.62%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.01
Jennings Randolph water supply account	0.00	0.00	0.00
Jennings Randolph water quality account	2.54	3.23	3.19
Patuxent Reservoir	0.00	0.00	1.69
Occoquan Reservoir	0.00	0.00	2.31
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.56	0.56	1.03
Edgemon Quarry	0.15	0.15	0.16
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.01
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.00	0.00	4.01
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.86	0.86	5.41
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	531	521	525
WMA July demand, no restrictions, with climate change impact, MGD	631	658	644
WMA annual demand, no restrictions, without climate change impact, MGD	525	--	--
WMA July demand, no restrictions, without climate change impact, MGD	603	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	490	300	396
Minimum average natural flow fall, MGD	392	301	327
Minimum average late summer flow downstream of intakes, MGD	257	140	156
Minimum average fall flow downstream of intakes, MGD	34	-111	157
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77 (96.25%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	79 (98.75%)	0 (0%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	74 (92.5%)	0 (0%)	1 (100%)

Table A.4-27: 2045 Lower Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	95.06%	74.06%	99.10%
Maximum number of days in a row of Potomac deficits	62	62	1
Number of days in which Potomac deficits must be allocated	103	95	3
Maximum amount of deficit allocated in a single day, MGD	-256	-256	-82
Average amount of deficit allocated, MGD	-127	-134	-45
Total amount of deficit allocated full simulation period, MG	-13061	-12668	-165
Number of Patuxent water supply shortfalls over simulation period	179	45	0
Number of Occoquan water supply shortfalls over simulation period	61	58	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	10.31%	7.38%	38.04%
Emergency restrictions	3.75%	36.41%	6.10%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.00
Jennings Randolph water supply account	0.00	0.00	0.00
Jennings Randolph water quality account	2.60	3.30	4.78
Patuxent Reservoir	0.00	0.00	1.63
Occoquan Reservoir	0.00	0.00	1.87
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.41	0.41	1.02
Edgemon Quarry	0.13	0.19	0.16
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.00
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.00	0.00	3.51
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.67	0.67	4.75
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	582	568	572
WMA July demand, no restrictions, with climate change impact, MGD	689	711	703
WMA annual demand, no restrictions, without climate change impact, MGD	576	--	--
WMA July demand, no restrictions, without climate change impact, MGD	662	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	490	300	396
Minimum average natural flow fall, MGD	392	301	327
Minimum average late summer flow downstream of intakes, MGD	241	72	93
Minimum average fall flow downstream of intakes, MGD	-28	-156	34
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	76 (95%)	1 (100%)	0 (0%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	79 (98.75%)	0 (0%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Savage Reservoir	74 (92.5%)	0 (0%)	1 (100%)

Table A.4-28: 2050 Higher Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	18	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	33.73%	10.94%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	1.15	1.29	1.16
Jennings Randolph water supply account	4.81	4.85	5.02
Jennings Randolph water quality account	2.54	4.30	3.18
Patuxent Reservoir	0.00	1.70	1.55
Occoquan Reservoir	2.04	2.61	2.88
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	0.97	0.97	1.06
Edgemon Quarry	0.15	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	6.01	6.13	6.18
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	10.96	10.96	12.32
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	12.28	12.28	14.29
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	486	486	486
WMA July demand, no restrictions, with climate change impact, MGD	569	593	583
WMA annual demand, no restrictions, without climate change impact, MGD	484	--	--
WMA July demand, no restrictions, without climate change impact, MGD	556	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	691	456	464
Minimum average natural flow fall, MGD	595	457	383
Minimum average late summer flow downstream of intakes, MGD	328	150	150
Minimum average fall flow downstream of intakes, MGD	252	146	173
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	78 (97.5%)	1 (100%)	1 (100%)

Table A.4-29: 2050 Higher Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	0	0	0
Maximum amount of deficit allocated in a single day, MGD	0	0	0
Average amount of deficit allocated, MGD	0	0	0
Total amount of deficit allocated full simulation period, MG	0	0	0
Number of Patuxent water supply shortfalls over simulation period	0	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	35.11%	16.17%
Emergency restrictions	0.00%	0.00%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.86	1.35	0.86
Jennings Randolph water supply account	3.91	4.49	3.94
Jennings Randolph water quality account	2.59	4.65	3.23
Patuxent Reservoir	0.48	2.03	1.89
Occoquan Reservoir	1.86	2.70	2.66
Savage Reservoir	0.63	0.65	0.65
Milestone Reservoir	1.05	1.05	1.07
Edgemon Quarry	0.16	0.18	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined	4.80	5.84	4.81
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	11.14	11.16	11.85
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	12.48	12.48	13.85
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	545	544	546
WMA July demand, no restrictions, with climate change impact, MGD	646	669	661
WMA annual demand, no restrictions, without climate change impact, MGD	538	--	--
WMA July demand, no restrictions, without climate change impact, MGD	618	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	745	495	499
Minimum average natural flow fall, MGD	646	496	411
Minimum average late summer flow downstream of intakes, MGD	336	141	144
Minimum average fall flow downstream of intakes, MGD	261	143	168
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	69 (86.25%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	80 (100%)	1 (100%)	1 (100%)

Table A.4-30: 2050 Higher Flows and High Demands.

Historical period for simulation of variability		1929-2009	1930	1966
Reliability, Vulnerability, Resiliency				
Percentage years with no Potomac deficits		100.00%	100.00%	100.00%
Maximum number of days in a row of Potomac deficits		0	0	0
Number of days in which Potomac deficits must be allocated		0	0	0
Maximum amount of deficit allocated in a single day, MGD		0	0	0
Average amount of deficit allocated, MGD		0	0	0
Total amount of deficit allocated full simulation period, MG		0	0	0
Number of Patuxent water supply shortfalls over simulation period		0	0	0
Number of Occoquan water supply shortfalls over simulation period		0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)				
Voluntary restrictions		5.85%	36.89%	32.94%
Emergency restrictions		0.16%	1.17%	0.00%
Minimum reservoir storage, BG				
Little Seneca Reservoir		0.21	0.25	0.29
Jennings Randolph water supply account		1.00	1.06	1.43
Jennings Randolph water quality account		2.59	4.78	3.31
Patuxent Reservoir		0.47	2.02	1.89
Occoquan Reservoir		1.24	1.48	2.74
Savage Reservoir		0.57	0.57	0.65
Milestone Reservoir		1.01	1.01	1.06
Edgemon Quarry		0.15	0.17	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined		1.22	1.31	1.72
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined		5.29	5.29	8.92
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined		6.61	6.61	10.90
Miscellaneous				
Number of years in simulation		80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD		598	595	595
WMA July demand, no restrictions, with climate change impact, MGD		708	729	723
WMA annual demand, no restrictions, without climate change impact, MGD		592	--	--
WMA July demand, no restrictions, without climate change impact, MGD		680	--	--
Minimum average flow				
Minimum average natural flow late summer, MGD		745	495	499
Minimum average natural flow fall, MGD		646	496	411
Minimum average late summer flow downstream of intakes, MGD		313	141	144
Minimum average fall flow downstream of intakes, MGD		247	140	168
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)				
Little Seneca Reservoir	78.68 (98.35%)	1 (100%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	69 (86.25%)	1 (100%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)	1 (100%)
Savage Reservoir	80 (100%)	1 (100%)	1 (100%)	1 (100%)

Table A.4-31: 2050 Medium Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	99.05%	99.77%	100.00%
Maximum number of days in a row of Potomac deficits	0	0	0
Number of days in which Potomac deficits must be allocated	1	1	0
Maximum amount of deficit allocated in a single day, MGD	-11	-11	0
Average amount of deficit allocated, MGD	-11	-11	0
Total amount of deficit allocated full simulation period, MG	-12	-12	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	3.75%	22.84%	20.68%
Emergency restrictions	2.10%	15.55%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.76
Jennings Randolph water supply account	0.33	0.33	3.82
Jennings Randolph water quality account	2.60	3.82	3.16
Patuxent Reservoir	0.00	1.32	1.51
Occoquan Reservoir	1.10	1.10	2.66
Savage Reservoir	0.35	0.35	0.65
Milestone Reservoir	0.87	0.87	1.06
Edgemon Quarry	0.16	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.33	0.33	4.58
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	3.01	3.01	10.02
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	4.33	4.33	11.98
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	491	488	491
WMA July demand, no restrictions, with climate change impact, MGD	584	608	600
WMA annual demand, no restrictions, without climate change impact, MGD	484	--	--
WMA July demand, no restrictions, without climate change impact, MGD	556	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	642	392	445
Minimum average natural flow fall, MGD	512	393	367
Minimum average late summer flow downstream of intakes, MGD	311	150	150
Minimum average fall flow downstream of intakes, MGD	232	91	176
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	78 (97.5%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-32: 2050 Medium Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.77%	97.19%	100.00%
Maximum number of days in a row of Potomac deficits	3	3	0
Number of days in which Potomac deficits must be allocated	10	10	0
Maximum amount of deficit allocated in a single day, MGD	-100	-100	0
Average amount of deficit allocated, MGD	-30	-30	0
Total amount of deficit allocated full simulation period, MG	-315	-315	0
Number of Patuxent water supply shortfalls over simulation period	38	0	0
Number of Occoquan water supply shortfalls over simulation period	0	0	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	6.80%	14.41%	35.36%
Emergency restrictions	2.51%	25.95%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.26
Jennings Randolph water supply account	0.00	0.00	1.40
Jennings Randolph water quality account	2.60	3.95	3.19
Patuxent Reservoir	0.00	0.51	1.52
Occoquan Reservoir	0.17	0.17	2.66
Savage Reservoir	0.19	0.19	0.65
Milestone Reservoir	0.70	0.70	1.04
Edgemon Quarry	0.15	0.19	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	1.66
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.91	0.91	7.11
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	2.12	2.12	9.04
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	545	539	541
WMA July demand, no restrictions, with climate change impact, MGD	646	668	660
WMA annual demand, no restrictions, without climate change impact, MGD	538	--	--
WMA July demand, no restrictions, without climate change impact, MGD	618	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	642	392	445
Minimum average natural flow fall, MGD	512	393	367
Minimum average late summer flow downstream of intakes, MGD	293	140	149
Minimum average fall flow downstream of intakes, MGD	183	0	173
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77.97 (97.46%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-33: 2050 Medium Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	97.63%	89.86%	99.99%
Maximum number of days in a row of Potomac deficits	18	18	0
Number of days in which Potomac deficits must be allocated	38	37	0
Maximum amount of deficit allocated in a single day, MGD	-159	-159	0
Average amount of deficit allocated, MGD	-44	-45	0
Total amount of deficit allocated full simulation period, MG	-1686	-1660	0
Number of Patuxent water supply shortfalls over simulation period	45	3	0
Number of Occoquan water supply shortfalls over simulation period	28	26	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	8.75%	10.45%	37.69%
Emergency restrictions	3.75%	31.13%	3.88%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.00
Jennings Randolph water supply account	0.00	0.00	0.00
Jennings Randolph water quality account	2.55	3.96	3.40
Patuxent Reservoir	0.00	0.00	2.37
Occoquan Reservoir	0.00	0.00	2.24
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.60	0.60	1.02
Edgemon Quarry	0.15	0.19	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.00
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.16	0.16	5.78
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.25	1.25	7.30
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	598	588	591
WMA July demand, no restrictions, with climate change impact, MGD	708	733	725
WMA annual demand, no restrictions, without climate change impact, MGD	592	--	--
WMA July demand, no restrictions, without climate change impact, MGD	680	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	642	392	445
Minimum average natural flow fall, MGD	512	393	367
Minimum average late summer flow downstream of intakes, MGD	272	139	149
Minimum average fall flow downstream of intakes, MGD	132	-59	146
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77 (96.25%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	80 (100%)	1 (100%)	1 (100%)
Patuxent Reservoir	66 (82.5%)	1 (100%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	77 (96.25%)	0 (0%)	1 (100%)

Table A.4-34: 2050 Lower Flows and Low Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	98.75%	81.68%	100.00%
Maximum number of days in a row of Potomac deficits	47	47	0
Number of days in which Potomac deficits must be allocated	67	67	0
Maximum amount of deficit allocated in a single day, MGD	-155	-155	0
Average amount of deficit allocated, MGD	-64	-64	0
Total amount of deficit allocated full simulation period, MG	-4289	-4289	0
Number of Patuxent water supply shortfalls over simulation period	142	9	0
Number of Occoquan water supply shortfalls over simulation period	13	12	0
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	7.09%	10.55%	39.10%
Emergency restrictions	2.51%	31.31%	0.00%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.24
Jennings Randolph water supply account	0.00	0.00	1.44
Jennings Randolph water quality account	2.41	3.10	2.94
Patuxent Reservoir	0.00	0.00	1.05
Occoquan Reservoir	0.00	0.00	2.58
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.76	0.76	1.05
Edgemon Quarry	0.15	0.18	0.18
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	1.68
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.07	0.07	5.30
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	1.20	1.20	7.13
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	487	479	484
WMA July demand, no restrictions, with climate change impact, MGD	579	605	595
WMA annual demand, no restrictions, without climate change impact, MGD	480	--	--
WMA July demand, no restrictions, without climate change impact, MGD	552	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	490	300	396
Minimum average natural flow fall, MGD	392	301	327
Minimum average late summer flow downstream of intakes, MGD	265	140	157
Minimum average fall flow downstream of intakes, MGD	84	-55	176
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	77.07 (96.34%)	1 (100%)	1 (100%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	79 (98.75%)	0 (0%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	74 (92.5%)	0 (0%)	1 (100%)

Table A.4-35: 2050 Lower Flows and Medium Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	97.26%	76.66%	100.00%
Maximum number of days in a row of Potomac deficits	61	61	0
Number of days in which Potomac deficits must be allocated	87	85	0
Maximum amount of deficit allocated in a single day, MGD	-228	-228	0
Average amount of deficit allocated, MGD	-108	-110	0
Total amount of deficit allocated full simulation period, MG	-9386	-9340	0
Number of Patuxent water supply shortfalls over simulation period	168	33	0
Number of Occoquan water supply shortfalls over simulation period	50	48	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	8.98%	8.01%	37.73%
Emergency restrictions	3.75%	35.12%	4.20%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.00
Jennings Randolph water supply account	0.00	0.00	0.00
Jennings Randolph water quality account	2.42	3.11	3.75
Patuxent Reservoir	0.00	0.00	1.85
Occoquan Reservoir	0.00	0.00	2.29
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.63	0.63	1.03
Edgemon Quarry	0.14	0.16	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.00
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.00	0.00	4.14
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.94	0.94	5.45
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	544	533	537
WMA July demand, no restrictions, with climate change impact, MGD	646	668	660
WMA annual demand, no restrictions, without climate change impact, MGD	538	--	--
WMA July demand, no restrictions, without climate change impact, MGD	618	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	490	300	396
Minimum average natural flow fall, MGD	392	301	327
Minimum average late summer flow downstream of intakes, MGD	253	139	155
Minimum average fall flow downstream of intakes, MGD	12	-128	143
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	76.98 (96.23%)	1 (100%)	0.98 (98%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	79 (98.75%)	0 (0%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	80 (100%)	1 (100%)	1 (100%)
Savage Reservoir	74 (92.5%)	0 (0%)	1 (100%)

Table A.4-36: 2050 Lower Flows and High Demands.

Historical period for simulation of variability	1929-2009	1930	1966
Reliability, Vulnerability, Resiliency			
Percentage years with no Potomac deficits	95.06%	72.50%	97.33%
Maximum number of days in a row of Potomac deficits	65	65	5
Number of days in which Potomac deficits must be allocated	117	100	10
Maximum amount of deficit allocated in a single day, MGD	-273	-273	-152
Average amount of deficit allocated, MGD	-136	-148	-80
Total amount of deficit allocated full simulation period, MG	-15976	-14883	-784
Number of Patuxent water supply shortfalls over simulation period	187	52	0
Number of Occoquan water supply shortfalls over simulation period	62	60	1
Percentage of years with restrictions (Percentage of days for 1930 and 1966)			
Voluntary restrictions	12.34%	6.95%	44.27%
Emergency restrictions	4.66%	37.19%	7.15%
Minimum reservoir storage, BG			
Little Seneca Reservoir	0.00	0.00	0.00
Jennings Randolph water supply account	0.00	0.00	0.00
Jennings Randolph water quality account	2.45	3.15	4.75
Patuxent Reservoir	0.00	0.00	1.41
Occoquan Reservoir	0.00	0.00	1.81
Savage Reservoir	0.00	0.00	0.65
Milestone Reservoir	0.49	0.49	1.02
Edgemon Quarry	0.15	0.19	0.17
Little Seneca Reservoir and Jennings Randolph water supply account, combined	0.00	0.00	0.00
Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.00	0.00	3.22
Milestone Reservoir, Edgemon, Patuxent, Occoquan, Little Seneca reservoirs and Jennings Randolph water supply, combined	0.76	0.76	4.47
Miscellaneous			
Number of years in simulation	80	1	1
WMA annual demand, no restrictions, with climate change impact, MGD	598	583	586
WMA July demand, no restrictions, with climate change impact, MGD	707	728	716
WMA annual demand, no restrictions, without climate change impact, MGD	592	--	--
WMA July demand, no restrictions, without climate change impact, MGD	680	--	--
Minimum average flow			
Minimum average natural flow late summer, MGD	490	300	396
Minimum average natural flow fall, MGD	392	301	327
Minimum average late summer flow downstream of intakes, MGD	228	-52	54
Minimum average fall flow downstream of intakes, MGD	-48	-173	-52
Number of years where reservoirs refill to 90 percent full by June 1 (percent probability)			
Little Seneca Reservoir	76 (95%)	1 (100%)	0 (0%)
Jennings Randolph water supply account	80 (100%)	1 (100%)	1 (100%)
Jennings Randolph water quality account	79 (98.75%)	0 (0%)	1 (100%)
Patuxent Reservoir	63 (78.75%)	0 (0%)	0 (0%)
Occoquan Reservoir	79 (98.75%)	1 (100%)	1 (100%)
Savage Reservoir	74 (92.5%)	0 (0%)	1 (100%)